HBA Lecture Notes in Mathematics
IMSc Lecture Notes in Mathematics

M. Ram Murty
Michael Dewar
Hester Graves

Problems in the Theory of Modular Forms

HINDUSTAN BOOK AGENCY
Springer

HBA Lecture Notes in Mathematics

IMSc Lecture Notes in Mathematics

Series Editor

Sanoli Gun, Institute of Mathematical Sciences, Chennai, Tamil Nadu, India

Editorial Board

R. Balasubramanian, Institute of Mathematical Sciences, Chennai
Abhay G. Bhatt, Indian Statistical Institute, New Delhi
Yuri F. Bilu, Université Bordeaux I, France
Partha Sarathi Chakraborty, Institute of Mathematical Sciences, Chennai
Carlo Gasbarri, University of Strasbourg, Germany
Anirban Mukhopadhyay, Institute of Mathematical Sciences, Chennai
V. Kumar Murty, University of Toronto, Toronto
D.S. Nagaraj, Institute of Mathematical Sciences, Chennai
Olivier Ramaré, Centre National de la Recherche Scientifique, France
Purusottam Rath, Chennai Mathematical Institute, Chennai
Parameswaran Sankaran, Institute of Mathematical Sciences, Chennai
Kannan Soundararajan, Stanford University, Stanford
V.S. Sunder, Institute of Mathematical Sciences, Chennai

About the Series

The *IMSc Lecture Notes in Mathematics* series is a subseries of the *HBA Lecture Notes in Mathematics* series. This subseries publishes high-quality lecture notes of the Institute of Mathematical Sciences, Chennai, India. Undergraduate and graduate students of mathematics, research scholars, and teachers would find this book series useful. The volumes are carefully written as teaching aids and highlight characteristic features of the theory. The books in this series are co-published with Hindustan Book Agency, New Delhi, India.

More information about this series at http://www.springer.com/series/15465

M. Ram Murty · Michael Dewar
Hester Graves

Problems in the Theory of Modular Forms

HINDUSTAN BOOK AGENCY

Springer

M. Ram Murty
Department of Mathematics and Statistics
Queen's University
Kingston, ON, Canada

Hester Graves
University of Michigan
Ann Arbor, MI, USA

Michael Dewar
Queen's University
Kingston, ON, Canada

This work is a co-publication with Hindustan Book Agency, New Delhi, licensed for sale in all countries in electronic form only. Sold and distributed in print across the world by Hindustan Book Agency, P-19 Green Park Extension, New Delhi 110016, India. ISBN: 978-93-80250-72-4 © Hindustan Book Agency 2016.

ISSN 2509-8071 (electronic)
HBA Lecture Notes in Mathematics
ISSN 2509-8098 (electronic)
IMSc Lecture Notes in Mathematics
ISBN 978-981-10-2651-5 (eBook)
DOI 10.1007/978-981-10-2651-5

Library of Congress Control Number: 2016952892

© Springer Science+Business Media Singapore 2016 and Hindustan Book Agency 2016
This work is subject to copyright. All rights in this online edition are reserved by the Publishers, whether the whole or part of the material is concerned, specifically the rights of reuse of illustrations, recitation, broadcasting, and transmission or information storage and retrieval, electronic adaptation, computer software, or by similar or dissimilar methodology now known or hereafter developed.
The use of general descriptive names, registered names, trademarks, service marks, etc. in this publication does not imply, even in the absence of a specific statement, that such names are exempt from the relevant protective laws and regulations and therefore free for general use.
The publishers, the authors and the editors are safe to assume that the advice and information in this book are believed to be true and accurate at the date of publication. Neither the publishers nor the authors or the editors give a warranty, express or implied, with respect to the material contained herein or for any errors or omissions that may have been made.

Printed on acid-free paper

This Springer imprint is published by Springer Nature
The registered company is Springer Nature Singapore Pte Ltd.
The registered company address is: 152 Beach Road, #22-06/08 Gateway East, Singapore 189721, Singapore

A great discovery solves a great problem but there is a grain of discovery in the solution of any problem.

- George Pólya, How to solve it.

Contents

Acknowledgements xi

Preface xiii

Part I Problems

1 Jacobi's *q*-series 3
1.1 The q-exponential function 3
1.2 Jacobi's Triple Product Identity 5
1.3 Jacobi's two-square theorem 7
1.4 Jacobi's four square theorem 9
1.5 Supplementary problems 14

2 The Modular Group 15
2.1 The full modular group 15
2.2 Subgroups of the modular group 16
2.3 The Hecke subgroups 18
2.4 Groups acting on topological spaces 19
2.5 Supplementary problems 22

3 The Upper Half-Plane 25
3.1 The group $\mathrm{GL}_n^+(\mathbb{R})$ and its subgroups 25
3.2 Fundamental domains 27
3.3 The extended upper half-plane 30
3.4 The Poincaré metric 32
3.5 Supplementary problems 33

4 Modular Forms of Level One 35
4.1 Fourier expansions and q-series 35
4.2 Eisenstein series for $\mathrm{SL}_2(\mathbb{Z})$ 37
4.3 The valence formula 41
4.4 The dimension formula 46

4.5 The j-function 48
4.6 Supplementary problems 50

5 The Ramanujan τ-function 53
5.1 The Eisenstein series E_2 53
5.2 Hecke operators of level one 60
5.3 Modular forms and Dirichlet series 64
5.4 Ramanujan congruences 67
5.5 Supplementary problems 69

6 Modular Forms of Higher Level 73
6.1 Modular forms for congruence subgroups 73
6.2 Eisenstein series for $\Gamma(N)$ 75
6.3 Fourier expansions of higher level Eisenstein series 76
6.4 The valence and dimension formulas 79
6.5 The Jacobi four-square theorem revisited 81
6.6 Supplementary problems 83

7 The Petersson Inner Product 85
7.1 The Hilbert space of cusp forms 85
7.2 Commutativity of the Hecke operators 87
7.3 Hecke operators as Hermitian operators 90
7.4 Basis of eigenforms 92
7.5 Supplementary problems 93

8 Hecke Operators of Higher Level 95
8.1 Hecke operators for congruence subgroups 95
8.2 Nebentypus 99
8.3 Oldforms and newforms 102
8.4 Supplementary problems 103

9 Dirichlet Series and Modular Forms 105
9.1 General Dirichlet series 105
9.2 The Poisson summation formula 112
9.3 L-functions attached to modular forms 123
9.4 Twists of L-series attached to modular forms 129
9.5 Supplementary problems 133

10 Special Topics 135
10.1 Elliptic functions 135
10.2 The modular invariant 140
10.3 Theta series and lattices 143
10.4 Special values of zeta and L-functions 147
10.5 Supplementary problems 149

Part II Solutions

1 Jacobi's q-series 153
1.1 The q-exponential function 153
1.2 Jacobi's Triple Product Identity 155
1.3 Jacobi's two-square theorem 156
1.4 Jacobi's four square theorem 156
1.5 Supplementary problems 158

2 The Modular Group 161
2.1 The full modular group 161
2.2 Subgroups of the modular group 162
2.3 The Hecke subgroups 167
2.4 Groups acting on topological spaces 168
2.5 Supplementary problems 170

3 The Upper Half-Plane 175
3.1 The group $\mathrm{GL}_n^+(\mathbb{R})$ and its subgroups 175
3.2 Fundamental domains 176
3.3 The extended upper-half plane 177
3.4 The Poincaré metric 180
3.5 Supplementary problems 182

4 Modular Forms of Level One 185
4.1 Fourier expansions and q-series 185
4.2 Eisenstein series for $\mathrm{SL}_2(\mathbb{Z})$ 187
4.3 The valence formula 189
4.4 The dimension formula 190
4.5 The j-function 191
4.6 Supplementary problems 192

5 The Ramanujan τ-function 199
5.1 The Eisenstein series E_2 199
5.2 Hecke operators of level one 203
5.3 Modular forms and Dirichlet series 205
5.4 Ramanujan congruences 207
5.5 Supplementary problems 209

6 Modular Forms of Higher Level 217
6.1 Modular forms for congruence subgroups 217
6.2 Eisenstein series for $\Gamma(N)$ 218
6.3 Fourier expansions of higher level Eisenstein series 220
6.4 The valence and dimension formulas 224
6.5 The Jacobi four-square theorem revisited 227
6.6 Supplementary problems 227

7 The Petersson Inner Product 231
7.1 The Hilbert space of cusp forms 231
7.2 Commutativity of the Hecke operators 234
7.3 Hecke operators as Hermitian operators 235
7.4 Basis of eigenforms 235
7.5 Supplementary problems 236

8 Hecke Operators of Higher Level 239
8.1 Hecke operators for congruence subgroups 239
8.2 Nebentypus 240
8.3 Oldforms and newforms 242
8.4 Supplementary problems 243

9 Dirichlet Series and Modular Forms 245
9.1 General Dirichlet series 245
9.2 The Poisson summation formula 251
9.3 L-functions attached to modular forms 260
9.4 Twists of L-series attached to modular forms 263
9.5 Supplementary problems 265

10 Special Topics 269
10.1 Elliptic functions 269
10.2 The modular invariant 273
10.3 Theta series and lattices 276
10.4 Special values of zeta and L-functions 279
10.5 Supplementary problems 281

A short guide for further reading 285
References 286

Index 289

Acknowledgements

This book is based on an upper level undergraduate course given at Queen's University in the fall semester of 2011. The class consisted of primarily undergraduates, several graduate students, and a few post-doctoral fellows. Since the topic of modular forms has become central not only in number theory, but also other parts of mathematics (the most spectacular being theoretical physics), I felt an urgency to deliver this beautiful theory to the sincere student as quickly and painlessly as possible. As the topic of modular forms can be approached from various directions, some balance had to be achieved where only the most minimal background is assumed. The great nineteenth century philosopher Vivekananda has said that "it is practice first, and knowledge afterwards." Somehow, knowledge is gained through doing, by applying our knowledge, though it may be incomplete and our understanding imperfect. From this perspective, I have always felt that the best way to learn anything is via a "hands on" approach where the student learns the nuances of the theory through practical problem solving.

My two earlier books "Problems in Analytic Number Theory" and "Problems in Algebraic Number Theory" published by Springer were written with this underlying philosophy. Judging from the feedback I have received from students and researchers around the world, this teaching philosophy is well-founded and I hope the same is true with this book.

I would like to thank my two post-doctoral fellows, Michael Dewar and Hester Graves, in assisting me in this endeavour. They not only transferred my hand-written notes into LaTeX, but added substantial sections to amplify and elucidate the material. I also thank V. Kumar Murty, Sanoli Gun, Purusottam Rath, Alia Hamieh, and Kannappan Sampath for their valuable comments on earlier versions of this book. I especially thank Kannappan for his help in the production of LaTeX diagrams included in this book and the referees for their helpful comments.

Ram Murty
Kingston, Ontario

Preface

The earliest murmurs of a theory of modular forms can be traced back to the work of Jacobi in 1829 when he wrote his famous treatise *Fundamenta Nova Theoriae Functionum Ellipticarum* dealing with q-series and elliptic functions. In some parenthetic sense, this was further developed by Riemann, Hurwitz, Dedekind, Eisenstein, and Kronecker. However, it is in the work of Ramanujan, in his celebrated paper [29] of 1916 in which he introduced the τ-function, where we find the seeds of a comprehensive theory. There, Ramanujan studied the infinite product (in the variable q) given by

$$q \prod_{n=1}^{\infty} (1-q^n)^{24} \tag{0.1}$$

which he expands as an infinite series and writes

$$\sum_{n=1}^{\infty} \tau(n) q^n.$$

Ramanujan computes by hand many of these coefficients and makes the following empirical observations about the integers $\tau(n)$:

(i) if $(m,n)=1$ then $\tau(mn)=\tau(m)\tau(n)$,
(ii) if p is prime and $\alpha \geq 0$ then $\tau(p^{\alpha+2}) = \tau(p)\tau(p^{\alpha+1}) - p^{11}\tau(p^{\alpha})$,
(iii) if p is prime then $|\tau(p)| \leq 2p^{11/2}$.

Items (i) and (ii) were proved by Mordell in 1917 [23], a year after Ramanujan's paper appeared. But conjecture (iii) defied attempts by many eminent mathematicians until 1974, when Deligne [8] resolved it as a corollary of his solution to the Weil conjectures. In fact, only after Ramanujan's conjecture was reformulated in the context of algebraic geometry and its connection to the Weil conjectures made explicit, did mathematicians realize its central place in number theory, and mathematics in general.

At first, Ramanujan's conjectures are amazing. Why should coefficents of the power series defined by the infinite product in (0.1) show such structure? Why should they be multiplicative?

These questions were first addressed in a serious and fundamental way by Erich Hecke, who in the 1930's wrote a sequence of papers enunciating what we now call the Hecke theory of modular forms. For the most part, his theory was quite satisfactory in the sense that it explained conceptually why (i) and (ii) should be true. However, when it came to (iii), the theory suggested only a general conjecture but could not "explain" why it should be true. Such an explanation had to await further developments in algebraic geometry, largely due to Weil and Grothendieck.

Afterwards, in the late 1940's, Hecke's theory was further developed on the one hand by Maass, who noticed the existence of a "real-analytic" counterpart, and by Rankin and Selberg, who developed a theory of L-series and obtained significant results towards conjecture (iii).

In the 1950's and 1960's, Harish-Chandra and, subsequently, Langlands reformulated the notion of a modular form in the larger framework of representation theory of Lie groups. This opened up a new universe linking number theory and representation theory, and subsequently led to the development of the Langlands program in the theory of automorphic representations. For a readable account of this connection, we refer the reader to Kumar Murty's article "Ramanujan and Harish-Chandra" [26].

In the 1960's and 1970's, Hecke's theory was extended to "higher levels," notably by Atkin and Lehner. At the same time, its connection to the theory of elliptic curves, and more generally abelian varieties, forged the link to arithmetic, algebraic geometry, and more precisely to the theory of Galois representations. Perhaps the most exciting event in this context was the insight of Hellegouarch and Frey relating Fermat's Last Theorem and modular forms. This led Serre to formulate precise conjectures that paved the way for a method of attack on Fermat's Last Theorem. The turning point was when Ribet showed that if every elliptic curve "arose from a modular form" (Taniyama's conjecture) then Fermat's Last Theorem follows.

Andrew Wiles recalls that when he heard this result, he set himself the task of proving Taniyama's conjecture. Wiles completed his proof in 1995. His solution of Fermat's Last Theorem required the full force of number theory, algebraic geometry, and representation theory.

Only a special case of Serre's conjecture was needed for Fermat's Last Theorem, but other cases had important consequences. A landmark theorem is the two-dimensional reciprocity law for Galois representations. In 2008, Khare and Wintenberger showed that every two-dimensional "odd" Galois representation "arises from a modular form."

This theorem can be viewed as a generalization of Wiles' work and at the same time, as the methods are different, it offers yet another (although equally difficult) resolution to Fermat's Last Theorem, which is only one of many conjectures to which the theory of modular forms has been applied.

A conjecture or a problem in mathematics acts like a muse that inspires further developments, new concepts, and a rich tapestry of fundamental ideas. The theory of modular forms is an essential part of mathematics and it is becoming increasingly clear that it will play a central role in the development of twenty-first century mathematics. This is our main motivation for the writing of this book. It is to acquaint the graduate student in a painless manner to the essential ideas of the theory. At the same time, as theory is sterile without practice, we have tried to invite and engage the student in this topic through the problem-solving approach. Along with my other two books, "Problems in Algebraic Number Theory" and 'Problems in Analytic Number Theory, " this book should serve as a practical guide for the serious student to teach herself or himself the rudiments of number theory and to embark in the exciting pursuit of research work in this area. At the end, I have listed some references for further study to assist the student in this lofty endeavour.

Kingston, Ontario *Ram Murty*
August 2014

About the Authors

M. Ram Murty is Professor at the Department of Mathematics and Statistics at Queen's University, Canada, where he is a Queen's Research Chair in Mathematics. He is also Professor of Philosophy at Queen's. He was elected a fellow of the Royal Society of Canada in 1990, the Indian National Science Academy in 2008, and won the Coxeter–James Prize, Jeffery–Williams Prize, the E.W.R. Steacie Fellowship, and the Killam Fellowship. His research areas include number theory, modular forms, elliptic curves, and sieve theory. An author of over five books with Springer, his book Non-vanishing of L-functions and Applications, coauthored by his brother V. Kumar Murty, won the 1996 Balaguer Prize and was published by Birkhauser. In addition, Ram is adjunct professor at McGill University; TIFR; IMSc; CMI; IIT Bombay; IISER, West Bengal; Vivekananda University; and Harish Chandra Research Institute, Uttar Pradesh.

Michael Dewar is a post-doctorate fellow at the Queen's University, Canada. He did his Ph.D. on "Ramanujan congruences in modular forms" from the University of Illinois at Urbana-Champaign, United States. His research interest lies in modular forms, Jacobi forms, and harmonic weak Maass forms.

Hester Graves is Professor at the Department of Mathematics, University of Michigan in Ann Arbor, United States.

Part I
Problems

Chapter 1
Jacobi's q-series

1.1 The q-exponential function

It is not clear exactly how to define a q-series. Some experts humorously suggest that it is any power series in q. To some extent this may be true. However, one can say that part of the theory is connected with modular forms and another part with combinatorics. Euler, Jacobi, Ramanujan and many others have made expert use of q-series to derive remarkable number theoretic results ranging from the study of the partition function, the number of representations of natural numbers as sums of squares to the development of exotic continued fractions as is evidenced by the recent book [7] on the subject. During the last fifty years, more connections have arisen, most notably with Lie theory and representation theory as well as theoretical physics. Since the subject is gaining prominence and significance, pointing to a parallel world of mathematics in which q-analogues of classical theories exist, and since the prerequisites for the theory are minimal, we found it fitting to introduce the reader to the world of modular forms through q-series. Already, one sees several classical results such as Jacobi's triple product identity and his celebrated formula for the number of ways of writing a natural number as a sum of four squares as immediate consequences. Moreover, as the problem of representing a natural number as a sum of k squares (with k even) is intimately tied to the theory of modular forms of integral weight, it seems fitting to begin with a study of how q-series can be used to study this problem. The number of ways of writing a number as a sum of k squares with k odd is related to the study of modular forms of half-integral weight, which is beyond the scope of this book.

To ease the reader into this fascinating world, we begin our study with the function

$$\mathcal{E}_q(x) = 1 + \sum_{n=1}^{\infty} \frac{x^n}{(q^n-1)(q^{n-1}-1)\cdots(q-1)},$$

© Springer Science+Business Media Singapore 2016 and Hindustan Book Agency 2016

M.R. Murty et al., *Problems in the Theory of Modular Forms*, HBA Lecture Notes in Mathematics, DOI 10.1007/978-981-10-2651-5_1

and view it as the q-analog of the classical exponential function. We work with $\mathcal{E}_q(x)$ as a formal Laurent series. If we define the empty product as 1, then the series can be written as

$$\mathcal{E}_q(x) = \sum_{n=0}^{\infty} \frac{x^n}{(q^n-1)(q^{n-1}-1)\cdots(q-1)},$$

It may not be clear to the reader why this should be considered as an analog of the exponential function. The underlying "philosophy" of moving from the world of natural numbers to the q-world seems to be partially based on the observation that

$$\lim_{q\to 1} \frac{q^n-1}{q-1} = n,$$

so that one views q^n-1 (or more precisely $(q^n-1)/(q-1)$) as the q-analog of the natural number n. Once this is understood, many of the functions in the q-world become meaningful and exhibit remarkable structural properties.

Exercise 1.1.1. *Show that*

$$\mathcal{E}_q(x) - \mathcal{E}_q\left(\frac{x}{q}\right) = \frac{x}{q}\mathcal{E}_q\left(\frac{x}{q}\right).$$

Exercise 1.1.2. *Prove that if* $|q| > 1$ *then*

$$\mathcal{E}_q(x) = \prod_{n=1}^{\infty}\left(1+\frac{x}{q^n}\right).$$

Exercise 1.1.3. *Show that if* $|q| < 1$ *then*

$$\prod_{n=0}^{\infty}(1+q^n x) = \sum_{n=0}^{\infty}\frac{q^{\binom{n}{2}}x^n}{(1-q)\cdots(1-q^n)}.$$

Exercise 1.1.4. *Prove that if* $|q| > 1$ *then*

$$(1+x)\mathcal{E}_{\frac{1}{q}}(x) = \frac{1}{\mathcal{E}_q(x)}.$$

Deduce that if $|q| < 1$ *then*

$$\prod_{n=0}^{\infty}(1+q^n x)^{-1} = \sum_{n=0}^{\infty}\frac{x^n}{(q^n-1)\cdots(q-1)}.$$

1.2 Jacobi's Triple Product Identity

We now give a simple proof of the celebrated identity of Jacobi:

Theorem 1.2.1 (Jacobi Triple Product). *If $|q| < 1$ and $x \neq 0$ then*

$$\prod_{n=0}^{\infty}(1-q^{2n+2})(1+q^{2n+1}x)\left(1+\frac{q^{2n+1}}{x}\right) = \sum_{n=-\infty}^{\infty} q^{n^2}x^n. \tag{1.1}$$

Remark. According to Askey, this was also contained in some unpublished work of Gauss (see [3]). In our proof of (1.1), we follow Andrews (see [1]).

Proof. By Exercise 1.1.3 we have, upon replacing q by q^2 and x by xq,

$$\begin{aligned}\prod_{n=0}^{\infty}(1+xq^{2n+1}) &= \sum_{n=0}^{\infty}\frac{q^{n^2}x^n}{(1-q^2)\cdots(1-q^{2n})}\\ &= \sum_{n=0}^{\infty} q^{n^2}x^n\frac{\prod_{j=0}^{\infty}(1-q^{2n+2j+2})}{\prod_{j=0}^{\infty}(1-q^{2j+2})}\\ &= \prod_{j=0}^{\infty}(1-q^{2j+2})^{-1}\sum_{n=-\infty}^{\infty} q^{n^2}x^n\prod_{j=0}^{\infty}(1-q^{2n+2j+2}),\end{aligned}$$

since for negative n, the product inside the sum is zero. Again by Exercise 1.1.3, replacing q by q^2 and replacing x by $-q^{2n+2}$, we can write the product inside the summation as

$$\sum_{m=0}^{\infty}\frac{(-1)^m q^{m^2+m+2nm}}{(1-q^2)\cdots(1-q^{2m})}.$$

The sum under consideration becomes

$$\sum_{n=-\infty}^{\infty} q^{n^2}x^n\sum_{m=0}^{\infty}\frac{(-1)^m q^{m^2+m+2nm}}{(1-q^2)\cdots(1-q^{2m})}.$$

Interchanging the sums, we get

$$\sum_{m=0}^{\infty}\frac{(-1)^m(\frac{q}{x})^m}{(1-q^2)\cdots(1-q^{2m})}\sum_{n=-\infty}^{\infty} q^{(m+n)^2}x^{n+m}.$$

The innermost sum is

$$\sum_{n=-\infty}^{\infty} q^{n^2}x^n$$

and can be factored out. The remaining sum is

$$\prod_{j=0}^{\infty}\left(1+\frac{q^{2j+1}}{x}\right)^{-1}$$

by Exercise 1.1.4 (where we have set q^2 for q and $\frac{q}{x}$ for x). Putting everything together gives the desired result. □

Exercise 1.2.2 (Euler's pentagonal number theorem). *Show that if* $|q|<1$ *then*

$$\prod_{n=1}^{\infty}(1-q^n)=\sum_{k=-\infty}^{\infty}(-1)^k q^{\frac{k(3k-1)}{2}}.$$

Theorem 1.2.3 (Jacobi's formula). *If* $|q|<1$ *then*

$$\prod_{n=1}^{\infty}(1-q^n)^3=\sum_{k=0}^{\infty}(-1)^k(2k+1)q^{\frac{k(k+1)}{2}}.$$

Proof. From the triple product identity, with x replaced by $-x$, we see that

$$\prod_{n=1}^{\infty}(1-q^{2n})(1-xq^{2n-1})(1-x^{-1}q^{2n-1})=\sum_{k=-\infty}^{\infty}(-x)^k q^{k^2}.$$

The left hand side has a factor $(1-xq)$ coming from $n=1$, and consequently vanishes when $x=\frac{1}{q}$. The same is true of the right hand side. Writing

$$\prod_{n=1}^{\infty}(1-xq^{2n-1})=(1-xq)\prod_{n=1}^{\infty}(1-xq^{2n+1}),$$

we obtain

$$\prod_{n=1}^{\infty}(1-q^{2n})(1-xq^{2n+1})(1-x^{-1}q^{2n-1})=\frac{1}{1-xq}\sum_{k=-\infty}^{\infty}(-x)^k q^{k^2}.$$

Putting $x=\frac{1}{q}$ in the left hand side gives

$$\prod_{n=1}^{\infty}(1-q^{2n})^3.$$

For the right hand side, we use l'Hopital's rule to take the limit as $x\to\frac{1}{q}$ to get

$$-\frac{1}{q}\sum_{k=-\infty}^{\infty}(-1)^k k q^{1-k+k^2}=-\sum_{k=-\infty}^{\infty}(-1)^k k q^{k^2-k}.$$

We observe that the function $f(k)=k^2-k$ has the property that $f(k)=f(-(k-1))$. Thus, pairing up k and $-(k-1)$ in the sum, we get that it is

$$-\sum_{k=1}^{\infty}\{(-1)^k k+(-1)^{k-1}(1-k)\}q^{k^2-k}=-\sum_{k=1}^{\infty}(-1)^k(2k-1)q^{k^2-k}.$$

We can rewrite this sum as

$$\sum_{k=1}^{\infty}(-1)^{k-1}(2(k-1)+1)q^{k(k-1)}=\sum_{k=0}^{\infty}(-1)^k(2k+1)q^{k(k+1)}.$$

Changing q by $q^{\frac{1}{2}}$ gives the desired result. □

Exercise 1.2.4. *Prove that*

$$\sum_{n=-\infty}^{\infty}(4n+1)q^{2n^2+n}=\sum_{k=0}^{\infty}(-1)^k(2k+1)q^{\frac{k(k+1)}{2}}.$$

1.3 Jacobi's two-square theorem

In this section, our goal is to obtain formulas for the number of ways a natural number m can be written as a sum of two squares. We recognize this as the computation of the m-th coefficient in the power series expansion

$$\left(\sum_{n=-\infty}^{\infty}q^{n^2}\right)^2.$$

We prove:

Theorem 1.3.1.

$$\left(\sum_{n=-\infty}^{\infty}q^{n^2}\right)^2=1+4\sum_{n=0}^{\infty}\left(\frac{q^{4n+1}}{1-q^{4n+1}}-\frac{q^{4n+3}}{1-q^{4n+3}}\right).$$

An immediate corollary is deduced by expanding the right hand side as a power series in q. Let $r_k(n)$ be the number of ways of writing n as a sum of k squares.

Corollary 1.3.2. *For $n\geq 1$, we have $r_2(n)=4(d_1(n)-d_3(n))$ where $d_i(n)$ is the number of divisors of n congruent to $i \pmod 4$.*

The following proof is due to M.D. Hirschhorn [15].

Proof of Theorem 1.3.1. In the Jacobi triple product identity, we put a^4q for x and q^2 for q to get

$$\sum_{n=-\infty}^{\infty}a^{4n}q^{2n^2+n}=\prod_{n=1}^{\infty}(1+a^4q^{4n-1})(1+a^{-4}q^{4n-3})(1-q^{4n}).$$

Multiply both sides by a and differentiate with respect to a, to obtain

$$\sum_{n=-\infty}^{\infty}(4n+1)a^{4n}q^{2n^2+n} = \prod_{n=1}^{\infty}(1+a^4q^{4n-1})(1+a^{-4}q^{4n-3})(1-q^{4n}) + a\frac{d}{da}\left\{\prod_{n=1}^{\infty}(1+a^4q^{4n-1})(1+a^{-4}q^{4n-3})(1-q^{4n})\right\}.$$

To differentiate the product, it is useful to observe that if

$$P(a) = \prod_{n=1}^{\infty} f_n(a),$$

then by taking the logarithmic derivative of both sides, we get

$$\frac{P'(a)}{P(a)} = \sum_{n=1}^{\infty}\frac{f_n'(a)}{f_n(a)}.$$

In our case, we get

$$\sum_{n=-\infty}^{\infty}(4n+1)a^{4n}q^{2n^2+n} = \prod_{n=1}^{\infty}(1+a^4q^{4n-1})(1+a^{-4}q^{4n-3})(1-q^{4n}) \times\left\{1+a\sum_{n=1}^{\infty}\frac{4a^3q^{4n-1}}{1+a^4q^{4n-1}}-\frac{4a^{-5}q^{4n-3}}{1+a^{-4}q^{4n-3}}\right\}.$$

We now put $a=1$ and get

$$\sum_{n=-\infty}^{\infty}(4n+1)q^{2n^2+n} = \left(\prod_{n=1}^{\infty}(1+q^{4n-1})(1+q^{4n-3})(1-q^{4n})\right) \times\left\{1-4\sum_{n=0}^{\infty}\left(\frac{q^{4n+1}}{1+q^{4n+1}}-\frac{q^{4n+3}}{1+q^{4n+3}}\right)\right\} \tag{1.2}$$

after changing the index of summation to start from $n=0$.

By Theorem 1.2.3 and Exercise 1.2.4, the sum on the left hand side is

$$\prod_{n=1}^{\infty}(1-q^n)^3.$$

This can be rewritten as

$$\prod_{n=1}^{\infty}(1-q^{2n-1})^3(1-q^{2n})^3.$$

The product on the right hand side of (1.2) can be written as

$$\prod_{n=1}^{\infty}\left(\frac{1-q^{4n-2}}{1-q^{2n-1}}\right)(1-q^{4n})=\prod_{n=1}^{\infty}\left(\frac{1-q^{2n}}{1-q^{2n-1}}\right),$$

so we now deduce that

$$\prod_{n=1}^{\infty}(1-q^{2n-1})^4(1-q^{2n})^2=1-4\sum_{n=0}^{\infty}\left(\frac{q^{4n+1}}{1+q^{4n+1}}-\frac{q^{4n+3}}{1+q^{4n+3}}\right).$$

But the left hand side is

$$\left(\sum_{n=-\infty}^{\infty}(-1)^n q^{n^2}\right)^2$$

from the triple product formula. Changing q to $-q$ gives

$$\left(\sum_{n=-\infty}^{\infty}q^{n^2}\right)^2=1+4\sum_{n=0}^{\infty}\left(\frac{q^{4n+1}}{1-q^{4n+1}}-\frac{q^{4n+3}}{1-q^{4n+3}}\right),$$

which is the desired result. □

Exercise 1.3.3. *Let $x_4(n)$ be defined as follows:*

$$x_4(n)=\begin{cases}0 & \text{if } n \text{ is even}\\ +1 & \text{if } n\equiv 1 \pmod 4\\ -1 & \text{if } n\equiv 3 \pmod 4\end{cases}$$

Prove that $x_4(n)$ is completely multiplicative. That is, show that $x_4(mn)=x_4(m)x_4(n)$.

Exercise 1.3.4. *Show that*

$$r_2(n)=4\sum_{d|n}x_4(d).$$

1.4 Jacobi's four square theorem

Using results obtained for $r_2(n)$, we will derive an explicit formula for $r_4(n)$, the number of ways of writing n as a sum of four squares. This formula is due to Jacobi who derived it using the theory of elliptic functions. Here, we will follow a method due to Ramanujan that is completely elementary and based on the following exercise.

Exercise 1.4.1. *Show that*

(a) $\dfrac{1}{2}\left(\cot\dfrac{\theta}{2}\right)\sin n\theta = \dfrac{1}{2} + \cos\theta + \cos 2\theta + \cdots + \cos(n-1)\theta + \dfrac{1}{2}\cos n\theta,$

(b) $2(\sin m\theta)(\sin n\theta) = \cos(m-n)\theta - \cos(m+n)\theta.$

Exercise 1.4.2. *For $|q| < 1$, we let $u_r = \frac{q^r}{1-q^r}$ so that*

$$\frac{q^r}{(1-q^r)^2} = u_r(1+u_r).$$

Prove that

$$\sum_{m=1}^{\infty} u_m(1+u_m) = \sum_{n=1}^{\infty} nu_n.$$

Exercise 1.4.3. *With notation as in the previous exercise, show that*

$$\sum_{m=1}^{\infty} (-1)^{m-1} u_{2m}(1+u_{2m}) = \sum_{n=1}^{\infty} (2n-1)u_{4n-2}.$$

We are now ready to prove the following theorem due to Ramanujan.

Theorem 1.4.4. *Let θ be real and not a multiple of π. Set*

$$L = L(q,\theta) = \frac{1}{4}\cot\frac{\theta}{2} + \sum_{n=1}^{\infty} u_n \sin n\theta,$$

$$T_1 = T_1(q,\theta) = \left(\frac{1}{4}\cot\frac{\theta}{2}\right)^2 + \sum_{n=1}^{\infty} u_n(1+u_n)\cos n\theta,$$

$$T_2 = T_2(q,\theta) = \frac{1}{2}\sum_{n=1}^{\infty} nu_n(1-\cos n\theta),$$

where $u_n = \frac{q^n}{1-q^n}$. Then $L^2 = T_1 + T_2$.

Before we begin the proof of this theorem, we indicate how it implies Jacobi's theorem on representing a natural number as a sum of four squares.

Corollary 1.4.5.

$$\left\{1 + 4\sum_{n=0}^{\infty}\left(\frac{q^{4n+1}}{1-q^{4n+1}} - \frac{q^{4n+3}}{1-q^{4n+3}}\right)\right\}^2 = 1 + 8\sum_{\substack{n=1 \\ n\not\equiv 0 \pmod 4}}^{\infty} \frac{nq^n}{1-q^n}.$$

Proof. Put $\theta = \frac{\pi}{2}$ in Theorem 1.4.4. Then

$$L = \frac{1}{4} + \sum_{n=1}^{\infty} \frac{q^n}{1-q^n} \sin n\frac{\pi}{2}.$$

If n is even, $\sin n\frac{\pi}{2} = 0$ so the right hand side is

$$\frac{1}{4} + \sum_{n=0}^{\infty} \frac{q^{2n+1}}{1-q^{2n+1}} \sin(2n+1)\frac{\pi}{2}.$$

Now, if n is even, $\sin(2n+1)\frac{\pi}{2} = 1$ and if n is odd, $\sin(2n+1)\frac{\pi}{2} = -1$. In other words,

$$4L = 1 + 4\sum_{n=0}^{\infty} \left(\frac{q^{4n+1}}{1-q^{4n+1}} - \frac{q^{4n+3}}{1-q^{4n+3}} \right).$$

Since $\cos n\frac{\pi}{2} = 0$ if n is odd and $(-1)^{\frac{n}{2}}$ if n is even, computing T_1 and T_2 in Theorem 1.4.4 gives

$$16T_1 = 1 - 16\sum_{m=1}^{\infty} (-1)^{m-1} u_{2m}(1+u_{2m}),$$
$$16T_2 = 8\sum_{m=1}^{\infty} (2m-1)u_{2m-1} + 32\sum_{m=1}^{\infty} (2m-1)u_{4m-2}.$$

Putting everything together and using Exercise 1.4.3 gives

$$16L^2 = 16(T_1+T_2) = 1 + 8\sum_{\substack{n=1 \\ n\not\equiv 0 \pmod 4}}^{\infty} nu_n,$$

which is the desired result. □

Corollary 1.4.6. *Let $r_4(n)$ be the number of ways of writing n as a sum of four squares. Then*

$$r_4(n) = 8\sum_{\substack{d\mid n \\ d\not\equiv 0 \pmod 4}} d.$$

In particular, $r_4(n) > 0$ for all $n \geq 1$.

Proof. We need only invoke Theorem 1.3.1 and Corollary 1.4.5 to deduce that

$$\sum_{n=0}^{\infty} r_4(n)q^n = \left(\sum_{n=-\infty}^{\infty} q^{n^2} \right)^4 = 1 + 8\sum_{\substack{n=1 \\ n\not\equiv 0 \pmod 4}}^{\infty} \frac{nq^n}{1-q^n}.$$

We write the right hand side as

$$1+8\sum_{\substack{n=1\\ n\not\equiv 0 \pmod 4}}^{\infty} n\sum_{m=1}^{\infty} q^{nm} = 1+8\sum_{n=1}^{\infty} q^n \left(\sum_{\substack{d\mid n\\ d\not\equiv 0 \pmod 4}} d\right),$$

from which we deduce the desired formula for $r_4(n)$. □

Proof of Theorem 1.4.4. Squaring L, we obtain

$$\begin{aligned} L^2 &= \left\{\frac{1}{4}\cot\frac{\theta}{2} + \sum_{n=1}^{\infty} u_n \sin n\theta\right\}^2 \\ &= \left(\frac{1}{4}\cot\frac{\theta}{2}\right)^2 + \frac{1}{2}\left(\cot\frac{\theta}{2}\right)\sum_{n=1}^{\infty} u_n \sin n\theta + \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} u_m u_n \sin m\theta \sin n\theta \\ &= \left(\frac{1}{4}\cot\frac{\theta}{2}\right)^2 + S_1 + S_2, \end{aligned}$$

where

$$S_1 = \frac{1}{2}\left(\cot\frac{\theta}{2}\right)\sum_{n=1}^{\infty} u_n \sin n\theta$$

and

$$S_2 = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} u_m u_n \sin m\theta \sin n\theta.$$

By Exercise 1.4.1,

$$\frac{1}{2}\left(\cot\frac{\theta}{2}\right)\sin n\theta = \frac{1}{2} + \cos\theta + \cos 2\theta + \cdots + \cos(n-1)\theta + \frac{1}{2}\cos n\theta$$

and

$$2\,(\sin m\theta)\,(\sin n\theta) = \cos(m-n)\theta - \cos(m+n)\theta,$$

so that

$$S_1 = \sum_{n=1}^{\infty} u_n \left\{\frac{1}{2} + \cos\theta + \cos 2\theta + \cdots + \cos(n-1)\theta + \frac{1}{2}\cos n\theta\right\}$$

and

$$S_2 = \frac{1}{2}\sum_{m=1}^{\infty}\sum_{n=1}^{\infty} u_m u_n \left\{\cos(m-n)\theta - \cos(m+n)\theta\right\}.$$

In other words,

$$L^2 = \left(\frac{1}{4}\cot\frac{\theta}{2}\right)^2 + c_0 + \sum_{k=1}^{\infty} c_k \cos k\theta$$

for certain constants c_k, after re-arranging $S_1 + S_2$ as a cosine series. Now

$$c_0 = \frac{1}{2}\sum_{n=1}^{\infty}\left(u_n + u_n^2\right) = \frac{1}{2}\sum_{n=1}^{\infty} n u_n,$$

by Exercise 1.4.2. For $k \geq 1$, we have

$$c_k = \frac{1}{2}u_k + \sum_{n=k+1}^{\infty} u_n + \frac{1}{2}\sum_{m-n=k} u_m u_n + \frac{1}{2}\sum_{n-m=k} u_m u_n - \frac{1}{2}\sum_{m+n=k} u_m u_n,$$

where $m, n \geq 1$ in all of the summations. We may rewrite this as

$$c_k = \frac{1}{2}u_k + \sum_{j=1}^{\infty} u_{k+j} + \sum_{j=1}^{\infty} u_j u_{j+k} - \frac{1}{2}\sum_{j=1}^{k-1} u_j u_{k-j}.$$

It is easily checked that

$$u_j u_{k-j} = u_k(1 + u_j + u_{k-j})$$

and

$$u_{k+j} + u_j u_{k+j} = u_k(u_j - u_{k+j})$$

so that

$$c_k = u_k\left\{\frac{1}{2} + \sum_{j=1}^{\infty}(u_j - u_{k+j}) - \frac{1}{2}\sum_{j=1}^{k-1}(1 + u_j + u_{k-j})\right\}.$$

The first sum telescopes and we obtain

$$\begin{aligned} c_k &= u_k\left\{\frac{1}{2} + (u_1 + u_2 + \cdots + u_k) - \frac{1}{2}(k-1) - (u_1 + u_2 + \cdots + u_{k-1})\right\} \\ &= u_k(1 + u_k - \frac{1}{2}k). \end{aligned}$$

Hence

$$\begin{aligned} L^2 &= \left(\frac{1}{4}\cot\frac{\theta}{2}\right)^2 + \frac{1}{2}\sum_{n=1}^{\infty} n u_n + \sum_{k=1}^{\infty} u_k(1 + u_k - \frac{1}{2}k)\cos k\theta \\ &= \left(\frac{1}{4}\cot\frac{\theta}{2}\right)^2 + \sum_{k=1}^{\infty} u_k(1 + u_k)\cos k\theta + \frac{1}{2}\sum_{k=1}^{\infty} k u_k(1 - \cos k\theta), \end{aligned}$$

which is the desired result. □

The methods utilised above to determine formulas for $r_2(n)$ and $r_4(n)$ can be extended for other values $r_{2k}(n)$ for certain values of k. We refer the

reader to consult [11] for more examples. In [11], the student will also find a discussion for finding formulas for $r_k(n)$ when k is odd and this is intimately connected with the theory of modular forms of half-integral weight which is beyond the scope of this book.

1.5 Supplementary problems

Exercise 1.5.1. *Define $\tau(n)$ by*

$$q\prod_{n=1}^{\infty}(1-q^n)^{24}=\sum_{n=1}^{\infty}\tau(n)q^n.$$

Show that $\tau(n)$ is odd if and only if $n=(2m+1)^2$ for some m.

Exercise 1.5.2. *Let $r_k(n)$ be the number of ways of writing n as a sum of k squares. Show that*

$$r_k(n)=\sum_{a=0}^{n}r_i(a)r_{k-i}(n-a)$$

for any i satisfying $1\leq i\leq k$.

Exercise 1.5.3. *Define the q-logarithm function $\mathcal{L}_q(x)$ as*

$$\mathcal{L}_q(x)=\sum_{n=1}^{\infty}\frac{x^n}{q^n-1}.$$

Suppose $|x|<|q|$ and $|q|>1$. Show that

$$\mathcal{L}_q(x)-\mathcal{L}_q\left(\frac{x}{q}\right)=\frac{x}{q-x}.$$

Deduce that

$$\mathcal{L}_q(x)=\sum_{n=1}^{\infty}\frac{x}{q^n-x}.$$

Exercise 1.5.4. *Show that if $|x|<|q|$ and $|q|>1$ then*

$$\mathcal{L}_q(x)=\frac{x\mathcal{E}_q'(-x)}{\mathcal{E}_q(-x)},$$

where the derivative is with respect to x.

Chapter 2
The Modular Group

2.1 The full modular group

Exercise 2.1.1. *Let R be a commutative ring with identity. Show that the set*

$$\mathrm{SL}_2(R) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : \begin{matrix} a, b, c, d \in R \\ ad - bc = 1 \end{matrix} \right\}$$

forms a group under matrix multiplication.

The *(full) modular group* $\mathrm{SL}_2(\mathbb{Z})$ plays a pivotal role in the theory of modular forms. One also considers $\mathrm{PSL}_2(\mathbb{Z}) = \mathrm{SL}_2(\mathbb{Z})/\{\pm I\}$. The relationship of $\mathrm{SL}_2(\mathbb{Z})$ to $\mathrm{SL}_2(\mathbb{R})$ is similar to the relationship of $\mathbb{Z}$ to $\mathbb{R}$ in the sense that $\mathbb{Z}$ is a discrete subgroup of $\mathbb{R}$ and $\mathrm{SL}_2(\mathbb{Z})$ is a discrete subgroup of $\mathrm{SL}_2(\mathbb{R})$. We will show below that $\mathrm{SL}_2(\mathbb{Z})$ is generated by the elements

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \text{ and } \ T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Theorem 2.1.2. *The matrices S and T generate* $\mathrm{SL}_2(\mathbb{Z})$.

Proof. Observe that

$$T^n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$$

so that

$$T^n \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a + nc & b + nd \\ c & d \end{pmatrix}. \tag{2.1}$$

Also $S^2 = -I$ and

$$S \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} -c & -d \\ a & b \end{pmatrix}.$$

© Springer Science+Business Media Singapore 2016 and Hindustan Book Agency 2016

M.R. Murty et al., *Problems in the Theory of Modular Forms*, HBA Lecture Notes in Mathematics, DOI 10.1007/978-981-10-2651-5_2

Now let $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ be any element of $\mathrm{SL}_2(\mathbb{Z})$. If $c = 0$, then $ad = 1$ implies that $a = d = \pm 1$. In this case

$$g = \pm \begin{pmatrix} 1 & b' \\ 0 & 1 \end{pmatrix} = \pm T^{b'}.$$

where $b' = \pm b$. Since $S^2 = -I$, either $g = T^{b'}$ or $S^2 T^{b'}$. If $c \neq 0$, we proceed as follows. Without loss of generality we may suppose $|a| \geq |c|$, for otherwise we can apply S to arrange this. By the Division Algorithm, we can write $a = cq + r$ with $0 \leq r < |c|$. Then $T^{-q} g$ has upper left entry $r = a - qc$ which is smaller than $|c|$. Applying S switches the rows (with a sign change) and so we can iterate the process if $r \neq 0$. After a finite number of steps, we are reduced to the case $c = 0$ and we are done. □

Exercise 2.1.3. *Show that S has order 4, ST has order 6, and T has infinite order.*

Exercise 2.1.4. *Show that $\mathrm{SL}_2(\mathbb{Z})$ is generated by two elements of finite order, namely S and ST of order 4 and 6 respectively.*

Exercise 2.1.5. *Show that any homomorphism*

$$\phi : \mathrm{SL}_2(\mathbb{Z}) \to \mathbb{C}^\times$$

has image contained in the finite subgroup of $\mathbb{C}^\times$ consisting of 12th roots of unity.

Exercise 2.1.6. *Show that $\mathrm{SL}_2(\mathbb{Z})$ is generated by*

$$T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \text{ and } U = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

Exercise 2.1.7. *Suppose that $(c, d, N) = 1$. Show that there are elements $c' = c + tN$ and $d' = d + sN$ for some integers s, t such that $(c', d') = 1$.*

2.2 Subgroups of the modular group

For each natural number N, the *principal congruence subgroup* of level N, denoted $\Gamma(N)$, is the group

$$\left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}) : \begin{pmatrix} a & b \\ c & d \end{pmatrix} \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \pmod{N} \right\}.$$

In particular, $\Gamma(1) = \mathrm{SL}_2(\mathbb{Z})$.

Exercise 2.2.1. *Show that the natural map*

$$\begin{aligned} \mathrm{SL}_2(\mathbb{Z}) &\to \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z}) \\ g &\mapsto g \pmod N \end{aligned}$$

is a surjective homomorphism.

Exercise 2.2.2. *Prove that* $\Gamma(N)$ *is a normal subgroup of* $\mathrm{SL}_2(\mathbb{Z})$ *of finite index.*

Exercise 2.2.3. *For any commutative ring* R *with identity, let* $\mathrm{GL}_2(R)$ *be the set of* 2×2 *invertible matrices with entries in* R*. Show that* $\mathrm{GL}_2(R)$ *is a group.*

Exercise 2.2.4. *Let* p *be a prime. Show that the order of* $\mathrm{GL}_2(\mathbb{Z}/p\mathbb{Z})$ *is given by* $(p^2-1)(p^2-p)$.

Exercise 2.2.5. *Let* p *be a prime. Show that* $\mathrm{SL}_2(\mathbb{Z}/p\mathbb{Z})$ *is a normal subgroup of* $\mathrm{GL}_2(\mathbb{Z}/p\mathbb{Z})$ *of index* $p-1$*. Deduce that* $\mathrm{SL}_2(\mathbb{Z}/p\mathbb{Z})$ *has order* $p(p^2-1)$.

Exercise 2.2.6. *Show that* $\mathrm{SL}_2(\mathbb{Z}/p^2\mathbb{Z})$ *has size* $p^4(p^2-1)$.

Exercise 2.2.7. *Apply an induction argument to show that*

$$\mathrm{SL}_2(\mathbb{Z}/p^n\mathbb{Z})$$

has size $p^{3n-2}(p^2-1)$.

Exercise 2.2.8. *Show that*

$$|\,\mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})| = N^3 \prod_{p|N} \left(1 - \frac{1}{p^2}\right).$$

Deduce that

$$[\mathrm{SL}_2(\mathbb{Z}) : \Gamma(N)] = N^3 \prod_{p|N} \left(1 - \frac{1}{p^2}\right).$$

A subgroup $\Gamma \subset \mathrm{SL}_2(\mathbb{Z})$ is called a *congruence subgroup* if $\Gamma(N) \subset \Gamma$ for some N. Since $\Gamma(N)$ is of finite index in $\mathrm{SL}_2(\mathbb{Z})$, it follows that any congruence subgroup is also of finite index in $\mathrm{SL}_2(\mathbb{Z})$. One can ask if the converse is true. Is every subgroup of finite index in $\mathrm{SL}_2(\mathbb{Z})$ a congruence subgroup? The answer is no and the first counter example was given by Fricke and Pick in 1887, independently. However, in the theory of modular forms, congruence subgroups will play a dominant role. If Γ is a congruence subgroup, the smallest N such that $\Gamma(N) \subset \Gamma$ is called the *level* of Γ.

It is clear from the previous discussion that more generally, we can consider $\mathrm{SL}_n(\mathbb{Z})$ and $\mathrm{SL}_n(\mathbb{Z}/N\mathbb{Z})$. One can prove that the natural map

$$\phi_N : \mathrm{SL}_n(\mathbb{Z}) \to \mathrm{SL}_n(\mathbb{Z}/N\mathbb{Z})$$

is again surjective and the index of $\ker \phi_N$ in $\mathrm{SL}_n(\mathbb{Z})$ can be given explicitly. Again a subgroup $\Gamma \subset \mathrm{SL}_n(\mathbb{Z})$ is called a *congruence subgroup* if $\ker \phi_N \subset \Gamma$

for some N. The group Γ is again of finite index. It is a celebrated theorem of Bass, Lazard, and Serre that for $n \geq 3$, every subgroup of finite index is a congruence subgroup. For $n \geq 3$, Bass, Milnor, and Serre extended this result in 1967 as follows. Let K be an algebraic number field with the ring of integers $\mathcal{O}_K$. All subgroups of finite index in $\mathrm{SL}_n(\mathcal{O}_K)(n \geq 3)$ are congruence subgroups if and only if K has at least one real embedding.

Exercise 2.2.9. *Show that* $\mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})$ *is generated by*

$$T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \text{ and } U = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix},$$

both of order N.

Exercise 2.2.10. *Show that* $\Gamma(2)$ *is generated by* $-I, T^2$, *and* U^2, *where*

$$T^2 = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \text{ and } U^2 = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}.$$

Exercise 2.2.11. *In* $\mathrm{SL}_2(\mathbb{Z})$, *show that the group* Γ *generated by* S *and* T^2 *is a congruence subgroup of index three.*

2.3 The Hecke subgroups

The *Hecke subgroups* are denoted $\Gamma_0(N)$ and defined by

$$\Gamma_0(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}) : c \equiv 0 \pmod{N} \right\}.$$

It is easy to see that this is a group. Clearly $\Gamma(N) \subset \Gamma_0(N)$ and so these are congruence subgroups.

Exercise 2.3.1. *Show that the map* $\Gamma_0(N) \to (\mathbb{Z}/N\mathbb{Z})^\times$ *given by*

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto d \pmod{N}$$

is a surjective homomorphism.

The kernel of the map in the previous exercise is denoted $\Gamma_1(N)$ and is given by

$$\Gamma_1(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}) : \begin{matrix} c \equiv 0 \pmod{N} \\ d \equiv 1 \pmod{N} \end{matrix} \right\}.$$

Thus we have

$$\Gamma(N) \subset \Gamma_1(N) \subset \Gamma_0(N) \subset \mathrm{SL}_2(\mathbb{Z}).$$

Exercise 2.3.2. *Show that the map* $\Gamma_1(N) \to \mathbb{Z}/N\mathbb{Z}$ *given by*

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto b \pmod{N}$$

is a surjective homomorphism with kernel $\Gamma(N)$.

Exercise 2.3.3. *Show that* $\Gamma_1(N)$ *has index* $N^2 \prod_{p|N} \left(1 - \frac{1}{p^2}\right)$ *in* $\mathrm{SL}_2(\mathbb{Z})$.

Exercise 2.3.4. *Show that* $\Gamma_0(N)$ *has index* $N \prod_{p|N} \left(1 + \frac{1}{p}\right)$ *in* $\mathrm{SL}_2(\mathbb{Z})$.

2.4 Groups acting on topological spaces

Let G be a group and X a topological space. We say that there is a *left action* of G on X if, for each element $g \in G$, we have a continuous map $x \mapsto gx \in X$ for $x \in X$ satisfying the conditions:

(a) $1x = x \quad \forall x \in X$
(b) $(gh)x = g(hx) \quad \forall g, h \in G, x \in X.$

From these axioms, we see that $x \mapsto g^{-1}x$ is the inverse map of $x \mapsto gx$. Thus each element $g \in G$ gives rise to a topological automorphism of X.

It may be clearer to define an action as a pairing $G \times X \to X$, where we use the notation $g \cdot x$ to denote the image of (g, x) under this mapping. We suppress the 'dot' for aesthetic reasons.

For each element $x \in X$, we define the *stabilizer subgroup* G_x as

$$G_x := \{g \in G : gx = x\}.$$

Exercise 2.4.1. *If* G *acts on* X *and* $g \in G$, $x \in X$, *show that*

$$G_{gx} = gG_xg^{-1}.$$

An element $x \in X$ is called a *fixed point* of $g \in G$ if $gx = x$. This is equivalent to saying $g \in G_x$.

For each element $x \in X$, we define the *G-orbit* of x to be

$$Gx := \{gx : g \in G\}.$$

The set of all G-orbits in X will be denoted $G \setminus X$. We say two elements $x, y \in X$ are *G-equivalent* if there is a $g \in G$ such that $gx = y$. In particular, $y \in Gx$. This is clearly an equivalence relation on X and we can decompose X as a union of disjoint orbits:

$$X = \coprod Gx. \tag{2.2}$$

If there is only one orbit, we say G *acts transitively* on X. This is equivalent to saying that any two elements of X are G-equivalent. That is, given any two elements $x, y \in X$, there is a $g \in G$ such that $gx = y$.

Exercise 2.4.2. *If G acts transitively on X, show that all stabilizers are conjugate.*

Exercise 2.4.3. *(Orbit-Stabilizer Formula) Suppose that G acts on X.*

(a) If $x \in X$ and $g, h \in G$, show that $gx = hx$ if and only if g and h lie in the same left coset of G_x.
(b) If G and X are both finite, show that $|Gx| = [G : G_x]$. Deduce that

$$|X| = \sum_i [G : G_{x_i}],$$

where the summation is over a complete set of representatives of the orbits of G.

We denote by π the surjective map $X \to G \setminus X$ given by

$$x \mapsto \pi(x) = Gx.$$

In other words, we associate to each x its G-orbit. We put the strongest topology on $G \setminus X$ such that π is continuous. More precisely, a subset U of $G \setminus X$ is open if and only if $\pi^{-1}(U)$ is open in X. The topological space $G \setminus X$ with this topology is called the *quotient space* of X by G. If U is any open set of X, we have

$$\pi^{-1}(\pi(U)) = \bigcup_{g \in G} gU$$

is again a union of open sets. Thus $\pi(U)$ is open so that π is an open continuous map of X onto $G \setminus X$.

Now suppose that G has the topological structure of a Hausdorff space and the map $G \times G \to G$ given by $(g, h) \mapsto gh^{-1}$ is a continuous map, we say G is a *topological group*.

We say that a topological group G acts on a topological space X if, in addition to conditions (a) and (b) defining group actions, we impose that the map $(g, x) \mapsto gx$ is a continuous map with respect to the product topology on $G \times X$.

We will not prove but simply state the following important theorem. (The reader can find a proof in Chapter 1 of [37].)

Theorem 2.4.4. *Let G be a topological group that acts transitively on a topological space X. If G is a locally compact group with a countable basis and X is a locally compact Hausdorff space, then for each $x \in X$, the space of right cosets G/G_x is homeomorphic to X, with the correspondence given by $gG_x \mapsto gx$.*

Remark. If G acts transitively on X, this theorem allows us to view X as a coset space G/G_x for any $x \in X$. Functions on X can be viewed as functions on G which are constant on G_x. This point of view has advantages in the higher dimensional theory of modular forms.

We also define a *right action* of a group G on a set X in which

(a) $x1 = x \quad \forall x \in X$
(b) $x(gh) = (xg)h \quad \forall g, h \in G, x \in X.$

The only difference between left and right actions is in the order in which group elements are applied: when gh acts from the left, $(gh)x = g(hx)$ with h being applied first, but when gh acts from the right, $x(gh) = (xg)h$ with g being applied first.

Suppose H and K are subgroups of a group G. The left action of H partitions G into right cosets: $G = \coprod_i Hg_i$. Now K acts from the right on the set $X = H\backslash G$ of cosets via the rule

$$(Hg_i)k = H(g_ik).$$

In particular, an element $k \in K$ simply permutes the right cosets.

More generally, suppose that X is a subset of G which is invariant under the left action of H and the right action of K. That is,

$$\forall h \in H, \forall k \in K, \quad hX = X = Xk.$$

Now as before, X is a disjoint union of its H orbits: $X = \coprod_i Hx_i$. Moreover, K acts on the set of orbits via

$$(Hx_i)k = H(x_ik),$$

where we use the fact that $x_ik \in X$ by the K-invariance of X. The upshot is that $k \in K$ simply permutes the orbits.

Example 2.4.5. For any $g \in G$ where G is a group with subgroups H and K, the set

$$X = \{hgk : h \in H, k \in K\}$$

is invariant under the left action of H and the right action of K. This set is called a *double coset* and is denoted HgK. Moreover, there exist $g_i \in G$ such that

$$HgK = \coprod_i Hg_i.$$

Exercise 2.4.6. *Let G be a group with subgroups H and K. Suppose that X is a subset of G which is invariant under the left action of H and the right action of K. Show that X is a disjoint union of double cosets HgK.*

Exercise 2.4.7. *Suppose a group H acts on a set Y and $|H\backslash Y| < \infty$. If $X \subseteq Y$ is invariant under the action of H and if H_1 is a finite-index subgroup of H, then prove that $|H_1\backslash X| < \infty$.*

2.5 Supplementary problems

Exercise 2.5.1. *If Γ is a congruence subgroup of level N and $\gamma \in \mathrm{SL}_2(\mathbb{Z})$, show that $\gamma\Gamma\gamma^{-1}$ is again a congruence subgroup of level N.*

Exercise 2.5.2. *Prove that*

$$\Gamma(M) \cap \Gamma(N) = \Gamma(\mathrm{lcm}(M, N)).$$

Exercise 2.5.3. *Prove that*

$$\Gamma(M)\Gamma(N) = \Gamma(\gcd(M, N)).$$

Exercise 2.5.4. *Define*

$$\Gamma^1(N) = \left\{ g \in \mathrm{SL}_2(\mathbb{Z}) : g \equiv \begin{pmatrix} 1 & 0 \\ \star & 1 \end{pmatrix} \pmod{N} \right\}.$$

Show that $\Gamma^1(N)$ is a congruence subgroup of level N.

Exercise 2.5.5. *If $\mathrm{SL}_2(\mathbb{Z})$ has a non-congruence subgroup, show that it has infinitely many.* (Hint: consider $\Gamma \cap \Gamma(p)$ for a non-congruence subgroup Γ and rational primes p.)

Exercise 2.5.6. *For each $g \in \mathrm{SL}_2(\mathbb{Z})$ and a subgroup Γ of finite index in $\mathrm{SL}_2(\mathbb{Z})$, we define $n_g(\Gamma)$ to be the least positive integer such that*

$$T^{n_g(\Gamma)} \in g\Gamma g^{-1}, \qquad T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \text{ as usual.}$$

We say Γ is of class N if the least common multiple of the set of numbers $\{n_g(\Gamma) : g \in \mathrm{SL}_2(\mathbb{Z})\}$ equals N. Show that $\Gamma(N)$ is of class N. Also show that the normal closure of T^N, namely

$$D(N) := \langle gT^N g^{-1} : g \in \mathrm{SL}_2(\mathbb{Z}) \rangle$$

is of class N. (This exercise extends the notion of congruence level defined for congruence subgroups to all subgroups of finite index, and is due to Fricke and Wohlfahrt.)

Exercise 2.5.7. *(a) Using the Chinese remainder theorem (or otherwise), show that for given x, y, the following congruence*

$$x(\alpha y + \beta) \equiv \gamma \mod m$$

has a solution in integers α, β, γ such that $(\alpha, \beta, m) = 1$.

(b) Using (a), show that given $g \in \Gamma(N)$, there exists

$$g' = \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix} \in \Gamma(N)$$

such that $b' \equiv c' \equiv 0 \mod mN$ and $g' \in D(N)gD(N)$ with $D(N)$ defined in Exercise 2.5.6. (Hint: consider conjugates of T by the transpose of T that is in $D(N)$.)

Exercise 2.5.8. *With notation as in Exercise 2.5.6, show that $D(N)\Gamma(MN) = \Gamma(N)$ for all positive integers M, N.*

Exercise 2.5.9. *Let Γ be a subgroup in $\mathrm{SL}_2(\mathbb{Z})$ of class N. Show that Γ is a congruence subgroup if and only if it is a congruence subgroup of level N. Deduce that if Γ is a subgroup of $\mathrm{SL}_2(\mathbb{Z})$ of class N and the index $[\mathrm{SL}_2(\mathbb{Z}) : \Gamma]$ does not divide $[\mathrm{SL}_2(\mathbb{Z}) : \Gamma(N)]$, then Γ is not a congruence subgroup.*

Exercise 2.5.10. *Let F be a free group on n generators, say $x_1, ..., x_n$. For a word $w \in F$, define $\ell_i(w)$ to be the sum of all the exponents of x_i occuring in w. For a subset I of $\{1, ..., n\}$, and a positive integer ℓ, define*

$$\Gamma_\ell(I) = \{w \in F : \ell_i(w) \equiv 0 \mod \ell \quad \forall i \in I\}.$$

Show that $\Gamma_\ell(I)$ is a normal subgroup of F of index $\ell^{|I|}$.

Exercise 2.5.11. *Recall that $\Gamma(2)$ is generated by $-I, T^2, U^2$ (see Exercise 2.2.10) and that the subgroup generated by $x_1 = T^2$ and $x_2 = U^2$ is free of index 2 in $\Gamma(2)$.*

(a) With notation as in the previous exercise, compute the class of the group $\Gamma_\ell(\{1\})$ and $\Gamma_\ell(\{1, 2\})$.

(b) Show that $\Gamma_\ell(\{1, 2\})$ is not a congruence subgroup if ℓ is not a power of 2.

Chapter 3
The Upper Half-Plane

3.1 The group $\mathrm{GL}_n^+(\mathbb{R})$ and its subgroups

Let $\mathbb{H}$ denote the upper half-plane,

$$\mathbb{H} = \{z \in \mathbb{C} : \mathrm{Im}(z) > 0\},$$

viewed as an open subset of $\mathbb{C}$ with the usual topology. We define an action of the group

$$\mathrm{GL}_2^+(\mathbb{R}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : \begin{matrix} a, b, c, d \in \mathbb{R} \\ ad - bc > 0 \end{matrix} \right\}$$

on $\mathbb{H}$ via the formula

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} z = \frac{az + b}{cz + d}. \tag{3.1}$$

Exercise 3.1.1. *Show that (3.1) defines an action of* $\mathrm{GL}_2^+(\mathbb{R})$ *on* $\mathbb{H}$.

The transformations

$$z \mapsto \frac{az + b}{cz + d}$$

are called *fractional linear transformations* or *Möbius transformations*. Note that the scalar matrices act trivially on $\mathbb{H}$.

Exercise 3.1.2. *Show that the group* $\mathrm{PSL}_2(\mathbb{R}) = \mathrm{SL}_2(\mathbb{R})/\pm I$ *acts* faithfully *on* $\mathbb{H}$. *That is,* $gz = z$ *for all* $z \in \mathbb{H}$ *implies* $g = 1 \in \mathrm{PSL}_2(\mathbb{R})$.

Exercise 3.1.3. *Show that the action of* $\mathrm{PSL}_2(\mathbb{R})$ *on* $\mathbb{H}$ *is* transitive, *i.e. there is only one orbit and for all* $z, w \in \mathbb{H}$, *there exists some* $g \in \mathrm{PSL}_2(\mathbb{R})$ *such that* $gz = w$.

Exercise 3.1.4. *Show that the map*

$$z \mapsto \frac{z - i}{z + i}$$

© Springer Science+Business Media Singapore 2016 and Hindustan Book Agency 2016

M.R. Murty et al., *Problems in the Theory of Modular Forms*, HBA Lecture Notes in Mathematics, DOI 10.1007/978-981-10-2651-5_3

is an analytic isomorphism of the upper half-plane $\mathbb{H}$ and the unit disc $D = \{w \in \mathbb{C} : |w| < 1\}$.

Theorem 3.1.5. *The group of analytic automorphisms of $\mathbb{H}$ is isomorphic to $\mathrm{PSL}_2(\mathbb{R})$.*

Proof. Let $\phi : \mathbb{H} \to \mathbb{H}$ be an analytic automorphism of $\mathbb{H}$. Write $\phi(i) = a + bi$ with $a, b \in \mathbb{R}$. Then the map

$$\psi(z) = \frac{\phi(z) - a}{b}$$

maps i to i. But now, let $h : \mathbb{H} \to D$ be given by

$$h(z) = \frac{z - i}{z + i} = \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix} z.$$

By Exercise 3.1.4, this is an analytic isomorphism of $\mathbb{H}$ onto the unit disc $D = \{z \in \mathbb{C} : |z| < 1\}$. Put $g(z) = (h \circ \psi \circ h^{-1})(z)$. Then $g : D \to D$ is analytic with $g(0) = 0$. By the Schwarz Lemma of complex analysis we deduce that $|g(z)| \leq |z|$. The Schwarz Lemma applied to the inverse map of g, namely $h \circ \psi^{-1} \circ h^{-1}$, shows that $|z| = |g^{-1}(g(z))| \leq |g(z)|$. Hence $|g(z)| = |z|$ and by the Schwarz Lemma $g(z) = e^{i\theta} z$ for some $\theta \in \mathbb{R}$. In particular,

$$g(z) = \begin{pmatrix} e^{i\theta} & 0 \\ 0 & 1 \end{pmatrix} z.$$

From the solution to Exercise 3.1.4,

$$h^{-1}(z) = \frac{i(1 + z)}{1 - z} = \begin{pmatrix} i & i \\ -1 & 1 \end{pmatrix} z,$$

and so

$$\begin{aligned} \psi(z) &= \left(h^{-1} \circ g \circ h\right)(z) \\ &= \begin{pmatrix} i & i \\ -1 & 1 \end{pmatrix} \begin{pmatrix} e^{i\theta} & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -i \\ 1 & i \end{pmatrix} z \\ &= \begin{pmatrix} ie^{i\theta} + i & e^{i\theta} - 1 \\ -e^{i\theta} + 1 & ie^{i\theta} + i \end{pmatrix} z. \end{aligned}$$

Matrices representing fractional linear transformations may be rescaled without changing the corresponding function. Multiply every entry by $e^{-i\theta/2}/2i$ to obtain

$$\psi(z) = \begin{pmatrix} \cos(\theta/2) & \sin(\theta/2) \\ -\sin(\theta/2) & \cos(\theta/2) \end{pmatrix} z.$$

Thus $\psi \in \mathrm{PSL}_2(\mathbb{R})$. On the other hand, $\phi(z)$ can be written as

$$\phi(z) = \begin{pmatrix} \sqrt{b} & \frac{a}{\sqrt{b}} \\ 0 & \frac{1}{\sqrt{b}} \end{pmatrix} \psi(z) = \begin{pmatrix} \sqrt{b} & \frac{a}{\sqrt{b}} \\ 0 & \frac{1}{\sqrt{b}} \end{pmatrix} \begin{pmatrix} \cos\left(\frac{\theta}{2}\right) & \sin\left(\frac{\theta}{2}\right) \\ -\sin\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right) \end{pmatrix} z,$$

and hence $\phi \in \mathrm{PSL}_2(\mathbb{R})$. □

If G is a topological group, a subgroup Γ of G is called a *discrete subgroup* if any sequence of elements $\gamma_n \in \Gamma$ with $\gamma_n \to 1$ is eventually constant. Alternately, every neighborhood of the identity contains only finitely many elements of Γ.

Exercise 3.1.6. *Show that* $\mathrm{SL}_2(\mathbb{Z})$ *is a discrete subgroup of* $\mathrm{SL}_2(\mathbb{R}) \subset \mathbb{R}^4$*, where* $\mathbb{R}^4$ *is endowed with the Euclidean topology.*

3.2 Fundamental domains

If Γ is a subgroup of $\mathrm{SL}_2(\mathbb{Z})$ and $\mathcal{F} \subset \mathbb{H}$ is a closed set with connected interior, we say that $\mathcal{F}$ is a *fundamental domain* for Γ (or $\Gamma\backslash\mathbb{H}$) if

a) any $z \in \mathbb{H}$ is Γ-equivalent to a point in $\mathcal{F}$;
b) no two interior points of $\mathcal{F}$ are Γ-equivalent;
c) the boundary of $\mathcal{F}$ is a finite union of smooth curves.

Example 3.2.1. Let Γ_∞ be the subgroup of $\mathrm{SL}_2(\mathbb{Z})$ given by

$$\Gamma_\infty = \left\{ \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} : n \in \mathbb{Z} \right\}.$$

A fundamental domain for $\Gamma_\infty\backslash\mathbb{H}$ is given by

$$\mathcal{F} = \left\{ z \in \mathbb{H} : -\frac{1}{2} \leq \mathrm{Re}(z) \leq \frac{1}{2} \right\}.$$

Indeed, given any $z \in \mathbb{H}$, we can apply a transformation γ of Γ_∞ to ensure $\gamma z \in \mathcal{F}$. If two interior points z, z' of $\mathcal{F}$ were equivalent, then $z' = z + n$ for some integer n. But then $\mathrm{Re}(z') = \mathrm{Re}(z) + n$ forces $n = 0$. Finally, the boundary of $\mathcal{F}$ in $\mathbb{H}$ consists of the two vertical lines $\mathrm{Re}(z) = -1/2$ and $\mathrm{Re}(z) = 1/2$, both of which are smooth curves.

We will now construct what is called the *standard fundamental domain* for the action of $\mathrm{SL}_2(\mathbb{Z})$ on $\mathbb{H}$.

Theorem 3.2.2. *Let*

$$\mathcal{F} = \left\{ z \in \mathbb{H} : |\mathrm{Re}(z)| \leq \frac{1}{2},\ |z| \geq 1 \right\}.$$

Then $\mathcal{F}$ *is a fundamental domain for the action of* $\mathrm{SL}_2(\mathbb{Z})$ *on* $\mathbb{H}$.

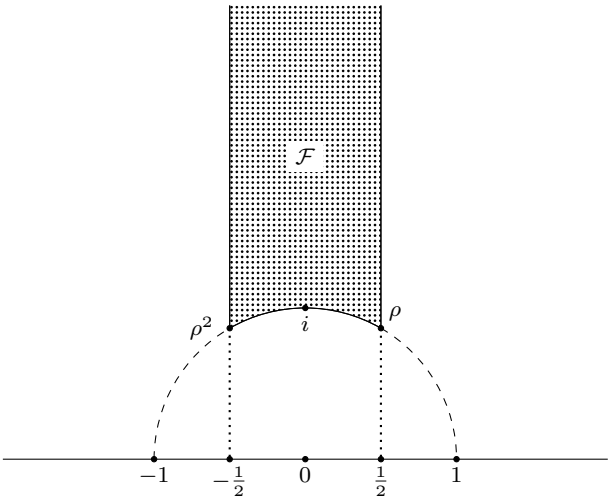


Fig. 3.1 Fundamental Domain for $\mathrm{SL}_2(\mathbb{Z})$

To prove this, we will first need the following lemma.

Lemma 3.2.3. *Fix $z \in \mathbb{H}$. The set of $(c,d) \in \mathbb{Z} \times \mathbb{Z}$ such that $(c,d) \neq (0,0)$ and $|cz+d| \leq 1$ is finite and non-empty.*

Proof. We can write z as $x+iy$. We know that $|cz+d| \leq 1$ if and only if $(cx+d)^2 + c^2y^2 \leq 1$, which implies that $c^2y^2 \leq 1$. Equivalently, $|c| \leq 1/y$, so there are only finitely many possibilities for c.

On the other hand, since $|cx+d| \leq 1$, we have $-1 \leq cx+d \leq 1$, implying that $-1-cx \leq d \leq 1-cx$. Thus we conclude there are only finitely many possibilities for d.

Finally, it is clear that $(c,d) = (0,1)$ satisfies the requirements. □

Proof of Theorem 3.2.2. The boundary of $\mathcal{F}$ is clearly a union of smooth curves and so condition (c) in the definition of a fundamental domain is satisfied. We need only check (a) and (b). To show (a), we fix $z \in \mathbb{H}$ and observe that for any given

$$\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}),$$

we have

$$\mathrm{Im}(\gamma z) = \frac{\mathrm{Im}(z)}{|cz+d|^2}. \tag{3.2}$$

Since $c, d \in \mathbb{Z}$ and are not both zero, the function $|cz+d|^2$ attains a positive minimum as γ ranges over elements of $\mathrm{SL}_2(\mathbb{Z})$. In fact, by Lemma 3.2.3 above, there are only finitely many $c, d \in \mathbb{Z}$ such that $|cz+d| \leq 1$. Thus we may choose $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ so that $|cz+d|$ is minimal and thus $\mathrm{Im}\,(\gamma z)$ is maximal. By translating z by an integer n (which is equivalent to $T^n z$), we may

suppose that $|\operatorname{Re}(z)| \leq 1/2$. We claim $|\gamma z| \geq 1$ so that $\gamma z \in \mathcal{F}$. Indeed, if we suppose otherwise, writing $w = \gamma z$, we have $|w| < 1$ so that $Sw = -1/w$ is such that $\operatorname{Im}(Sw) = \frac{\operatorname{Im}(w)}{|w|^2} > \operatorname{Im}(w)$. This means $\operatorname{Im}(S\gamma z) > \operatorname{Im}(\gamma z)$, which is a contradiction to our choice of γ. This proves that every element of $\mathbb{H}$ is $\mathrm{SL}_2(\mathbb{Z})$-equivalent to some point in $\mathcal{F}$.

It only remains to prove (b), i.e. to show that no two interior points of $\mathcal{F}$ are $\mathrm{SL}_2(\mathbb{Z})$-equivalent. Indeed, suppose that $z, w \in \mathcal{F}$ are interior points and that $\gamma z = w$ for some $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$. Without loss of generality, we may suppose that $\operatorname{Im}(w) \geq \operatorname{Im}(z)$. This implies $|cz + d| \leq 1$. In particular, $|\operatorname{Im}(cz + d)| = |c| \operatorname{Im}(z) \leq 1$. Since $\operatorname{Im}(z) \geq \sqrt{3}/2$ for all $z \in \mathcal{F}$, this means that $|c| \leq 2/\sqrt{3}$. Thus $c = 0$ or $c = \pm 1$.

If $c = 0$, then $ad = 1$. Multiplying γ by $-I$ if necessary, we may suppose $a = d = 1$. Thus γ is a power of T and it is clear that if $z \in \mathcal{F}$, then $\gamma z \notin \mathcal{F}$ unless z lies on the boundary. If $c = \pm 1$, we may multiply γ by $-I$ if necessary and assume $c = 1$. Then, if $z = x + iy$, we have

$$1 \geq |cz + d|^2 = (x + d)^2 + y^2 > (x + d)^2 + \frac{3}{4}$$

so that $|x+d| < 1/2$. Since $|x| < 1/2$, this means $d = 0$. But now $1 \geq |cz+d| = |z|$ which implies that z is not an interior point of $\mathcal{F}$. □

Exercise 3.2.4. *Suppose that $z, w \in \mathcal{F}$ are distinct points which are $\mathrm{SL}_2(\mathbb{Z})$-equivalent. Show that $\operatorname{Im}(z) = \operatorname{Im}(w)$ and $\operatorname{Re}(z) = -\operatorname{Re}(w)$.*

Exercise 3.2.5. *If $\mathcal{F}$ is the standard fundamental domain for $\mathrm{SL}_2(\mathbb{Z})$ and $\gamma \in \mathrm{SL}_2(\mathbb{Z})$, show that $\gamma\mathcal{F}$ is again a fundamental domain for $\mathrm{SL}_2(\mathbb{Z})$.*

One can prove that any discrete subgroup Γ of $\mathrm{SL}_2(\mathbb{R})$ has a fundamental domain (see Theorem 1.6.2 of [22]). The next theorem allows us to construct fundamental domains in certain cases.

Theorem 3.2.6. *Let $\mathcal{F}$ be a fundamental domain for $\mathrm{SL}_2(\mathbb{Z})$ and let Γ be a congruence subgroup with $-I \in \Gamma$. If $g_1, \ldots, g_r$ are coset representatives of Γ such that $\mathcal{D} = \bigcup_{i=1}^{r} g_i^{-1}\mathcal{F}$ has a connected interior, then $\mathcal{D}$ is a fundamental domain for Γ.*

Proof. It is clear that the boundary of each $g_i^{-1}\mathcal{F}$ consists of a finite number of smooth curves so that condition (c) in the definition of a fundamental domain is satisfied. We need to verify (a) and (b). Now any $z \in \mathbb{H}$ is $\mathrm{SL}_2(\mathbb{Z})$-equivalent to some point $w \in \mathcal{F}$. Thus $w = \tau z$ with $\tau \in \mathrm{SL}_2(\mathbb{Z})$. We can write $\tau = g_i\gamma$ for some i and $\gamma \in \Gamma$. Hence $\gamma z \in g_i^{-1}\mathcal{F}$ so that $\gamma z \in \mathcal{D}$ and (a) is satisfied.

Now suppose two interior points of $\mathcal{D}$ are Γ-equivalent. In other words, there is some $\gamma \in \Gamma$, and some interior point $z \in \mathcal{D}$ such that $\gamma z \in \mathcal{D}$ is also an interior point. The action of γ is a homeomorphism and so there exists an open set $\mathcal{U} \subset \mathcal{D}$ containing z such that $\gamma z \in \gamma\mathcal{U} \subset \mathcal{D}$. By the construction of $\mathcal{D}$, there exist indices $1 \leq i, j \leq r$ such that $g_i z, g_j\gamma z \in \mathcal{F}$. Let $\mathcal{V} = g_i\mathcal{U} \cap \mathcal{F}^\circ$,

where $\mathcal{F}^\circ$ denotes the interior of $\mathcal{F}$. The set $\mathcal{V}$ is open and non-empty. Take any $x \in \mathcal{V}$. Then

$$\left(\gamma g_i^{-1}\right) x \in \gamma g_i^{-1} \mathcal{V} \subseteq \gamma \mathcal{U} \subset \mathcal{D}.$$

Let k be such that $\left(\gamma g_i^{-1}\right) x \in g_k^{-1} \mathcal{F}$. Now x and $g_k \gamma g_i^{-1} x$ are $\mathrm{SL}_2(\mathbb{Z})$-equivalent interior points of $\mathcal{F}$. Hence they are equal. The map $w \mapsto g_k \gamma g_i^{-1} w$ is an automorphism of $\mathbb{H}$. Thus, there is an open set $\mathcal{W} \subset \mathcal{F}^\circ$ containing x such that $g_k \gamma g_i^{-1} \mathcal{W} \subset \mathcal{F}^\circ$. But then every element of $\mathcal{W}$ is $\mathrm{SL}_2(\mathbb{Z})$-equivalent to some element of $\mathcal{F}^\circ$. This can only happen if $g_k \gamma g_i^{-1}$ acts trivially on $\mathcal{W}$. As any two holomorphic functions agreeing on an open set must agree everywhere, we deduce that $g_k \gamma g_i^{-1} = \pm I$ by Exercise 3.1.2. Now

$$g_i \Gamma = g_k \gamma \{\pm I\} \Gamma = g_k \Gamma.$$

As the g_i are distinct coset representatives, we conclude that $g_i = g_k$ and so $\gamma = \pm I$. In particular, $z = \gamma z$. This proves (b). □

Exercise 3.2.7. *Show that $I, S, T^{-1}S$ are coset representatives for $\Gamma_0(2)$ in $\mathrm{SL}_2(\mathbb{Z})$. Deduce that a fundamental domain for $\Gamma_0(2)$ is given by $\mathcal{F} \cup S\mathcal{F} \cup ST\mathcal{F}$, where $\mathcal{F}$ is the standard fundamental domain for $\mathrm{SL}_2(\mathbb{Z})$.*

3.3 The extended upper half-plane

We define the extended upper half-plane $\mathbb{H}^\star$ as

$$\mathbb{H}^\star = \mathbb{H} \cup \mathbb{Q} \cup \{i\infty\}.$$

That is, $\mathbb{H}^\star$ is obtained by adjoining all the rational numbers and $i\infty$, called the *cusps*. We topologize $\mathbb{H}^\star$ in the following way. For $z \in \mathbb{H}$, we take the usual fundamental system of neighborhoods. For a cusp $s \neq i\infty$, we take all sets of the form

$$\{s\} \cup C^\circ,$$

where C° is the interior of a circle in $\mathbb{H}$ tangent to the real axis at s. If $s = i\infty$, we take all sets of the form

$$\{i\infty\} \cup \{z \in \mathbb{H} : \mathrm{Im}(z) > c\},$$

for all $c > 0$, as a fundamental system of open neighborhoods of $i\infty$. This makes $\mathbb{H}^\star$ into a Hausdorff space.

We define an action of $\mathrm{SL}_2(\mathbb{Z})$ on $\mathbb{H}^\star$ as follows. For $z \in \mathbb{H}$, the action is defined via the usual formula

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} z = \frac{az+b}{cz+d}.$$

For the cusps $\mathbb{Q} \cup \{i\infty\}$, we identify $i\infty$ with $\frac{1}{0}$ and define

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \frac{r}{s} = \frac{ar+bs}{cr+ds}.$$

The reader may verify directly that this is indeed an action.

Exercise 3.3.1. *Show that every cusp is* $\mathrm{SL}_2(\mathbb{Z})$*-equivalent to* $i\infty$*.*

For any subgroup Γ of $\mathrm{SL}_2(\mathbb{Z})$ of finite index, we can define a topology on the orbit space $\Gamma\backslash\mathbb{H}^\star$. Indeed, we have the natural map

$$\pi : \mathbb{H}^\star \longrightarrow \Gamma\backslash\mathbb{H}^\star,$$

sending each point $z \in \mathbb{H}^\star$ to its orbit under Γ. We impose the strongest topology to ensure π is continuous. In other words, the open sets of $\Gamma\backslash\mathbb{H}^\star$ are

$$\{U \subseteq \Gamma\backslash\mathbb{H}^\star : \pi^{-1}(U) \text{ is open in } \mathbb{H}^\star\}.$$

This makes $\Gamma\backslash\mathbb{H}^\star$ into a Hausdorff space (see Theorem 1.28 of [37], p. 12). If $\Gamma = \mathrm{SL}_2(\mathbb{Z})$, then $\Gamma\backslash\mathbb{H}^\star$ has the complex analytic structure of the Riemann sphere. More generally, if Γ is a congruence subgroup, $\Gamma\backslash\mathbb{H}^\star$ is a compact Riemann surface.

Exercise 3.3.2. *If* Γ *is of finite index in* $\mathrm{SL}_2(\mathbb{Z})$*, show that the number of* Γ*-inequivalent cusps is* $\leq [\mathrm{SL}_2(\mathbb{Z}) : \Gamma]$*.*

Exercise 3.3.3. *Show that* $\{0, i\infty\}$ *is a complete set of inequivalent cusps of* $\Gamma_0(2)$*.*

Exercise 3.3.4. *With the action of* $\mathrm{SL}_2(\mathbb{Z})$ *on* $\mathbb{H}^\star$*, show that the stabilizer* $\Gamma_{i\infty}$ *of* $i\infty$ *is given by*

$$\left\{ \pm \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} : n \in \mathbb{Z} \right\}.$$

Exercise 3.3.5. *Let* p *be prime. Show that the matrices*

$$ST^k = \begin{pmatrix} 0 & -1 \\ 1 & k \end{pmatrix}, \qquad 0 \leq k \leq p-1,$$

along with $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$*, give a complete set of right coset representatives for* $\Gamma_0(p)$ *in* $\mathrm{SL}_2(\mathbb{Z})$*.*

Exercise 3.3.6. *Let* p *be prime. Show that the set of matrices*

$$\begin{pmatrix} k & 1 \\ -1 & 0 \end{pmatrix}, \qquad 0 \leq k \leq p-1,$$

along with $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$*, is a complete set of left coset representatives for* $\Gamma_0(p)$*.*

Exercise 3.3.7. *Let* p *be prime. Show that a complete set of inequivalent cusps for* $\Gamma_0(p)$ *is given by* $\{0, i\infty\}$*.*

3.4 The Poincaré metric

Since the topology on $\mathbb{H}$ is the subspace topology inherited from $\mathbb{C}$, and as $\mathbb{C}$ is a metric space with the usual metric, $\mathbb{H}$ is also a metric space. However, this metric cannot be lifted to $\Gamma\backslash\mathbb{H}$ with Γ being a subgroup of $\mathrm{PSL}_2(\mathbb{R})$ since it is not $\mathrm{PSL}_2(\mathbb{R})$ invariant. Poincaré defined a metric on $\mathbb{H}$ which is $\mathrm{SL}_2(\mathbb{R})$-invariant and lifts to $\Gamma\backslash\mathbb{H}$. The Poincaré metric on $\mathbb{H}$ is given by the distance function (also called the hyperbolic metric)

$$d(z,w) = \cosh^{-1}\left(1 + \frac{|z-w|^2}{2\,\mathrm{Im}(z)\,\mathrm{Im}(w)}\right),$$

where $z, w \in \mathbb{H}$ and $\cosh^{-1}$ is the inverse function of the hyperbolic cosine. Notice that this gives rise to the same topology on $\mathbb{H}$ as before. The fact that $d(z,w)$ is a metric on $\mathbb{H}$ is easily seen from simple differential geometric considerations which we shall not discuss here.

Exercise 3.4.1. *Prove that $d(z,w)$ is $\mathrm{SL}_2(\mathbb{R})$-invariant. That is, show that $d(gz, gw) = d(z,w)$ for all $g \in \mathrm{SL}_2(\mathbb{R})$.*

Exercise 3.4.2. *Show that for $x \geq 1$,*

$$\cosh^{-1} x = \log(x + \sqrt{x^2 - 1}).$$

Exercise 3.4.3. *Show that for $y > 0$,*

$$d(i, iy) = |\log y|.$$

One can also define an $\mathrm{SL}_2(\mathbb{R})$-invariant measure on $\mathbb{H}$ which lifts to $\Gamma\backslash\mathbb{H}^\star$ for any subgroup Γ of $\mathrm{SL}_2(\mathbb{R})$. This is given by

$$d\mu := \frac{dxdy}{y^2}.$$

Theorem 3.4.4. *The measure $d\mu$ is $\mathrm{SL}_2(\mathbb{R})$-invariant on $\mathbb{H}$.*

Proof. We need to check that

$$d\mu(gz) = d\mu(z) \text{ for all } g \in \mathrm{SL}_2(\mathbb{R}).$$

Indeed, let $z = x + iy$ and $g = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{R})$ so that

$$\begin{aligned} gz &= \frac{(ad+bc)x + bd + ac(x^2+y^2) + iy}{c^2(x^2+y^2) + 2cdx + d^2} \\ &= u(x,y) + iv(x,y) \qquad \text{(say)}. \end{aligned}$$

By the change of variable formula in $\mathbb{R}^n$ (see p. 67 of [42]), we have

$$dudv = (detJ)dxdy,$$

where J is the Jacobian matrix

$$\begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix}.$$

A tedious calculation shows that $\det J = (1/y)^2$, from which the result follows. □

Exercise 3.4.5. *Let $\mathcal{F}$ be the standard fundamental domain for $\mathrm{SL}_2(\mathbb{Z})$. Show that*

$$\int\int_{\mathcal{F}} \frac{dxdy}{y^2} = \frac{\pi}{3}.$$

3.5 Supplementary problems

Exercise 3.5.1. *Show that a set of coset representatives for Γ_∞ in $\mathrm{SL}_2(\mathbb{Z})$ is given by*

$$\left\{ \begin{pmatrix} \star & \star \\ c & d \end{pmatrix} : (c,d) = 1, c, d \in \mathbb{Z} \right\},$$

where the asterisks indicate that we can take any a, b such that $ad - bc = 1$.

Exercise 3.5.2. *Let $\mathcal{F}$, Γ, $g_1, \ldots, g_r$ and $\mathcal{D}$ be as in Theorem 3.2.6. Show that*

$$\int\int_{\mathcal{D}} \frac{dxdy}{y^2} = [\mathrm{SL}_2(\mathbb{Z}) : \Gamma]\frac{\pi}{3}.$$

Exercise 3.5.3. *Let $G = \mathrm{SL}_2(\mathbb{R})$ act on $\mathbb{H}$. Show that K, the stabilizer of i, consists of matrices*

$$\begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \qquad \text{for } \theta \in \mathbb{R}.$$

Then show that the upper half-plane is homeomorphic to G/K.

Chapter 4
Modular Forms of Level One

4.1 Fourier expansions and q-series

We are interested in holomorphic functions $f(z) : \mathbb{H} \to \mathbb{C}$ for which

$$f\left(\frac{az+b}{cz+d}\right) = (cz+d)^k f(z), \qquad \forall \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}). \tag{4.1}$$

We will use the topology on the extended upper half-plane $\mathbb{H}^\star$ to define what it means for $f(z)$ to be holomorphic at the cusp $i\infty$.

Exercise 4.1.1. *Show that the punctured fundamental neighborhood*

$$U_c = \{z \in \mathbb{H} : \mathrm{Im}(z) > c\}$$

of $i\infty$ is mapped via $z \mapsto e^{2\pi i z}$ into the punctured open disc centered at zero, of radius $e^{-2\pi c}$.

The function $q(z) = e^{2\pi i z}$ from Exercise 4.1.1 can be extended (continuously) by sending $i\infty$ to 0. Hence, the fundamental neighborhoods of $i\infty$ are mapped to the fundamental neigborhoods of the origin.

Since $T = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$, condition (4.1) implies that $f(z+1) = f(z)$ for all $z \in \mathbb{H}$. Hence, there is a well-defined induced map, which we also call f, from the unit disc to $\mathbb{C}$:

$$q \mapsto f(z),$$

where $z \in \mathbb{H}$ is any point with $q = e^{2\pi i z}$. If $f(z) : \mathbb{H} \to \mathbb{C}$ is holomorphic, then $f(q)$ will be holomorphic on the punctured unit disc and hence will have a Laurent expansion centered at $q = 0$,

$$f(q) = \sum_{n=-\infty}^{\infty} a_n q^n,$$

which is called a *q-expansion*. We call

© Springer Science+Business Media Singapore 2016 and Hindustan Book Agency 2016

M.R. Murty et al., *Problems in the Theory of Modular Forms*, HBA Lecture Notes in Mathematics, DOI 10.1007/978-981-10-2651-5_4

$$f(z) = \sum_{n=-\infty}^{\infty} a_n e^{2\pi i z n}$$

the *Fourier series at* $i\infty$. We define the *order of* f *at* $i\infty$ to be the index of the first non-zero coefficient in the q-series:

$$v_{i\infty}(f) := \inf \{n \in \mathbb{Z} : a_n \neq 0\}.$$

If $-\infty < v_{i\infty}(f) < 0$, then f is said to be *meromorphic at* $i\infty$ with a pole of order $|v_{i\infty}(f)|$. If $v_{i\infty}(f) \geq 0$ then f is said to be *holomorphic at* $i\infty$. If $v_{i\infty}(f) > 0$ then we say that f *vanishes at* $i\infty$ and has a zero of order $v_{i\infty}(f)$.

A (holomorphic) *modular form of weight* $k \in \mathbb{Z}$ *for the full modular group* $\mathrm{SL}_2(\mathbb{Z})$ is a holomorphic function $f(z) : \mathbb{H} \to \mathbb{C}$ which satisfies (4.1) and which is holomorphic at $i\infty$. A modular form on $\mathrm{SL}_2(\mathbb{Z})$ is called a *cusp form* if it vanishes at the cusp $i\infty$, i.e. $a_0 = 0$ in the Fourier expansion at $i\infty$. In this chapter, we will write $f(i\infty) = a_0$. Unfortunately, the value at $i\infty$ will not be well-defined for the modular forms of higher level which we will meet in Chapter 6.

Exercise 4.1.2. *Let* $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{GL}_2^+(\mathbb{R})$ *and* $z \in \mathbb{H}$. *Define* $j(\gamma, z) = cz + d$. *Show that*

$$j(\gamma_1\gamma_2, z) = j(\gamma_1, \gamma_2 z) j(\gamma_2, z).$$

For $\gamma \in \mathrm{GL}_2^+(\mathbb{R})$, it is convenient to introduce the "slash" notation: for any holomorphic function f on $\mathbb{H}$,

$$(f|\gamma)(z) := (\det \gamma)^{k/2} j(\gamma, z)^{-k} f(\gamma z). \tag{4.2}$$

Then the modularity condition (4.1) is equivalent to

$$f|\gamma = f, \qquad \forall \gamma \in \mathrm{SL}_2(\mathbb{Z}).$$

Exercise 4.1.3. *Show that* $f|(\gamma_1\gamma_2) = (f|\gamma_1)|\gamma_2$ *for all* $\gamma_1, \gamma_2 \in \mathrm{GL}_2^+(\mathbb{R})$.

Exercise 4.1.4. *Prove that if* $f : \mathbb{H} \to \mathbb{C}$ *is holomorphic, satisfies*

(i) $f(z+1) = f(z)$
(ii) $f\left(-\frac{1}{z}\right) = z^k f(z)$

for all $z \in \mathbb{H}$, *and is holomorphic at* $i\infty$, *then* f *is a modular form of weight* k *for* $\mathrm{SL}_2(\mathbb{Z})$.

Let $M_k(\mathrm{SL}_2(\mathbb{Z}))$ denote the set of all modular forms of weight k on $\mathrm{SL}_2(\mathbb{Z})$. Let $S_k(\mathrm{SL}_2(\mathbb{Z})) \subseteq M_k(\mathrm{SL}_2(\mathbb{Z}))$ denote the set of cusp forms of weight k on $\mathrm{SL}_2(\mathbb{Z})$. The use of the letter S for the space of cusp forms comes from the German word "Spitzenform" which means cusp form.

Exercise 4.1.5. *Let* k_1, $k_2 \in \mathbb{Z}$. *Show that:*

(a) Both $M_{k_1}(\mathrm{SL}_2(\mathbb{Z}))$ and $S_{k_1}(\mathrm{SL}_2(\mathbb{Z}))$ are $\mathbb{C}$-vector spaces.
(b) If $f \in M_{k_1}(\mathrm{SL}_2(\mathbb{Z}))$ and $g \in M_{k_2}(\mathrm{SL}_2(\mathbb{Z}))$, then $fg \in M_{k_1+k_2}(\mathrm{SL}_2(\mathbb{Z}))$.
(c) If $f \in M_{k_1}(\mathrm{SL}_2(\mathbb{Z}))$ and $g \in S_{k_2}(\mathrm{SL}_2(\mathbb{Z}))$, then $fg \in S_{k_1+k_2}(\mathrm{SL}_2(\mathbb{Z}))$.

Later, we will see that $M_k(\mathrm{SL}_2(\mathbb{Z}))$ and $S_k(\mathrm{SL}_2(\mathbb{Z}))$ are finite-dimensional vector spaces. Consequently, methods of linear algebra can be applied to study them and to construct "nice" bases for $M_k(\mathrm{SL}_2(\mathbb{Z}))$ and $S_k(\mathrm{SL}_2(\mathbb{Z}))$. The previous exercise also allows us to speak about a graded $\mathbb{C}$-algebras of holomorphic modular forms and holomorphic cusp forms.

Exercise 4.1.6. *Show that $M_k(\mathrm{SL}_2(\mathbb{Z})) = \{0\}$ for k odd.*

4.2 Eisenstein series for $\mathrm{SL}_2(\mathbb{Z})$

We begin with a very simple observation. Suppose we are given a binary quadratic form

$$Q(m,n) = Am^2 + 2Bmn + Cn^2, \qquad A, B, C \in \mathbb{R},$$

which is positive definite. That is, $A > 0$, and $AC - B^2 > 0$. We may write $Q(m,n)$ in matrix form:

$$Q(m,n) = (m,n) \begin{pmatrix} A & B \\ B & C \end{pmatrix} \begin{pmatrix} m \\ n \end{pmatrix}.$$

Since the matrix $\left(\begin{smallmatrix} A & B \\ B & C \end{smallmatrix}\right)$ is a real symmetric matrix, we can diagonalize it. Thus, we can find an orthogonal matrix P such that

$$P \begin{pmatrix} A & B \\ B & C \end{pmatrix} P^t = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix},$$

and if we put

$$\begin{pmatrix} u \\ v \end{pmatrix} = P \begin{pmatrix} m \\ n \end{pmatrix},$$

then we can write $Q(m,n) = \lambda_1 u^2 + \lambda_2 v^2$. Since Q is positive definite, the eigenvalues λ_1, λ_2 are real and positive. We see that

$$Q(m,n) \geq \lambda(u^2 + v^2),$$

where

$$\lambda = \min(\lambda_1, \lambda_2) > 0.$$

Moreover, P is an orthogonal transformation and thus preserves length. Consequently, $u^2 + v^2 = m^2 + n^2$. This proves that

$$Q(m,n) \geq \lambda(m^2 + n^2)$$

for some $\lambda > 0$. If z is a complex number, then $|mz+n|^2$ is a positive definite quadratic form in m, n.

These elementary observations from linear algebra allow us to construct an important family of modular forms. For $k > 2$, define

$$G_k(z) = \sum_{m,n}{}' \frac{1}{(mz+n)^k}, \qquad z \in \mathbb{H},$$

where the summation is over all $(m, n) \in \mathbb{Z}^2$, with $(m, n) \neq (0, 0)$. The dash on the summation indicates this latter condition.

Exercise 4.2.1. *Show that the series for $G_k(z)$ converges absolutely for all $z \in \mathbb{H}$ and for $k > 2$.*

By the Weierstrass M-test, we see that $G_k(z)$ defines a holomorphic function for $z \in \mathbb{H}$. Its behavior at $i\infty$ is easily seen to come from the terms when $m = 0$. Thus

$$G_k(i\infty) = \sum_{\substack{n\in\mathbb{Z}\\ n\neq 0}} \frac{1}{n^k}.$$

Observe that $G_k(z)$ is identically zero if k is odd as is easily seen by pairing up the summands corresponding to (m, n) and $(-m, -n)$. However, for k even, $G_k(z)$ is a non-trivial function and

$$G_k(i\infty) = 2\zeta(k),$$

where $\zeta(s)$ denotes the Riemann zeta function. In addition, $G_k(z)$ is a modular form of weight k since

$$\begin{aligned} G_k(z+1) &= G_k(z), \\ G_k\left(-\frac{1}{z}\right) &= z^k G_k(z), \end{aligned}$$

as is easily verified. Indeed,

$$G_k(z+1) = \sum_{m,n}{}' \frac{1}{(mz+m+n)^k},$$

and as (m, n) ranges over elements of $\mathbb{Z}^2 \setminus \{(0, 0)\}$, so does $(m, m + n)$. Similarly,

$$G_k\left(-\frac{1}{z}\right) = z^k \sum_{m,n}{}' \frac{1}{(nz-m)^k} = z^k G_k(z)$$

since $(n, -m)$ ranges over elements of $\mathbb{Z}^2 \setminus \{(0, 0)\}$ as (m, n) ranges over $\mathbb{Z}^2 \setminus \{(0, 0)\}$. By Exercise 4.1.4, we see that $G_k(z)$ is a modular form of weight k for $\mathrm{SL}_2(\mathbb{Z})$.

To study the behavior of $G_k(z)$ near $i\infty$, we now derive its q-expansion. Recall

$$\sin \pi z = \pi z \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right).$$

Taking the logarithmic derivative gives

$$\pi \cot \pi z = \frac{1}{z} + \sum_{n=1}^{\infty} \left(\frac{1}{z+n} + \frac{1}{z-n}\right), \tag{4.3}$$

which is easily verified to be an absolutely convergent series. On the other hand, for any $z \in \mathbb{H}$ we have

$$\begin{aligned} \pi \cot \pi z = \pi \frac{\cos \pi z}{\sin \pi z} &= \pi i \frac{e^{i\pi z} + e^{-i\pi z}}{e^{i\pi z} - e^{-i\pi z}} \\ &= \pi i \frac{q+1}{q-1} = \pi i - \frac{2\pi i}{1-q} = \pi i - 2\pi i \sum_{n=0}^{\infty} q^n, \end{aligned} \tag{4.4}$$

where $q = e^{2\pi i z}$. By successive differentiation of (4.3), we deduce:

Theorem 4.2.2 (Lipschitz formula). *For $k \geq 2$ and for $q = e^{2\pi i z}$ where $z \in \mathbb{H}$, we have*

$$\sum_{n \in \mathbb{Z}} \frac{1}{(z+n)^k} = \frac{(-2\pi i)^k}{(k-1)!} \sum_{n=1}^{\infty} n^{k-1} q^n.$$

This allows us to prove:

Theorem 4.2.3. *For every even $k \geq 4$,*

$$G_k(z) = 2\zeta(k) + \frac{2(-2\pi i)^k}{(k-1)!} \sum_{n=1}^{\infty} \sigma_{k-1}(n) q^n,$$

where $q = e^{2\pi i z}$ and $\sigma_s(n) = \sum_{d|n} d^s$.

Proof. We have, by separating $m = 0$ from the sum defining $G_k(z)$,

$$G_k(z) = 2\zeta(k) + 2 \sum_{m=1}^{\infty} \sum_{n \in \mathbb{Z}} \frac{1}{(mz+n)^k}.$$

We can expand the inner sum using the Lipschitz formula (Theorem 4.2.2):

$$G_k(z) = 2\zeta(k) + 2\frac{(-2\pi i)^k}{(k-1)!} \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} n^{k-1} q^{mn}.$$

Collecting terms for which $mn = r$, we see the coefficient of q^r is $\sigma_{k-1}(r)$.

□

Exercise 4.2.4. *Define the Bernoulli numbers B_k by the power series*

$$\frac{x}{e^x - 1} = \sum_{k=0}^{\infty} \frac{B_k x^k}{k!},$$

so that $B_0 = 1$, $B_1 = -\frac{1}{2}$, $B_2 = \frac{1}{6}$, $B_3 = 0$, $B_4 = -\frac{1}{30}$, ... Show that

$$2\zeta(2k) = -\frac{(2\pi i)^{2k} B_{2k}}{(2k)!}, \qquad k \geq 1.$$

Exercise 4.2.5. *Prove that the Bernoulli numbers satisfy the recursion formula:*

$$(m+1)B_m = -\sum_{k=0}^{m-1} \binom{m+1}{k} B_k.$$

Exercise 4.2.6. *Define the n-th Bernoulli polynomial by*

$$B_n(x) = \sum_{k=0}^{n} \binom{n}{k} B_k x^{n-k}.$$

Prove that

$$\frac{te^{xt}}{e^t - 1} = \sum_{n=0}^{\infty} B_n(x) \frac{x^n}{n!}.$$

Using Exercise 4.2.4, we obtain:

Corollary 4.2.7. *For every even $k \geq 4$,*

$$E_k(z) = 1 - \frac{2k}{B_k} \sum_{n=1}^{\infty} \sigma_{k-1}(n) q^n$$

is a modular form of weight k for the full modular group $\mathrm{SL}_2(\mathbb{Z})$ satisfying $E_k(i\infty) = 1$.

Proof. Since $G_k(z) = 2\zeta(k)E_k(z)$, we can use Exercise 4.2.4 and Theorem 4.2.3 and the result is immediate. □

Here are some examples:

$$E_4(z) = 1 + 240 \sum_{n=1}^{\infty} \sigma_3(n) q^n, \qquad E_{10}(z) = 1 - 264 \sum_{n=1}^{\infty} \sigma_9(n) q^n,$$

$$E_6(z) = 1 - 504 \sum_{n=1}^{\infty} \sigma_5(n) q^n, \qquad E_{12}(z) = 1 + \frac{65520}{691} \sum_{n=1}^{\infty} \sigma_{11}(n) q^n,$$

$$E_8(z) = 1 + 480 \sum_{n=1}^{\infty} \sigma_7(n) q^n, \qquad E_{14}(z) = 1 - 24 \sum_{n=1}^{\infty} \sigma_{13}(n) q^n.$$

Exercise 4.2.8. *Prove that* $M_k(\mathrm{SL}_2(\mathbb{Z})) = \mathbb{C}E_k \oplus S_k(\mathrm{SL}_2(\mathbb{Z}))$ *for all* $k \geq 4$. *Deduce that* $\dim M_k(\mathrm{SL}_2(\mathbb{Z})) = 1 + \dim S_k(\mathrm{SL}_2(\mathbb{Z}))$.

It is easily verified that

$$\Delta := \frac{E_4^3 - E_6^2}{1728}$$

is a non-trivial cusp form of weight 12 for the full modular group. The Fourier coefficients of Δ define the celebrated Ramanujan τ-function:

$$\Delta(z) = \sum_{n=1}^{\infty} \tau(n) q^n = q - 24q^2 + 252q^3 + \cdots .$$

Later, we will see that

$$\Delta(z) = q \prod_{n=1}^{\infty} (1 - q^n)^{24}.$$

4.3 The valence formula

Given a meromorphic function $f : \mathbb{H} \to \mathbb{C}$, not identically zero, and $z_0 \in \mathbb{H}$, there is a unique integer n such that $f(z)/(z - z_0)^n$ is holomorphic and non-zero at z_0. We call n the *order* of f at z_0 and denote it by $v_{z_0}(f)$. If f is holomorphic, then $v_{z_0}(f)$ is the order of the zero of f at z_0.

If f is a modular form of weight k for the full modular group, the identity

$$f(z) = (cz + d)^{-k} f\left(\frac{az + b}{cz + d}\right)$$

shows that $v_{z_0}(f) = v_{gz_0}(f)$ for all $g \in \mathrm{SL}_2(\mathbb{Z})$. In other words, $v_{z_0}(f)$ depends only on the orbit of z_0 under $\mathrm{SL}_2(\mathbb{Z})$ so that we need only study $v_z(f)$ for z in a fundamental domain of $\mathrm{SL}_2(\mathbb{Z})$. Recall that we have defined $v_{i\infty}(f)$ to be the order of the zero at $q = 0$ in the q-expansion

$$f(z) = \sum_{n=0}^{\infty} a_n q^n.$$

That is, $v_{i\infty}(f)$ is the smallest value of n such that $a_n \neq 0$.

Let us note that $f \in M_k(\mathrm{SL}_2(\mathbb{Z}))$ has only a finite number of zeros in the standard fundamental domain $\mathcal{F}$ since f being holomorphic on $\mathbb{H}$ implies that the zeros are isolated. Moreover, for some $c > 0$, the region $\{z \in \mathcal{F} : \mathrm{Im}(z) > c\} \cup \{i\infty\}$, being a fundamental neighborhood of $i\infty$, contains no zero of f except possibly $i\infty$. All remaining zeros are contained

in the compact region $\{z \in \mathcal{F} : \mathrm{Im}(z) \leq c\}$ and this number is finite (see, for example, Theorem 10.18 on page 208 of [33]).

Theorem 4.3.1 (The valence formula). *Let f be a modular form, not identically zero, of weight k for the full modular group $\mathrm{SL}_2(\mathbb{Z})$. Then*

$$v_{i\infty}(f) + \frac{1}{2}v_i(f) + \frac{1}{3}v_{\rho^2}(f) + {\sum_{\substack{z \neq i, \rho^2 \\ z \in \mathcal{F}}}}' v_z(f) = \frac{k}{12},$$

where the primed summation excludes points with $\mathrm{Re}\, z = \frac{1}{2}$ and points with both $|z| = 1$ and $\mathrm{Re}\, z > 0$. Here $i = \sqrt{-1}$ and $\rho = e^{\frac{\pi i}{3}}$.

We use primed summation in the valence formula to account for the fact that half of the boundary of $\mathcal{F}$ is $\mathrm{SL}_2(\mathbb{Z})$-equivalent to the other half.

Proof. We first give the proof in the case that f has no zeros on the boundary of the standard fundamental domain $\mathcal{F}$ in $\mathbb{H}$. Then, we indicate how to modify the proof so as to treat the case when some zeros lie on the boundary.

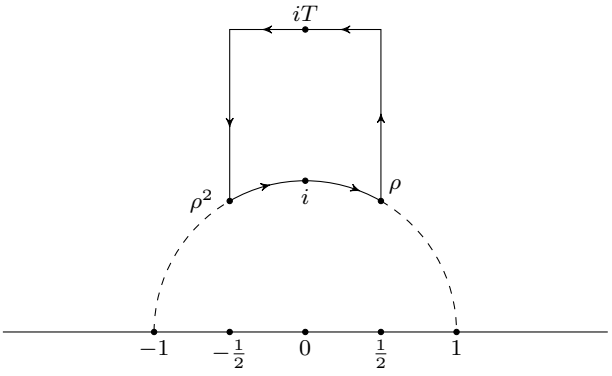


Fig. 4.1 Contour without zeros on $\partial\mathcal{F}$

We choose T large enough so that all zeros z_0 of f have $\mathrm{Im}(z_0) \leq T$, apart from possibly $z_0 = i\infty$. We integrate f'/f around the contour $\mathcal{C}$ indicated in Figure 4.1 . By Cauchy's argument principle,

$$\frac{1}{2\pi i}\int_{\mathcal{C}} \frac{f'(z)}{f(z)}dz = \sum_{z \in \mathcal{F}} v_z(f)$$

The contour has several segments. Since $f(z+1) = f(z)$, the two vertical segments cancel each other. The horizontal segment maps (under $z \mapsto q = e^{2\pi i z}$) into the circle of radius $e^{-2\pi T}$ about $q = 0$ oriented clockwise and thus

contributes $-v_{i\infty}(f)$ to the integral. The map $z \mapsto -1/z$ maps the arc from ρ^2 to i to the arc from ρ to i. Since

$$f\left(-\frac{1}{z}\right) = z^k f(z),$$

we find upon logarithmic differentiation (actually, logarithmic differentialization!) that

$$\frac{f'\left(-\frac{1}{z}\right)}{f\left(-\frac{1}{z}\right)} d\left(-\frac{1}{z}\right) = \left(\frac{k}{z} + \frac{f'(z)}{f(z)}\right) dz.$$

Hence the integral along the circular arc from ρ^2 to ρ is

$$\begin{aligned}\frac{1}{2\pi i}\left[\int_{\rho^2}^{i} \frac{f'(z)}{f(z)}dz + \int_{i}^{\rho} \frac{f'(z)}{f(z)}dz\right] &= \frac{1}{2\pi i}\left[\int_{\rho}^{i} \left(\frac{k}{z} + \frac{f'(z)}{f(z)}\right) dz + \int_{i}^{\rho} \frac{f'(z)}{f(z)}dz\right] \\ &= \frac{k}{2\pi i}\int_{\rho}^{i} \frac{dz}{z}.\end{aligned}$$

But this is easily parametrized by $z = e^{i\theta}$:

$$\frac{k}{2\pi i}\int_{\rho}^{i} \frac{dz}{z} = \frac{k}{2\pi i}\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{d(e^{i\theta})}{e^{i\theta}} = \frac{k}{2\pi}\left(\frac{\pi}{2} - \frac{\pi}{3}\right) = \frac{k}{12}.$$

Hence

$$\sum_{z\in\mathcal{F}} v_z(f) = -v_{i\infty}(f) + \frac{k}{12}.$$

This gives the result in the case that f has no zeros on the boundary of $\mathcal{F}$. We now indicate how to modify this proof to allow for zeros on the boundary.

We modify $\mathcal{C}$ to create $\mathcal{C}'$ as follows. If there is a zero z_1 with $\operatorname{Re} z_1 = -1/2$ and $z_1 \neq \rho^2$ then we deform the contour as indicated in Figure 4.2. The arc around Tz_1 is the transform by T of the arc around z_1. If there is a zero z_2 with $|z_2| = 1$ and $-\frac{1}{2} < \operatorname{Re} z_2 < 0$, then we deform the contour as indicated in Figure 4.3. The arc around Sz_2 is the image under S of the arc around z_2. We make as many such deformations as are necessary. Finally, we deform the contour near the points ρ^2, i, and ρ so that it moves along circular arcs of small enough radius $\epsilon > 0$. See Figure 4.4. The argument principle now gives

$$\frac{1}{2\pi i}\int_{\mathcal{C}'} \frac{f'(z)}{f(z)}dz = {\sum_{\substack{z\neq i,\rho^2 \\ z\in\mathcal{F}}}}' v_z(f).$$

The modified vertical contours still cancel each other since their orientations are opposite. Moreover, the arcs along $|z| = 1$ including the deformations around points z_2 and Sz_2, but excluding the arcs around ρ^2, i, and ρ, will contribute $k/12$ once we take the limit $\epsilon \to 0$. To determine the contributions

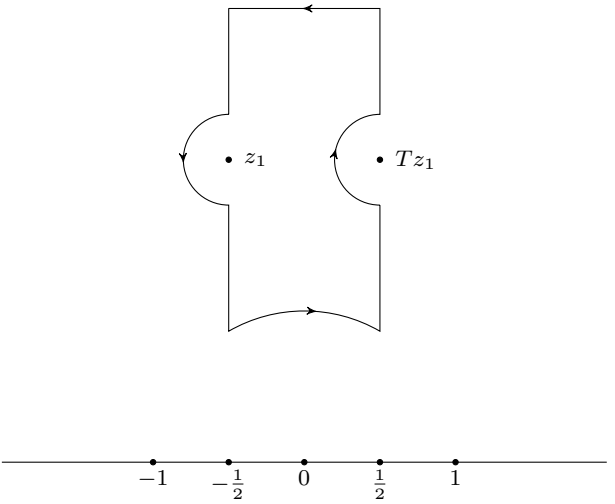


Fig. 4.2 Contour with zeros on vertical edge of $\mathcal{F}$

of the arcs centered at ρ^2, i, and ρ, we take the limit $\epsilon \to 0$ and proceed as follows.

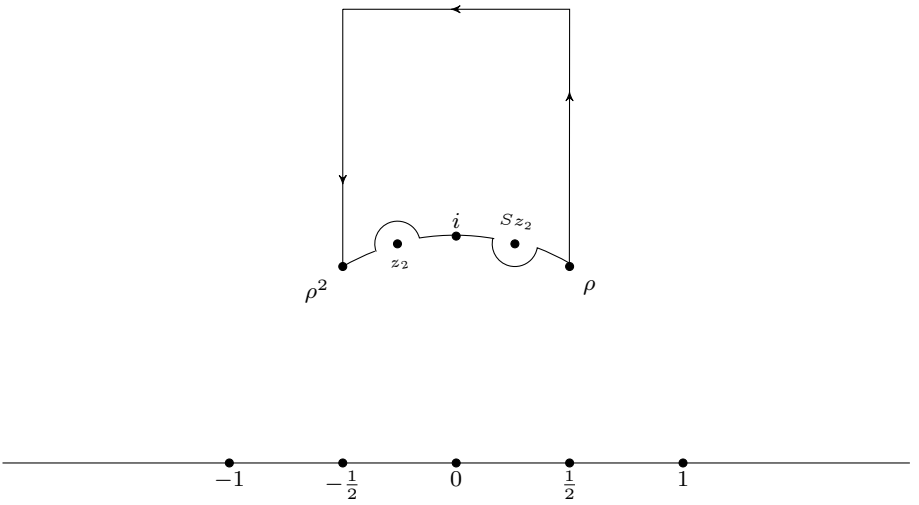


Fig. 4.3 Contour with zeros on lower edge of $\mathcal{F}$

We consider i first. If the arc were the full circle (oriented clockwise) around i, the contribution would be $-v_i(f)$. Since it is only a half-circle, the arc is easily parametrized with the angle ranging from $-\pi$ to 0 so that the final contribution is $-\frac{1}{2}v_i(f)$. The same is true of the arcs centered at ρ

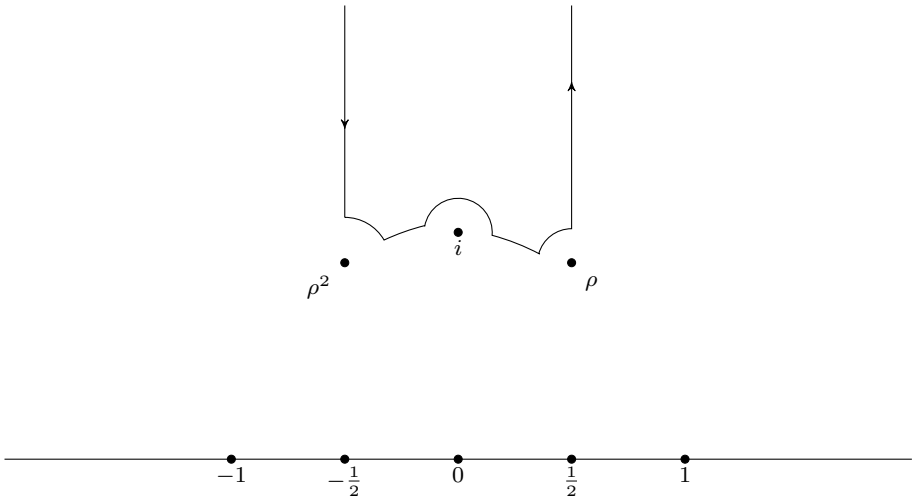


Fig. 4.4 Contour with zeros at i and ρ

and ρ^2. The arc around ρ is parametrized with the angle ranging from $\pi/2$ to $2\pi/3$ (see Figure 4.5). The integral around the arc centered at ρ contributes $-v_\rho(f)/6$. The same is true of the contribution from the arc centered at ρ^2. But ρ and ρ^2 are $\mathrm{SL}_2(\mathbb{Z})$-equivalent so the final contribution is $-v_\rho(f)/3$. □

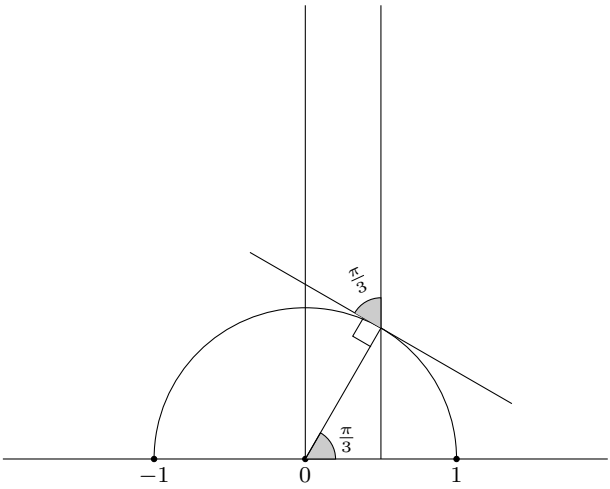


Fig. 4.5 Tangential angle at ρ

Exercise 4.3.2. *If $k < 0$, show that $M_k(\mathrm{SL}_2(\mathbb{Z})) = \{0\}$.*

Exercise 4.3.3. *Show that* $M_0(\mathrm{SL}_2(\mathbb{Z})) = \mathbb{C}$.

Exercise 4.3.4. *Show that* $M_2(\mathrm{SL}_2(\mathbb{Z})) = \{0\}$.

Exercise 4.3.5. *Show that* $E_4(z)$ *has a simple zero at* $z = \rho$.

Exercise 4.3.6. *Show that* $E_6(z)$ *has a simple zero at* $z = i$.

Exercise 4.3.7. *Show that* $S_k(\mathrm{SL}_2(\mathbb{Z})) = \{0\}$ *for* $k \leq 10$.

Exercise 4.3.8. *Show that* $M_k(\mathrm{SL}_2(\mathbb{Z})) = \mathbb{C}E_k$ *for* $k = 4, 6, 8,$ *and* 10.

Exercise 4.3.9. *Show that* $E_4^2 = E_8$. *Deduce that*

$$\sigma_7(n) = \sigma_3(n) + 120 \sum_{m=1}^{n-1} \sigma_3(m)\sigma_3(n-m).$$

Exercise 4.3.10. *Show that* $E_4 E_6 = E_{10}$. *Deduce that*

$$11\sigma_9(n) = 21\sigma_5(n) - 10\sigma_3(n) + 5040 \sum_{m=1}^{n-1} \sigma_3(n)\sigma_5(n-m).$$

4.4 The dimension formula

Ramanujan's cusp form

$$\Delta = \frac{E_4^3 - E_6^2}{1728}$$

plays a special role in the theory of modular forms. Observe that the valence formula shows that $\Delta(z) \neq 0$ for all $z \in \mathbb{H}$. It only has a simple zero at $i\infty$. This observation allows us to prove:

Theorem 4.4.1. *The following statements hold:*

(i) $M_k(\mathrm{SL}_2(\mathbb{Z})) = \{0\}$ *for all* $k < 0$ *and* $k = 2$;

(ii) For $k = 0, 4, 6, 8,$ *and* 10, $M_k(\mathrm{SL}_2(\mathbb{Z}))$ *is one-dimensional and spanned by* $1, E_4, E_6, E_8,$ *and* E_{10}, *respectively;*

(iii) Multiplication by Δ *defines an isomorphism of* $M_{k-12}(\mathrm{SL}_2(\mathbb{Z}))$ *onto* $S_k(\mathrm{SL}_2(\mathbb{Z}))$ *for all* $k \in \mathbb{Z}$.

Proof. We have already noted that (i) and (ii) are immediate by the valence formula (see Exercises 4.3.2, 4.3.3, 4.3.4, and 4.3.8). Clearly, if $g \in M_{k-12}(\mathrm{SL}_2(\mathbb{Z}))$, then $g\Delta \in S_k(\mathrm{SL}_2(\mathbb{Z}))$. If $f \in S_k(\mathrm{SL}_2(\mathbb{Z}))$, then $g = f/\Delta$ is a modular form of weight $k - 12$ since Δ does not vanish on the upper half-plane, implying g is analytic. This proves (iii). □

Corollary 4.4.2 (The dimension formula). *For $k \geq 0$,*

$$\dim M_k(\mathrm{SL}_2(\mathbb{Z})) = \begin{cases} \lfloor \frac{k}{12} \rfloor & \text{if } k \equiv 2 \pmod{12} \\ \lfloor \frac{k}{12} \rfloor + 1 & \text{if } k \not\equiv 2 \pmod{12} \end{cases}$$

Proof. The formula is true for $k \leq 10$ by (i) and (ii) of Theorem 4.4.1. By Exercise 4.2.8 and by part (iii) of Theorem 4.4.1, for $k \geq 12$ we have

$$\begin{aligned} \dim M_k(\mathrm{SL}_2(\mathbb{Z})) &= 1 + \dim S_k(\mathrm{SL}_2(\mathbb{Z})) \\ &= 1 + \dim M_{k-12}(\mathrm{SL}_2(\mathbb{Z})). \end{aligned}$$

This proves the result by induction. □

Exercise 4.4.3. *For $k \geq 4$, show that*

$$\dim(S_k(\mathrm{SL}_2(\mathbb{Z})) = \begin{cases} \lfloor \frac{k}{12} \rfloor - 1 & \text{if } k \equiv 2 \pmod{12} \\ \lfloor \frac{k}{12} \rfloor & \text{if } k \not\equiv 2 \pmod{12} \end{cases}$$

Corollary 4.4.4. *The modular forms $E_4^\alpha E_6^\beta$ with $4\alpha + 6\beta = k$, and α, β integers ≥ 0, form a basis for $M_k(\mathrm{SL}_2(\mathbb{Z}))$.*

Proof. We first show that any modular form f of weight k for $\mathrm{SL}_2(\mathbb{Z})$ can be written as a linear combination of such elements. We proceed by induction on k. This is clear for $k \leq 10$ by Exercises 4.3.8-4.3.10. For general k, choose positive integers α, β such that $4\alpha+6\beta = k$. This is clearly possible since k is even and we can find positive integers α, β such that $2\alpha+3\beta = k/2$. (Indeed, if $k/2$ is even, we write $2\alpha = k/2$. If $k/2$ is odd, we write $2(\alpha-1)+3 = k/2$.) Then $g = E_4^\alpha E_6^\beta$ has weight k and is non-zero at $i\infty$. So there exists $\lambda \in \mathbb{C}$ such that $f - \lambda g$ vanishes at $i\infty$. Thus we can write $f = \lambda g + \Delta h$ for some $h \in M_{k-12}(\mathrm{SL}_2(\mathbb{Z}))$ by (iii) of Theorem 4.4.1. Applying the induction hypothesis to h shows that the monomials $E_4^\alpha E_6^\beta$ span $M_k(\mathrm{SL}_2(\mathbb{Z}))$. To show that these elements are linearly independent, we suppose that

$$\sum_{4\alpha+6\beta=k} c_{\alpha\beta} E_4^\alpha E_6^\beta = 0.$$

Since $E_4(\rho) = 0$ and $E_6(\rho) \neq 0$, we see that $c_{\alpha\beta}$ with $\alpha = 0$ must vanish. The same argument shows that $c_{\alpha\beta}$ with $\beta = 0$ also vanishes, using the fact $E_6(i) = 0$ and $E_4(i) \neq 0$. Thus, we have $\alpha \geq 1, \beta \geq 1$ in the summation. Hence

$$\sum_{4(\alpha-1)+6(\beta-1)=k-10} c_{\alpha\beta} E_4^{\alpha-1} E_6^{\beta-1} E_4 E_6 = 0.$$

Dividing through by $E_4 E_6$ and repeating the process completes the proof by induction. □

Exercise 4.4.5. *Prove that* $\Delta = \dfrac{E_4^3 - E_6^2}{1728}$ *has integer Fourier coefficients.*

4.5 The j-function

The *j-function* is defined by $j = E_4^3/\Delta$. Clearly

$$j\left(\frac{az+b}{cz+d}\right) = j(z) \qquad \forall \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}).$$

It is not holomorphic at $i\infty$ since Δ vanishes there to first order. Consequently, it has a simple pole there. Since Δ has integer coefficients (Exercise 4.4.5) with lead coefficient 1,

$$\Delta = q - 24q^2 + \dots,$$

we see that the Fourier coefficients of the j-function's expansion at $i\infty$ are also integers. We have

$$j(z) = \frac{1}{q} + 744 + 196884q + 21493760q^2 + \dots \tag{4.5}$$

If we write $c(n)$ for the n-th Fourier coefficient of the j-function, then Petersson [28] showed in 1932 that

$$c(n) \sim \frac{e^{4\pi\sqrt{n}}}{\sqrt{2}n^{3/4}},$$

as $n \to \infty$. Thus the integer coefficients $c(n)$ grow at an exponential rate. These coefficients are related to the representation theory of a sporadic simple group known as the Monster group that played a decisive role in the classification of all finite simple groups. The j-function has many more remarkable properties that we cannot adumbrate here. It is ubiquitous in mathematics and one is compelled to study its beauty and character irrespective of any subdiscipline. We will content ourselves here with the following.

Theorem 4.5.1. *Let $\mathcal{F}$ denote the fundamental domain from Theorem 3.2.2 and let $\mathcal{F}'$ denote the set obtained by removing points with $\operatorname{Re} z = \frac{1}{2}$ and points with both $|z| = 1$ and $\operatorname{Re} z > 0$. The j-function $z \mapsto j(z)$ defines a bijection from $\mathcal{F}'$ onto $\mathbb{C}$.*

Proof. We must show that given $\lambda \in \mathbb{C}$, there is a unique $z \in \mathcal{F}'$ such that $j(z) = \lambda$. In other words, we have to show that $E_4^3 - \lambda\Delta$ has a unique zero in $\mathcal{F}'$. But this is clear from the valence formula. Indeed, $f = E_4^3 - \lambda\Delta$ has weight 12 and the only decomposition of

$$a + \frac{b}{2} + \frac{c}{3} = 1$$

with a, b, c, integers ≥ 0 arises from

$$(a, b, c) = (1, 0, 0), (0, 2, 0), \text{ or } (0, 0, 3).$$

This means f has a unique zero in $\mathcal{F}'$. □

Exercise 4.5.2. *Show that $j(\rho) = 0$ and $j(i) = 1728$. Moreover, show that $v_\rho\Big(j(z)\Big) = 3$, that $v_i\Big(j(z) - 1728\Big) = 2$, and that $v_{z_0}\Big(j(z) - j(z_0)\Big) = 1$ for all $z_0 \in \mathbb{H}$ which are not $\mathrm{SL}_2(\mathbb{Z})$-equivalent to either ρ or i.*

A *modular function of level one* is a meromorphic function $f : \mathbb{H} \to \mathbb{C}$ that satisfies

$$f\left(\frac{az+b}{cz+d}\right) = f(z) \qquad \forall \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$$

and is meromorphic at $i\infty$. In particular, it admits a q-expansion with only finitely many negative powers of q.

Exercise 4.5.3. *Show that any modular function f has a finite number of zeros and poles in $\mathcal{F}$.*

It turns out that every modular function of level one is a rational function in j as is proved in the following theorem.

Theorem 4.5.4. *Let f be a meromorphic function on $\mathbb{H}$. The following are equivalent:*

i) f is a modular function;
ii) f is a quotient of two modular forms of the same weight;
iii) f is a rational function of j.

Proof. The implications $(iii) \Rightarrow (ii) \Rightarrow (i)$ are immediate since $j = E_4^3/\Delta$. It therefore suffices to prove $(i) \Rightarrow (iii)$. Let f be a modular function. Suppose that it has a pole at z_0 of order r. Then

$$(j(z) - j(z_0))^r f(z)$$

is again a modular function without a pole at z_0. In this way, we can multiply f by a suitable polynomial in j and assume f is holomorphic on $\mathbb{H}$. Since Δ has a simple zero at $i\infty$, we can multiply f by a suitable power of Δ and assume that $\Delta^n f$ is holomorphic at $i\infty$, for some positive integer n. But now, $\Delta^n f$ is holomorphic and is a modular form of weight $12n$. By Corollary 4.4.4, it can be written as a linear combination of monomials $E_4^\alpha E_6^\beta$. We are therefore reduced to showing $E_4^\alpha E_6^\beta/\Delta^n$ is a polynomial in j. But $4\alpha + 6\beta = 12n$, so that $3|\alpha$ and $2|\beta$. Thus, we can write this as $(E_4^3)^{\alpha/3}(E_6^2)^{\beta/2}/\Delta^n = (E_4^3/\Delta)^{\alpha/3}(E_6^2/\Delta)^{\beta/2}$. Since $E_4^3 - E_6^2 = 1728\Delta$ we see that $E_6^2/\Delta = j - 1728$. As $j = E_4^3/\Delta$, we are done. □

The reader may easily check that the valence formula also holds for modular functions, the proof being essentially identical. This permits an alternative proof of $(i) \Rightarrow (iii)$ that treats zeros and poles on the same footing. Let f be a modular function, not identically zero. Assume first that $v_i(f) = v_\rho(f) = 0$. Let $z_1, z_2, \ldots, z_r$ and $w_1, w_2, \ldots, w_s$ denote, respectively, the zeros and poles in $\mathbb{H}$ of $f(z)$, listed according to multiplicity. Set

$$g(z) = \frac{\prod_{i=1}^{r} (j(z) - j(z_i))}{\prod_{i=1}^{s} (j(z) - j(w_i))}.$$

Then f/g is a modular function with no zeros or poles in $\mathbb{H}$. By the valence formula, $v_{i\infty}(f/g) = 0$. Hence, f/g is a non-vanishing, holomorphic modular form of weight 0. By Exercise 4.3.3, $f = cg$ for some $c \in \mathbb{C}$, as desired. If $v_i(f) \neq 0$, then an analysis using the valence formula shows that $2|v_i(f)$. By Exercise 4.5.2, $j(z) - j(i)$ has a double zero at $z = i$. Hence, i should only appear in the list of zeros and poles $|v_i(f)|/2$ many times. Similarly, ρ should appear $|v_\rho(f)|/3$ times.

4.6 Supplementary problems

Exercise 4.6.1. *Show that the Bernoulli numbers B_k vanish for odd $k \geq 3$.*

Exercise 4.6.2. *Show that for k even and positive, $(-1)^{k/2} B_k < 0$.*

Exercise 4.6.3. *Prove that*

$$1^k + 2^k + \cdots + n^k = \frac{B_{k+1}(n+1) - B_{k+1}}{k+1},$$

for every natural number k.

Exercise 4.6.4. *Prove that $\zeta(k)$ tends to 1 as k tends to infinity and use this to show that*

$$|B_{2k}| \sim 4\pi\sqrt{e}(k/\pi e)^{2k+1/2}$$

as k tends to infinity.

Exercise 4.6.5. *Show that if the Fourier coefficients of $f \in M_k(\mathrm{SL}_2(\mathbb{Z}))$ are integers, then f can be written as a polynomial in E_4, E_6, and Δ with integer coefficients.*

Exercise 4.6.6. *Suppose $f = \sum_{n=0}^{\infty} a_n q^n \in M_k(\mathrm{SL}_2(\mathbb{Z}))$. Show that for any $x \in \mathbb{R}$ we have $\lim_{y\to\infty} f(x+iy) = a_0$. Show that for any $x \in \mathbb{Q}$, if f is not a cusp form, then $\lim_{y\to 0} f(x+iy) = \infty$, and if f is a cusp form, then $\lim_{y\to 0} f(x+iy) = 0$. Deduce that the real line is a natural boundary for any modular form.*

Exercise 4.6.7. *Prove that $E_4^2 E_6 = E_{14}$.*

Exercise 4.6.8. *Let $r_k = \dim M_k(SL_2(\mathbb{Z}))$. Put $\ell = k - 12r_k + 12$ and $E_0 = 1$. Show that*

$$E_\ell E_{14-\ell} = E_{14}$$

for $\ell = 0, 4, 6, 8, 10$ and 14.

Exercise 4.6.9. *Let f and g be modular forms of weight k and ℓ respectively, for the full modular group. Show that*

$$kfg' - \ell f'g$$

is a modular form of weight $k + \ell + 2$ for the full modular group.

Exercise 4.6.10. *Let $j(z) = E_4^3/\Delta$. Prove that*

$$\frac{dj}{dz} = -2\pi i E_{14}/\Delta.$$

Exercise 4.6.11. *Let $r_k = \dim M_k(SL_2(\mathbb{Z}))$ and $\ell = k - 12r_k + 12$. For any modular form $f \in M_k(SL_2(\mathbb{Z}))$, show that*

$$fE_\ell^{-1} \in M_{12r_k-12}(SL_2(\mathbb{Z})).$$

Exercise 4.6.12. *With notation as in the previous exercise, show that*

$$fE_\ell^{-1}\Delta^{1-r_k}$$

is a modular function and hence a polynomial in j.

Exercise 4.6.13. *Let f be a modular form of weight k for the full modular group whose Fourier expansion at $i\infty$ is given by*

$$f(z) = \sum_{n=0}^{\infty} a_n e^{2\pi i n z}, \qquad a_n \in \mathbb{C}.$$

Recall that by Corollary 4.4.4, there are unique c_{ab}'s such that

$$f = \sum_{4a+6b=k} c_{ab} E_4^a E_6^b, \qquad c_{ab} \in \mathbb{C}.$$

Let F be an algebraic extension of $\mathbb{Q}$. Suppose that

$$\#\{n : a_n \in F\} \geq \dim M_k(SL_2(\mathbb{Z})).$$

Then, show that c_{ab} belong to F, whence all the Fourier coefficients belong to F.

Exercise 4.6.14. *If f is a modular form whose Fourier coefficients at $i\infty$ are algebraic numbers, then show that the smallest exension field of the rationals containing all the Fourier coefficients of f is a finite extension of $\mathbb{Q}$.*

Exercise 4.6.15. *Suppose that f is a modular form of weight k whose Fourier expansion at $i\infty$ is given by*

$$f(z) = \sum_{n=0}^{\infty} a_n e^{2\pi i n z}, \qquad a_n \in \overline{\mathbb{Q}}.$$

Show that, for every $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, the series

$$f^{\sigma}(z) := \sum_{n=0}^{\infty} \sigma(a_n) e^{2\pi i n z},$$

defines a modular form of weight k.

Chapter 5
The Ramanujan τ-function

5.1 The Eisenstein series E_2

In his fundamental paper of 1916, Ramanujan introduced the τ-function as being the coefficients in the power series expansion of the infinite product

$$q\prod_{n=1}^{\infty}(1-q^n)^{24}=\sum_{n=1}^{\infty}\tau(n)q^n. \tag{5.1}$$

He made three conjectures regarding $\tau(n)$:

(i) $\tau(mn)=\tau(m)\tau(n)$ for $(m,n)=1$
(ii) if p is prime, then $\tau(p^{\alpha+1})=\tau(p)\tau(p^{\alpha})-p^{11}\tau(p^{\alpha-1})$ for $\alpha\geq 1$;
(iii) if p is prime, then $|\tau(p)|\leq 2p^{11/2}$.

Our goal in this chapter is to prove the first two of these conjectures in the setting of the theory of modular forms and develop what is now called Hecke theory of level one.

Ramanujan discovered an intimate connection between the function $\tau(n)$ and $r_{24}(n)$ which counts the number of ways of expressing n as a sum of 24 squares. More precisely, let $\sigma_v^*(n)$ be $\sigma_v(n)$ if n is odd and $\sigma_v^e(n)-\sigma_v^o(n)$ if n is even with $\sigma_v^e(n),\sigma_v^o(n)$ being the sumof the v-th powers of the even and odd divisors of n respectively. Ramanujan proved that

$$r_{24}(n)=\frac{16}{691}\sigma_{11}^*(n)+e_{24}(n),$$

where

$$e_{24}(n)=\frac{128}{691}\left\{(-1)^{n-1}259\tau(n)-512\tau(n/2)\right\},$$

with the last term appearing only if n is even. We will not prove this formula in this book but refer the reader to Chapters 9 and 10 of [13] and [11] for further details.

© Springer Science+Business Media Singapore 2016 and Hindustan Book Agency 2016

M.R. Murty et al., *Problems in the Theory of Modular Forms*, HBA Lecture Notes in Mathematics, DOI 10.1007/978-981-10-2651-5_5

Our first goal is to prove that the left hand side of (5.1) is a cusp form of weight 12 for the full modular group for $q = e^{2\pi iz}$ and $z \in \mathbb{H}$. We follow a method due to Hurwitz and study the Eisenstein series E_2 (defined below) to realize this goal.

The double series

$$G_2(z) := \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty}{}' \frac{1}{(mz+n)^2}, \qquad z \in \mathbb{H} \tag{5.2}$$

can be shown to converge by using a technique due to Hurwitz. The primed summation again means that we omit $(m,n) = (0,0)$. We introduce

$$a_{m,n}(z) = \frac{1}{mz+n-1} - \frac{1}{mz+n}$$

for $(m,n) \neq (0,0), (0,1)$. Let us observe that

$$\frac{1}{(mz+n)^2} - a_{m,n}(z) = \frac{-1}{(mz+n)^2(mz+n-1)},$$

and that the double series

$$\sum\sum_{(m,n)\neq(0,0),(0,1)} \frac{1}{(mz+n)^2(mz+n-1)}$$

converges absolutely. Thus,

$$\begin{aligned} G_2(z) &= 2\zeta(2) + \sum_{m\neq 0} \sum_{n=-\infty}^{\infty} \frac{1}{(mz+n)^2} \\ &= 2\zeta(2) + \sum_{m\neq 0} \sum_{n=-\infty}^{\infty} \left(\frac{1}{(mz+n)^2} - a_{m,n}(z) \right) + \sum_{m\neq 0} \sum_{n=-\infty}^{\infty} a_{m,n}(z), \end{aligned}$$

where now the first double sum in the last equality is absolutely convergent and the inner sum in the second double sum telescopes to zero. Therefore

$$G_2(z) = 2\zeta(2) + \sum_{m\neq 0} \sum_{n=-\infty}^{\infty} \left(\frac{1}{(mz+n)^2} - a_{m,n}(z) \right).$$

Because the double sum is now absolutely convergent, we may interchange the order of summation to get

$$G_2(z) = 2\zeta(2) + \sum_{n=-\infty}^{\infty} \sum_{m\neq 0} \left(\frac{1}{(mz+n)^2} - a_{m,n}(z) \right). \tag{5.3}$$

From (5.2), we have

$$G_2(z) = 2\zeta(2) + \sum_{m\neq 0}\sum_{n=-\infty}^{\infty} \frac{1}{(mz+n)^2}$$

so that

$$\begin{aligned} G_2\left(-\frac{1}{z}\right) &= 2\zeta(2) + \sum_{m\neq 0}\sum_{n=-\infty}^{\infty} \frac{z^2}{(nz-m)^2} \\ &= 2\zeta(2)(1+z^2) + \sum_{m\neq 0}\sum_{n\neq 0} \frac{z^2}{(nz-m)^2} \\ &= 2\zeta(2)z^2 + \sum_{m=-\infty}^{\infty}\sum_{n\neq 0} \frac{z^2}{(nz-m)^2}. \end{aligned}$$

Re-labelling our index of summation, we obtain

$$G_2\left(-\frac{1}{z}\right) = 2\zeta(2)z^2 + \sum_{n=-\infty}^{\infty}\sum_{m\neq 0} \frac{z^2}{(mz+n)^2}.$$

From (5.3), we get

$$G_2(z) = 2\zeta(2) + \left(G_2\left(-\frac{1}{z}\right) - 2\zeta(2)z^2\right) z^{-2} - \sum_{n=-\infty}^{\infty}\sum_{m\neq 0} a_{m,n}(z).$$

We now proceed to evaluate the last sum:

$$\sum_{n=-\infty}^{\infty}\sum_{m\neq 0} a_{m,n}(z) = \sum_{n=-\infty}^{\infty}\sum_{m\neq 0}\left(\frac{1}{mz+n-1} - \frac{1}{mz+n}\right).$$

Recall the cotangent expansion (4.3):

$$\pi\cot(\pi z) - \frac{1}{z} = \sum_{m=1}^{\infty}\left(\frac{1}{z+m} + \frac{1}{z-m}\right).$$

Using this, we see that the sum in question is

$$\begin{aligned} \frac{1}{z}\sum_{\substack{n\neq -\infty \\ n\neq 0,1}}^{\infty}\left(\pi\cot\left(\frac{\pi(n-1)}{z}\right) - \frac{z}{n-1} - \pi\cot\left(\frac{\pi n}{z}\right) + \frac{z}{n}\right) \\ + \sum_{m\neq 0}\left(\frac{1}{mz-1} - \frac{1}{mz}\right) + \sum_{m\neq 0}\left(\frac{1}{mz} - \frac{1}{mz+1}\right). \end{aligned}$$

The last two sums combine to give

$$\sum_{m\neq 0}\left(\frac{1}{mz-1}-\frac{1}{mz+1}\right)=-\frac{1}{z}\sum_{m\neq 0}\left(\frac{1}{(1/z)+m}+\frac{1}{(1/z)-m}\right)$$
$$=-\frac{2}{z}\left(\pi\cot\left(\frac{\pi}{z}\right)-z\right)$$
$$=\frac{-2\pi}{z}\cot\left(\frac{\pi}{z}\right)+2.$$

We now examine

$$\frac{1}{z}\sum_{\substack{n\neq-\infty\\ n\neq 0,1}}^{\infty}\left(\pi\cot\left(\frac{\pi(n-1)}{z}\right)-\frac{z}{n-1}-\pi\cot\left(\frac{\pi n}{z}\right)+\frac{z}{n}\right)$$
$$=\frac{1}{z}\lim_{N\to\infty}\sum_{\substack{n\neq-N\\ n\neq 0,1}}^{N}\left(\pi\cot\left(\frac{\pi(n-1)}{z}\right)-\pi\cot\left(\frac{\pi n}{z}\right)+\frac{z}{n}-\frac{z}{n-1}\right).$$

Keeping in mind that cotangent is an odd function, we easily see that this telescopes to

$$\frac{1}{z}\lim_{N\to\infty}\left\{2\pi\cot\left(\frac{\pi}{z}\right)-\pi\cot\left(\frac{\pi(N+1)}{z}\right)-\pi\cot\left(\frac{\pi N}{z}\right)+\frac{z}{N}+\frac{z}{N+1}-2z\right\}$$
$$=\frac{2\pi}{z}\cot\left(\frac{\pi}{z}\right)-2-\frac{2\pi}{z}\lim_{N\to\infty}\cot\left(\frac{\pi N}{z}\right).$$

Now, as $z\in\mathbb{H}$, by (4.4) we have

$$\lim_{N\to\infty}\cot\left(\frac{\pi N}{z}\right)=i\lim_{N\to\infty}\frac{e^{2\pi i(N/z)}+1}{e^{2\pi i(N/z)}-1}=i+2i\lim_{N\to\infty}\frac{1}{e^{2\pi i(N/z)}-1}=i.$$

Putting everything together, we obtain that

$$\sum_{n=-\infty}^{\infty}\sum_{m\neq 0}a_{m,n}(z)=-\frac{2\pi i}{z}.$$

This proves:

Theorem 5.1.1. *For all $z\in\mathbb{H}$,*

$$G_2\left(-\frac{1}{z}\right)=z^2G_2(z)-2\pi iz.$$

Thus G_2 is not a modular form, but rather an example of a "quasi"-modular form. We can normalize G_2 and consider

$$E_2(z) = 1 + \frac{1}{2\zeta(2)} \sum_{m \neq 0} \sum_{n=-\infty}^{\infty} \frac{1}{(mz+n)^2}.$$

Exercise 5.1.2. *Show that $E_2(z)$ has the Fourier expansion at $i\infty$ given by*

$$E_2(z) = 1 - 24 \sum_{n=1}^{\infty} \sigma_1(n) q^n,$$

where $q = e^{2\pi i z}$ and $\sigma_1(n) = \sum_{d|n} d$.

Exercise 5.1.3. *Show that for all $z \in \mathbb{H}$,*

$$E_2\left(-\frac{1}{z}\right) = z^2 E_2(z) + \frac{6}{\pi i} z.$$

Theorem 5.1.4 (Jacobi). *The cusp form $\Delta = \dfrac{E_4^3 - E_6^2}{1728} \in S_{12}(\mathrm{SL}_2(\mathbb{Z}))$ is Ramanujan's infinite product*

$$\Delta = q \prod_{n=1}^{\infty} (1-q^n)^{24}.$$

Proof. Take $q = e^{2\pi i z}$ and let

$$F(z) = q \prod_{n=1}^{\infty} (1-q^n)^{24}.$$

Clearly $F(z+1) = F(z)$ for all $z \in \mathbb{H}$. If we can show that

$$F\left(-\frac{1}{z}\right) = z^{12} F(z),$$

then by Exercises 4.1.4 and 4.4.3, we conclude that $F = c\Delta$ for some $c \in \mathbb{C}$. Comparing Fourier coefficients, we easily see that $c = 1$. Now, logarithmic differentiation with respect to z gives

$$\frac{F'(z)}{F(z)} = 2\pi i - 48\pi i \sum_{n=1}^{\infty} \frac{n q^n}{1-q^n},$$

Noting that

$$\frac{q^n}{1-q^n} = \sum_{m=1}^{\infty} q^{mn},$$

we see from Exercise 5.1.2 that

$$\frac{F'(z)}{F(z)} = 2\pi i E_2(z).$$

Replacing z by $-1/z$ and using Exercise 5.1.3, we obtain

$$\frac{F'(-1/z)}{F(-1/z)} = 2\pi i \left(z^2 E_2(z) + \frac{6z}{\pi i} \right) = z^2 \frac{F'(z)}{F(z)} + 12z.$$

Writing this equation as

$$\frac{F'(-1/z)}{F(-1/z)} \cdot \frac{1}{z^2} = \frac{F'(z)}{F(z)} + \frac{12}{z},$$

we immediately see that $F(-1/z)$ and $z^{12}F(z)$ have the same logarithmic derivative. Thus $F(-1/z) = \lambda z^{12} F(z)$ for some constant λ. Putting $z = i$, we have $\lambda = 1$, as $F(z) \neq 0$ for $z \in \mathbb{H}$. This completes the proof. □

Exercise 5.1.5. *For $z \in \mathbb{H}$, we define the* Dedekind η-function *by*

$$\eta(z) = e^{2\pi i z/24} \prod_{n=1}^{\infty} (1 - e^{2\pi i n z}).$$

Show that

$$\eta\left(-\frac{1}{z}\right) = \sqrt{\frac{z}{i}}\eta(z),$$

where the square root is the branch with non-negative real part.

Exercise 5.1.6. *Show that*

$$\frac{\Delta'(z)}{\Delta(z)} = 2\pi i E_2(z).$$

Exercise 5.1.7. *Prove that for all $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$ and all $z \in \mathbb{H}$, we have*

$$E_2\left(\frac{az+b}{cz+d}\right) = (cz+d)^2 E_2(z) + \frac{6c(cz+d)}{\pi i}.$$

Exercise 5.1.8. *The* Serre derivative *of a modular form $f \in M_k(\mathrm{SL}_2(\mathbb{Z}))$ is defined as*

$$\theta_k f := \frac{1}{2\pi i} f' - \frac{k}{12} E_2 f.$$

Prove that $\theta_k : M_k(\mathrm{SL}_2(\mathbb{Z})) \to M_{k+2}(\mathrm{SL}_2(\mathbb{Z}))$.

Theorem 5.1.9 (Ramanujan). *Let*

$$P = 1 - 24\sum_{n=1}^{\infty}\sigma_1(n)q^n,$$
$$Q = 1 + 240\sum_{n=1}^{\infty}\sigma_3(n)q^n,$$
$$R = 1 - 504\sum_{n=1}^{\infty}\sigma_5(n)q^n.$$

For the differential operator $D = q\frac{d}{dq}$, we have

$$DP = \frac{1}{12}(P^2 - Q),$$
$$DQ = \frac{1}{3}(PQ - R),$$
$$DR = \frac{1}{2}(PR - Q^2).$$

Remark. The reader will note that if we replace q by $e^{2\pi iz}$, then P, Q, and R become $E_2, E_4,$ and E_6 respectively. The differential equations stated in the theorem were first proved by Ramanujan in his 1916 paper using the theory of elliptic functions. The differential equations have now become central in the theory of modular forms and have even found application in Nesterenko's work in transcendental number theory. Our proof here will involve only the basic property of the Serre derivative stated in Exercise 5.1.8.

Proof. Noting that if $q = e^{2\pi iz}$, then

$$D = q\frac{d}{dq} = \frac{1}{2\pi i}\frac{d}{dz},$$

we see that the Serre derivative can be written as

$$\theta_k f = Df - \frac{k}{12}E_2 f.$$

By Exercise 5.1.8, we have $\theta_4 E_4 \in M_6(\mathrm{SL}_2(\mathbb{Z}))$ and $\theta_6 E_6 \in M_8(\mathrm{SL}_2(\mathbb{Z}))$. Since these vector spaces are 1-dimensional and spanned by E_6 and E_8 respectively, we see that $\theta_4 E_4 = \lambda_1 E_6$ and $\theta_6 E_6 = \lambda_2 E_8$ for appropriate constants $\lambda_1, \lambda_2 \in \mathbb{C}$. A quick check of the q-expansion of both sides shows $\lambda_1 = -1/3$ and $\lambda_2 = -1/2$. Since $E_8 = E_4^2$, this proves that $DQ = \frac{1}{3}(PQ - R)$, and $DR = \frac{1}{2}(PR - Q^2)$. To show that $DP = \frac{1}{12}(P^2 - Q)$, we need only note that

$$\frac{1}{2\pi i}E_2'(z) - \frac{1}{12}E_2^2(z) \in M_4(\mathrm{SL}_2(\mathbb{Z})),$$

by a straightforward computation using Exercise 5.1.7. Since $M_4(\mathrm{SL}_2(\mathbb{Z}))$ is also one-dimensional and spanned by E_4, we quickly find on comparing

coefficients that

$$DP - \frac{1}{12}P^2 = -\frac{1}{12}Q. \qquad \square$$

5.2 Hecke operators of level one

Let $\mathbf{X}_m$ be the set of 2×2 matrices with integer entries and determinant m, with m a natural number. The group $\mathrm{SL}_2(\mathbb{Z})$ acts on $\mathbf{X}_m$ via left multiplication. We first show that the number of orbits is finite and then find orbit representatives. Using this result, we will construct special linear operators $T_m : M_k(\mathrm{SL}_2(\mathbb{Z})) \to M_k(\mathrm{SL}_2(\mathbb{Z}))$ called Hecke operators and proceed to study their eigenfunctions and eigenvalues. This study will lead to a proof of the first two conjectures of Ramanujan concerning the τ-function.

Theorem 5.2.1. *The number of orbits for the action of* $\mathrm{SL}_2(\mathbb{Z})$ *on* $\mathbf{X}_m$ *is* $\sigma_1(m)$*, and a set of orbit representatives can be taken as*

$$\left\{ \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} : \quad ad = m, \quad 0 \le b \le d-1 \right\}.$$

That is,

$$\mathbf{X}_m = \coprod_{\substack{ad=m \\ d>0}} \coprod_{b \,(\mathrm{mod}\, d)} \mathrm{SL}_2(\mathbb{Z}) \begin{pmatrix} a & b \\ 0 & d \end{pmatrix}.$$

Proof. First we show that every matrix in $\mathbf{X}_m$ appears in some orbit as given in the theorem. Indeed, let

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \in \mathbf{X}_m.$$

We want to find matrices

$$\begin{pmatrix} r & s \\ t & u \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}) \quad \text{and} \quad \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \in \mathbf{X}_m$$

such that

$$\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} = \begin{pmatrix} r & s \\ t & u \end{pmatrix} \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} = \begin{pmatrix} ra & rb+sd \\ ta & tb+ud \end{pmatrix}.$$

This suggests we take $a = (\alpha, \gamma)$, and $r = \alpha/a, t = \gamma/a$. As $(r,t) = 1$, we can find integers s, u such that

$$ru - ts = 1.$$

Then

$$\begin{pmatrix} a & b \\ 0 & d \end{pmatrix} = \begin{pmatrix} r & s \\ t & u \end{pmatrix}^{-1} \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} = \begin{pmatrix} u & -s \\ -t & r \end{pmatrix} \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$$

so that

$$b = \beta u - \delta s$$
$$d = -\beta t + \delta r.$$

Note that $ad = m$ simply by taking determinants. To ensure $0 \leq b \leq d-1$, we note that

$$\begin{pmatrix} 1 & -k \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} = \begin{pmatrix} a & b-kd \\ 0 & d \end{pmatrix}$$

may also be taken as a representative of the orbit $\mathrm{SL}_2(\mathbb{Z}) \begin{pmatrix} a & b \\ 0 & d \end{pmatrix}$ so we can ensure $0 \leq b \leq d-1$.

Finally, we need to show these representatives are $\mathrm{SL}_2(\mathbb{Z})$-inequivalent. Indeed, if $g = \left(\begin{smallmatrix} r & s \\ t & u \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$ is such that

$$g \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} = \begin{pmatrix} a' & b' \\ 0 & d' \end{pmatrix},$$

then we get

$$\begin{pmatrix} ar & br+sd \\ at & bt+ud \end{pmatrix} = \begin{pmatrix} a' & b' \\ 0 & d' \end{pmatrix}.$$

Thus, $t = 0$ and $ru = \det g = 1$ so that $r = \pm 1$. As $a, a' > 0$, we must have $r = 1$ so that $a = a'$. But then $d = d'$ since $ad = a'd' = m$. Finally, $b + sd = b'$ leads to $s = 0$ since $0 \leq b \leq d-1$ and $0 \leq b' \leq d-1$. This completes the proof. □

This theorem allows us to define the m-th Hecke operator T_m as follows. For $f \in M_k(\mathrm{SL}_2(\mathbb{Z}))$, let

$$T_m(f) := m^{k/2-1} \sum_{\substack{ad=m \\ d>0}} \sum_{b \pmod d} f \left| \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \right..$$

Theorem 5.2.2. *Let m be a positive integer. If $f \in M_k(\mathrm{SL}_2(\mathbb{Z}))$, then $T_m(f) \in M_k(\mathrm{SL}_2(\mathbb{Z}))$. If $f \in S_k(\mathrm{SL}_2(\mathbb{Z}))$, then $T_m(f) \in S_k(\mathrm{SL}_2(\mathbb{Z}))$. In other words, T_m is a linear transformation of the vector spaces $M_k(\mathrm{SL}_2(\mathbb{Z}))$ and $S_k(\mathrm{SL}_2(\mathbb{Z}))$.*

Proof. First we need to show that $T_m(f)|\gamma = T_m(f)$ for all $\gamma \in \mathrm{SL}_2(\mathbb{Z})$. Fix $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ and an orbit representative $\left(\begin{smallmatrix} a & b \\ 0 & d \end{smallmatrix}\right)$. Since $\mathrm{SL}_2(\mathbb{Z})\, \mathbf{X}_m = \mathbf{X}_m\, \mathrm{SL}_2(\mathbb{Z})$ (simply by taking determinants), we see that

$$\begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \gamma = \gamma' \begin{pmatrix} a' & b' \\ 0 & d' \end{pmatrix}$$

for some $\gamma' \in \mathrm{SL}_2(\mathbb{Z})$ and $a'd' = m$, $0 \leq b' \leq d'-1$. But $f|\gamma' = f$ and so

$$f\left|\begin{pmatrix} a & b \\ 0 & d \end{pmatrix}\gamma = f\right|\gamma'\begin{pmatrix} a' & b' \\ 0 & d' \end{pmatrix} = f\left|\begin{pmatrix} a' & b' \\ 0 & d' \end{pmatrix}\right..$$

Moreover, the new representatives $\left(\begin{smallmatrix} a' & b' \\ 0 & d' \end{smallmatrix}\right)$ obtained this way are again distinct as we range over $ad = m$, $0 \le b \le d-1$. Indeed, suppose also that

$$\begin{pmatrix} a_1 & b_1 \\ 0 & d_1 \end{pmatrix}\gamma = \gamma''\begin{pmatrix} a' & b' \\ 0 & d' \end{pmatrix}$$

for another $a_1 d_1 = m$ and $0 \le b_1 \le d_1 - 1$. Now

$$\begin{pmatrix} a_1 & b_1 \\ 0 & d_1 \end{pmatrix}\gamma = \gamma''(\gamma')^{-1}\begin{pmatrix} a & b \\ 0 & d \end{pmatrix}\gamma.$$

After cancelling the γ from both sides, we conclude (as in the proof of Theorem 5.2.1) that this implies that $\gamma''(\gamma')^{-1}$ is the identity and that $\left(\begin{smallmatrix} a & b \\ 0 & d \end{smallmatrix}\right) = \left(\begin{smallmatrix} a_1 & b_1 \\ 0 & d_1 \end{smallmatrix}\right)$.

The fact that $T_m(f)$ is holomorphic at $i\infty$ when $f \in M_k(\mathrm{SL}_2(\mathbb{Z}))$ and that $T_m(f)$ vanishes at $i\infty$ when $f \in S_k(\mathrm{SL}_2(\mathbb{Z}))$ will follow from the explicit q-expansion which we develop in Theorem 5.2.4 below. □

Exercise 5.2.3. *Show that*

$$\sum_{b=0}^{d-1} e^{2\pi i n b/d} = \begin{cases} d & \text{if } d \mid n \\ 0 & \text{otherwise} \end{cases}$$

Theorem 5.2.4. *If $f \in M_k(\mathrm{SL}_2(\mathbb{Z}))$ has q-expansion*

$$f = \sum_{n=0}^{\infty} \lambda(n) q^n$$

at $i\infty$, then the q-expansion at $i\infty$ of $T_m(f)$ is

$$T_m(f) = \sum_{n=0}^{\infty}\left(\sum_{d \mid n,m} d^{k-1}\lambda\left(\frac{mn}{d^2}\right)\right) q^n.$$

Proof. Writing

$$f(z) = \sum_{n=0}^{\infty} \lambda(n) e^{2\pi i n z},$$

and recalling the definition (4.2) of the slash operator for matrices in $\mathrm{GL}_2^+(\mathbb{R})$, we see that

$$T_m(f)(z) = \frac{1}{m}\sum_{\substack{ad=m \\ d>0}}\left(\frac{m}{d}\right)^k \sum_{b=0}^{d-1}\sum_{n=0}^{\infty} \lambda(n) e^{2\pi i n(az+b)/d}.$$

Interchanging the two inner sums and using Exercise 5.2.3, we obtain

$$T_m(f)(z) = \sum_{\substack{ad=m \\ d>0}} \left(\frac{m}{d}\right)^{k-1} \sum_{\substack{n=0 \\ d|n}}^{\infty} \lambda(n) e^{2\pi i n a z/d}.$$

Writing $n = dr$, we can write the inner sum as

$$\sum_{r=0}^{\infty} \lambda(dr) e^{2\pi i a r z}.$$

Collecting terms with $ar = s$, we get

$$T_m(f)(z) = \sum_{s=0}^{\infty} \left(\sum_{\substack{ad=m \\ ar=s}} \left(\frac{m}{d}\right)^{k-1} \lambda(dr) \right) e^{2\pi i s z}.$$

The coefficient of $e^{2\pi i s z}$ can be re-written as

$$\sum_{a|m,s} a^{k-1} \lambda\left(\frac{ms}{a^2}\right),$$

as claimed. □

Exercise 5.2.5. *Prove that $T_m(\Delta) = \tau(m)\Delta$. Deduce that $\tau(mn) = \tau(m)\tau(n)$ for $(m, n) = 1$.*

Exercise 5.2.6. *Prove that for any $m, n \geq 1$,*

$$\tau(m)\tau(n) = \sum_{d|m,n} d^{11} \tau\left(\frac{mn}{d^2}\right).$$

Exercise 5.2.7. *Let p be prime and $a \geq 1$. Show that*

$$\tau(p^{a+1}) = \tau(p)\tau(p^a) - p^{11}\tau(p^{a-1}).$$

Thus, the theory of Hecke operators resolves the first two of Ramanujan's conjectures.

Exercise 5.2.8. *Let f be $E_4\Delta$, $E_6\Delta$, $E_8\Delta$, $E_{10}\Delta$, or $E_{14}\Delta$. Show that the Fourier coefficients of f are multiplicative.*

Exercise 5.2.9. *Let p be a prime number. Show that*

$$\sum_{n=0}^{\infty} \tau(p^n) x^n = \frac{1}{1 - \tau(p)x + p^{11}x^2}.$$

Exercise 5.2.10. *Prove that for $n > 1$,*

$$(1-n)\tau(n) = 24\sum_{j=1}^{n-1}\sigma_1(j)\tau(n-j).$$

Exercise 5.2.11. *If n is even, prove that $8|\tau(n)$.*

Exercise 5.2.12. *If p is prime, show that $\tau(p)$ is even.*

5.3 Modular forms and Dirichlet series

Ramanujan's third conjecture that $|\tau(p)| \leq 2p^{11/2}$ was resolved in 1974 by P. Deligne as a consequence of his solution to the celebrated Weil conjectures. In this section, we will establish a much weaker bound (due to Hecke) and initiate a study of Dirichlet series attached to modular forms. In his 1916 paper, Ramanujan suggested the study of the associated Dirichlet series

$$L_\Delta(s) := \sum_{n=1}^{\infty}\frac{\tau(n)}{n^s}$$

as a means to understand the nature of $\tau(n)$. Indeed, Ramanujan noted that his conjectures (i) and (ii) imply that we can write

$$\sum_{n=1}^{\infty}\frac{\tau(n)}{n^s} = \prod_p\left(1-\frac{\tau(p)}{p^s}+\frac{1}{p^{2s-11}}\right)^{-1}. \tag{5.4}$$

We will see below that $L_\Delta(s)$ converges absolutely for $\mathrm{Re}(s) > 7$ and extends to an entire function satisfying the functional equation

$$(2\pi)^{-s}\Gamma(s)L_\Delta(s) = (2\pi)^{-(12-s)}\Gamma(12-s)L_\Delta(12-s), \tag{5.5}$$

where

$$\Gamma(s) = \int_0^\infty e^{-t}t^{s-1}dt$$

is Euler's Γ-function.

In his lectures on the work of Ramanujan, Hardy [13, p. 185] writes "The functional equation (5.5) must have been familiar to Ramanujan, but I cannot find an explicit statement of it either in his collected papers or in the notebooks." Apparently, equation (5.5) was proved by Wilton in 1929, but it was Erich Hecke who recognized (5.5) and (5.4) as a special case of a general theory of modular forms, Dirichlet series, and Euler products. This theory we now call Hecke theory. In this section, we discuss some aspects of this theory pertaining to level one.

Exercise 5.3.1. *Let p be prime. Prove that Ramanujan's conjecture*

$$|\tau(p)| \leq 2p^{11/2}$$

is equivalent to the assertion that the roots of the quadratic polynomial $1 - \tau(p)x + p^{11}x^2$ are non-real.

Exercise 5.3.2. *Assuming Ramanujan's conjecture that $|\tau(p)| \leq 2p^{11/2}$, prove that $|\tau(n)| \leq n^{11/2}d(n)$ for every natural number n. Here $d(n) = \sigma_0(n)$ is the number of divisors of n.*

Exercise 5.3.3. *Let $f \in M_k(\mathrm{SL}_2(\mathbb{Z}))$. Show that $y^{k/2}|f(z)|$ is invariant under $\mathrm{SL}_2(\mathbb{Z})$, where $y = \mathrm{Im}(z)$.*

Theorem 5.3.4 (Hecke). *If $f \in S_k(\mathrm{SL}_2(\mathbb{Z}))$ has Fourier expansion*

$$f(z) = \sum_{n=1}^{\infty} a_f(n)q^n$$

at $i\infty$, then $a_f(n) = O\big(n^{k/2}\big)$.

Proof. By the previous exercise, the continuous function $\phi(z) = y^{k/2}|f(z)|$ is invariant under the modular group. Hence we consider $z \in \mathcal{F}$, the standard fundamental domain. As $\mathrm{Im}\, z \to \infty$, $q \to 0$ and so since f is a cusp form,

$$|f(z)| = O(q) = O\big(e^{-2\pi y}\big)\,. \tag{5.6}$$

Thus $\phi(z) = O\big(y^{k/2}e^{-2\pi y}\big)$ and so $\phi(z) \to 0$ as $y \to \infty$. This implies that ϕ is bounded in $\mathcal{F}$ and hence in $\mathbb{H}$. In other words, there is a constant c such that

$$y^{k/2}|f(z)| \leq c, \qquad \forall z \in \mathbb{H}. \tag{5.7}$$

Now fix y and vary x between 0 and 1. Then $q = e^{2\pi i(x+iy)}$ traverses counterclockwise the circle C_y of radius $e^{-2\pi y}$, centered at zero. By Cauchy's residue theorem,

$$a_f(n) = \frac{1}{2\pi i}\int_{C_y} f(z)q^{-n-1}dq = \int_0^1 f(x+iy)q^{-n}dx.$$

Using (5.7) we obtain

$$|a_f(n)| \leq cy^{-k/2}e^{-2\pi ny}$$

for any $y > 0$. Choosing $y = 1/n$ gives $a_f(n) = O\big(n^{k/2}\big)$. □

Corollary 5.3.5. *We have $\tau(n) = O\big(n^6\big)$.*

Corollary 5.3.6. *Suppose $f = \sum_{n=1}^{\infty} a_f(n)q^n \in S_k(\mathrm{SL}_2(\mathbb{Z}))$. The Dirichlet series*

$$L_f(s) = \sum_{n=1}^{\infty} \frac{a_f(n)}{n^s}$$

converges absolutely for $\mathrm{Re}(s) > k/2 + 1$.

Proof. This is immediate from Hecke's estimate for $a_f(n)$ given by Theorem 5.3.4. □

We now proceed to derive the analytic continuation and functional equation for $L_f(s)$, following Hecke.

Theorem 5.3.7. *Let* $f \in S_k(\mathrm{SL}_2(\mathbb{Z}))$. *The function* $L_f(s)$ *extends to an entire function and satisfies the functional equation*

$$(2\pi)^{-s}\Gamma(s)L_f(s) = i^k(2\pi)^{-(k-s)}\Gamma(k-s)L_f(k-s).$$

Proof. The estimate (5.6) shows that the integral (for fixed $y_0 > 0$)

$$\int_{y_0}^{\infty} f(iy)y^{s-1}dy$$

converges absolutely for all complex values of s. Since f is of exponential decay on the imaginary axis, we may differentiate under the integral sign and deduce that the integral defines an entire function of s. Moreover, by modularity $f(iy) = i^{-k}y^{-k}f(i/y) = O\big(y^{-k}e^{-2\pi/y}\big)$ as $y \to 0$. Hence the limit as $y_0 \to 0$ of the integral exists and defines an entire function. Thus we have the entire function:

$$\int_0^{\infty} f(iy)y^{s-1}dy = \sum_{n=1}^{\infty} a_f(n) \int_0^{\infty} e^{-2\pi ny}y^{s-1}dy = (2\pi)^{-s}\Gamma(s)L_f(s),$$

after a change of variable ($t = 2\pi ny$) in the integral. We split the integral on the left hand side as

$$\int_0^1 f(iy)y^{s-1}dy + \int_1^{\infty} f(iy)y^{s-1}dy.$$

We change variables ($y \mapsto 1/y$) in the first integral and the integral becomes

$$\int_1^{\infty} f\left(\frac{i}{y}\right) y^{-s-1}dy.$$

Since $i/y = -1/iy$, and f is a modular form of weight k,

$$f\left(-\frac{1}{iy}\right) = (iy)^k f(iy),$$

so that

$$\int_1^\infty f\left(\frac{i}{y}\right) y^{-s-1} dy = i^k \int_1^\infty f(iy) y^{k-s-1} dy.$$

Thus

$$(2\pi)^{-s}\Gamma(s)L_f(s) = \int_1^\infty f(iy)\left(y^s + i^k y^{k-s}\right)\frac{dy}{y},$$

from which the functional equation is immediate. □

Exercise 5.3.8. *Suppose that $f \in M_k(\mathrm{SL}_2(\mathbb{Z}))$ with Fourier expansion at $i\infty$ given by*

$$f(z) = \sum_{n=0}^{\infty} a_f(n) q^n,$$

and that $a_f(0) \neq 0$. Show that $L_f(s) = \sum_{n=1}^{\infty} \frac{a_f(n)}{n^s}$ has an analytic continuation to the entire complex plane, except for a simple pole at $s = 0$ and $s = k$ and satisfies the functional equation

$$(2\pi)^{-s}\Gamma(s)L_f(s) = i^k(2\pi)^{-(k-s)}\Gamma(k-s)L_f(k-s).$$

5.4 Ramanujan congruences

There are many interesting arithmetical properties of the τ-function, too numerous to expound here. We will content ourselves with a sampling of such results and invite the reader to consult the article [45] by H.P.F. Swinnerton-Dyer. Here is an example of an elementary congruence for the τ-function.

Exercise 5.4.1. *Prove that $\tau(n) \equiv n\sigma_1(n) \pmod 5$.*

Some of these congruences are intertwined with yet another interesting arithmetical function, the partition function. In fact, in many of the proofs of Ramanujan's congruences (to be stated below), Theorem 5.1.9 plays a fundamental role. We illustrate this through the following discussion.

Let $p(n)$ be defined by the series

$$\sum_{n=0}^{\infty} p(n) q^n = \prod_{n=1}^{\infty} (1-q^n)^{-1}.$$

The function $p(n)$ is the partition function and counts the number of ways of writing $n = \lambda_1 + \lambda_2 + \cdots + \lambda_r$ with $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_r \geq 0, \lambda_i \in \mathbb{Z}$. The partition function appears in many branches of mathematics ranging from number theory, combinatorics, group theory, representation theory, to mathematical physics. Ramanujan discovered that

$$p(5n+4) \equiv 0 \pmod 5$$
$$p(7n+5) \equiv 0 \pmod 7$$
$$p(11n+6) \equiv 0 \pmod{11},$$

and these were proved by him using the theory of elliptic functions. These are often called Ramanujan congruences. We show below how the theory of modular forms leads to "quick" proofs of these congruences. We follow the treatment of Berndt [4].

Theorem 5.4.2. *For all $n \geq 0$, we have $p(5n+4) \equiv 0 \pmod 5$.*

Proof. We have already seen in Exercise 5.4.1 that $Q \equiv 1 \pmod 5$ and $R \equiv P \pmod 5$. Thus

$$1728\Delta = Q^3 - R^2 \equiv Q - P^2 \pmod 5,$$

and from Theorem 5.1.9,

$$3DP \equiv Q - P^2 \pmod 5.$$

Now by Theorem 5.1.4,

$$1728\Delta = 1728q \prod_{n=1}^{\infty}(1-q^n)^{24} \equiv 3q \prod_{n=1}^{\infty} \frac{(1-q^n)^{25}}{1-q^n} \pmod 5$$

and $(1-q^n)^{25} \equiv 1 - q^{25n} \pmod 5$. Hence

$$3q \prod_{n=1}^{\infty}(1-q^n)^{-1} \prod_{n=1}^{\infty}(1-q^{25n}) \equiv 3DP \pmod 5.$$

Therefore,

$$q\left(\sum_{n=0}^{\infty} p(n)q^n\right) \equiv DP \prod_{n=1}^{\infty}(1-q^{25n})^{-1} \pmod 5.$$

We compare the coefficient of q^{5m} of both sides of the congruence and note that $DP \equiv \sum_{n=1}^{\infty} n\sigma(n)q^n \pmod 5$ so that the $5m$-th coefficient of the right hand side is also divisible by 5. This yields $p(5m-1) \equiv 0 \pmod 5$ as required. □

Exercise 5.4.3. *Prove that $Q^2 \equiv P \pmod 7$ and $R \equiv 1 \pmod 7$.*

Exercise 5.4.4. *Prove that*

$$(Q^3 - R^2)^2 \equiv (QP-1)^2 \equiv -q\frac{dP^2}{dq} - 3q\frac{dQ}{dq} \pmod 7.$$

Exercise 5.4.5. *Prove that*

$$(Q^3 - R^2)^2 \equiv q^2 \left(\sum_{n=0}^{\infty} p(n) q^n \right) \prod_{n=1}^{\infty} (1 - q^{49n}) \pmod 7.$$

Deduce that $p(7n + 5) \equiv 0 \pmod 7$.

We close this section by mentioning Lehmer's conjecture that $\tau(n) \neq 0$. Recently, work of Brunier, Ono, and Rhoades [6] relates the question to irrationality of Fourier coefficients of harmonic weak Maass forms.

5.5 Supplementary problems

Exercise 5.5.1. *Prove that* $E_6^2 = E_{12} - \frac{762048}{691}\Delta$.

Exercise 5.5.2. *Show that* $E_{12} = 1 + \frac{65520}{691} \sum_{n=1}^{\infty} \sigma_{11}(n) q^n$.

Exercise 5.5.3. *Prove that* $\tau(n) \equiv \sigma_{11}(n) \pmod{691}$.

Exercise 5.5.4. *Prove that* $E_{14} \equiv E_2 \pmod{13}$.

Exercise 5.5.5. *Prove that* $\tau(n) \equiv n\sigma_3(n) \pmod 7$.

Exercise 5.5.6. *Prove that*

$$\sum_{j=1}^{n-1} \sigma_5(j)\sigma_5(n-j) = \frac{65}{(252)(691)} \sigma_{11}(n) + \frac{\sigma_5(n)}{252} - \frac{3\tau(n)}{691}.$$

Deduce that

$$\sum_{j=1}^{n-1} \sigma_5(j)\sigma_5(n-j) = \frac{65}{(252)(691)} \sigma_{11}(n) + O\left(n^6\right).$$

Exercise 5.5.7. *If* $k_1 > k_2 \geq 4$ *are even, prove that for any* $\epsilon > 0$,

$$\sum_{j=1}^{n-1} \sigma_{k_1-1}(j)\sigma_{k_2-1}(n-j) = c(k_1, k_2)\sigma_{k_1+k_2-1}(n) + O\left(n^{k_1-1+\epsilon}\right),$$

where $c(k_1, k_2) \in \mathbb{Q}$ *is a constant depending on* k_1 *and* k_2.

Exercise 5.5.8. *Let* f_k *be a sequence of complex numbers satisfying the recurrence with* $f_0 = 1$, *and*

$$f_{k+1} = f f_k - g f_{k-1}, \qquad k \geq 1$$

for some constants f *and* g. *Write*

$$1 - fx + gx^2 = (1 - \alpha x)(1 - \beta x).$$

Show that

$$f_k = \frac{\alpha^{k+1} - \beta^{k+1}}{\alpha - \beta}.$$

If $\alpha = \beta$, the right hand side is viewed as $\alpha^k + \alpha^{k-1}\beta + \cdots + \alpha\beta^{k-1} + \beta^k = (k+1)\alpha^k$.

Exercise 5.5.9. *Let g be a completely multiplicative function. An arithmetical function f is called g-*multiplicative *if the equation*

$$f(m)f(n) = \sum_{d|m,n} g(d) f\left(\frac{mn}{d^2}\right) \tag{5.8}$$

holds for all natural numbers m and n. Show that f is g-multiplicative if and only if f is multiplicative and for all primes p and all natural numbers $k \geq 1$, we have

$$f\left(p^{k+1}\right) = f(p)f(p^k) - g(p)f(p^{k-1}). \tag{5.9}$$

Exercise 5.5.10. *Let $g_k(n) = n^k$. Show that the Ramanujan τ-function is g_{11}-multiplicative.*

Exercise 5.5.11. *Let $g_s(n) = n^s$. Show that $\sigma_s(n) := \sum_{d|n} d^s$ is g_s-multiplicative.*

Exercise 5.5.12. *Let F and G be two complex-valued functions defined on $\mathbb{N} \times \mathbb{N}$. Suppose that*

$$F(m,n) = \sum_{d|n} G(m/d, n/d).$$

Show that

$$G(m,n) = \sum_{d|m,n} \mu(d) F(m/d, n/d)$$

and conversely. (Here μ denotes the Möbius function.)

Exercise 5.5.13. *Show that the Ramanujan τ-function satisfies*

$$\tau(mn) = \sum_{d|m,n} d^{11} \mu(d) \tau(m/d) \tau(n/d)$$

for any two natural numbers m, n.

Exercise 5.5.14. *Let A be the $n \times n$ matrix whose (i,j) entry is $\tau(ij)$. Show that the determinant of A is given by $n!^{11}\mu(1)\mu(2)\cdots\mu(n)$. In particular, the determinant is zero if $n \geq 4$.*

Exercise 5.5.15. *Show that $\tau(n)$ is odd if and only if n is an odd square.*

Exercise 5.5.16. *Let* $r = \dim M_k(SL_2(\mathbb{Z}))$. *Put*

$$F_k := E_{12r-k+2}\Delta^{-r}.$$

Show that F_k *has q-expansion*

$$C_{kr}q^{-r} + \cdots + C_{k1}q^{-1} + C_{k0} + \cdots$$

with $C_{kr} = 1$ *and* $C_{kj} \in \mathbb{Z}$ *for all* j.

Exercise 5.5.17. *Let* j *denote the* j*-function. Show that for any integer* $m \geq 0$,

$$j^m \frac{dj}{dz}$$

has a q-expansion without a constant term.

Exercise 5.5.18. *Let* f *be a modular form of weight* k *for the full modular group with q-expansion*

$$f(z) = a_0 + a_1 q + a_2 q^2 + \cdots$$

With C_{kj} *defined as in Exercise 5.5.16, show that*

$$C_{k0}a_0 + C_{k1}a_1 + \cdots + C_{kr}a_r = 0.$$

Exercise 5.5.19. *Suppose that* $k \equiv 2 \pmod 4$. *With notation as in the previous exercise, show that* $C_{k0} \neq 0$.

Exercise 5.5.20. *Suppose that* $k \equiv 0 \pmod 4$. *With notation as in the previous exercise, show that* $C_{k0} \neq 0$.

Chapter 6
Modular Forms of Higher Level

6.1 Modular forms for congruence subgroups

Let Γ be a congruence subgroup of $\mathrm{SL}_2(\mathbb{Z})$ of level N. Recall that $\mathbb{H}^\star$ is the extended upper half-plane:

$$\mathbb{H}^\star = \mathbb{H} \cup \mathbb{Q} \cup \{i\infty\}.$$

We are interested in holomorphic functions $f(z) : \mathbb{H} \to \mathbb{C}$ for which

$$f\left(\frac{az+b}{cz+d}\right) = (cz+d)^k f(z), \qquad \forall \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma. \tag{6.1}$$

The reader may wish to stop and review Section 4.1 before proceeding.

The punctured fundamental neighborhood

$$U_c = \{z \in \mathbb{H} : \mathrm{Im}(z) > c\}$$

of $i\infty$ is mapped via $z \mapsto e^{2\pi iz/N}$ onto the punctured open disc centered at zero, of radius $e^{-2\pi c/N}$. We may write $q^{1/N} = e^{2\pi iz/N}$ for a typical point in this punctured disc.

Let h be the least positive integer such that $\left(\begin{smallmatrix} 1 & h \\ 0 & 1 \end{smallmatrix}\right) \in \Gamma$. Since Γ is of level N, we have $h \leq N$. (In fact, a straightforward $\gcd$ argument shows that $h|N$.) For any function satisfying (6.1) we have $f(z+h) = f(z)$. Hence we have a well-defined map, which we also call f, from the unit disc to $\mathbb{C}$:

$$q^{1/h} \mapsto f(z)$$

where $z \in \mathbb{H}$ is any point such that $q^{1/h} = e^{2\pi iz/h}$. If $f(z) : \mathbb{H} \to \mathbb{C}$ is holomorphic, then $f(q^{1/h})$ will be holomorphic on the punctured unit disc and hence will have a Laurent expansion centered at $q^{1/h} = 0$,

© Springer Science+Business Media Singapore 2016 and Hindustan Book Agency 2016

M.R. Murty et al., *Problems in the Theory of Modular Forms*, HBA Lecture Notes in Mathematics, DOI 10.1007/978-981-10-2651-5_6

$$f(q^{1/h}) = \sum_{n=-\infty}^{\infty} a_n q^{n/h}.$$

We call

$$f(z) = \sum_{n=-\infty}^{\infty} a_n e^{2\pi i z n/h}$$

a *Fourier series at* $i\infty$.

More generally, for any $\gamma = \left(\begin{smallmatrix} r & s \\ t & u \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$, the function

$$(f|\gamma)(z) = (tz+u)^{-k} f\left(\frac{rz+s}{tz+u}\right)$$

will again be holomorphic on $\mathbb{H}$ and will now be modular under $\gamma^{-1}\Gamma\gamma$. That is

$$(f|\gamma) \,|\, \begin{pmatrix} a & b \\ c & d \end{pmatrix} = f|\gamma, \qquad \forall \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \gamma^{-1}\Gamma\gamma.$$

Since $\Gamma(N)$ is normal in $\mathrm{SL}_2(\mathbb{Z})$ by Exercise 2.2.2, the group $\gamma^{-1}\Gamma\gamma$ will also be a level N congruence subgroup. Thus we let h be the least positive integer such that $\left(\begin{smallmatrix} 1 & h \\ 0 & 1 \end{smallmatrix}\right) \in \gamma^{-1}\Gamma\gamma$ so that $f|\gamma$ will be h-periodic:

$$(f|\gamma)(z+h) = (f|\gamma)(z).$$

As above, we consider the holomorphic function on the unit disc given by

$$q^{1/h} \mapsto (f|\gamma)(z)$$

where $z \in \mathbb{H}$ is any point such that $q^{1/h} = e^{2\pi i z/h}$. Now $(f|\gamma)$ will have a Laurent expansion

$$(f|\gamma)(q^{1/h}) = \sum_{n=-\infty}^{\infty} b_n q^{n/h}$$

centered at $q^{1/h} = 0$. We say that

$$(f|\gamma)(z) = \sum_{n=-\infty}^{\infty} b_n e^{2\pi i z n/h} \tag{6.2}$$

is a *Fourier expansion of* f *at* $\gamma(i\infty)$.

It is tempting to define the order of f at $\gamma(i\infty)$ to be the index of the least non-zero coefficient in (6.2). However, there is a technical complication which can occur when $\left(\begin{smallmatrix} -1 & 0 \\ 0 & -1 \end{smallmatrix}\right) \notin \Gamma$. It is possible that there is some $\left(\begin{smallmatrix} -1 & h' \\ 0 & -1 \end{smallmatrix}\right) \in \gamma^{-1}\Gamma\gamma$ with $h' \geq 1$ but $\left(\begin{smallmatrix} 1 & h' \\ 0 & 1 \end{smallmatrix}\right) \notin \gamma^{-1}\Gamma\gamma$. In this case, $h = 2h'$. The theory of Riemann surfaces (which we sidestep because the reader may not yet be familiar with Riemann surfaces) suggests that the Laurent

series (6.2) be considered as an expansion in terms of $q^{1/h'}$ rather than $q^{1/h} = \left(q^{1/h'}\right)^{1/2}$. This leads to the possibility of half-integer orders at these so-called *irregular cusps*. Therefore, we define the *width* of the cusp $\gamma\,(i\infty)$ to be the least positive integer h' such that at least one of

$$\begin{pmatrix} 1 & h' \\ 0 & 1 \end{pmatrix} \in \gamma^{-1}\Gamma\gamma$$

or

$$\begin{pmatrix} -1 & h' \\ 0 & -1 \end{pmatrix} \in \gamma^{-1}\Gamma\gamma$$

holds and write the expansion of $f|\gamma$ as

$$(f|\gamma)(z) = \sum_{n=-\infty}^{\infty} c_n e^{2\pi i z n/h'}. \tag{6.3}$$

Now we define the *order of f at $\gamma(i\infty)$* to be the index of the least non-zero coefficient in (6.3):

$$v_{\gamma(i\infty)}(f) := \inf\left\{n \in \frac{1}{2}\mathbb{Z} : c_n \neq 0\right\}.$$

Exercise 6.1.1. *Suppose $f : \mathbb{H} \to \mathbb{C}$ is holomorphic and satisfies (6.1) for a congruence subgroup Γ of level N. Suppose $\gamma, \tau \in \mathrm{SL}_2(\mathbb{Z})$ and that $\gamma(i\infty)$ and $\tau(i\infty)$ are Γ-equivalent. Prove that $v_{\gamma(i\infty)}(f) = v_{\tau(i\infty)}(f)$. That is, the order of f at a Γ-equivalence class of cusps is well-defined.*

If $-\infty < v_{\gamma(i\infty)}(f) < 0$, then f is said to be *meromorphic at $\gamma(i\infty)$* with a pole of order $|v_{\gamma(i\infty)}(f)|$. If $v_{\gamma(i\infty)}(f) \geq 0$ then f is said to be *holomorphic at $\gamma(i\infty)$*. If $v_{\gamma(i\infty)}(f) > 0$ then we say that *f vanishes at $\gamma(i\infty)$* and has a zero of order $v_{\gamma(i\infty)}(f)$.

A *modular form of weight k for Γ* is a holomorphic function $f : \mathbb{H} \to \mathbb{C}$ satisfying (6.1) which is holomorphic at every cusp. We denote by $M_k(\Gamma)$ the $\mathbb{C}$-vector space of modular forms of weight k for Γ. If $f \in M_k(\Gamma)$ is such that f vanishes at all the cusps, we say f is a *cusp form* of weight k for Γ and denote by $S_k(\Gamma)$ the $\mathbb{C}$-vector space of such forms.

6.2 Eisenstein series for $\Gamma(N)$

Exercise 6.2.1. *Let $k \geq 3$ and N a natural number. Let $g \in (\mathbb{Z}/N\mathbb{Z})^2$. Define the Eisenstein series*

$$G_{k,g}(z) = \sum_{(m,n)\equiv g \pmod N}{}' \frac{1}{(mz+n)^k}, \qquad z \in \mathbb{H}$$

where the summation is over all integers m, n satisfying the congruence condition $(m,n) \equiv g \pmod N$ and the dash on the summation means we exclude $(m,n) = (0,0)$ in the case $g = (0,0)$. For any $\gamma \in \mathrm{SL}_2(\mathbb{Z})$, show that $G_{k,g}|\gamma = G_{k,g\gamma}$. Conclude that $G_{k,g} \in M_k(\Gamma(N))$.

Exercise 6.2.2. *Show that $G_{k,g} = (-1)^k G_{k,-g}$. Deduce that if k is odd and $2g \equiv (0,0) \pmod N$, then $G_{k,g} = 0$.*

Exercise 6.2.3. *Let Γ be a congruence subgroup of $\mathrm{SL}_2(\mathbb{Z})$ of level N. If $\gamma_1, \ldots, \gamma_r$ are distinct right coset representatives of $\Gamma(N)$ in Γ, show that*

$$f := \sum_{j=1}^{r} G_{k,g\gamma_j} \in M_k(\Gamma).$$

Exercise 6.2.4. *Let Γ_1, Γ_2 be congruence subgroups of $\mathrm{SL}_2(\mathbb{Z})$ such that $\Gamma_1 \subset \Gamma_2$. If $f \in M_k(\Gamma_1)$ and $\{\gamma_1, \ldots, \gamma_r\}$ is a complete set of right coset representatives of Γ_1 in Γ_2, show that*

(a) $\displaystyle\sum_{j=1}^{r} f|\gamma_j \in M_k(\Gamma_2)$,

(b) $\displaystyle\prod_{j=1}^{r} (f|\gamma_j) \in M_{kr}(\Gamma_2)$.

Exercise 6.2.5. *Show that $\Gamma_0(4)$ is generated by $T = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, by $\left(\begin{smallmatrix} 1 & 0 \\ 4 & 1 \end{smallmatrix}\right)$, and by $\left(\begin{smallmatrix} -1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$.*

Exercise 2.3.3 shows that $[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(4)] = 6$ and the reader may easily verify that

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ 1 & 2 \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ 1 & 3 \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ 1 & 4 \end{pmatrix} \tag{6.4}$$

is a set of right coset representatives.

Exercise 6.2.6. *Show that there are exactly three inequivalent cusps for the group $\Gamma_0(4)$. Find the widths of each cusp.*

6.3 Fourier expansions of higher level Eisenstein series

As in the previous section, let $g = (a_1, a_2) \in (\mathbb{Z}/N\mathbb{Z})^2$ and consider $G_{k,g}(z)$. We derive the q-expansion of $G_{k,g}(z)$. Let us write

$$G_{k,g}(z) = b_{k,g}(0) + \sum_{n=1}^{\infty} b_{k,g}(n) e^{2\pi i n z/N}. \tag{6.5}$$

Theorem 6.3.1. *The Fourier coefficients $b_{k,g}(n)$ in (6.5) are given as follows:*

$$b_{k,g}(0) = \begin{cases} 0 & \text{if } a_1 \not\equiv 0 \pmod N, \\ \sum_{m \equiv a_2 \pmod N} m^{-k} & \text{if } a_1 \equiv 0 \pmod N, \end{cases}$$

and

$$b_{k,g}(n) = \frac{(-2\pi i)^k}{N^k (k-1)!} \sum_{\substack{d \mid n \\ n/d \equiv a_1 \pmod N}} d^{k-1} (\operatorname{sgn} d) e^{2\pi i a_2 d/N}$$

where the summation is over all divisors of n (positive and negative) and

$$\operatorname{sgn} d = \begin{cases} +1 & \text{if } d > 0, \\ -1 & \text{if } d < 0. \end{cases}$$

Proof. The computation of $b_{k,g}(0)$ was done at the end of the first paragraph of the solution to Exercise 6.2.1. We separate this term out and write

$$G_{k,g}(z) = b_{k,g}(0) + \sum_{\substack{m \equiv a_1 \pmod N \\ m \neq 0}} \sum_{n \equiv a_2 \pmod N} \left(\frac{1}{mz+n} \right)^k.$$

Writing $n = Nj + a_2$, we can write the inner sum as

$$\sum_{j \in \mathbb{Z}} (mz + Nj + a_2)^{-k} = N^{-k} \sum_{j \in \mathbb{Z}} \left(\frac{mz + a_2}{N} + j \right)^{-k}.$$

As in the level one case, we use Lipschitz's formula (Theorem 4.2.2) which states that for $k \geq 2$ and $z \in \mathbb{H}$, we have

$$\sum_{n \in \mathbb{Z}} \frac{1}{(z+n)^k} = \frac{(-2\pi i)^k}{(k-1)!} \sum_{n=1}^{\infty} n^{k-1} e^{2\pi i n z}.$$

Since $(mz + a_2)/N \in \mathbb{H}$ if and only if $m > 0$, we first consider the cases with $m > 0$ for which we have

$$\sum_{j \in \mathbb{Z}} \left(\frac{mz + a_2}{N} + j \right)^{-k} = \frac{(-2\pi i)^k}{(k-1)!} \sum_{d=1}^{\infty} d^{k-1} e^{2\pi i d a_2/N} e^{2\pi i d m z/N}.$$

If $m < 0$, we can re-write $m = -m'$ with $m' > 0$ and get

$$(-1)^k \sum_{j\in\mathbb{Z}} \left(\frac{m'z-a_2}{N}-j\right)^{-k} = (-1)^k \frac{(-2\pi i)^k}{(k-1)!} \sum_{d=1}^{\infty} d^{k-1} e^{-2\pi i d a_2/N} e^{2\pi i d m' z/N}.$$

We can combine both Fourier expansions (by grouping together $dm = n$ and $dm' = n$) to get

$$b_{k,g}(n) = \frac{(-2\pi i)^k}{N^k(k-1)!} \sum_{\substack{d\mid n \\ n/d\equiv a_1 \pmod N}} d^{k-1} (\operatorname{sgn} d)\, e^{2\pi i a_2 d/N}$$

allowing for both positive and negative integers d that divide n in the summation. □

Exercise 6.3.2. *For $g = (a_1, 0) \in (\mathbb{Z}/N\mathbb{Z})^2$ and $n \neq 0$, show that*

$$b_{k,g}(n) = \frac{(-2\pi i)^k}{N^k(k-1)!} \sum_{\substack{d\mid n \\ n/d\equiv a_1 \pmod N}} d^{k-1}(\operatorname{sgn} d).$$

Exercise 6.3.3. *If $g = (0, a_2) \in (\mathbb{Z}/N\mathbb{Z})^2$, show that $b_{k,g}(n) = 0$ if $N \nmid n$, and*

$$b_{k,g}(Nn) = \frac{(-2\pi i)^k}{N^k(k-1)!} \sum_{d\mid n} d^{k-1}(\zeta^{da_2} + (-1)\zeta^{-da_2}),$$

where $\zeta = e^{2\pi i/N}$. Deduce that $G_{k,g}(z)$, with $g = (0, a_2)$, has a q-expansion of the form

$$\sum_{n=0}^{\infty} a_n q^n, \qquad q = e^{2\pi i z}.$$

Exercise 6.3.4. *Prove that if $g = (0, a_2) \in (\mathbb{Z}/N\mathbb{Z})^2$, then $G_{k,g} \in M_k(\Gamma_1(N))$ for $k \geq 3$.*

Exercise 6.3.5. *Let*

$$E_{2,N}(z) := E_2(z) - NE_2(Nz).$$

Show that $E_{2,N} \in M_2(\Gamma_0(N))$.

Exercise 6.3.6. *If $f \in M_k(\mathrm{SL}_2(\mathbb{Z}))$, show that*

$$f_N(z) := f(z) - Nf(Nz) \in M_k(\Gamma_0(N)).$$

In particular, deduce that for $k \geq 3$,

$$E_{k,N}(z) := E_k(z) - NE_k(Nz) \in M_k(\Gamma_0(N)).$$

6.4 The valence and dimension formulas

It is possible to derive the analogs of the valence and dimension formulas for any congruence subgroup Γ of $\mathrm{SL}_2(\mathbb{Z})$. The general method used to derive these formulas is via the Riemann-Roch theorem, which is usually a topic in Riemann surface theory and algebraic geometry. Thus, we content ourselves by merely stating the results and referring the interested reader to [9] for further details. Here are the analogues of Theorem 4.3.1 and Corollary 4.4.2, for an arbitrary congruence subgroup Γ of $\mathrm{SL}_2(\mathbb{Z})$.

We first need to define elliptic points. A point $z \in \mathbb{H}$ is called an *elliptic point* for Γ if $\{\pm I\}\,\Gamma_z$ is strictly larger than $\{\pm I\}$, where $\Gamma_z = \{\gamma \in \Gamma : \gamma z = z\}$ is the stabilizer subgroup of z. In other words, z is an elliptic point for Γ if and only if $\{\pm I\}\Gamma_z \neq \{\pm I\}$. We define the *order of an elliptic point* to be $|\{\pm I\}\Gamma_z/\{\pm I\}|$.

Exercise 6.4.1. *Show that the only elliptic points for* $\mathrm{SL}_2(\mathbb{Z})$ *lying in the standard fundamental domain* $\mathcal{F}$ *are* i, $\rho = e^{\pi i/3}$, *and* ρ^2.

Exercise 6.4.2. *Find the stabilizer subgroups* $\mathrm{SL}_2(\mathbb{Z})_i$ *and* $\mathrm{SL}_2(\mathbb{Z})_\rho$ *in* $\mathrm{SL}_2(\mathbb{Z})$ *and show that* $\mathrm{SL}_2(\mathbb{Z})_i \,/\, \{\pm I\}$ *and* $\mathrm{SL}_2(\mathbb{Z})_\rho \,/\, \{\pm I\}$ *have orders two and three, respectively.*

Exercise 6.4.3. *If* Γ *has finite index in* $\mathrm{SL}_2(\mathbb{Z})$, *show that there are only finitely many* Γ*-inequivalent elliptic points for* Γ.

Theorem 6.4.4 (The valence formula). *Let* Γ *be a congruence subgroup of* $\mathrm{SL}_2(\mathbb{Z})$ *and* $0 \neq f \in M_k(\Gamma)$. *Then*

$$\sum_{z \in \Gamma\backslash\mathbb{H}^\star} \frac{v_z(f)}{|\Gamma_z|} = \frac{k}{2}\left(\frac{\epsilon_2}{2} + \frac{2\epsilon_3}{3} + \epsilon_\infty + 2g - 2\right),$$

where the sum is over Γ*-equivalence classes of* $z \in \mathbb{H}^*$, $v_z(f)$ *denotes the order of* f *at* z, Γ_z *is the stabilizer of* z *in* $\{\pm I\}\Gamma/\{\pm I\}$, g *is the genus of* $\Gamma\backslash\mathbb{H}^\star$, ϵ_2 *is the number of elliptic points of order* 2, ϵ_3 *is the number of elliptic points of order* 3, *and* ϵ_∞ *is the number of* Γ*-inequivalent cusps.*

Exercise 6.4.5. *Verify that Theorem 6.4.4 reduces to Theorem 4.3.1 in the case* $\Gamma = \mathrm{SL}_2(\mathbb{Z})$.

Theorem 6.4.6. *(The dimension formula) Let* Γ *be a congruence subgroup of* $\mathrm{SL}_2(\mathbb{Z})$ *and* k *an even non-negative integer. Then*

$$\dim M_k(\Gamma) = \begin{cases} (k-1)(g-1) + [\frac{k}{4}]\epsilon_2 + [\frac{k}{3}]\epsilon_3 + \frac{k}{2}\epsilon_\infty & \text{if } k \geq 2 \\ 1 & \text{if } k = 0 \end{cases}$$

and

$$\dim S_k(\Gamma) = \begin{cases} \dim M_k(\Gamma) - \epsilon_\infty & \text{if } k \geq 4 \\ g & \text{if } k = 2\,, \\ 0 & \text{if } k = 0 \end{cases}$$

where $g, \epsilon_2, \epsilon_3, \epsilon_\infty$ have the same meaning as in Theorem 6.4.4.

For most applications, it is not essential to have results as precise as Theorems 6.4.4 and 6.4.6. Often, crude bounds suffice. To keep this monograph self-contained, we therefore take an alternate route using an idea of Sturm that allows us to reduce to the level one case.

Theorem 6.4.7 (Sturm's bound). *Let Γ be a congruence subgroup and $f \in M_k(\Gamma)$. Let $r_1, \ldots, r_t$ be the Γ-inequivalent cusps of Γ. If*

$$\sum_{i=1}^{t} v_{r_i}(f) > \frac{k[\mathrm{SL}_2(\mathbb{Z}) : \{\pm I\}\,\Gamma]}{12},$$

then $f = 0$.

Proof (Sketch). If $\{\pm I\}\,\Gamma = \mathrm{SL}_2(\mathbb{Z})$, the result is clear from the valence formula (Theorem 4.3.1). Let $M = [\mathrm{SL}_2(\mathbb{Z}) : \{\pm I\}\,\Gamma]$ and $1 = \gamma_1, \ldots, \gamma_M$ be a complete set of coset representatives for $\{\pm I\}\,\Gamma$ in $\mathrm{SL}_2(\mathbb{Z})$. Then, the function

$$F(z) = \prod_{j=1}^{M} (f|\gamma_j)(z)$$

is clearly an element of $M_{kM}(\mathrm{SL}_2(\mathbb{Z}))$ since for $\gamma \in \mathrm{SL}_2(\mathbb{Z})$, the products $\gamma_j\gamma$, $1 \leq j \leq M$, are again a complete set of coset representatives so that $F|\gamma = F$ for all $\gamma \in \mathrm{SL}_2(\mathbb{Z})$.

Consider one of the Γ-inequivalent cusps r_j. If this cusp has width h, then there are h coset representatives $\left\{\gamma_{r_j}^{(1)}, \ldots, \gamma_{r_j}^{(h)}\right\} \subseteq \{\gamma_1, \ldots, \gamma_M\}$ such that $r_j = \gamma_{r_j}^{(s)}(i\infty)$. Moreover, each of the corresponding Fourier expansions will have the same order:

$$(f|\gamma_{r_j}^{(s)})(z) = b_s q^{v_{r_j}(f)/h} + \cdots\,.$$

Hence the product of these will have order $v_{r_j}(f)$. Thus,

$$v_\infty(F) = \sum_{i=1}^{t} v_{r_i}(f) > \frac{kM}{12}.$$

By the level one case, we deduce that $F = 0$ and hence some $f|\gamma_j = 0$. But then $f = (f|\gamma_j)|\gamma_j^{-1} = 0|\gamma_j^{-1} = 0$. □

Exercise 6.4.8. *Let* $f = \sum_{n=0}^{\infty} a_n q^n$ *and* $g = \sum_{n=0}^{\infty} b_n q^n \in M_k(\Gamma)$ *for some congruence subgroup* Γ *of* $\mathrm{SL}_2(\mathbb{Z})$. *Suppose that* $a_n = b_n$ *for all* $n \leq k[\mathrm{SL}_2(\mathbb{Z}) : \{\pm\}\Gamma]/12$. *Show that* $f = g$.

Exercise 6.4.9. *For any congruence subgroup* Γ *of* $\mathrm{SL}_2(\mathbb{Z})$, *prove that*

$$\dim M_k(\Gamma) \leq \frac{k}{12}[\mathrm{SL}_2(\mathbb{Z}) : \{\pm\}\Gamma] + 1.$$

Exercise 6.4.10. *Prove that*

$$\dim M_4(\Gamma_0(4)) = 3.$$

(Hint: consider $E_4, E_{4,2}$, *and* $E_{4,4}$ *from Exercise 6.3.6.)*

Exercise 6.4.11. *Prove that*

$$M_2(\Gamma_0(2)) = \mathbb{C}E_{2,2},$$

where $E_{2,2}(z) = E_2(z) - 2E_2(2z)$.

Exercise 6.4.12. *Show that* $\dim M_2(\Gamma_0(4)) = 2$ *and that* $\{E_{2,2}, E_{2,4}\}$ *is a basis for* $M_2(\Gamma_0(4))$.

6.5 The Jacobi four-square theorem revisited

In Chapter 1, Corollary 1.4.6, we derived a formula for $r_4(n)$, the number of ways of writing n as a sum of four squares. In this section, we will use the theory of modular forms to derive the same formula. The method will also apply to derive formulas for $r_k(n)$, the number of ways of writing n as a sum of k squares, for other values of k.

Exercise 6.5.1. *Let* $\eta(z)$ *denote the Dedekind* η*-function. Prove that*

$$\theta(z) := \sum_{n=-\infty}^{\infty} q^{n^2} = \frac{\eta^5(2z)}{\eta^2(z)\eta^2(4z)},$$

where $q = e^{2\pi i z}$.

Theorem 6.5.2. *Let*

$$\theta(z) = \sum_{n=-\infty}^{\infty} e^{2\pi i n^2 z}.$$

Then

$$\theta\left(\frac{z}{4z+1}\right) = \sqrt{4z+1}\ \theta(z),$$

where the branch of the square root is the principal branch (whose image is contained in the right half-plane).

Proof. We use Exercises 6.5.1 and 5.1.5. From the former we have

$$\theta(z) = \frac{\eta^5(2z)}{\eta^2(z)\eta^2(4z)}$$

and from the latter we have

$$\eta\left(-\frac{1}{z}\right) = \sqrt{\frac{z}{i}}\eta(z).$$

It is immediate from the definition that $\eta(z+1) = e^{\frac{\pi i}{12}}\eta(z)$. Now,

$$\begin{aligned}\eta\left(\frac{4z}{4z+1}\right) &= \eta\left(-\frac{1}{4z+1}+1\right) = e^{\frac{\pi i}{12}}\eta\left(-\frac{1}{4z+1}\right)\\ &= e^{\frac{\pi i}{12}}\sqrt{\frac{4z+1}{i}}\eta(4z+1) = e^{\frac{\pi i}{6}}\sqrt{\frac{4z+1}{i}}\eta(4z).\end{aligned}$$

Also,

$$\begin{aligned}\eta\left(\frac{2z}{4z+1}\right) &= \eta\left(-\left(-\frac{1}{2z}-2\right)^{-1}\right) = \sqrt{i\left(\frac{4z+1}{2z}\right)}\eta\left(-\frac{1}{2z}-2\right)\\ &= e^{\frac{-\pi i}{6}}\sqrt{i\left(\frac{4z+1}{2z}\right)}\eta\left(-\frac{1}{2z}\right)\\ &= e^{\frac{-\pi i}{6}}\sqrt{i\left(\frac{4z+1}{2z}\right)}\sqrt{\frac{2z}{i}}\eta(2z).\end{aligned}$$

Finally,

$$\begin{aligned}\eta\left(\frac{z}{4z+1}\right) &= \eta\left(-\left(-\frac{1}{z}-4\right)^{-1}\right) = \sqrt{i\left(\frac{4z+1}{z}\right)}\eta\left(-\frac{1}{z}-4\right)\\ &= e^{\frac{-\pi i}{3}}\sqrt{i\left(\frac{4z+1}{z}\right)}\eta\left(-\frac{1}{z}\right)\\ &= e^{\frac{-\pi i}{3}}\sqrt{i\left(\frac{4z+1}{z}\right)}\sqrt{\frac{z}{i}}\eta(z).\end{aligned}$$

Putting everything together gives

$$\theta\left(\frac{z}{4z+1}\right) = \sqrt{4z+1}\ \theta(z). \qquad \square$$

Theorem 6.5.3 (Jacobi's four square theorem). *We have* $\theta^4 \in M_2(\Gamma_0(4))$ *and*

$$r_4(n) = 8\sum_{\substack{d|n\\ 4\nmid d}} d.$$

Proof. By Theorem 6.5.2, $\theta^4\left(\frac{z}{4z+1}\right) = (4z+1)^2\theta^4(z)$. Since $\theta^4(z+1) = \theta^4(z)$ and since $\Gamma_0(4)$ is generated by $\left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right)$ and $\left(\begin{smallmatrix}1&0\\4&1\end{smallmatrix}\right)$ and $-I$ by Exercise 6.2.5, we deduce $\theta^4 \in M_2(\Gamma_0(4))$. By Exercise 6.4.12, $E_{2,2}$ and $E_{2,4}$ comprise a basis for $M_2(\Gamma_0(4))$. Thus $\theta^4 = aE_{2,2} + bE_{2,4}$. A quick calculation comparing coefficients of q-expansions of both sides of this equation leads to

$$\theta^4 = -\frac{1}{3}E_{2,4},$$

which leads to the formula

$$r_4(n) = 8\Big(\sigma(n) - 4\sigma(n/4)\Big),$$

where $\sigma(n/4) = 0$ if $4 \nmid n$. If $4|n$, we write $n = 4n_1$ and $4\sigma(n_1)$ is the sum of the divisors d of n with $4|d$. This observation leads to the desired formula. $\square$

6.6 Supplementary problems

Exercise 6.6.1. *Show that* $\eta^{12}(2z) \in S_6(\Gamma_0(4))$.

Exercise 6.6.2. *Show that* $S_6(\Gamma_0(4)) = \eta^{12}(2z)\mathbb{C}$.

Exercise 6.6.3. *Let* $k \geq 6$ *be even. Show that* $S_k(\Gamma_0(4)) = \eta^{12}(2z)M_{k-6}(\Gamma_0(4))$.

Exercise 6.6.4. *For even* $k \geq 0$, *prove that* $\dim M_k(\Gamma_0(4)) = \frac{k}{2} + 1$.

Exercise 6.6.5. *Let* $r_{12}(n)$ *be the number of ways of writing* n *as a sum of* 12 *squares and let* $\eta^{12}(2z) = \sum_{n=1}^{\infty} a_n q^n$. *Show that*

$$r_{12}(n) = 8\sigma_5(n) - 512\sigma_5\left(\frac{n}{4}\right) + 16a_n.$$

Exercise 6.6.6. *Let* $f(z) \in M_k\left(\Gamma_1(N)\right)$ *and suppose* $r \in \mathbb{Z}$ *is positive. Show that* $f(rz) \in M_k\left(\Gamma_1(rN)\right)$. *Moreover, if* $f(z)$ *is a cusp form, then so is* $f(rz)$.

Chapter 7
The Petersson Inner Product

7.1 The Hilbert space of cusp forms

Let Γ be a congruence subgroup of $\mathrm{SL}_2(\mathbb{Z})$ and $\mathcal{F}$ any fundamental domain for Γ. Let $\overline{\Gamma} = \Gamma/\{\pm I\}$. For $f, g \in S_k(\Gamma)$, we define the **Petersson inner product** as

$$(f,g) = \frac{1}{\left[\mathrm{SL}_2(\mathbb{Z}) : \overline{\Gamma}\right]} \iint_{\mathcal{F}} y^k f(z)\overline{g(z)}\frac{dxdy}{y^2},$$

where $z = x+iy$. Since $\frac{dxdy}{y^2}$ and $y^k f(z)\overline{g(z)}$ are both Γ-invariant (by (3.2) and Theorem 3.4.4), the integral is well-defined and independent of the choice of fundamental domain. Moreover, it is easily seen to converge since both f and g are cusp forms and $|f(z)| = |g(z)| = O\big(e^{-2\pi y}\big)$ as $y \to \infty$.

Exercise 7.1.1. *Let $f, g \in M_k(\Gamma)$ and let $\mathcal{F}$ be any fundamental domain for the congruence subgroup Γ. Show that the integral*

$$\iint_{\mathcal{F}} y^k f(z)\overline{g(z)}\frac{dxdy}{y^2} < \infty$$

as long as one of $f, g \in S_k(\Gamma)$.

Exercise 7.1.2. *Prove that*

(i) (f,g) is linear in f,
(ii) (f,g) is conjugate symmetric. That is, $(f,g) = \overline{(g,f)}$,
(iii) $(f,f) > 0$ for $f \neq 0$.

Therefore, the Petersson inner product defines a Hermitian inner product on $S_k(\Gamma)$.

Theorem 7.1.3. *Suppose that $\Gamma_1 \subset \Gamma_2$ are congruence subgroups. Let $f, g \in S_k(\Gamma_2)$. Then (f,g) is independent of whether f, g are considered as elements of $S_k(\Gamma_1)$ or $S_k(\Gamma_2)$.*

© Springer Science+Business Media Singapore 2016 and Hindustan Book Agency 2016

M.R. Murty et al., *Problems in the Theory of Modular Forms*, HBA Lecture Notes in Mathematics, DOI 10.1007/978-981-10-2651-5_7

Proof. Let $\mathcal{F}_2$ be a fundamental domain for $\overline{\Gamma}_2$, and take coset representatives α_j such that $\overline{\Gamma}_2 = \bigcup_j \alpha_j \overline{\Gamma}_1$ and $\mathcal{F}_1 = \bigcup_j \alpha_j^{-1}(\mathcal{F}_2)$ is a fundamental domain for $\overline{\Gamma}_1$. Now

$$\begin{aligned}
&\frac{1}{\left[\mathrm{SL}_2(\mathbb{Z}) : \overline{\Gamma}_1\right]} \iint_{\mathcal{F}_1} y^k f(z)\overline{g(z)}\frac{dxdy}{y^2} \\
&= \frac{1}{\left[\mathrm{SL}_2(\mathbb{Z}) : \overline{\Gamma}_2\right] \cdot \left[\overline{\Gamma}_2 : \overline{\Gamma}_1\right]} \sum_j \iint_{\alpha_j^{-1}(\mathcal{F}_2)} y^k f(z)\overline{g(z)}\frac{dxdy}{y^2} \\
&= \frac{1}{\left[\mathrm{SL}_2(\mathbb{Z}) : \overline{\Gamma}_2\right] \cdot \left[\overline{\Gamma}_2 : \overline{\Gamma}_1\right]} \sum_j \iint_{\mathcal{F}_2} (\operatorname{Im} \alpha_j^{-1} z)^k f(\alpha_j^{-1} z)\overline{g(\alpha_j^{-1} z)}\frac{dxdy}{y^2}.
\end{aligned}$$

There are $\left[\overline{\Gamma}_2 : \overline{\Gamma}_1\right]$ summands and, since $f, g \in S_k(\Gamma_2)$, we find that they are all equal to

$$\iint_{\mathcal{F}_2} y^k f(z)\overline{g(z)}\frac{dxdy}{y^2},$$

so that the original integral is

$$\frac{1}{\left[\mathrm{SL}_2(\mathbb{Z}) : \overline{\Gamma}_2\right]} \iint_{\mathcal{F}_2} y^k f(z)\overline{g(z)}\frac{dxdy}{y^2},$$

as desired. □

Exercise 7.1.4. *Suppose $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$ has integer entries. Let $D = \det \alpha$. If $\Gamma(N) \subseteq \Gamma$ for some N, show that $\Gamma(ND) \subseteq \alpha^{-1}\Gamma\alpha$ and $\Gamma(ND) \subseteq \alpha\Gamma\alpha^{-1}$.*

Exercise 7.1.5. *Let $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$ have integer entries. Show that there exists $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ such that*

$$\gamma^{-1}\alpha = \begin{pmatrix} a & b \\ 0 & d \end{pmatrix},$$

where a and d are positive integers and $b \in \mathbb{Z}$.

Exercise 7.1.6. *Let $f \in M_k(\Gamma)$ with Γ a congruence subgroup of $\mathrm{SL}_2(\mathbb{Z})$. Let $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$. Show that $f|\alpha$ is holomorphic at every cusp $s \in \mathbb{Q} \cup \{i\infty\}$.*

Exercise 7.1.7. *Let $f \in S_k(\Gamma)$ with Γ a congruence subgroup of $\mathrm{SL}_2(\mathbb{Z})$. Let $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$. Show that $f|\alpha$ vanishes at every cusp $s \in \mathbb{Q} \cup \{i\infty\}$.*

Theorem 7.1.8. *Let Γ be a congruence subgroup of $\mathrm{SL}_2(\mathbb{Z})$. Let $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$ have integer entries and put $\Gamma' = \alpha^{-1}\Gamma\alpha \cap \Gamma$. Then Γ' is a congruence subgroup and the map*

$$f \mapsto f|\alpha$$

takes $M_k(\Gamma)$ to $M_k(\Gamma')$ and $S_k(\Gamma)$ to $S_k(\Gamma')$.

Proof. If $D = \det \alpha$, then by Exercise 7.1.4, $\Gamma(ND) \subseteq \alpha^{-1}\Gamma\alpha$, where N is the level of Γ. Thus $\Gamma(ND) \subset \Gamma \cap \alpha^{-1}\Gamma\alpha \subseteq \mathrm{SL}_2(\mathbb{Z})$ so that Γ' is a congruence subgroup of $\mathrm{SL}_2(\mathbb{Z})$.

If $\tau \in \Gamma'$, we can write $\tau = \alpha^{-1}\gamma\alpha$ for some $\gamma \in \Gamma$ so that

$$(f|\alpha)|\tau = f|(\alpha\tau) = f|(\gamma\alpha) = (f|\gamma)|\alpha = f|\alpha$$

since $f|\gamma$ for all $\gamma \in \Gamma$. By Exercise 7.1.6, $f|\alpha$ is holomorphic at the cusps for Γ'. The analogous result for $S_k(\Gamma')$ follows similarly from Exercise 7.1.7. □

Exercise 7.1.9. *Let Γ' be a subgroup of $\mathrm{GL}_2^+(\mathbb{R})$ and $\mathcal{F}'$ a fundamental domain for the action of Γ' on $\mathbb{H}$. If $\alpha \in \mathrm{GL}_2^+(\mathbb{R})$, show that $\alpha^{-1}\mathcal{F}'$ is a fundamental domain for the action of $\alpha^{-1}\Gamma'\alpha$ on $\mathbb{H}$.*

Theorem 7.1.10. *Let $f, g \in S_k(\Gamma)$ with Γ a congruence subgroup of $\mathrm{SL}_2(\mathbb{Z})$. Let $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$. Then*

$$(f, g) = (f|\alpha, g|\alpha),$$

where the latter inner product is with respect to $\Gamma' = \alpha^{-1}\Gamma\alpha \cap \Gamma$.

Proof. By Theorem 7.1.8, $f|\alpha, g|\alpha \in S_k(\Gamma')$ where $\Gamma' = \alpha^{-1}\Gamma\alpha \cap \Gamma$. Let $\Gamma'' = \Gamma \cap \alpha\Gamma\alpha^{-1}$. Then $\alpha^{-1}\Gamma''\alpha = \Gamma'$, and so if $\mathcal{F}''$ is a fundamental domain for Γ'', then $\alpha^{-1}\mathcal{F}''$ is a fundamental domain for Γ', by Exercise 7.1.9. Thus,

$$\begin{aligned}(f|\alpha, g|\alpha) &= \frac{1}{\left[\mathrm{SL}_2(\mathbb{Z}) : \overline{\Gamma'}\right]} \iint_{\alpha^{-1}\mathcal{F}''} (f|\alpha)(z)\overline{(g|\alpha)(z)}y^k \frac{dxdy}{y^2} \\ &= \frac{1}{\left[\mathrm{SL}_2(\mathbb{Z}) : \overline{\Gamma'}\right]} \iint_{\mathcal{F}''} f(z)\overline{g(z)}y^k \frac{dxdy}{y^2} \\ &= (f, g)\end{aligned}$$

by the invariance of the inner product and the fact that

$$\left[\mathrm{SL}_2(\mathbb{Z}) : \overline{\Gamma'}\right] = \left[\mathrm{SL}_2(\mathbb{Z}) : \overline{\alpha^{-1}\Gamma''\alpha}\right] = \left[\mathrm{SL}_2(\mathbb{Z}) : \overline{\Gamma''}\right]. \qquad \square$$

Exercise 7.1.11. *Let $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$ have integer entries and determinant D. Let $\alpha' = D\alpha^{-1}$. Show that for $f, g \in S_k(\Gamma)$, with Γ a congruence subgroup of $\mathrm{SL}_2(\mathbb{Z})$, we have*

$$(f|\alpha, g) = (f, g|\alpha').$$

7.2 Commutativity of the Hecke operators

Let T_n denote the nth Hecke operator for $\mathrm{SL}_2(\mathbb{Z})$, as developed in Section 5.2. We prove that $T_nT_m = T_mT_n$ so that the T_n's form a commuting family of linear transformations on $M_k(\mathrm{SL}_2(\mathbb{Z}))$ and $S_k(\mathrm{SL}_2(\mathbb{Z}))$. We will use this fact along with the multiplicative properties of the Hecke operators in the next section to prove that the T_n's are Hermitian with respect to the Pettersson inner product.

Theorem 7.2.1. *Let $(m,n)=1$. Then $T_nT_m = T_{mn} = T_mT_n$.*

Proof. Let $f \in M_k(\mathrm{SL}_2(\mathbb{Z}))$ have q-expansion

$$f(z) = \sum_{r=0}^{\infty} \lambda(r) q^r.$$

By Theorem 5.2.4, we have $T_n(f) = \sum_{r=0}^{\infty} b(r)q^r$, where

$$b(r) = \sum_{d|(n,r)} d^{k-1} \lambda\left(\frac{nr}{d^2}\right).$$

Writing $T_m\big(T_n(f)\big) = \sum_{r=0}^{\infty} c(r)q^r$, we have

$$c(r) = \sum_{e|(m,r)} e^{k-1} b\left(\frac{mr}{e^2}\right) = \sum_{e|(m,r)} e^{k-1} \sum_{d|\left(n,\frac{mr}{e^2}\right)} d^{k-1}\lambda\left(\frac{mnr}{d^2e^2}\right).$$

As $(m,n)=1$, the inner sum over divisors d of $(n, mr/e^2)$ is really a sum over divisors d of $(n, r/e)$ as $(n, m/e) = 1$. Therefore,

$$\begin{aligned} c(r) &= \sum_{e|(m,r)} e^{k-1} \sum_{d|\left(n,\frac{r}{e}\right)} d^{k-1}\lambda\left(\frac{mnr}{d^2e^2}\right) \\ &= \sum_{e|(m,r)} \sum_{d|(n,r)} (ed)^{k-1}\lambda\left(\frac{mnr}{d^2e^2}\right). \end{aligned}$$

As d runs over divisors of (n,r) and e runs over divisors of (m,r), de runs over divisors of (mn,r). Thus

$$c(r) = \sum_{t|(mn,r)} t^{k-1}\lambda\left(\frac{mnr}{t^2}\right)$$

so that $T_mT_n = T_{mn}$ when $(m,n)=1$. □

Theorem 7.2.2. *Let p be a prime number. For all $s \geq 1$,*

$$T_{p^s}T_p = T_{p^{s+1}} + p^{k-1}T_{p^{s-1}}.$$

Proof. Since the Hecke operators are defined in terms of the slash operator, it is notationally convenient to introduce

$$\widetilde{T_{p^s}} = \sum_{j=0}^{s} \sum_{b=0}^{p^j-1} \begin{bmatrix} p^{s-j} & b \\ 0 & p^j \end{bmatrix},$$

where we view the right hand side as a "formal sum" of matrices. Then

$$p^{s(1-\frac{k}{2})}T_{p^s}(f) = f|\widetilde{T_{p^s}}.$$

Thus,

$$\widetilde{T_p} = \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} + \sum_{c=0}^{p-1} \begin{pmatrix} 1 & c \\ 0 & p \end{pmatrix}.$$

Observe that for all a, b, $d \in \mathbb{Z}$ and all $f \in M_k(\mathrm{SL}_2(\mathbb{Z}))$, by the definition of the slash operator (4.2) we have

$$f\left|\begin{pmatrix} a & b \\ 0 & d \end{pmatrix}\right. = \left(\frac{a}{d}\right)^{k/2} f\left(\frac{a}{d}z + \frac{b}{d}\right), \tag{7.1}$$

$$f\left|\begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix}\right. = f(z). \tag{7.2}$$

Since $f \in M_k(\mathrm{SL}_2(\mathbb{Z}))$ is periodic, the right hand side of (7.1) depends only on $b \pmod d$, rather than b itself. In particular, we actually view $\widetilde{T_{p^s}}$ as an equivalence class of formal sums of matrices in which matrices are equivalent if their corresponding slash operators are equal.

Although notationally T_{p^s} is a left action, $\widetilde{T_{p^s}}$ acts from the right. Hence we consider

$$\widetilde{T_p}\widetilde{T_{p^s}} = \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} \sum_{j=0}^{s} \sum_{b=0}^{p^j-1} \begin{pmatrix} p^{s-j} & b \\ 0 & p^j \end{pmatrix} + \sum_{c=0}^{p-1} \sum_{j=0}^{s} \sum_{b=0}^{p^j-1} \begin{pmatrix} 1 & c \\ 0 & p \end{pmatrix} \begin{pmatrix} p^{s-j} & b \\ 0 & p^j \end{pmatrix}. \tag{7.3}$$

Now

$$\begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} p^{s-j} & b \\ 0 & p^j \end{pmatrix} = \begin{pmatrix} p^{s+1-j} & bp \\ 0 & p^j \end{pmatrix}$$

and

$$\begin{pmatrix} 1 & c \\ 0 & p \end{pmatrix} \begin{pmatrix} p^{s-j} & b \\ 0 & p^j \end{pmatrix} = \begin{pmatrix} p^{s-j} & b + p^j c \\ 0 & p^{j+1} \end{pmatrix}$$

so that the first sum of (7.3) can be rewritten as

$$\begin{pmatrix} p^{s+1} & 0 \\ 0 & 1 \end{pmatrix} + \sum_{j=1}^{s} \sum_{b=0}^{p^j-1} \begin{pmatrix} p^{s+1-j} & bp \\ 0 & p^j \end{pmatrix}, \tag{7.4}$$

while the second sum of (7.3) is

$$\sum_{j=0}^{s} \sum_{c=0}^{p-1} \sum_{b=0}^{p^j-1} \begin{pmatrix} p^{s-j} & b + p^j c \\ 0 & p^{j+1} \end{pmatrix}.$$

In this last sum, the two inner summations can be combined to give

$$\sum_{b'=0}^{p^{j+1}-1} \begin{pmatrix} p^{s-j} & b' \\ 0 & p^{j+1} \end{pmatrix}.$$

This combines with the first term in equation (7.4) to give $\widetilde{T_{p^{s+1}}}$. In equation (7.4), the effect of the matrix $\left(\begin{smallmatrix} p^{s+1-j} & bp \\ 0 & p^j \end{smallmatrix}\right)$ is the same as $\left(\begin{smallmatrix} p^{s-j} & b \\ 0 & p^{j-1} \end{smallmatrix}\right)$ by (7.2). By (7.1), as the b's range over residue classes $\pmod{p^j}$, we effectively get p copies of each matrix

$$\begin{pmatrix} p^{s-j} & b' \\ 0 & p^{j-1} \end{pmatrix}, \qquad \text{with } 0 \le b' \le p^{j-1}-1,$$

so that these combine to give $p\widetilde{T_{p^{s-1}}}$. Thus,

$$\widetilde{T_p}\widetilde{T_{p^s}} = \widetilde{T_{p^{s+1}}} + p\widetilde{T_{p^{s-1}}}.$$

Since $\widetilde{T_{p^s}} = T_{p^s}/p^{s(\frac{k}{2}-1)}$, we immediately deduce the theorem. □

Exercise 7.2.3. *Let p be prime. Show that for all r, $s \ge 1$, we have*

$$T_{p^r}T_{p^s} = T_{p^s}T_{p^r}.$$

It may be easier to prove Exercise 7.2.4 before Exercise 7.2.3. In fact, after Exercise 7.2.4, Exercise 7.2.3 becomes a trivial statement about commuting polynomials.

Exercise 7.2.4. *Let p be prime. Show that there is a polynomial $P_s(x) \in \mathbb{Z}[x]$ such that $T_{p^s} = P_s(T_p)$.*

Exercise 7.2.5. *Prove that for all m, $n \ge 1$, the Hecke operators commute:*

$$T_nT_m = T_mT_n.$$

7.3 Hecke operators as Hermitian operators

We prove that Hecke operators are Hermitian.

Theorem 7.3.1. *For any f, $g \in S_k(\mathrm{SL}_2(\mathbb{Z}))$ and any $n \ge 1$,*

$$(T_n(f), g) = (f, T_n(g)).$$

Proof. By Theorem 7.2.1 and Exercise 7.2.4 we see that the algebra of Hecke operators are generated by the T_p's, with p prime. Thus, to prove the theorem, it suffices to show $(T_p(f), g) = (f, T_p(g))$ for all primes p. By the definition of T_p,

$$T_p(f) = p^{\frac{k}{2}-1}\left\{ f\left|\begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} + \sum_{b=0}^{p-1} f\left|\begin{pmatrix} 1 & b \\ 0 & p \end{pmatrix}\right.\right\}$$

and so by linearity

$$(T_p(f), g) = p^{\frac{k}{2}-1}\left\{ \left(f\left|\begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix}, g\right.\right) + \sum_{b=0}^{p-1}\left(f\left|\begin{pmatrix} 1 & b \\ 0 & p \end{pmatrix}, g\right.\right)\right\}$$

By Exercise 7.1.11, we have

$$\left(f\left|\begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix}, g\right.\right) = \left(f, g\left|\begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}\right.\right)$$

and

$$\left(f\left|\begin{pmatrix} 1 & b \\ 0 & p \end{pmatrix}, g\right.\right) = \left(f, g\left|\begin{pmatrix} p & -b \\ 0 & 1 \end{pmatrix}\right.\right).$$

For each $1 \le b \le p-1$, we observe that $\alpha := \left(\begin{smallmatrix} 0 & -1 \\ 1 & b \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$ and $\beta := \left(\begin{smallmatrix} 0 & 1 \\ -1 & -b \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$. Now by Theorem 7.1.10, we have

$$\left(f, g\left|\begin{pmatrix} p & -b \\ 0 & 1 \end{pmatrix}\right.\right) = \left(f|\beta, g\left|\begin{pmatrix} p & -b \\ 0 & 1 \end{pmatrix}\beta\right.\right)$$

Since $f|\beta = f$ and $g|\alpha = g$, we have

$$\left(f|\beta, g\left|\begin{pmatrix} p & -b \\ 0 & 1 \end{pmatrix}\beta\right.\right) = \left(f, g\left|\alpha\begin{pmatrix} p & -b \\ 0 & 1 \end{pmatrix}\beta\right.\right) = \left(f, g\left|\begin{pmatrix} 1 & b \\ 0 & p \end{pmatrix}\right.\right).$$

Putting all of these together shows that $(T_p(f), g) = (f, T_p(g))$. □

An immediate consequence of this theorem is:

Corollary 7.3.2. *The eigenvalues of T_n are real algebraic numbers.*

Proof. If $T_n(f) = \lambda_n f$, then by the theorem $(T_n(f), f) = (f, T_n(f))$ so that $(\lambda_n f, f) = (f, \lambda_n f)$ which implies $\lambda_n(f, f) = \overline{\lambda_n}(f, f)$ so that $\lambda_n = \overline{\lambda_n}$. Hence the eigenvalues are real.

To see that they are algebraic numbers, we recall that by Corollary 4.4.4 any element of $S_k(\mathrm{SL}_2(\mathbb{Z}))$ is a polynomial in E_4 and E_6. Therefore, $S_k(\mathrm{SL}_2(\mathbb{Z}))$ has a basis $g_1, \ldots, g_r$ with the g_i having rational Fourier coefficients in their q-expansions. Since $T_n(g_i)$ has again rational Fourier coefficients (see Theorem 5.2.4), we can represent T_n by a rational matrix and so its characteristic polynomial has rational coefficients. Therefore, the eigenvalues of T_n are algebraic numbers. □

Exercise 7.3.3. *Suppose $f = \sum_{n=1}^{\infty} a_n q^n \in S_k(\mathrm{SL}_2(\mathbb{Z}))$ is an eigenfunction for all T_m. Prove that $a_1 \neq 0$ and that the eigenvalue of T_m is a_m/a_1.*

By the previous exercise, we can speak of a "normalized" eigenfunction, where we have $a_1 = 1$. The Fourier coefficients of a normalized eigenfunction are the eigenvalues of the T_n's.

Exercise 7.3.4. *Let $f, g \in S_k(\mathrm{SL}_2(\mathbb{Z}))$ be normalized eigenfunctions for all the Hecke operators. Show that $f = g$ or $(f, g) = 0$.*

From this exercise, we can deduce that if $f_1, \ldots, f_r$ is a maximal set of distinct normalized eigenfunctions, then they are linearly independent. To see this, suppose that j is the smallest index such that we can write $f_{j+1} = \sum_{i=1}^{j} \alpha_i f_i$. Then

$$0 \neq (f_{j+1}, f_{j+1}) = \left(\sum_{i=1}^{j} \alpha_i f_i, f_{j+1} \right) = \sum_{i=1}^{j} \alpha_i (f_i, f_{j+1}) = 0.$$

Hence, $r \leq \dim S_k(\mathrm{SL}_2(\mathbb{Z}))$. Applying a basic theorem of linear algebra, we show in the next section that $S_k(\mathrm{SL}_2(\mathbb{Z}))$ always has a basis of simultaneous eigenfunctions of all Hecke operators. Normalized eigenfunctions are also called *Hecke eigenforms*.

7.4 Basis of eigenforms

We begin with a basic theorem from linear algebra.

Theorem 7.4.1. *Let R be a commutative ring of Hermitian operators on a finite dimensional Hilbert space V. Then V has an orthogonal basis $f_1, f_2, \ldots, f_r$ of eigenvectors of R.*

Proof. We induct on the dimension of V. Since V is finite dimensional, the space of linear operators on V is also finite dimensional, and so we can find a finite set of operators, say $\{S_1, S_2, \ldots, S_m\}$, that span R.

Assume $\dim V \neq 0$. We first show by induction on m that V contains one eigenvector f_1 of $S_1, S_2, \ldots, S_m$. If $m = 1$, the result is clear. Let λ_1 be an eigenvalue of S_1 and let

$$V_1 = \{f \in V : S_1 f = \lambda_1 f\}$$

be the corresponding eigenspace. Since $S_j S_1 = S_1 S_j$, we see that $S_j V_1 \subseteq V_1$, for $j = 2, \ldots m$. Therefore, V_1 contains an eigenvector f_1 of $S_2 \ldots, S_m$ by the induction hypothesis. Then

$$V = \mathbb{C} f_1 \oplus (\mathbb{C} f_1)^{\perp},$$

where

$$(\mathbb{C} f_1)^{\perp} = \{g \in V : (f_1, g) = 0\}.$$

Clearly, the S_j's preserve this space since for $g \in (\mathbb{C}f_1)^\perp$, we have $(f_1, S_j g) = (S_j f_1, g) = 0$ since f_1 is an eigenfunction for the S_j's. By the induction hypothesis on the dimension of V, we see that $(\mathbb{C}f_1)^\perp$ has an orthogonal basis $f_2, \ldots, f_r$. This completes the proof. □

Theorem 7.4.2. *The space $S_k(\mathrm{SL}_2(\mathbb{Z}))$ has an orthogonal basis of Hecke eigenforms.*

Proof. We apply Theorem 7.4.1 to the commutative ring of Hecke operators acting on the finite dimensional vector space $S_k(\mathrm{SL}_2(\mathbb{Z}))$ to deduce the result. □

Exercise 7.4.3. *Show that E_k is an eigenfunction for all the Hecke operators acting on $M_k(\mathrm{SL}_2(\mathbb{Z}))$.*

Exercise 7.4.4. *Show that $(E_k, f) = 0$ for any $f \in S_k(\mathrm{SL}_2(\mathbb{Z}))$.*

Exercise 7.4.5. *Show that Δ^k is not a Hecke eigenform for any $k \geq 2$.*

7.5 Supplementary problems

Exercise 7.5.1. *Let $12 \mid k$ and $d = \dim S_k(\mathrm{SL}_2(\mathbb{Z}))$. For each $1 \leq j \leq d$, define*

$$f_j := \Delta^j E_6^{2(d-j)} = \sum_{n=1}^{\infty} a_n^{(j)} q^n.$$

Verify that $a_n^{(j)} = 0$ for $n < j$ and $a_j^{(j)} = 1$. Conclude that the f_j form a basis for $S_k(\mathrm{SL}_2(\mathbb{Z}))$.

Exercise 7.5.2. *Let $k \geq 4$ be even and $d = \dim S_k(\mathrm{SL}_2(\mathbb{Z}))$. Choose non-negative integers a, b such that $12 \neq 4a + 6b \leq 14$ and $4a + 6b \equiv k \pmod{12}$. For each $1 \leq j \leq d$, define*

$$f_j := \Delta^j E_6^{2(d-j)+b} E_4^a = \sum_{n=1}^{\infty} a_n^{(j)} q^n.$$

Verify that $a_n^{(j)} = 0$ for $n < j$ and $a_j^{(j)} = 1$. Conclude that the f_j form a basis for $S_k(\mathrm{SL}_2(\mathbb{Z}))$. (This basis is called the Miller basis *in the literature.)*

Exercise 7.5.3. *Let k, d, a, and b be as in Exercise 7.5.2. Verify that there exists a basis $g_0, g_1, \ldots, g_d$ for $M_k(\mathrm{SL}_2(\mathbb{Z}))$ such that*

$$g_i = q^i + \sum_{n=d+1}^{\infty} b_n^{(i)} q^n$$

with $b_n^{(i)} \in \mathbb{Z}$ for all n and all i. Conclude that if

$$f = \sum_{n=0}^{\infty} c_n q^n \in M_k\left(\mathrm{SL}_2(\mathbb{Z})\right)$$

has $c_0, c_1, \ldots, c_d \in \mathbb{Z}$*, then* $c_n \in \mathbb{Z}$ *for all* n*.*

Exercise 7.5.4. *Let* $d = \dim S_k(\mathrm{SL}_2(\mathbb{Z}))$ *and fix a positive integer* m*. Suppose*

$$f = \sum_{n=0}^{\infty} c_n q^n \in M_k\left(\mathrm{SL}_2(\mathbb{Z})\right)$$

is such that for $i = 0, 1, \ldots, d$*, we have that* $c_i \in \mathbb{Z}$ *and that* $c_i \equiv 0 \pmod{m}$*. Prove that* $c_i \equiv 0 \pmod{m}$ *for all* $i \geq 0$*.*

Exercise 7.5.5. *Let* f *be a Hecke eigenform for the full modular group. Show that the Fourier coefficients of* f *are algebraic integers.*

Chapter 8
Hecke Operators of Higher Level

8.1 Hecke operators for congruence subgroups

Let Γ_1 and Γ_2 be subgroups of $\mathrm{GL}_2^+(\mathbb{Q})$ and let $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$. Recall the double coset

$$\Gamma_1 \alpha \Gamma_2 = \{\gamma_1 \alpha \gamma_2 : \gamma_1 \in \Gamma_1, \gamma_2 \in \Gamma_2\}$$

from Example 2.4.5. The group Γ_1 acts on the double coset from the left and Γ_2 acts from the right. Moreover, we can decompose the double coset as a disjoint union of Γ_1-orbits:

$$\Gamma_1 \alpha \Gamma_2 = \coprod_j \Gamma_1 \alpha \gamma_j$$

with suitable γ_j in Γ_2.

Exercise 8.1.1. *Let p be prime and $\Gamma = \mathrm{SL}_2(\mathbb{Z})$. Let $\mathbf{X}_p$ be the set of matrices with integer entries of determinant p. Show that*

$$\mathbf{X}_p = \Gamma \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} \Gamma.$$

Fix $N \in \mathbb{N}$. Let S^+ be an additive subgroup of $\mathbb{Z}$, i.e. $S^+ = M\mathbb{Z}$ for some integer M. Let $S^\times$ be a multiplicative subgroup of $(\mathbb{Z}/N\mathbb{Z})^\times$. We shall also use $S^\times$ to indicate the pre-image of $S^\times$ under the natural map $\mathbb{Z} \to S^\times$. Let

$$\begin{aligned} X_n &= X_n(N, S^\times, S^+) \\ &:= \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, b, c, d \in \mathbb{Z}, ad - bc = n, N|c, a \in S^\times, b \in S^+ \right\}. \end{aligned}$$

Here are some examples:

Example 8.1.2. Take $N = 1$, $S^+ = \mathbb{Z}$, and $S^\times = \{1\}$. Then $X_n(1, S^\times, S^+) = \mathbf{X}_n$ from Section 5.2.

© Springer Science+Business Media Singapore 2016 and Hindustan Book Agency 2016

M.R. Murty et al., *Problems in the Theory of Modular Forms*, HBA Lecture Notes in Mathematics, DOI 10.1007/978-981-10-2651-5_8

Example 8.1.3. The standard Hecke congruence subgroups can all be written in this form:

$$\begin{aligned}\Gamma_1(N) &= X_1(N, 1, \mathbb{Z}) \\ \Gamma(N) &= X_1(N, 1, N\mathbb{Z}) \\ \Gamma_0(N) &= X_1(N, (\mathbb{Z}/N\mathbb{Z})^\times, \mathbb{Z}).\end{aligned}$$

Exercise 8.1.4. *For fixed N, S^+, and $S^\times$, show that*

$$X_m X_n \subseteq X_{mn}.$$

Exercise 8.1.5. *Show that*

$$X_1(N, S^\times, S^+)$$

is a congruence subgroup of $\mathrm{SL}_2(\mathbb{Z})$.

In particular, Exercise 8.1.5 shows that X_1 is a group. Exercise 8.1.4 shows that the set X_n is invariant under the left and the right action of X_1 since $X_1 X_n \subseteq X_n$ and $X_n X_1 \subseteq X_n$. We apply Exercise 2.4.7 with $H = \mathrm{SL}_2(\mathbb{Z})$, $H_1 = X_1$, $Y = \mathbf{X}_n$ and $X = X_n$ along with Theorem 5.2.1 to deduce that $|X_1 \backslash X_n| < \infty$. Thus there is a finite set of orbit representatives $\alpha_i \in X_n$ such that

$$X_n = \coprod_i X_1 \alpha_i.$$

Moreover, multiplying the orbits on the right by any fixed $\gamma \in X_1$ simply permutes the orbits:

$$\coprod_i X_1 \alpha_i \gamma = \coprod_i X_1 \alpha_i. \tag{8.1}$$

We define the nth *Hecke operator* on $M_k(X_1)$ as the map which sends $f \in M_k(X_1)$ to the finite sum

$$T_n(f) := n^{\frac{k}{2}-1} \sum_i f|\alpha_i. \tag{8.2}$$

Theorem 8.1.6. *Let $\Gamma = X_1(N, S^\times, S^+)$ as above. For any k, $n \geq 1$, the Hecke operator in (8.2) is a well-defined linear map from $M_k(\Gamma)$ to $M_k(\Gamma)$ and from $S_k(\Gamma)$ to $S_k(\Gamma)$.*

Proof. Suppose $f \in M_k(\Gamma)$ and $\gamma \in \Gamma$. Then for any α_i, since $f|\gamma\alpha_i = f|\alpha_i$, we see that the map does not depend on the choice of orbit representatives and is thus well-defined. Moreover,

$$\begin{aligned}\Big(T_n(f)\Big)\Big|\gamma &= \left(n^{\frac{k}{2}-1}\sum_i f|\alpha_i\right)\Bigg|\gamma \\ &= n^{\frac{k}{2}-1}\sum_i f|\alpha_i\gamma \\ &= T_n(f),\end{aligned}$$

where the final step follows from (8.1) which shows that γ simply permutes the orbits. By Exercise 7.1.6, each $f|\alpha_i$ is holomorphic at every cusp $\mathbb{Q}\cup\{i\infty\}$ and hence the finite sum $T_n(f)$ is also holomorphic at every cusp. We conclude that $T_n(f)\in M_k(\Gamma)$. If in addition $f\in S_k(\Gamma)$, then by Exercise 7.1.7 we conclude that $T_n(f)\in S_k(\Gamma)$. The linearity of the map follows from the linearity of the slash operator. □

We write down the Hecke operators explicitly in the case $\Gamma=\Gamma_1(N)=X_1(N,1,\mathbb{Z})$ in the following theorem.

Theorem 8.1.7. *For each $a\in(\mathbb{Z}/N\mathbb{Z})^\times$, fix $\sigma_a\in\mathrm{SL}_2(\mathbb{Z})$ such that*

$$\sigma_a\equiv\begin{pmatrix}a^{-1}&0\\0&a\end{pmatrix}\pmod N,$$

where a^{-1} denotes the multiplicative inverse of $a\pmod N$. Then $X_n=X_n(N,1,\mathbb{Z})$ has the decomposition

$$X_n=\coprod_{\substack{ad=n\\a>0\\(a,N)=1}}\coprod_{b=0}^{d-1}\Gamma_1(N)\sigma_a\begin{pmatrix}a&b\\0&d\end{pmatrix},$$

where the first disjoint union is taken over all positive integers a dividing n that are coprime to N.

Proof. The terms on the right hand side are clearly contained in X_n. We first show the union is a disjoint union. Suppose we had

$$\gamma_1\sigma_{a_1}\begin{pmatrix}a_1&b_1\\0&d_1\end{pmatrix}=\gamma_2\sigma_{a_2}\begin{pmatrix}a_2&b_2\\0&d_2\end{pmatrix},\qquad\gamma_1,\gamma_2\in\Gamma_1(N).$$

Then $\mathrm{SL}_2(\mathbb{Z})$ contains the matrix

$$\begin{pmatrix}a_1&b_1\\0&d_1\end{pmatrix}\begin{pmatrix}a_2&b_2\\0&d_2\end{pmatrix}^{-1}=\begin{pmatrix}a_1/a_2&\star\\0&d_1/d_2\end{pmatrix}$$

implying that $a_1=a_2$ and $d_1=d_2$ since $a_i>0$. Thus,

$$\begin{pmatrix}a_1&b_1\\0&d_1\end{pmatrix}=\begin{pmatrix}1&j\\0&1\end{pmatrix}\begin{pmatrix}a_2&b_2\\0&d_2\end{pmatrix}$$

for some j. Since b_1 and b_2 are both $\leq d-1$, we must have $j=0$, which means $b_1 = b_2$.

To complete the proof, we need to show that any element of X_n is contained in the right hand side. Let $\left(\begin{smallmatrix}\alpha & \beta\\ \gamma & \delta\end{smallmatrix}\right) \in X_n$. Choose coprime integers g, h such that $g\alpha + h\gamma = 0$. (This can be done as the reader can directly verify.) Let x, y be integers such that $xh - gy = 1$. Then set

$$A = \begin{pmatrix} x & y \\ g & h \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}).$$

Now

$$A \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} = \begin{pmatrix} \star & \star \\ 0 & \star \end{pmatrix}$$

and has determinant n. Replacing A by $\pm\left(\begin{smallmatrix}1 & j\\ 0 & 1\end{smallmatrix}\right)A$ if necessary, we may suppose that

$$A \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} = \begin{pmatrix} a & b \\ 0 & d \end{pmatrix}, \qquad \begin{array}{c} a > 0, \\ 0 \leq b \leq d-1. \end{array}$$

Since $\left(\begin{smallmatrix}\alpha & \beta\\ \gamma & \delta\end{smallmatrix}\right) \in X_n$, considered modulo N this equality gives

$$A \begin{pmatrix} 1 & \star \\ 0 & n \end{pmatrix} \equiv \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \pmod{N}$$

which implies that A^{-1} has the form

$$\begin{pmatrix} a^{-1} & \star \\ 0 & a \end{pmatrix} \pmod{N},$$

which can be written as

$$\begin{pmatrix} 1 & \star \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a^{-1} & 0 \\ 0 & a \end{pmatrix} \pmod{N}.$$

Therefore we can find a matrix $B \in \Gamma_1(N)$ such that

$$A^{-1} = B\sigma_a.$$

This completes the proof. □

This gives, for $(n, N) = 1$, an explicit description of T_n which is

$$T_n(f) = n^{k/2-1} \sum_{\substack{ad=n \\ a>0}} \sum_{b=0}^{d-1} f \Big| \sigma_a \begin{pmatrix} a & b \\ 0 & d \end{pmatrix}.$$

8.2 Nebentypus

A *Dirichlet character modulo N* is a homomorphism $\chi : (\mathbb{Z}/N\mathbb{Z})^\times \to \mathbb{C}^\times$. This implies $\chi(1) = 1$.

Exercise 8.2.1. *Let χ be a Dirichlet character modulo N. For each $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma_0(N)$, define $\psi(\gamma) = \chi(d)$. Show that $\psi(\gamma_1\gamma_2) = \psi(\gamma_1)\psi(\gamma_2)$.*

Exercise 8.2.2. *Show that the set of Dirichlet characters modulo N forms a group of order $\phi(N)$ under multiplication, where ϕ denotes the Euler function.*

Exercise 8.2.3. *Show that*

$$\sum_{\chi} \chi(d) = \begin{cases} \phi(N) & \text{if } d \equiv 1 \pmod{N} \\ 0 & \text{otherwise} \end{cases}$$

where the sum is over all Dirichlet characters modulo N.

Exercise 8.2.4. *If $\chi \neq 1$ is a Dirichlet character modulo N, show that*

$$\sum_{a \in (\mathbb{Z}/N\mathbb{Z})^\times} \chi(a) = 0.$$

For any Dirichlet character χ modulo N, we define the following vector subspace of $M_k(\Gamma_1(N))$:

$$M_k(\Gamma_0(N), \chi) := \left\{ f \in M_k(\Gamma_1(N)) : f|\gamma = \chi(d)f, \quad \forall \gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma_0(N) \right\}.$$

In particular, if χ is the trivial character, then $M_k(\Gamma_0(N), \chi) = M_k(\Gamma_0(N))$.

Theorem 8.2.5. *We have the decomposition*

$$M_k(\Gamma_1(N)) = \bigoplus_{\chi} M_k(\Gamma_0(N), \chi),$$

where the direct sum is over all Dirichlet characters modulo N.

Proof. We first show that any $f \in M_k(\Gamma_1(N))$ can be written as a sum of functions $f_\chi \in M_k(\Gamma_0(N), \chi)$. Let

$$f_\chi = \frac{1}{\phi(N)} \sum_{d \in (\mathbb{Z}/N\mathbb{Z})^\times} \overline{\chi}(d) f|\gamma_d,$$

where $\gamma_d \in \Gamma_0(N)$ such that the lower right entry of γ_d is congruent to $d \pmod{N}$. This is well-defined since $f \in M_k(\Gamma_1(N))$. We now check that we indeed have $f_\chi \in M_k(\Gamma_0(N), \chi)$. For any $\gamma = \left(\begin{smallmatrix} \alpha & \beta \\ \gamma & \delta \end{smallmatrix}\right) \in \Gamma_0(N)$, we have

$$f_\chi|\gamma = \frac{1}{\phi(N)} \sum_{d\in(\mathbb{Z}/N\mathbb{Z})^\times} \overline{\chi}(d) f|\gamma_d\gamma = \frac{1}{\phi(N)} \sum_{d\in(\mathbb{Z}/N\mathbb{Z})^\times} \overline{\chi}(d) f|\gamma_{d\delta}$$
$$= \frac{1}{\phi(N)} \sum_{d\in(\mathbb{Z}/N\mathbb{Z})^\times} \overline{\chi}(d\delta)\chi(\delta) f|\gamma_{d\delta}.$$

Now as d runs through coprime residue classes modulo N, so does $d\delta$. Thus, we can change variables in the sum and deduce that

$$f_\chi|\gamma = \chi(\delta) f_\chi.$$

Moreover, f_χ is holomorphic at all cusps by Exercise 7.1.6. In other words, $f \in M_k(\Gamma_0(N), \chi)$. Now

$$\sum_\chi f_\chi = \sum_{d\in(\mathbb{Z}/N\mathbb{Z})^\times} \left(\frac{1}{\phi(N)} \sum_\chi \overline{\chi}(d) \right) f|\gamma_d,$$

and the inner sum is zero by Exercise 8.2.3 unless $d = 1$, in which case it is 1 so that

$$f = \sum_\chi f_\chi.$$

To show the sum is a direct sum, we consider $g \in M_k(\Gamma_0(N), \psi)$. For any character χ, we compute

$$g_\chi = \left(\frac{1}{\phi(N)} \sum_{d\in(\mathbb{Z}/N\mathbb{Z})^\times} \psi\overline{\chi}(d) \right) g = \begin{cases} g & \text{if } \chi = \psi \\ 0 & \text{otherwise,} \end{cases}$$

by Exercise 8.2.4. In particular, if

$$g \in M_k(\Gamma_0(N), \psi) \cap \bigoplus_{\chi\neq\psi} M_k(\Gamma_0(N), \chi),$$

then

$$g = g_\psi = \left(\sum_{\chi\neq\psi} g_\chi \right)_\psi = 0. \qquad \square$$

In the previous section, we defined Hecke operators for $\Gamma_1(N)$. Theorem 8.1.7 allows us to give the general formula for any n as

$$T_n(f) = n^{\frac{k}{2}-1} \sum_{\substack{ad=n \\ a>0 \\ (a,N)=1}} \sum_{b=0}^{d-1} f|\sigma_a \begin{pmatrix} a & b \\ 0 & d \end{pmatrix}. \tag{8.3}$$

If $f \in M_k(\Gamma_0(N), \chi)$, we have $f|\sigma_a = \chi(a)f$ so that

$$T_n(f) = n^{\frac{k}{2}-1} \sum_{\substack{ad=n \\ a>0}} \chi(a) \sum_{b=0}^{d-1} f \left| \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \right. .$$

We drop the condition $(a, N) = 1$ since $\chi(a) = 0$ if $(a, N) \neq 1$.

Exercise 8.2.6. *Let $f \in M_k(\Gamma_0(N), \chi)$. If $f(z) = \sum_{m=0}^{\infty} \lambda(m) q^m$, then prove that*

$$T_n(f) = \sum_{m=0}^{\infty} \left(\sum_{a|m,n} \chi(a) a^{k-1} \lambda\left(\frac{mn}{a^2}\right) \right) q^m.$$

If a modular form is an eigenvector of a Hecke operator, then we call it an *eigenfunction* (or eigenform).

Exercise 8.2.7. *If $f = \sum_{m=0}^{\infty} \lambda(m) q^m \in M_k(\Gamma_0(N), \chi)$ is an eigenfunction for all T_n's, show that $\lambda(1) \neq 0$ and $\lambda(mn)\lambda(1) = \lambda(m)\lambda(n)$ for $(m, n) = 1$.*

In light of Exercise 8.2.7, we say that an eigenvector $f = \sum_{m=0}^{\infty} \lambda(m) q^m$ is *normalized* if $\lambda(1) = 1$.

Exercise 8.2.8. *For $p|N$, and $f \in M_k(\Gamma_0(N), \chi)$, show that*

$$T_p(f) = p^{\frac{k}{2}-1} \sum_{b=0}^{p-1} f \left| \begin{pmatrix} 1 & b \\ 0 & p \end{pmatrix} \right. .$$

Exercise 8.2.9. *Let $f = \sum_{n=0}^{\infty} a_n q^n \in M_k(\Gamma_0(N), \chi)$ and then write $T_p(f) = \sum_{n=0}^{\infty} b_n q^n$. Show that*

$$b_n = a_{pn} + \chi(p) p^{k-1} a_{n/p},$$

where we interpret $a_{n/p} = 0$ when $p \nmid n$.

Exercise 8.2.10. *Let $f = \sum_{m=0}^{\infty} \lambda(m) q^m \in M_k(\Gamma_0(N), \chi)$ be a normalized eigenfunction for all the T_n's. Show that for p prime,*

$$\lambda(p^{a+1}) = \lambda(p)\lambda(p^a) - \chi(p) p^{k-1} \lambda(p^{a-1})$$

for $a \geq 1$.

As before, we deduce that the T_n's form a commutative algebra of operators. We now state (without proof) the action of these Hecke operators on the Petersson inner product.

Theorem 8.2.11. *Let $f \in M_k(\Gamma_1(N))$, and $g \in S_k(\Gamma_1(N))$. For each $a \in (\mathbb{Z}/N\mathbb{Z})^\times$, fix $\sigma_a \in \mathrm{SL}_2(\mathbb{Z})$ such that*

$$\sigma_a \equiv \begin{pmatrix} a^{-1} & 0 \\ 0 & a \end{pmatrix} \pmod{N},$$

where a^{-1} denotes the multiplicative inverse of $a \pmod N$. If $(n,N)=1$, then $(T_n(f),g) = (f|\sigma_n, T_n(g))$. In particular, if $f \in M_k(\Gamma_0(N),\chi)$, then $(T_n(f),g) = \chi(n)(f,T_n(g))$.

This allows us to prove:

Corollary 8.2.12. *Let $(n,N)=1$ and let χ be a Dirichlet character modulo N. Let c_n be a square root of $\overline{\chi(n)}$. Then c_nT_n is a Hermitian operator on $S_k(\Gamma_0(N),\chi)$. That is, $(c_nT_n(f),g) = (f,c_nT_n(g))$.*

Proof. This is clear from the theorem. Indeed, since c_n is a root of unity, we have $(c_nT_n(f),g) = c_n(T_n(f),g) = c_n\chi(n)(f,T_n(g)) = c_n\overline{c_n}^2(f,T_n(g)) = \overline{c_n}(f,T_n(g)) = (f,c_nT_n(g))$. □

Exercise 8.2.13. *Show that $S_k(\Gamma_0(N),\chi)$ has a basis of eigenforms for all the T_n with $(n,N)=1$.*

8.3 Oldforms and newforms

Exercise 8.2.13 only gives us a basis of eigenforms for all the T_n's under the restriction that $(n,N)=1$. One would like a basis for all T_p's, including $p|N$. Since all the T_p's commute, we know that the T_p's preserve each eigenspace. If each eigenspace were one-dimensional, then we would deduce that the basis is also a basis of eigenforms for T_p's with $p|N$. However, this need not be the case. The problem arises from eigenforms arising from lower levels, as the following examples show.

Example 8.3.1. Let $f \in S_k(\Gamma_0(N),\chi)$ and $r|N$. If f is an eigenform for all T_n with $(n,N)=1$, show that $f|\left(\begin{smallmatrix} r & 0 \\ 0 & 1 \end{smallmatrix}\right)$ is also an eigenform for all T_n with $(n,N)=1$ with the same eigenvalues.

Solution. We have

$$T_n\left(f\left|\begin{pmatrix} r & 0 \\ 0 & 1 \end{pmatrix}\right.\right) = n^{\frac{k}{2}-1}\sum_{\substack{ad=n \\ a>0}} \chi(a) \sum_{b \pmod d} f\left|\begin{pmatrix} r & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} a & b \\ 0 & d \end{pmatrix}\right..$$

Now

$$\begin{pmatrix} r & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} a & b \\ 0 & d \end{pmatrix} = \begin{pmatrix} a & br \\ 0 & d \end{pmatrix}\begin{pmatrix} r & 0 \\ 0 & 1 \end{pmatrix}$$

so that the sum becomes

$$n^{\frac{k}{2}-1}\sum_{\substack{ad=n \\ a>0}} \chi(a) \sum_{b \pmod d} f\left|\begin{pmatrix} a & br \\ 0 & d \end{pmatrix}\begin{pmatrix} r & 0 \\ 0 & 1 \end{pmatrix}\right..$$

As b runs through residue classes modulo d, so does br since $(r,n) = 1$. Hence

$$T_n\left(f\left|\begin{pmatrix} r & 0 \\ 0 & 1 \end{pmatrix}\right.\right) = T_n(f)\left|\begin{pmatrix} r & 0 \\ 0 & 1 \end{pmatrix}\right. = \lambda_n f\left|\begin{pmatrix} r & 0 \\ 0 & 1 \end{pmatrix}\right.,$$

where $T_n(f) = \lambda_n f$. □

Exercise 8.3.2. *Show that $\Delta(z)$ and $\Delta(2z)$ are both eigenforms for the T_n's (n odd) acting on $S_{12}(\Gamma_0(2))$, with the same eigenvalues.*

Exercise 8.3.3. *Show that $\Delta(z)$ and $\Delta(2z)$ are linearly independent in $S_{12}(\Gamma_0(2))$.*

In 1970, Atkin and Lehner realized how to resolve this difficulty by focusing on forms "genuinely" of level N. Indeed, let $d_1 d_2 = N$. Suppose $f \in M_k(\Gamma_1(d_1))$. Then $f \in M_k(\Gamma_1(N))$. If $g(z) := f(d_2 z)$, then $g \in M_k(\Gamma_1(N))$ by Exercise 6.6.6. The subspace of $S_k(\Gamma_1(N))$ spanned by forms obtained in these two ways from $f \in S_k(\Gamma_1(d))$, as we range over proper divisors $d|N$ is called the space of *oldforms*, denoted $S_k^{\text{old}}(\Gamma_1(N))$. One can show that the T_n's preserve this space. The orthogonal complement of this space with respect to the Petersson inner product is called the space of *newforms*, denoted $S_k^{\text{new}}(\Gamma_1(N))$. It turns out that this space has the *multiplicity one* property: the intersection of the eigenspaces for all the T_n's with $(n,N) = 1$ is one-dimensional and so is also an eigenspace for the T_p's with $p|N$. This fact allows us to construct a complete theory of L-series and functional equations analogous to the level one case.

8.4 Supplementary problems

Exercise 8.4.1. *Let $w_N = \left(\begin{smallmatrix} 0 & -1 \\ N & 0 \end{smallmatrix}\right)$. Prove that $w_N \Gamma_0(N) w_N^{-1} \subset \Gamma_0(N)$.*

The action of w_N on modular forms is called the Atkin-Lehner involution.

Exercise 8.4.2. *Show that if $f \in M_k(\Gamma_0(N))$, then $f|w_N \in M_k(\Gamma_0(N))$.*

Exercise 8.4.3. *Show that if $f \in S_k(\Gamma_0(N))$, then $f|w_N \in S_k(\Gamma_0(N))$.*

Exercise 8.4.4. *Show that if $f \in M_k(\Gamma_0(N))$, then $(f|w_N)\,|w_N = f$. Thus w_N is, in fact, an involution.*

Chapter 9
Dirichlet Series and Modular Forms

9.1 General Dirichlet series

The general Dirichlet series is a discrete analog of the more familiar Laplace transform of a function in the theory of complex variables. It is of the form

$$\sum_{n=1}^{\infty} a_n e^{-\lambda_n s} \tag{9.1}$$

where $s \in \mathbb{C}$, $\lambda_n \in \mathbb{R}$ with λ_n monotonically increasing to infinity, and the a_n's are complex numbers. Such series give us a convenient way to package the sequence of a_n's for further study. For example, if $\lambda_n = n$, then equation (9.1) becomes a power series

$$\sum_{n=1}^{\infty} a_n x^n$$

with $x = e^{-s}$. The reader is probably familiar with the theory of power series from a first course in calculus. Power series have a wide variety of applications, most notably to combinatorial problems.

Ordinary Dirichlet series (which we will now consider) arise with $\lambda_n = \log n$. Then (9.1) becomes

$$\sum_{n=1}^{\infty} \frac{a_n}{n^s} \tag{9.2}$$

and such series are useful in analytic number theory, especially in the context of multiplicative number theory. Indeed, series of the form (9.2) were introduced in 1839 by P.G.L. Dirichlet to prove his celebrated theorem on the infinitude of primes in a given arithmetic progression. However, Dirichlet studied such series only for a real variable s. It was Riemann who initiated in 1860 an ardent study of such series for complex variables of s. In his famous memoir, he focused on the special case of (9.2) with $a_n = 1$ for all n

© Springer Science+Business Media Singapore 2016 and Hindustan Book Agency 2016

M.R. Murty et al., *Problems in the Theory of Modular Forms*, HBA Lecture Notes in Mathematics, DOI 10.1007/978-981-10-2651-5_9

and introduced (what we now call) the Riemann zeta function $\zeta(s)$, which for $\mathrm{Re}(s) > 1$ is given by the Dirichlet series

$$\sum_{n=1}^{\infty} \frac{1}{n^s}. \tag{9.3}$$

Exercise 9.1.1. *Show that the series (9.3) converges absolutely for* $\mathrm{Re}(s) > 1$.

Exercise 9.1.2. *Given a sequence of complex numbers* a_n*, we suppose that for any* $\epsilon > 0$*, there is a constant* $C(\epsilon)$ *such that* $|a_n| \leq C(\epsilon)n^{\epsilon}$ *for all natural numbers* n*. Show that*

$$\sum_{n=1}^{\infty} \frac{a_n}{n^s}$$

converges absolutely for $\mathrm{Re}(s) > 1$.

Riemann showed that $\zeta(s)$ can be analytically continued to the entire complex plane, apart from a simple pole at $s = 1$ and that it satisfies a functional equation relating $\zeta(s)$ with $\zeta(1-s)$. More importantly, Riemann noted that (9.3) can be rewritten as an infinite product over prime numbers:

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_{p} \left(1 - \frac{1}{p^s}\right)^{-1} \tag{9.4}$$

for $\mathrm{Re}(s) > 1$. It is this connection that relates the distribution of prime numbers with the zeros of the Riemann zeta function.

Exercise 9.1.3. *Prove the identity (9.4) for* $\mathrm{Re}(s) > 1$.

Exercise 9.1.4. *Let* $f(n)$ *be a* multiplicative *function. That is,* $f(mn) = f(m)f(n)$ *for* $(m, n) = 1$*. Assume that for any* $\epsilon > 0$*, there is a constant* $C(\epsilon)$ *such that* $|f(n)| \leq C(\epsilon)n^{\epsilon}$*. Show that for* $\mathrm{Re}(s) > 1$*, we have*

$$\sum_{n=1}^{\infty} \frac{f(n)}{n^s} = \prod_{p} \left(1 + \frac{f(p)}{p^s} + \frac{f(p^2)}{p^{2s}} + \cdots\right),$$

where the product is over the prime numbers.

In our study of Dirichlet series, it is useful to begin with several results from classical analysis.

Proposition 9.1.5 (Abel's Lemma). *Let* $m \in \mathbb{Z}$ *and suppose* a_n *and* b_n *are sequences of complex numbers. Let*

$$A(r) = \sum_{m \leq n \leq r} a_n.$$

Then

$$\sum_{m\le n\le r} a_n b_n = A(r)b_r - A(m-1)b_m - \sum_{m\le n<r} A(n)(b_{n+1}-b_n).$$

Proof. We have

$$\begin{aligned}\sum_{m\le n\le r} a_n b_n &= \sum_{m\le n\le r} \Big(A(n)-A(n-1)\Big)b_n \\ &= \sum_{m\le n\le r} A(n)b_n - \sum_{m-1\le n\le r-1} A(n)b_{n+1},\end{aligned}$$

after a change of the index of summation. The two sums can be combined to give

$$A(r)b_r - A(m-1)b_m - \sum_{m\le n<r} A(n)\left(b_{n+1}-b_n\right). \qquad \square$$

Exercise 9.1.6. *Suppose that $\{a_n\}_{n=1}^{\infty}$ is a sequence of complex numbers and $f(u)$ is a continuously differentiable function on the interval $[1,x]$. Let $A(t)=\sum_{n\le t} a_n$. Prove that*

$$\sum_{n\le x} a_n f(n) = A(x)f(x) - \int_1^x A(t)f'(t)dt.$$

Exercise 9.1.7. *Let $\alpha>\beta>0$. Write $z=x+iy$, with $x,y\in\mathbb{R}$, $i=\sqrt{-1}$, and $x>0$. Show that*

$$\left|e^{-\alpha z}-e^{-\beta z}\right| \le \frac{|z|}{x}\left(e^{-\alpha x}-e^{-\beta x}\right).$$

Here is the main result regarding the region of convergence for a general Dirichlet series.

Theorem 9.1.8. *Let $s_0\in\mathbb{C}$ and suppose $\{a_n\}_{n=1}^{\infty}$ is a sequence of complex numbers. If the series*

$$\sum_{n=1}^{\infty} a_n e^{-\lambda_n s}$$

converges for $s=s_0$, then it converges uniformly in any region of the form $\mathrm{Re}(s-s_0)\ge 0$, $\mathrm{Arg}(s-s_0)\le\alpha$ with $\alpha<\pi/2$.

Proof. Without loss of any generality, we may assume $s_0=0$ and that $\lambda_n>0$ (since by our monotonicity condition for the λ_n's, only a finite number can be negative and these terms converge for all values of s). We now apply Abel's lemma (Proposition 9.1.5). Since

$$\sum_{n=1}^{\infty} a_n$$

converges, we have that for any $\epsilon > 0$, there is an $N(\epsilon)$ such that for

$$A(r) = \sum_{n \leq r} a_n,$$

we have

$$|A(r_1) - A(r_2)| < \epsilon \text{ for } r_1, r_2 > N(\epsilon).$$

Applying Abel's lemma with $b_n = e^{-\lambda_n s}$, we get

$$\sum_{m \leq n \leq r} a_n e^{-\lambda_n s} = A(r)e^{-\lambda_r s} - A(m-1)e^{-\lambda_m s} - \sum_{m \leq n < r} A(n)\left(e^{-\lambda_{n+1} s} - e^{-\lambda_n s}\right).$$

Writing $s = \sigma + it$, with $\sigma, t \in \mathbb{R}$ and applying Exercise 9.1.7, we get for $r, m > N(\epsilon) + 1$,

$$\begin{aligned}\left|\sum_{m \leq n \leq r} a_n e^{-\lambda_n s}\right| &\leq \epsilon \left(e^{-\lambda_r \sigma} + e^{-\lambda_m \sigma} + \sum_{m \leq n < r} \left|e^{-\lambda_{n+1} s} - e^{-\lambda_n s}\right|\right) \\ &\leq \epsilon \left(e^{-\lambda_r \sigma} + e^{\lambda_m \sigma} + \frac{|s|}{\sigma} \sum_{m \leq n < r} \left(e^{-\lambda_n \sigma} - e^{-\lambda_{n+1} \sigma}\right)\right).\end{aligned}$$

The last sum telescopes to give

$$e^{-\lambda_m \sigma} - e^{-\lambda_r \sigma}$$

so that

$$\left|\sum_{m \leq n \leq r} a_n e^{-\lambda_n s}\right| \leq 2\epsilon \left(1 + \frac{|s|}{\sigma}\right)$$

since $e^{-\lambda_n \sigma} \leq 1$ for $\sigma > 0$. Thus, in every domain where

$$\frac{|s|}{\sigma} \leq M,$$

we have for $m, r \geq N(\epsilon)$ that

$$\left|\sum_{m \leq n \leq r} a_n e^{-\lambda_n s}\right| \leq 2(M+1)\epsilon.$$

In other words, in such a region, the series converges uniformly. But the region defined by the condition $|s|/\sigma \leq M$ is precisely the region $\cos \arg(s) \geq 1/M$. Since M can be arbitrarily large, this means $\arg(s) \leq \alpha$ with $\alpha < \pi/2$.

□

Exercise 9.1.9. *Suppose that*

$$A(x) = \sum_{n \leq x} a_n = O(x^\delta)$$

for some $\delta > 0$. Show that

$$\sum_{n=1}^{\infty} \frac{a_n}{n^s} = s \int_1^{\infty} \frac{A(t)dt}{t^{s+1}}$$

and that the integral converges absolutely for $\mathrm{Re}(s) > \delta$.

The following exercise gives the analytic continuation of $\zeta(s)$ to the region $\mathrm{Re}(s) > 0$.

Exercise 9.1.10. *Show that*

$$\zeta(s) = \frac{s}{s-1} - s \int_1^{\infty} \frac{\{t\}}{t^{s+1}} dt,$$

where $\{t\} = t - \lfloor t \rfloor$ and the integral converges absolutely for $\mathrm{Re}(s) > 0$. Deduce that $\lim_{s \to 1^+} (s-1)\zeta(s) = 1$.

Given a general Dirichlet series $f(s) = \sum_{n=1}^{\infty} a_n e^{-\lambda_n s}$, Theorem 9.1.8 shows that if it converges at $s = s_0$, then it converges for any s with $\mathrm{Re}(s) > \mathrm{Re}(s_0)$, and the function thus defined is holomorphic in this region. Thus, the set of values of s for which the Dirichlet series $f(s)$ converges contains a maximal open half-plane, called the *half-plane of convergence*. It is convenient to view the empty set $\varnothing$ and the entire complex plane $\mathbb{C}$ as open half-planes. If the half-plane of convergence is given by $\mathrm{Re}(s) > \sigma_0$, we call σ_0 the *abscissa of convergence*. The half-plane of convergence of the Dirichlet series

$$\sum_{n=1}^{\infty} |a_n| e^{-\lambda_n s}$$

is called the *half-plane of absolute convergence* of f. If we denote by σ_1 its abscissa of convergence, it is clear that

$$\sigma_0 \leq \sigma_1.$$

Exercise 9.1.11. *Show that the series*

$$1 - \frac{1}{2^s} + \frac{1}{3^s} - \frac{1}{4^s} + \frac{1}{5^s} - \cdots$$

has abscissa of convergence $\sigma_0 = 0$ and abscissa of absolute convergence $\sigma_1 = 1$.

Since the Dirichlet series

$$f(s) = \sum_{n=1}^{\infty} a_n e^{-\lambda_n s}$$

converges for any s with $\mathrm{Re}(s) > \sigma_0$, we can apply Theorem 9.1.8 to deduce that if $f(s)$ converges at s_0, then $f(s) \to f(s_0)$ when $s \to s_0$ in the region $\mathrm{Re}(s) \geq \mathrm{Re}(s_0)$ and $\arg(s - s_0) \leq \alpha$ with $\alpha < \frac{\pi}{2}$. This follows from the uniform convergence and the fact that $e^{-\lambda_n s}$ tends to $e^{-\lambda_n s_0}$ as $s \to s_0$.

Exercise 9.1.12. *Show that*

$$f(s) = \sum_{n=1}^{\infty} a_n e^{-\lambda_n s}$$

can be identically zero only if all its coefficients a_n are zero.

A fundamental theorem in the theory of general Dirichlet series is the following due to Landau.

Theorem 9.1.13 (Landau). *Let*

$$f(s) = \sum_{n=1}^{\infty} a_n e^{-\lambda_n s}$$

be a Dirichlet series with non-negative coefficients. Then $f(s)$ has a singularity at its abscissa of convergence.

Proof. Let σ_0 be its abscissa of convergence. Without loss of any generality, we may suppose $\sigma_0 = 0$ (since we can replace a_n by $a_n e^{-\lambda_n \sigma_0}$). If $f(s)$ is analytic at $s = 0$, we can find $\epsilon > 0$ so that $f(s)$ is analytic in the disc $|s-1| \leq 1 + \epsilon$, centered at 1. In particular, its Laurent series converges there:

$$f(s) = \sum_{k=0}^{\infty} \frac{f^{(k)}(1)}{k!}(s-1)^k.$$

Now

$$f^{(k)}(1) = (-1)^k \sum_{n=1}^{\infty} a_n \lambda_n^k e^{-\lambda_n},$$

since the Dirichlet series is uniformly convergent in its half-plane of absolute convergence. Thus, our Laurent series becomes

$$f(s) = \sum_{k=0}^{\infty} \frac{1}{k!} \left(\sum_{n=1}^{\infty} a_n \lambda_n^k e^{-\lambda_n} \right) (1-s)^k,$$

valid for $|s - 1| \leq 1 + \epsilon$. In particular, this is valid for $s = -\epsilon$. Thus

$$f(-\epsilon) = \sum_{k=0}^{\infty} \frac{1}{k!} \sum_{n=1}^{\infty} a_n \lambda_n^k e^{-\lambda_n} (1+\epsilon)^k$$

is convergent. This being a series of non-negative terms, we can re-arrange the sums to get

$$f(-\epsilon) = \sum_{n=1}^{\infty} a_n e^{-\lambda_n} \left(\sum_{k=0}^{\infty} \frac{\lambda_n^k (1+\epsilon)^k}{k!} \right).$$

The inner sum is $e^{\lambda_n(1+\epsilon)}$ so that we deduce

$$\sum_{n=1}^{\infty} a_n e^{\lambda_n \epsilon}$$

converges. Thus, our Dirichlet series converges for $s = -\epsilon$ and hence for $\mathrm{Re}(s) > -\epsilon$, by our earlier result. Hence the abscissa of convergence cannot be zero.

Exercise 9.1.14. *Prove the identity*

$$\sum_{n=0}^{\infty} \left(\frac{\alpha^{n+1} - \beta^{n+1}}{\alpha - \beta} \right) \left(\frac{\gamma^{n+1} - \delta^{n+1}}{\gamma - \delta} \right) T^n = \frac{1 - \alpha\beta\gamma\delta T^2}{(1-\alpha\gamma T)(1-\alpha\delta T)(1-\beta\gamma T)(1-\beta\delta T)}.$$

Exercise 9.1.15. *A function $a : \mathbb{N} \to \mathbb{C}$ is called an* arithmetical function. *Given two arithmetical functions a and b, we define the* Dirichlet convolution *$a * b$, to be the arithmetical function given by*

$$(a * b)(n) = \sum_{d|n} a(d) b\left(\frac{n}{d}\right)$$

where the sum is over positive divisors d of n. Show that

$$\sum_{n=1}^{\infty} \frac{(a*b)(n)}{n^s} = \left(\sum_{n=1}^{\infty} \frac{a(n)}{n^s} \right) \left(\sum_{n=1}^{\infty} \frac{b(n)}{n^s} \right).$$

Exercise 9.1.16. *An arithmetical function $a : \mathbb{N} \to \mathbb{C}$ is said to be* completely multiplicative *if $a(mn) = a(m)a(n)$ for all m, n. Show that*

$$\sum_{n=1}^{\infty} \frac{a(n)}{n^s} = \prod_{p} \left(1 - \frac{a(p)}{p^s} \right)^{-1}.$$

Exercise 9.1.17. *Given any arithmetical function $a : \mathbb{N} \to \mathbb{C}$, we will write*

$$L(s, a) := \sum_{n=1}^{\infty} \frac{a(n)}{n^s}.$$

If a, b, c, d are completely multiplicative functions, show that

$$L(s,(a*b)(c*d)) = \frac{L(s,ac)L(s,ad)L(s,bc)L(s,bd)}{L(2s,abcd)}.$$

Here ac denotes the arithmetical function given by $(ac)(n) := a(n)c(n)$.

Exercise 9.1.18. *Let*

$$\sigma_u(n) = \sum_{d|n} d^u.$$

Show that for any complex numbers a, b,

$$\sum_{n=1}^{\infty} \frac{\sigma_a(n)\sigma_b(n)}{n^s} = \frac{\zeta(s)\zeta(s-a)\zeta(s-b)\zeta(s-a-b)}{\zeta(2s-a-b)}.$$

Exercise 9.1.19. *Let $t_0 \in \mathbb{R}$. With $a = it_0$, $b = -it_0$ in the previous exercise, show that*

$$\sum_{n=1}^{\infty} \frac{(\sigma_{it_0}(n))^2}{n^s} = \frac{\zeta(s)^2\zeta(s-it_0)\zeta(s+it_0)}{\zeta(2s)}.$$

Using Landau's theorem (Theorem 9.1.13), show that $\zeta(1+it_0) \neq 0$.

9.2 The Poisson summation formula

In section 5.3, we introduced the Dirichlet series associated to a modular form of weight k on the full modular group. There we extended this Dirichlet series to the entire complex plane and derived a functional equation relating its value at s to $k-s$. This result is prototypical of a larger theme. One can say that this theme begins with Riemann's celebrated memoir of 1860 in which he introduced his zeta function and derived its analytic continuation and functional equation using the rudimentary ideas from the theory of Fourier transforms, most notably the Poisson summation formula. These ideas are highlighted in the following theorem.

Let us suppose we are given two sequences of complex numbers $\{a_n\}_{n=0}^{\infty}$ and $\{b_n\}_{n=0}^{\infty}$. We suppose that these sequences have polynomial growth, i.e. $a_n = O(n^c)$ and $b_n = O(n^c)$ for some $c > 0$. To each of these sequences, we associate for $t \geq 0$,

$$f(t) = \sum_{n=0}^{\infty} a_n e^{-\pi n t}$$

$$g(t) = \sum_{n=0}^{\infty} b_n e^{-\pi n t}.$$

We also associate the following Dirichlet series:

$$L_f(s) = \sum_{n=1}^{\infty} \frac{a_n}{n^s},$$
$$L_g(s) = \sum_{n=1}^{\infty} \frac{b_n}{n^s}.$$

Theorem 9.2.1. *With f and g as above, assume that for $t > 0$,*

$$f\left(\frac{1}{t}\right) = wt^k g(t)$$

for some complex number w and $k \in \mathbb{R}$. Then $L_f(s)$ and $L_g(s)$ extend analytically to the entire complex plane apart from possible simple poles at $s = 0$ and k and satisfy the functional equation

$$\pi^{-s}\Gamma(s)L_f(s) = w\pi^{-(k-s)}\Gamma(k-s)L_g(k-s).$$

The proof will show that if $a_0 = b_0 = 0$, there are no poles.

Proof. By the assumption of polynomial growth on the coefficients a_n, we see that

$$\int_0^{\infty} (f(t) - a_0)t^s \frac{dt}{t} = \int_0^{\infty} \left(\sum_{n=1}^{\infty} a_n e^{-\pi nt}\right) t^s \frac{dt}{t}$$

is an absolutely convergent integral. Moreover, $L_f(s)$ converges absolutely for $\mathrm{Re}(s)$ sufficiently large and so in this region the integral can be re-written as

$$\sum_{n=1}^{\infty} a_n \int_0^{\infty} e^{-\pi nt} t^{s-1} dt = \pi^{-s}\Gamma(s)L_f(s)$$

after an easy change of variable. Now

$$\int_0^{\infty} (f(t) - a_0)t^{s-1}dt = \int_0^1 (f(t) - a_0)t^{s-1}dt + \int_1^{\infty} (f(t) - a_0)t^{s-1}dt.$$

The first integral is

$$-\frac{a_0}{s} + \int_0^1 f(t)t^{s-1}dt.$$

We change t to $1/t$ in this integral expression to get

$$-\frac{a_0}{s} + \int_1^{\infty} f\left(\frac{1}{t}\right) t^{-s-1}dt.$$

Since $f(1/t) = wt^k g(t)$, this becomes

$$-\frac{a_0}{s}+w\int_1^\infty g(t)t^{k-s}\frac{dt}{t}=-\frac{a_0}{s}+w\int_1^\infty (g(t)-b_0)t^{k-s}\frac{dt}{t}-\frac{wb_0}{k-s},$$

for $\mathrm{Re}(s)$ sufficiently large. All of this can be combined to give

$$\pi^{-s}\Gamma(s)L_f(s)=-\frac{a_0}{s}-\frac{wb_0}{k-s}+\int_1^\infty (f(t)-a_0)t^s\frac{dt}{t}+\int_1^\infty (g(t)-b_0)t^{k-s}\frac{dt}{t}.$$

The right hand side is analytic for all $s\in\mathbb{C}$ apart from a simple pole at $s=0$ with residue $-a_0$ and $s=k$ with residue $-wb_0$.

Changing f to g in the above identity shows that, upon using $g(1/t)=(1/w)t^kf(t)$,

$$\pi^{-s}\Gamma(s)L_g(s)=-\frac{b_0}{s}-\frac{(1/w)a_0}{k-s}+\int_1^\infty (g(t)-b_0)t^s\frac{dt}{t}+\int_1^\infty (f(t)-a_0)t^{k-s}\frac{dt}{t},$$

from which the functional equation is immediate. □

A wealth of examples of f,g satisfying the conditions of Theorem 9.2.1 arise from the theory of modular forms as was evident from Theorem 5.3.7 and Exercise 5.3.8. More examples can be derived by applying the Poisson summation formula, which we now discuss.

Let $F:\mathbb{R}\to\mathbb{C}$ be a smooth function (that is, infinitely differentiable). We define the Schwartz space S as the set of smooth functions F such that for every pair of integers $m,n\geq 0$,

$$\sup_{x\in\mathbb{R}}\left|x^mF^{(n)}(x)\right|<\infty.$$

It is clear that S is a vector space over the field of complex numbers.

Exercise 9.2.2. *If $F\in S$, show that*

$$\left|\int_{-\infty}^\infty t^mF^{(n)}(t)dt\right|<\infty$$

for all pairs of integers $m,n\geq 0$.

For $F\in S$, we define the *Fourier transform* $\hat{F}$ by setting

$$\hat{F}(x)=\int_{-\infty}^\infty F(t)e^{-2\pi itx}dt.$$

Exercise 9.2.3. *If $F\in S$, show that $\hat{F}\in S$.*

The Fourier inversion formula shows that for $F\in S$, we have

$$F(t)=\int_{-\infty}^\infty \hat{F}(x)e^{2\pi itx}dx.$$

In other words, $\hat{\hat{F}}(x) = F(-x)$.

Exercise 9.2.4. *Show that*

$$\int_{-\infty}^{\infty} e^{-\pi x^2} dx = 1.$$

Exercise 9.2.5. *For $F(x) = e^{-\pi x^2}$, show that $\hat{F}(u) = e^{-\pi u^2}$. In other words, F is its own Fourier transform.*

Exercise 9.2.6. *Let $F \in S$ and define $G(t) = F(At + B)$ for real constants A, B, with $A \neq 0$. Show that $A\hat{G}(t) = e^{2\pi i B t/A}\hat{F}(t/A)$.*

Theorem 9.2.7 (Poisson summation formula). *Let $F \in S$. Then*

$$\sum_{n\in\mathbb{Z}} F(n) = \sum_{n\in\mathbb{Z}} \hat{F}(n).$$

Proof. Let

$$G(v) = \sum_{n\in\mathbb{Z}} F(n+v).$$

It is easily checked that G is a continuous function of v with period one. We can compute the Fourier coefficients of G:

$$\begin{aligned} c_m :&= \int_0^1 G(v)e^{-2\pi imv}dv = \int_0^1 \left(\sum_{n\in\mathbb{Z}} F(n+v)\right) e^{-2\pi imv}dv \\ &= \sum_{n\in\mathbb{Z}} \int_n^{n+1} F(x)e^{-2\pi imx}dx = \int_{-\infty}^{\infty} F(x)e^{-2\pi imx}dx = \hat{F}(m). \end{aligned}$$

Since $\hat{F} \in S$,

$$\sum_{m\in\mathbb{Z}} |\hat{F}(m)| < \infty$$

so

$$G(v) = \sum_{m\in\mathbb{Z}} \hat{F}(m)e^{2\pi imv}$$

is the Fourier series representing G. Setting $v = 0$ gives the Poisson summation formula. □

Exercise 9.2.8. *Show that if $F(x) = e^{-(Ax+B)^2\pi}$ with $A, B \in \mathbb{R}$ and $A \neq 0$, then*

$$A\hat{F}(x) = e^{2\pi iBx/A}e^{-\pi(x+B)^2}.$$

Exercise 9.2.9. *For any $a \in \mathbb{R}$, show that*

$$\sum_{n\in\mathbb{Z}} e^{-\pi(n+a)^2/x} = x^{1/2}\sum_{n\in\mathbb{Z}} e^{-\pi n^2 x + 2\pi ina}.$$

Exercise 9.2.10. *Show that for $x > 0$,*

$$\sum_{n\in\mathbb{Z}} e^{-\pi n^2 x} = x^{-1/2} \sum_{n\in\mathbb{Z}} e^{-\pi n^2/x}.$$

Following Riemann, we now derive the analytic continuation and functional equation for the function $\zeta(s)$. Indeed, let

$$\vartheta(z) = \sum_{n\in\mathbb{Z}} e^{\pi i n^2 z},$$

so that $f(t) = \vartheta(it)$ satisfies

$$f\left(\frac{-1}{t}\right) = t^{1/2} f(t)$$

by virtue of Exercise 9.2.10. We can then apply Theorem 9.2.1 with $k = 1/2$ to deduce that $\zeta(2s)$ extends analytically to the entire complex plane apart from a possible simple pole at $s = 0$ and $s = 1/2$ and satisfies the functional equation

$$\pi^{-s}\Gamma(s)\zeta(2s) = \pi^{-(\frac{1}{2}-s)}\Gamma\left(\frac{1}{2} - s\right)\zeta(1 - 2s).$$

Changing s to $s/2$, we deduce the following.

Theorem 9.2.11 (Riemann, 1860). *The function*

$$\int_0^\infty (\vartheta(it) - 1)\, t^s \frac{dt}{t} = 2\pi^{-s}\Gamma(s)\zeta(2s)$$

defined originally for $\mathrm{Re}(s) > 1/2$, extends analytically to the entire complex plane via the formula

$$2\pi^{-s}\Gamma(s)\zeta(2s) = -\frac{1}{s} - \frac{1}{\frac{1}{2} - s} + \int_1^\infty (\vartheta(it) - 1)\,(t^2 + t^{\frac{1}{2}-s})\frac{dt}{t}.$$

Changing s to $s/2$ and dividing by 2 gives

$$\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s) = \frac{1}{s-1} - \frac{1}{s} + \frac{1}{2}\int_1^\infty (\vartheta(it) - 1)\left(t^{s/2} + t^{\frac{1-s}{2}}\right)\frac{dt}{t}$$

and the right hand side is invariant under the map $s \to 1 - s$. Moreover, the expression is analytic for all $s \in \mathbb{C}$ except at $s = 0$ and $s = 1$ where it has a simple pole.

Exercise 9.2.12. *Show that $(s - 1)\zeta(s)$ extends to an entire function.*

Exercise 9.2.13. *Show that $\zeta(s)$ has simple zeros at $s = -2n$, for each positive integer n.*

Exercise 9.2.14. *Show that* $\zeta(0) = -1/2$.

Exercise 9.2.15. *Let* $f \in S$. *For* $x > 0$ *show that*

$$\sum_{n\in\mathbb{Z}} f(nx)$$

converges absolutely and satisfies

$$x \sum_{n\in\mathbb{Z}} f(nx) = \sum_{n\in\mathbb{Z}} \hat{f}\left(\frac{n}{x}\right).$$

Exercise 9.2.16. *Let* $f \in S$ *be an even function. That is,* $f(x) = f(-x)$. *For any* $x > 0$, *show that*

$$\sum_{n=1}^{\infty} f\left(\frac{n}{x}\right) = x \sum_{n=1}^{\infty} \hat{f}(nx) + \frac{1}{2}\left(x\hat{f}(0) - f(0)\right).$$

Exercise 9.2.17. *Let* $f \in S$. *Set* $\zeta(s, f) = \int_0^\infty f(x)x^s \frac{dx}{x}$. *Show that the integral converges for all* s *with* $\mathrm{Re}(s) > 1$.

Exercise 9.2.18. *Let* $f \in S$ *be an even function. With* $\zeta(s, f)$ *defined as in the previous exercise, show that* $\zeta(s, f)$ *extends for all* $s \in \mathbb{C}$, *apart from possible simple poles at* $s = 0, 1$ *and satisfies the functional equation*

$$\zeta^\star(s, f) = \zeta^\star(1 - s, \hat{f}),$$

where $\zeta^\star(s, f) = \zeta(s)\zeta(s, f)$.

It is clear from the previous discussion that the essential tool in deriving the analytic continuation and functional equation for the Riemann zeta function is the transformation law of Exercise 9.2.10. The function

$$\Theta(z) = \sum_{n\in\mathbb{Z}} e^{2\pi i n^2 z}$$

is a classical theta-function. It is a modular form of weight $1/2$ in the following sense. Let d be an odd integer and c any integer. If d is a positive odd prime number, the familiar Legendre symbol $\left(\frac{c}{d}\right)$ is defined as 0 if $d|c$, $+1$ if c is a square $(\mathrm{mod}\, d)$ and -1 otherwise. This is extended by multiplicativity for positive odd d. That is, if $d = \prod_j p_j$ with p_j odd primes (not necessarily distinct) then

$$\left(\frac{c}{d}\right) = \prod_j \left(\frac{c}{p_j}\right).$$

For $d = \pm 1$, we adopt the convention $\left(\frac{0}{\pm 1}\right) = 1$. For d negative, we define

$$\left(\frac{c}{d}\right) = \begin{cases} \left(\frac{c}{|d|}\right) & \text{if } c > 0 \\ -\left(\frac{c}{|d|}\right) & \text{if } c < 0 \end{cases}$$

Exercise 9.2.19. *Show that the symbol* $\left(\frac{c}{d}\right)$ *is completely multiplicative in each of the variables.*

To describe the modular transformation law of the Θ function, we define $\sqrt{z}$ for $z \in \mathbb{C}$ to be the branch whose argument is in the interval $(-\pi/2, \pi/2]$. Finally, for any $z \in \mathbb{H}$ and $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, we set

$$\varepsilon_d = \begin{cases} 1 \text{ if } d \equiv 1 \pmod 4 \\ i \text{ if } d \equiv 3 \pmod 4, \end{cases}$$

and

$$j(\gamma, z) := \left(\frac{c}{d}\right) \varepsilon_d^{-1} \sqrt{cz + d}.$$

Then

$$\Theta(\gamma z) = j(\gamma, z)\Theta(z) \qquad \forall \gamma \in \Gamma_0(4).$$

We will not prove this fact here, but only present it to indicate to the student that our study has a natural entry into the theory of modular forms of half-integral weight.

However, Riemann's method of deriving the analytic continuation and functional equation for $\zeta(s)$ extends to similar results for the Dirichlet L-functions $L(s, \chi)$ defined as follows. Let $\chi : (\mathbb{Z}/N\mathbb{Z})^\star \to \mathbb{C}$ be a Dirichlet character. Then

$$L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s}.$$

Riemann's method can be adapted to derive the analytic continuation and functional equation of $L(s, \chi)$. The discussion splits into two parts, according as χ is *even* (that is, $\chi(-1) = 1$) or χ is *odd* (that is, $\chi(-1) = -1$).

We begin with a preliminary discussion of Dirichlet characters. Let χ_0 denote the *trivial character* modulo N. Thus $\chi_0(n) = 1$ if $(n, N) = 1$ and 0 otherwise. If $d|N$ and ψ is a Dirichlet character modulo d, then $\chi_0\psi$ is a Dirichlet character modulo N. A Dirichlet character modulo N obtained in this way from a proper divisor d of N (that is, $d|N$, $d < N$) is called *imprimitive*. Otherwise we call the character *primitive*. Primitive characters modulo N are periodic with period N and no smaller period. By definition, every imprimitive character χ of modulus N is induced from a character whose modulus d is a proper divisor of N. The *conductor* of a character χ is the least modulus d such that χ is induced from a (necessarily primitive) character of modulus d.

Exercise 9.2.20. *Let* $d|N$ *and let* ψ *be a character modulo* d*. Prove that*

$$L(s, \chi_0\psi) = L(s, \psi) \prod_{p|N} \left(1 - \frac{\psi(p)}{p^s}\right).$$

The previous exercise shows that to derive the analytic continuation and functional equation for $L(s, \chi)$, we need only focus our attention on the primitive characters $\chi \pmod N$. The following exercises reduce the problem to a simple application of the Poisson summation formula. To this end, we introduce for every character $\chi \pmod N$, the *Gauss sum* $\tau(\chi)$, defined as

$$\tau(\chi) = \sum_{m=1}^{N} \chi(m) e\left(\frac{m}{N}\right), \qquad \text{where } e(t) = e^{2\pi i t}.$$

This sum plays a fundamental role in number theory.

Exercise 9.2.21. *If $(n, N) = 1$, show that*

$$\chi(n)\tau(\overline{\chi}) = \sum_{m=1}^{N} \overline{\chi}(m) e\left(\frac{mn}{N}\right),$$

where $\overline{\chi}$ denotes the complex conjugate of χ.

Theorem 9.2.22. *If χ is a primitive non-trivial character modulo N, show that*

$$\chi(n)\tau(\overline{\chi}) = \sum_{m=1}^{N} \overline{\chi}(m) e\Big(\frac{mn}{N}\Big).$$

That is, the formula in the previous exercise holds without the condition $(n, N) = 1$.

Proof. By virtue of Exercise 9.2.21, we need to show the identity also holds for $(n, N) > 1$. In this case, the left hand side is zero and so it suffices to show the right hand size is zero. Write $n/N = n_1/N_1$ with $(n, N_1) = 1$. Thus $N_1 | N$ and $N_1 < N$. If $N_1 = 1$, then n is a multiple of N so that $e(mn/N) = 1$ and the right hand side is zero since

$$\sum_{m=1}^{N} \overline{\chi(m)} = 0$$

as χ is a non-trivial character. So we may suppose $1 < N_1 < N$. We have to prove that

$$\sum_{m=1}^{N} \overline{\chi}(m) e\left(\frac{mn_1}{N_1}\right) = 0.$$

Write $N = N_1 N_2$ and put $m = aN_1 + b$ with $0 \le a < N_2$, $1 \le b \le N_1$. The sum in question can be rewritten as

$$\sum_{1\le b\le N_1} e\left(\frac{bn_1}{N_1}\right) \sum_{0\le a<N_2} \overline{\chi}(aN_1+b)$$

and it suffices to prove that the inner sum is zero. Let us write

$$S(b) = \sum_{0\le a<N_2} \overline{\chi}(aN_1+b).$$

Let us note that

$$\begin{aligned} S(b+N_1) &= \sum_{a=1}^{N_2} \overline{\chi}(aN_1+b) = \overline{\chi}(b) + \sum_{1\le a<N_2} \overline{\chi}(aN_1+b) \\ &= S(b). \end{aligned}$$

Thus $S(b)$ is periodic with period N_1.

We see that for any integer c satisfying $(c,N)=1$ and $c\equiv 1 \pmod{N_1}$,

$$\chi(c)S(b) = \sum_{0\le a<N_2} \overline{\chi}(caN_1+cb)$$

and as a ranges over $0\le a<N_2$, we see that ca ranges over the same set $\pmod{N_2}$. Thus,

$$\chi(c)S(b) = S(bc).$$

Since $c\equiv 1 \pmod{N_1}$, we write $c=qN_1+1$ and so $S(bc)=S(bqN_1+b)=S(b)$ as $S(b)$ is periodic with period N_1. As χ is a primitive character modulo N, it is not periodic to any modulus N_1 with N_1 a proper factor of N. Thus, there are integers c_1, c_2 such that $(c_1,N)=(c_2,N)=1$, $c_1\equiv c_2 \pmod{N_1}$ and $\chi(c_1)\neq\chi(c_2)$. From the preceding, upon setting $c\equiv c_1c_2^{-1} \pmod{N}$, we have that $c\equiv 1 \pmod{N_1}$ so that $\chi(c)S(b)=S(b)$. But $\chi(c)=\chi(c_1)\overline{\chi}(c_2)\neq 1$ and we conclude $S(b)=0$ as desired. □

Exercise 9.2.23. *If χ is a primitive character modulo N, then $|\tau(\chi)|^2=N$.*

The previous exercise shows that if χ is a primitive character modulo N then $|\tau(\chi)|=\sqrt{N}$.

Now suppose χ is an even character (that is, $\chi(-1)=1$). We have

$$\Gamma(s) = \int_0^\infty e^{-x}x^s\frac{dx}{x}$$

so that after a simple change of variables:

$$\pi^{-s/2}N^{s/2}\Gamma\left(\frac{s}{2}\right)n^{-s} = \int_0^\infty e^{-n^2\pi x/N}x^{s/2}\frac{dx}{x}.$$

We multiply this equation by $\chi(n)$ and sum over $n\ge 1$ to get

$$\pi^{-s/2}N^{s/2}\Gamma\left(\frac{s}{2}\right)L(s,\chi)=\int_0^\infty\left(\sum_{n=1}^\infty\chi(n)e^{-n^2\pi x/N}\right)x^{s/2}\frac{dx}{x}$$

for $\mathrm{Re}(s)>1$. Since $\chi(-1)=1$ and $\chi(0)=0$, we can rewrite the sum in the integrand as

$$\theta(x,\chi)=\sum_{n=-\infty}^\infty\chi(n)e^{-n^2\pi x/N}.$$

Theorem 9.2.24. *If χ is a primitive character modulo N is even, then*

$$\tau(\overline{\chi})\theta(x,\chi)=\left(\frac{N}{\chi}\right)^{1/2}\theta\left(\frac{1}{x},\overline{\chi}\right).$$

Proof. The left hand side is

$$\sum_{n=-\infty}^\infty\tau(\overline{\chi})\chi(n)e^{-n^2\pi x/N}.$$

By Theorem 9.2.22,

$$\chi(n)\tau(\overline{\chi})=\sum_{m=1}^N\overline{\chi}(m)e\left(\frac{mn}{N}\right)$$

so that

$$\tau(\overline{\chi})\theta(x,\chi)=\sum_{m=1}^N\overline{\chi}(m)\sum_{n=-\infty}^\infty e^{-n^2\pi x/N+2\pi imn/N}.$$

By Exercise 9.2.9, the inner sum is equal to

$$\left(\frac{N}{x}\right)^{1/2}\sum_{n=-\infty}^\infty e^{-(n+m/N)^2\pi N/x}$$

so that

$$\begin{aligned}\tau(\overline{\chi})\theta(x,\chi)&=\left(\frac{N}{x}\right)^{1/2}\sum_{m=1}^N\overline{\chi}(m)\left(\sum_{n=-\infty}^\infty e^{-(nN+m)^2\pi/Nx}\right)\\&=\left(\frac{N}{x}\right)^{1/2}\sum_{r\in\mathbb{Z}}\overline{\chi}(r)e^{-\pi r^2/Nx}\\&=\left(\frac{N}{x}\right)^{1/2}\theta\left(\frac{1}{x},\overline{\chi}\right).\end{aligned}$$

□

We are now ready to derive the analytic continuation and functional equation for $L(s,\chi)$ with χ even.

Theorem 9.2.25. *Let χ be a primitive character modulo N. Suppose χ is even. Then*

$$2\pi^{-s/2}N^{s/2}\Gamma\left(\frac{s}{2}\right)L(s,\chi)=\int_1^\infty \theta(x,\chi)x^{s/2}\frac{dx}{x}+\frac{N^{1/2}}{\tau(\overline{\chi})}\int_1^\infty \theta(x,\overline{\chi})x^{\frac{1-s}{2}}\frac{dx}{x}.$$

Both integrals converge for all $s\in\mathbb{C}$ and thus we obtain analytic continuation of $L(s,\chi)$ to the entire complex plane. Moreover,

$$\pi^{-s/2}N^{s/2}\Gamma\left(\frac{s}{2}\right)L(s,\chi)=w_\chi\pi^{-(1-s)/2}N^{(1-s)/2}\Gamma\left(\frac{1-s}{2}\right)L(1-s,\overline{\chi})$$

where $w_\chi=N^{1/2}/\tau(\chi)$.

Proof. Since

$$\begin{aligned}2\pi^{-s/2}N^{s/2}\Gamma\left(\frac{s}{2}\right)L(s,\chi)&=\int_0^\infty \theta(x,\chi)x^{s/2}\frac{dx}{x}\\&=\int_1^\infty \theta(x,\chi)x^{s/2}\frac{dx}{x}+\int_1^\infty \theta\left(\frac{1}{x},\chi\right)x^{-s/2}\frac{dx}{x}\end{aligned}$$

after a simple change of variables. By Theorem 9.2.24,

$$\theta\left(\frac{1}{x},\chi\right)=\tau(\chi)\theta(x,\overline{\chi})\left(\frac{x}{N}\right)^{1/2}.$$

Therefore

$$2\pi^{-s/2}N^{s/2}\Gamma\left(\frac{s}{2}\right)L(s,\chi)=\int_1^\infty \theta(x,\chi)x^{s/2}\frac{dx}{x}+\int_1^\infty \frac{N^{1/2}}{\tau(\overline{\chi})}\theta(x,\overline{\chi})x^{\frac{1-s}{2}}\frac{dx}{x}.$$

□

Exercise 9.2.26. *Prove that*

$$\sum_{n=-\infty}^{\infty} ne^{-n^2\pi x/N+2\pi imn/N}=i\left(\frac{N}{x}\right)^{3/2}\sum_{n=-\infty}^{\infty}\left(n+\frac{m}{N}\right)e^{-\pi N(n+\frac{m}{N})^2/x}.$$

Exercise 9.2.27. *Let χ be an odd Dirichlet character modulo N. Let*

$$\vartheta_1(x,\chi)=\sum_{n=-\infty}^{\infty} n\chi(n)e^{-\pi n^2x/N}.$$

Show that

$$2\pi^{-(\frac{s+1}{2})}\Gamma\left(\frac{s+1}{2}\right)N^{\frac{s+1}{2}}L(s,\chi)=\int_0^\infty \vartheta_1(x,\chi)x^{\frac{s+1}{2}}\frac{dx}{x}.$$

Exercise 9.2.28. *Prove that if χ is an odd primitive Dirichlet character modulo N, then*

$$\vartheta_1\left(\frac{1}{x},\chi\right) = x^{3/2}\left(\frac{\tau(\chi)}{iN^{1/2}}\right)\vartheta_1(x,\overline{\chi})$$

Exercise 9.2.29. *Let χ be an odd primitive Dirichlet character modulo N. Let*

$$\xi(s,\chi) := \pi^{-s/2}N^{s/2}\Gamma\left(\frac{s+1}{2}\right)L(s,\chi).$$

Then $\xi(s,\chi)$ is entire and satisfies the functional equation

$$\xi(s,\chi) = w_\chi\xi(1-s,\overline{\chi}),$$

where $w_\chi = \tau(\chi)/iN^{1/2}$.

9.3 L-functions attached to modular forms

In Chapter 6, we defined the notion of the Fourier expansion of a modular form at each cusp. Here we will study the nature of these expansions in the context of the Dirichlet series attached to them. More precisely, let $f \in S_k(\Gamma_0(N),\chi)$ be a cusp form and suppose that

$$f(z) = \sum_{n=1}^{\infty} a_n e^{2\pi inz}$$

is its Fourier expansion at the cusp $i\infty$. Let

$$w = \begin{pmatrix} 0 & -1 \\ N & 0 \end{pmatrix}$$

and put $g = f|w$ so that, by a simple extension of Exercise 8.4.3, $g \in S_K(\Gamma_0(N),\chi)$. Let $g(z) = \sum_{n=1}^{\infty} b_n e^{2\pi inz}$ be its Fourier expansion at $i\infty$. Thus, this is the Fourier expansion of f at the cusp 0. Since

$$g(z) = N^{k/2}(Nz)^{-k}f\left(-\frac{1}{Nz}\right)$$

we see that by putting $z = it, t > 0$, we have

$$g(it) = i^{-k}N^{-k/2}t^{-k}f\left(\frac{i}{Nt}\right).$$

We can apply the method of Theorem 9.2.1 to deduce:

Theorem 9.3.1. *Let f and g be as above and let*

$$L_f(s) = \sum_{n=1}^{\infty} \frac{a_n}{n^s},$$
$$L_g(s) = \sum_{n=1}^{\infty} \frac{b_n}{n^s}.$$

Then $L_f(s)$ and $L_g(s)$ extend to entire functions and satisfy the functional equation

$$\Lambda_f(s) = i^k \Lambda_g(k-s),$$

where

$$\Lambda_f(s) = \left(\frac{\sqrt{N}}{2\pi}\right)^s \Gamma(s) L_f(s)$$
$$\Lambda_g(s) = \left(\frac{\sqrt{N}}{2\pi}\right)^s \Gamma(s) L_g(s).$$

Proof. We have

$$(2\pi)^{-s}\Gamma(s)L_g(s) = \int_0^{\infty} g(it)t^{s-1}dt.$$

Since $g(it) = i^{-k}N^{-k/2}t^{-k}f(i/Nt)$, the integral is

$$i^{-k}N^{-k/2}\int_0^{\infty} f\left(\frac{i}{N}\right)t^{-k+s-1}dt.$$

We put $v = 1/Nt$ and change variables to get

$$i^{-k}N^{-k/2}\int_0^{\infty} f(iv)N^{k-s}v^{k-s-1}dv = i^{-k}N^{k/2-s}\int_0^{\infty} f(iv)v^{k-s-1}dv$$
$$= i^{-k}N^{k/2-s}(2\pi)^{-(k-s)}\Gamma(k-s)L_f(k-s)$$

from which the theorem follows. □

The assumption that f and g are cusp forms is not essential in the previous theorem, as shown in the following exercise.

Exercise 9.3.2. *Let $f \in M_k(\Gamma_0(N), \chi)$ and $g = f|w$ with $w = \begin{pmatrix} 0 & -1 \\ N & 0 \end{pmatrix}$. Supposing that*

$$f(z) = \sum_{n=0}^{\infty} a_n e^{2\pi i n z}$$
$$g(z) = \sum_{n=0}^{\infty} b_n e^{2\pi i n z}$$

are the Fourier expansions of f and g at $i\infty$, prove that

$$\Lambda_f(s) := \left(\frac{\sqrt{N}}{2\pi}\right)^s \Gamma(s) L_f(s)$$
$$\Lambda_g(s) := \left(\frac{\sqrt{N}}{2\pi}\right)^s \Gamma(s) L_g(s),$$

where $L_f(s) = \sum_{n=1}^{\infty} \frac{a_n}{n^s}$ and $L_g(s) = \sum_{n=1}^{\infty} \frac{b_n}{n^s}$ extend analytically to the entire complex plane with poles at $s = 0$ and $s = k$. Moreover,

$$\Lambda_f(s) + \frac{a_0}{s} + \frac{b_0 i^k}{k-s}$$

and

$$\Lambda_g(s) + \frac{b_0}{s} + \frac{a_0 i^k}{k-s}$$

are entire and satisfy the functional equation

$$\Lambda_f(s) = i^k \Lambda_g(k-s).$$

Exercise 9.3.3. *With f, g as in the previous exercise, show that $L_f(0) = -a_0, L_g(0) = -b_0$,*

$$\lim_{s \to k} (s-k) L_f(s) = b_0 \left(\frac{2\pi i}{\sqrt{N}}\right)^k / \Gamma(k)$$

and

$$\lim_{s \to k} (s-k) L_g(s) = a_0 \left(\frac{2\pi i}{\sqrt{N}}\right)^k / \Gamma(k).$$

Theorem 9.3.1 and Exercise 9.3.2 have a converse. The reader will note that the integral expressions for

$$\Lambda_f(s) + \frac{a_0}{s} + \frac{b_0 i^k}{k-s}$$

and

$$\Lambda_g(s) + \frac{b_0}{s} + \frac{a_0 i^k}{k-s}$$

are bounded in any vertical strip of the form $\sigma_1 \leq \mathrm{Re}(s) \leq \sigma_2$.

Using some fundamental results from complex analysis, we proceed in the converse direction. The first result concerns the Γ function, defined for $\mathrm{Re}(s) > 0$ by the integral,

$$\Gamma(s) = \int_0^\infty e^{-t} t^{s-1} dt.$$

A simple integration by parts shows that $\Gamma(s+1) = s\Gamma(s)$ and this allows us to extend $\Gamma(s)$ analytically to the entire complex plane with only simple poles at $s = 0, -1, -2, \ldots$

Stirling's formula for the asymptotic behavior of $\Gamma(s)$ as $|s| \to \infty$ implies that if σ is fixed, and $|t| \to \infty$, then

$$|\Gamma(\sigma + it)| \sim \sqrt{2\pi} e^{-\frac{\pi}{2}|t|} |t|^{\sigma - 1/2}$$

(see, for example, Exercise 6.3.15 of [24]). This asymptotic formula allows us to deduce that if $\Lambda_f(s)$ and $\Lambda_g(s)$ are bounded in every vertical strip, then for any fixed interval $\sigma_1 \leq \mathrm{Re}(s) \leq \sigma_2$, the series $L_f(s)$ and $L_g(s)$ are $O\Big(|t|^{\frac{1}{2}-\sigma} e^{\pi|t|/2}\Big)$ as $|t| \to \infty$. A second result from complex analysis, namely the Phragmén-Lindelöf theorem, allows us to sharpen this estimate. We state this without proof but refer the reader to Theorem 8.1.13 of [24].

Theorem 9.3.4 (Phragmén-Lindelöf). *Suppose that $f(s)$ is analytic in the vertical strip $a \leq \mathrm{Re}(s) \leq b$, and that for some $\alpha \geq 1$,*

$$|f(s)| = O\Big(e^{|t|^\alpha}\Big)$$

as $|t| \to \infty$. If $f(s)$ is bounded on the two vertical lines $\mathrm{Re}(s) = a$ and $\mathrm{Re}(s) = b$, then $f(s)$ is bounded in the entire vertical strip.

We apply this theorem in our context via the following exercise.

Exercise 9.3.5. *Suppose that $f(s)$ is analytic in the vertical strip defined by $a \leq \mathrm{Re}(s) \leq b$ and that for some $\alpha \geq 1$,*

$$f(s) = O\Big(e^{|t|^\alpha}\Big)$$

as $|t| \to \infty$. If for some $A > 0$, $f(s) = O\big(|t|^A\big)$ on $\mathrm{Re}(s) = a$ and $\mathrm{Re}(s) = b$, then show that $f(s) = O\big(|t|^A\big)$ for all s in the vertical strip $a \leq \mathrm{Re}(s) \leq b$.

In our context, both $L_f(s)$ and $L_g(s)$ are bounded in some fixed half-plane. Moreover $s(s-k)L_f(s)$ and $s(s-k)L_g(s)$ are entire. By the functional equation relating $\Lambda_f(s)$ to $\Lambda_g(k-s)$ and Stirling's formula, we see that for some $A > 0$,

$$s(s-k)L_f(s) = O\big(|t|^A\big),$$
$$s(s-k)L_g(s) = O\big(|t|^A\big),$$

in any vertical strip.

Exercise 9.3.6. *Show that for any $\sigma > 0$,*

$$e^{-x} = \frac{1}{2\pi i} \int_{\sigma - i\infty}^{\sigma + i\infty} \Gamma(s) x^{-s} ds.$$

Exercise 9.3.7. *With f as in Exercise 9.3.2, show that for $\sigma > 0$ we have*

$$f\left(\frac{iy}{\sqrt{N}}\right) - a_0 = \frac{1}{2\pi i} \int_{\sigma - i\infty}^{\sigma + i\infty} \Lambda_f(s) y^{-s} ds.$$

Using Exercise 9.3.7 and moving the line of integration to $\mathrm{Re}(s) = -U$ (where U is positive and not an integer), we see by Cauchy's Theorem that

$$f\left(\frac{iy}{\sqrt{N}}\right) - a_0 = \frac{1}{2\pi i} \int_{-U - i\infty}^{-U + i\infty} \Lambda_f(s) y^{-s} ds - a_0 - i^k b_o y^{-k}.$$

By the functional equation

$$\Lambda_f(s) = i^k \Lambda_g(k - s),$$

we have

$$f\left(\frac{iy}{\sqrt{N}}\right) = \frac{1}{2\pi i} \int_{-U - i\infty}^{-U + i\infty} i^k \Lambda_g(k - s) y^{-s} ds - i^k b_0 y^{-k}.$$

We replace $k - s$ by s in the integral and obtain

$$f\left(\frac{iy}{\sqrt{N}}\right) = \frac{1}{2\pi i} \int_{k - U - i\infty}^{k + U + i\infty} i^k \Lambda_g(s) y^{s-k} ds - i^k b_0 y^{-k}.$$

By Exercise 9.3.7, the integral is easily seen to be

$$i^k \left(g\left(\frac{iy}{\sqrt{N}}\right) - b_0 \right) y^{-k}.$$

We deduce

$$f\left(\frac{iy}{\sqrt{N}}\right) = i^k y^{-k} g\left(\frac{iy}{\sqrt{N}}\right).$$

Changing y to $1/\sqrt{N}t$, we get

$$f\left(\frac{i}{Nt}\right) = i^k N^{k/2} t^k g(it).$$

In other words, for $z = it$,

$$f\left(-\frac{1}{Nz}\right) = N^{k/2} z^k g(z),$$

which now holds for all $z \in \mathbb{H}$ by the principle of analytic continuation. Thus $f|w_N = g$. This proves the converse of Theorem 9.3.1 and Exercise 9.3.2.

Our discussion can be summarized by the following theorem, due essentially to E. Hecke.

Theorem 9.3.8 (Hecke). *Suppose f and g are given by the Fourier series*

$$f(z) = \sum_{n=0}^{\infty} a_n e^{2\pi i n z}$$
$$g(z) = \sum_{n=0}^{\infty} b_n e^{2\pi i n z}$$

with a_n, b_n bounded by $O(n^\alpha)$ for some $\alpha \geq 0$. Put $L_f(s) = \sum_{n=1}^{\infty} \frac{a_n}{n^s}$, $L_g(s) = \sum_{n=1}^{\infty} \frac{b_n}{n^s}$ and

$$\Lambda_f(s) = \left(\frac{\sqrt{N}}{2\pi}\right)^s \Gamma(s) L_f(s)$$
$$\Lambda_g(s) = \left(\frac{\sqrt{N}}{2\pi}\right)^s \Gamma(s) L_g(s)$$

with N a positive number (not necessarily an integer). Set

$$w = \begin{pmatrix} 0 & -1 \\ N & 0 \end{pmatrix}$$

and

$$f|w = \left(\sqrt{N}z\right)^{-k} f\left(-\frac{1}{Nz}\right)$$

where k is a fixed positive integer. Then the following are equivalent:

(a) $g = f|w$

(b) $\Lambda_f(s)$ and $\Lambda_g(s)$ extend to the entire complex plane and both

$$\Lambda_f(s) + \frac{a_0}{s} + \frac{b_0 i^k}{k-s}$$

and

$$\Lambda_g(s) + \frac{b_0}{s} + \frac{a_0 i^{-k}}{k-s}$$

are entire and bounded in every vertical strip and satisfy the functional equation

$$\Lambda_f(s) = i^k \Lambda_g(k-s).$$

9.4 Twists of L-series attached to modular forms

Given L_f as in the previous section, and χ a primitive Dirichlet character modulo q, then it is natural to consider

$$L_f(s,\chi) = \sum_{n=1}^{\infty} \frac{a_n \chi(n)}{n^s}.$$

Such a Dirichlet series is called a *twist* of $L_f(s)$.

Exercise 9.4.1. *Verify the matrix identity*

$$\begin{pmatrix} 1 & u/r \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & -d^2u/r \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} a+uc/r & b-bcdu/r-cd^2u^2/r^2 \\ c & d-cd^2u/r \end{pmatrix}$$

where $ad-bc=1$.

We use this to prove:

Theorem 9.4.2. *Let $f \in M_k(\Gamma_0(q),\chi)$ where χ is a Dirichlet character of conductor $q^\star$ with $q^\star | q$. Let ψ be a primitive Dirichlet character modulo r. If*

$$f(z) = \sum_{n=0}^{\infty} a_n e^{2\pi i n z},$$

then the "twisted series"

$$f_\psi(z) = \sum_{n=0}^{\infty} a_n \psi(n) e^{2\pi i n z}$$

is an element of $M_k(\Gamma_0(N),\chi\psi^2)$, where N is the least common multiple of q, $q^\star r$, and r^2. Moreover, if f is a cusp form, then so is f_ψ.

Proof. Let $\tau(\psi)$ be the Gauss sum

$$\tau(\psi) = \sum_{u \,(\mathrm{mod}\, r)} \psi(u) e\left(\frac{u}{r}\right).$$

Since ψ is a primitive Dirichlet character modulo r, we have $|\tau(\psi)|^2 = r$ by Exercise 9.2.23. Also,

$$\tau(\overline{\psi})\psi(n) = \sum_{u \,(\mathrm{mod}\, r)} \overline{\psi}(u) e\left(\frac{un}{r}\right)$$

by Theorem 9.2.22. Thus we can write f_ψ in terms of f as

$$\tau(\overline{\psi}) f_\psi = \sum_{u \,(\mathrm{mod}\, r)} \overline{\psi}(u) f \left| \begin{pmatrix} 1 & u/r \\ 0 & 1 \end{pmatrix} \right. . \tag{9.5}$$

By Exercise 9.4.1,

$$\alpha := \begin{pmatrix} 1 & u/r \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & -d^2u/r \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} a + uc/r & b - bcdu/r - cd^2u^2/r^2 \\ c & d - cd^2u/r \end{pmatrix}$$

so that if $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma_0(N)$, then $\alpha \in \Gamma_0(N)$ also, because N is the least common multiple of q, $q^\star r$, and r^2 and $N|c$ (hence $r^2|c$). Also

$$\alpha \equiv \begin{pmatrix} a & \star \\ 0 & d \end{pmatrix} \pmod{q^\star}.$$

Thus,

$$\tau(\overline{\psi}) f_\psi \mid \gamma = \sum_{u \,(\mathrm{mod}\, r)} \overline{\psi}(u)\chi(\alpha) f \left| \begin{pmatrix} 1 & d^2u/r \\ 0 & 1 \end{pmatrix} \right.$$

since

$$\begin{pmatrix} 1 & u/r \\ 0 & 1 \end{pmatrix}^{-1} \alpha \begin{pmatrix} 1 & -d^2u/r \\ 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \gamma$$

and $f|\alpha = f$. As $(d, r) = 1$, we see that $d^2u \pmod r$ forms a complete set of residue classes modulo r as u ranges over all residue classes modulo r. Thus,

$$\begin{aligned} \tau(\overline{\psi}) f_\psi|\gamma &= \chi(d)\psi(d)^2 \sum_{u \,(\mathrm{mod}\, r)} \overline{\psi}(u) f \left| \begin{pmatrix} 1 & u/r \\ 0 & 1 \end{pmatrix} \right. \\ &= \chi(d)\psi^2(d)\tau(\overline{\psi}) f_\psi \end{aligned}$$

by (9.5). Since $\tau(\overline{\psi}) \neq 0$, we deduce $f_\psi \in M_k(\Gamma_0(N), \chi\psi^2)$ as claimed. Moreover, if f is a cusp form, then from (9.5), we see that f_ψ is also a cusp form. □

Exercise 9.4.3. *Let $(q, r) = 1$ and $(u, r) = 1$. Let d and w be integers such that $dr - quw = 1$. Verify the matrix identity with $N = qr^2$,*

$$\begin{pmatrix} 1 & u/r \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ N & 0 \end{pmatrix} = r \begin{pmatrix} 0 & -1 \\ q & 0 \end{pmatrix} \begin{pmatrix} r & -w \\ -qu & d \end{pmatrix} \begin{pmatrix} 1 & w/r \\ 0 & 1 \end{pmatrix}.$$

Exercise 9.4.4. *Let*

$$w_N = \begin{pmatrix} 0 & -1 \\ N & 0 \end{pmatrix}, \qquad w_q = \begin{pmatrix} 0 & -1 \\ q & 0 \end{pmatrix},$$

and $f \in S_k(\Gamma_0(q), \chi)$ where χ is a Dirichlet character modulo q. Let ψ be a primitive Dirichlet character modulo r with $(q, r) = 1$. With f_ψ as in Theorem 9.4.2, $N = qr^2$, $g = f|w_q$, show that

$$f_\psi|w_N = w(\psi) g_{\overline{\psi}}$$

where

$$w(\psi) = \chi(r)\psi(q)\tau(\psi)^2/r.$$

Given $f \in S_k(\Gamma_0(q), \chi)$ and ψ a primitive Dirichlet character modulo r, we associate the "twisted" L-series:

$$L_f(s, \psi) = \sum_{n=1}^{\infty} \frac{a_n \psi(n)}{n^s}$$

where the a_n's are the Fourier coefficients of f at $i\infty$. We put

$$\Lambda_f(s, \psi) = \left(\frac{\sqrt{N}}{2\pi}\right)^s \Gamma(s) L_f(s, \psi).$$

By Theorem 9.3.8 and Exercise 9.4.4, we deduce:

Theorem 9.4.5. *Let $f \in S_k(\Gamma_0(q), \chi)$, ψ a primitive Dirichlet character modulo r with $(q, r) = 1$ and $N = qr^2$. Then $\Lambda_f(s, \psi)$ is an entire function, bounded in vertical strips and satisfying the functional equation*

$$\Lambda_f(s, \psi) = i^k w(\psi) \Lambda_g(k - s, \overline{\psi})$$

where

$$w(\psi) = \chi(r)\psi(q)\tau(\psi)^2/r$$

and $g = f|w_q$, with $w_q = \left(\begin{smallmatrix} 0 & -1 \\ q & 0 \end{smallmatrix}\right)$.

There is a family of results that aim at deriving a sort of converse to Theorem 9.4.5. This program of research goes under the name of "converse theory" in the theory of modular forms (and more generally the theory of automorphic forms). Theorem 9.3.8 shows that for two functions f, g with Fourier expansions

$$f(z) = \sum_{n=0}^{\infty} a_n e^{2\pi i n z}$$
$$g(z) = \sum_{n=0}^{\infty} b_n e^{2\pi i n z},$$

with $a_n, b_n = O(n^\alpha)$ for some $\alpha \geq 0$ and L-series

$$L_f(s) = \sum_{n=1}^{\infty} \frac{a_n}{n^s}$$
$$L_g(s) = \sum_{n=1}^{\infty} \frac{b_n}{n^s}$$

related by the functional equation

$$\left(\frac{\sqrt{N}}{2\pi}\right)^s \Gamma(s)L_f(s) = i^k \left(\frac{\sqrt{N}}{2\pi}\right)^{k-s} \Gamma(k-s)L_g(k-s),$$

we have $g = f\left|\left(\begin{smallmatrix} 0 & -1 \\ N & 0 \end{smallmatrix}\right)\right.$. In the special case $N = 1$, the functional equation implies that $g(z) = f\left|\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)\right.$. In particular if $f = g$, we see that the functional equation implies $f\left|\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)\right. = f$. Since $f\left|\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)\right. = f$, we deduce that $f|\gamma = f$ for all γ in the group generated by $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ which is the full modular group $\mathrm{SL}_2(\mathbb{Z})$ by Theorem 2.1.2. If $N > 1$, one functional equation is not sufficient. However, in a celebrated paper A. Weil [46] showed the following holds, which we state without proof. An elegant and simple proof of this theorem in the English language has been found by Razar [32] and we invite the student to study it.

Theorem 9.4.6 (Weil). *Let k be a positive even integer and χ a character modulo q. Suppose f and g are given by the Fourier series*

$$f(z) = \sum_{n=0}^{\infty} a_n e^{2\pi i n z}$$
$$g(z) = \sum_{n=0}^{\infty} b_n e^{2\pi i n z},$$

with a_n, $b_n = O(n^\alpha)$ for a fixed $\alpha \geq 0$. Let

$$\Lambda_f(s) = \left(\frac{\sqrt{q}}{2\pi}\right)^s \Gamma(s)L_f(s)$$
$$\Lambda_g(s) = \left(\frac{\sqrt{q}}{2\pi}\right)^s \Gamma(s)L_g(s).$$

Suppose that $\Lambda_f(s)$ and $\Lambda_g(s)$ have meromorphic continuations for all $s \in \mathbb{C}$ and

$$\Lambda_f(s) + \frac{a_0}{s} + \frac{b_0 i^k}{k-s},$$
$$\Lambda_g(s) + \frac{b_0}{s} + \frac{a_0 i^k}{k-s}$$

are entire and bounded in vertical strips and satisfy $\Lambda_f(s) = i^k \Lambda_g(k-s)$. Further, let R be a set of prime numbers coprime with q which meets every coprime residue class modulo c for any $c \geq 1$. For each $r \in R$ and every primitive character ψ modulo r, suppose that $\Lambda_f(s,\psi)$ and $\Lambda_g(s,\psi)$ are entire and bounded in vertical strips and satisfy the functional equation given as in Theorem 9.4.5 with $w(\psi)$ as specified there. Then $f \in M_k(\Gamma_0(q),\chi)$, $g \in M_k(\Gamma_0(q),\overline{\chi})$ and $g = f|w_q$. Moreover, if $L_f(s)$ and $L_g(s)$ converge absolutely on some line $\mathrm{Re}(s) = \sigma$ with $0 < \sigma < k$, then f and g are cusp forms.

9.5 Supplementary problems

Exercise 9.5.1. *If $\tau(\chi)$ denotes the Gauss sum, prove that*

$$\overline{\tau(\chi)} = \chi(-1)\tau(\overline{\chi}).$$

Exercise 9.5.2. *For $\mathrm{Re}(c) > 0$, let $F(x) = e^{-c|x|}$. Show that*

$$\hat{F}(u) = \frac{2c}{c^2 + 4\pi^2 u^2}.$$

Exercise 9.5.3. *Show that for $\mathrm{Re}(c) > 0$,*

$$\frac{e^c + 1}{e^c - 1} = \sum_{n=-\infty}^{\infty} \frac{2c}{c^2 + 4\pi^2 n^2}.$$

Exercise 9.5.4. *From the previous exercise, deduce Euler's formula that*

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Exercise 9.5.5. *By differentiating the identity in Exercise 9.5.3, deduce that*

$$\sum_{n=1}^{\infty} \frac{1}{n^{2k}} \in \pi^{2k}\mathbb{Q}.$$

Exercise 9.5.6. *For $\mathrm{Re}(c) > 0$, and $0 \leq v < 1$, show that*

$$\sum_{n\in\mathbb{Z}} \frac{e^{2\pi i n v}}{c^2 + 4\pi^2 n^2} = \frac{1}{2c}\left(e^{-cv} + \frac{2\cosh cv}{e^c - 1}\right).$$

Exercise 9.5.7. *Prove that for $0 \leq v < 1$,*

$$\sum_{n\neq 0} \frac{e^{2\pi i n v}}{n^2} = 2\left(v^2 - v + \frac{1}{6}\right)\pi^2.$$

Chapter 10
Special Topics

10.1 Elliptic functions

Let ω_1 and ω_2 be two non-zero complex numbers which are linearly independent over $\mathbb{R}$. They generate a *lattice* L which is the additive group consisting of elements of the form

$$m\omega_1 + n\omega_2, \qquad m, n \in \mathbb{Z}.$$

We denote by $\widehat{\mathbb{C}}$ the extended complex plane $\mathbb{C} \cup \{\infty\}$. A meromorphic function $f : \mathbb{C} \to \widehat{\mathbb{C}}$ is said to be an *elliptic function* with respect to L if

$$f(z + \omega) = f(z) \qquad \forall z \in \mathbb{C},\ \forall \omega \in L.$$

Sometimes, such functions are also called *doubly periodic* in the literature. We can identify the space of such functions with meromorphic functions on the complex torus $\mathbb{C}/L$. The values of any such function are completely determined by its values on the *fundamental parallelogram* which we take to be the (closed) region in the complex plane enclosed by the parallelogram $\overline{P}$ with vertices 0, ω_1, ω_2, and $\omega_1 + \omega_2$:

$$\overline{P} = \{r\omega_1 + s\omega_2 : 0 \leq r, s \leq 1\}.$$

It will be convenient to let P denote the interior of $\overline{P}$ along with the two sides incident to the origin, including the origin itself:

$$P = \{r\omega_1 + s\omega_2 : 0 \leq r, s < 1\}.$$

Sometimes we take any translate of P as a fundamental parallelogram (see Exercise 10.1.4 below). It is easily verified that the set of all elliptic functions for a lattice L, denoted $\mathcal{E}(L)$, forms a field.

Exercise 10.1.1. *If f is a holomorphic elliptic function, then f is constant.*

© Springer Science+Business Media Singapore 2016 and Hindustan Book Agency 2016

M.R. Murty et al., *Problems in the Theory of Modular Forms*, HBA Lecture Notes in Mathematics, DOI 10.1007/978-981-10-2651-5_10

Exercise 10.1.2. *If two elliptic functions (for a fixed lattice L) have the same poles with the same principal parts, show that they differ by a constant.*

Exercise 10.1.3. *If f is an elliptic function with no poles on the boundary of the fundamental parallelogram P, show that the sum of the residues of f in P is zero.*

Exercise 10.1.4. *Let f be an elliptic function with respect to a lattice L. Show that there is an $\alpha \in \mathbb{C}$ such that f is holomorphic on the boundary of the translate $\alpha + P$, where P is the fundamental parallelogram of L.*

Exercise 10.1.5. *Show that there is no non-constant elliptic function having exactly one simple pole in the fundamental parallelogram P.*

Exercise 10.1.6. *Let f be an elliptic function. Show that the number of zeros of f in the fundamental parallelogram P is equal to the number of poles of f (counted with multiplicity).*

The *order* of an elliptic function f is defined as the number of poles of f (counted with multiplicity) in the fundamental parallelogram. By the previous exercise, the order of f is also equal to the number of zeros of f (counted with multiplicity) in P.

Exercise 10.1.7. *If f is an elliptic function of order m, then for any $c \in \mathbb{C}$, show that the number of solutions of $f(z) = c$ with $z \in P$ is exactly m (counted with multiplicities).*

The simplest non-constant elliptic function with respect to the lattice L is the celebrated *Weierstrass $\wp$ function* defined by

$$\wp(z) := \frac{1}{z^2} + \sum_{0 \neq \omega \in L} \left[\frac{1}{(z-\omega)^2} - \frac{1}{\omega^2} \right].$$

One might naively hope for a simpler elliptic function like $\sum_{\omega \in L} 1/(z-\omega)$, but this does not converge. Moreover, Exercise 10.1.5 implies that 2 is the smallest possible order for a non-constant elliptic function. We now address the question of convergence of the series defining $\wp(z)$.

Exercise 10.1.8. *For any $s > 2$, show that the series*

$$\sum_{\substack{\omega \in L \\ \omega \neq 0}} \frac{1}{\omega^s}$$

is absolutely convergent.

Exercise 10.1.9. *Show that the sum*

$$\sum_{0 \neq \omega \in L} \left[\frac{1}{(z-\omega)^2} - \frac{1}{\omega^2} \right]$$

is absolutely convergent for any $z \in \mathbb{C} \setminus L$.

By the solution to Exercise 10.1.9, the infinite sum defining $\wp(z)$ converges uniformly on compact subsets of $\mathbb{C} \setminus L$. Thus the sum defines a meromorphic function on $\mathbb{C}$. Now

$$\wp'(z) = -2 \sum_{\omega \in L} \frac{1}{(z-\omega)^3}$$

and it is clear that $\wp'(z)$ is a doubly periodic function, with periods ω_1 and ω_2. Hence for any $\omega \in L$,

$$\wp'(z+\omega) = \wp'(z) \qquad \forall z \in \mathbb{C} \setminus L.$$

Hence

$$\wp(z+\omega) - \wp(z)$$

is a constant function of z. But for $z = -\omega/2$, this term is $\wp(\omega/2) - \wp(-\omega/2)$ and since $\wp(z)$ is an even function (an easy consequence of Exercise 10.1.9), this term is zero. Hence we have proved:

Theorem 10.1.10. *The Weierstrass $\wp$-function is an elliptic function of the lattice L. Moreover, it is an even function of z. The derivative $\wp'(z)$ is an odd elliptic function. Moreover, $\wp(z)$ is holomorphic for all $z \in P$ with $z \neq 0$. Its only poles are at elements of L and these are all double order poles.*

We now show that the field of elliptic functions on L is generated by $\wp(z)$ and $\wp'(z)$.

Theorem 10.1.11. *Any even elliptic function on L is a rational function of $\wp(z)$. Any elliptic function f on L can be written as*

$$f(z) = g\Big(\wp(z)\Big) + \wp'(z) h\Big(\wp(z)\Big)$$

where g and h are rational functions.

Proof. It suffices to prove the first assertion because any function $f(z)$ can be written as $f(z) = f_e(z) + f_o(z)$ with

$$f_e(z) = \frac{f(z) + f(-z)}{2},$$

which is even, and

$$f_o(z) = \frac{f(z) - f(-z)}{2},$$

which is odd. Now

$$f_o(z) = \wp'(z) \left(\frac{f_o(z)}{\wp'(z)} \right)$$

and $f_0/\wp'$ is an even elliptic function.

Now suppose f is even. Then $\operatorname{ord}_{z_0}(f) = \operatorname{ord}_{-z_0}(f)$ for any $z_0 \in \mathbb{C}$. We now show that if $2z_0 \in L$, then $\operatorname{ord}_{z_0}(f)$ is even. To see this we differentiate the equation $f(z) = f(-z)$ to get

$$f^{(i)}(z) = (-1)^i f^{(i)}(-z),$$

so that if $2z_0 \in L$, then $f^{(i)}(z_0) = f^{(i)}(z_0 - 2z_0) = f^{(i)}(-z_0)$ and this implies $f^{(i)}(z_0) = 0$ for all odd i. In other words, if $2z_0 \in L$, then $\mathrm{ord}_{z_0}(f)$ is even.

Now let $P_e = P/\pm 1$ be "half" of the fundamental parallelogram. Consider the function

$$g(z) = \prod_{\substack{v \in P_e \\ v \neq 0}} \Big(\wp(z) - \wp(v)\Big)^{n_v}$$

where

$$n_v = \begin{cases} \mathrm{ord}_v(f) & \text{if } 2v \notin L \\ \mathrm{ord}_v(f)/2 & \text{if } 2v \in L. \end{cases}$$

For $2v \notin L$, the term $\wp(z) - \wp(v)$ has zeros at the distinct points $z = \pm v$, and a pole of order 2 at $z = 0$. Hence the zeros are simple and so

$$\mathrm{ord}_v \Big(\wp(z) - \wp(v)\Big)^{n_v} = \mathrm{ord}_v(f).$$

If $v \in 2L$, then $\wp(z) - \wp(v)$ has a double zero at the point $z = v = -v$. In this way, we see that $f(z)$ and $g(z)$ have exactly the same zeros and poles (with the same orders) except possibly at $z = 0$. But by Exercise 10.1.6, they have the same order at $z = 0$ also. Therefore $f(z)/g(z)$ is a holomorphic elliptic function which, by Exercise 10.1.1, is a constant. □

Thus, the field $\mathcal{E}(L)$ of elliptic functions associated to a lattice L is generated by $\wp$ and $\wp'$.

Exercise 10.1.12. *Let*

$$G_k = \sum_{\substack{\omega \in L \\ \omega \neq 0}} \omega^{-k}$$

Prove that

$$\wp(z) = z^{-2} + \sum_{\substack{k=2 \\ k \text{ even}}}^{\infty} (k+1) G_{k+2} z^k$$

is the Laurent expansion of $\wp(z)$ at $z = 0$.

Observe that if k is odd, then $G_k = 0$ because the ω and $-\omega$ terms cancel.

Exercise 10.1.13. *With notation as in the previous exercise, show that for $z \in \mathbb{C}\backslash L$,*

$$\wp'(z)^2 = 4\wp(z)^3 - 60G_4\wp(z) - 140G_6.$$

The previous exercise shows that the points $\Big(\wp(z), \wp'(z)\Big)$ for $z \notin L$ lie on the curve

$$y^2 = 4x^3 - g_2 x - g_3 \tag{10.1}$$

where

$$g_2 = 60G_4, \qquad g_3 = 140G_6$$

Curves of the form (10.1) are called *elliptic curves*. The cubic polynomial on the right hand side has discriminant given by

$$\Delta = g_2^3 - 27g_3^2$$

Exercise 10.1.14. *Let L be a lattice and g_2, g_3 defined as above. Show that all of the complex solutions of the equation*

$$y^2 = 4x^3 - g_2x - g_3$$

are given by $\Big(\wp(z), \wp'(z)\Big)$ where $\wp(z)$ is the Weierstrass $\wp$-function attached to L and z ranges over all complex numbers in $\mathbb{C} \setminus L$.

The Weierstrass $\wp$-function satisfies a remarkable addition formula which we record in the following theorem.

Theorem 10.1.15. *Suppose z, w, $z + w \in \mathbb{C} \setminus L$. If $z \neq w$, we have*

$$\wp(z+w) = -\wp(z) - \wp(w) + \frac{1}{4}\left(\frac{\wp'(z) - \wp'(w)}{\wp(z) - \wp(w)}\right)^2.$$

When $z = w$, we have

$$\wp(2z) = -2\wp(z) + \frac{1}{4}\left(\frac{\wp''(z)}{\wp'(z)}\right)^2.$$

Proof. Let $(x_1, y_1) = (\wp(z), \wp'(z))$ and $(x_2, y_2) = (\wp(w), \wp'(w))$ be the corresponding points on the elliptic curve $y^2 = 4x^3 - g_2x - g_3$. Let $y = ax + b$ be the line through these two points. Then

$$\wp'(z) = a\wp(z) + b$$
$$\wp'(w) = a\wp(w) + b.$$

Now the elliptic function

$$\wp'(u) - a\wp(u) - b$$

has poles in the fundamental parallelogram P only at $u = 0$ and this pole has order 3. By Exercise 10.1.6, it has three zeroes in P (counting multiplicities). Working modulo L, we already have two of these zeros at $u = z$ and $u = w$. Let $u = t$ be the third zero. Then

$$\wp'(t) = a\wp(t) + b.$$

Now, for any elliptic function f it is a fact that

$$\sum_{w\in P} w \operatorname{ord}_w f \in L$$

(see Exercise 10.5.1). Thus

$$z + w + t \in L.$$

Also,

$$\wp'(t)^2 = 4\wp(t)^3 - g_2\wp(t) - g_3.$$

Thus, the cubic equation

$$(ax+b)^2 = 4x^3 - g_2x - g_3$$

has roots $x = \wp(z)$, $\wp(w)$, and $\wp(t)$. Since the sum of the roots of the cubic is $a^2/4$, we get

$$\wp(z) + \wp(w) + \wp(t) = \frac{a^2}{4}.$$

Since a is the slope of the line joining (x_1, y_1) and (x_2, y_2), we get

$$\wp(z) + \wp(w) + \wp(t) = \frac{1}{4}\left(\frac{\wp'(z) - \wp'(w)}{\wp(z) - \wp(w)}\right)^2.$$

Since $t = -z - w \pmod{L}$ and $\wp(-t) = \wp(t)$, we deduce the first assertion of the theorem. The second assertion is obtained by taking the limit as $z \to w$. □

Exercise 10.1.16. *Let L be the lattice spanned by ω_1, ω_2 and let $\wp(z)$ be the associated Weierstrass function. Show that $\wp\left(\frac{\omega_1}{2}\right)$, $\wp\left(\frac{\omega_2}{2}\right)$, and $\wp\left(\frac{\omega_1+\omega_2}{2}\right)$ are the roots of the cubic $4x^3 - g_2x - g_3$ with g_2 and g_3 defined by (10.1).*

Exercise 10.1.17. *Suppose that ω_1, ω_2 are linearly independent over $\mathbb{R}$ and that L is the lattice spanned by ω_1, ω_2. Let $\wp(z)$ be the associated $\wp$-function. Show that $\wp\left(\frac{\omega_1}{2}\right)$, $\wp\left(\frac{\omega_2}{2}\right)$, and $\wp\left(\frac{\omega_1+\omega_2}{2}\right)$ are all distinct.*

10.2 The modular invariant

For a complex number $z \in \mathbb{H}$, let L_z be the lattice spanned by 1 and z. We denote the corresponding g_2, g_3 associated to L_z by $g_2(z)$ and $g_3(z)$ respectively. Thus

$$g_2(z) = 60 \sum_{(m,n)\neq(0,0)} (mz+n)^{-4}$$
$$g_3(z) = 140 \sum_{(m,n)\neq(0,0)} (mz+n)^{-6}.$$

Consistent with our notation introduced in Chapter 4, these equations are $g_2(z) = 60G_4(z)$ and $g_3(z) = 140G_6(z)$. These are modular forms of weight 4 and 6, respectively, for the full modular group $\mathrm{SL}_2(\mathbb{Z})$. We define the *discriminant function*

$$\Delta_0(z) = g_2(z)^3 - 27g_3(z)^2.$$

Theorem 10.2.1. *For $z \in \mathbb{H}$, we have $\Delta_0(z) \neq 0$.*

Proof. This statement is equivalent to showing that the roots of the associated cubic $4x^3 - g_2(z)x - g_3(z)$ are all distinct. But this follows from Exercise 10.1.17. □

It is clear that $\Delta_0(z)$ is a modular form of weight 12 for the full modular group. It is related to the $\Delta(z)$ introduced in Chapter 4. Indeed, it is easily verified that

$$\Delta_0(z) = (2\pi)^{12}\,\Delta(z)$$

using our explicit knowledge of $\zeta(2k)$ given in Exercise 4.2.4. As before, we let

$$j(z) = \frac{E_4^3(z)}{\Delta(z)}$$

and it is immediately checked that in our notation above,

$$j(z) = 1728\frac{g_2(z)^3}{g_2(z)^3 - 27g_3(z)^2} \tag{10.2}$$

which by the previous theorem is well-defined for $z \in \mathbb{H}$. The j-function is also called the *modular invariant* for several reasons. The first is that

$$j\left(\frac{az+b}{cz+d}\right) = j(z) \qquad \forall z \in \mathbb{H}.$$

However, now that we have related the j-function to elliptic curves, we will elaborate on this relationship.

Exercise 10.2.2. *If ω_1, ω_2 are linearly independent over $\mathbb{R}$ and $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$, show that the lattice generated by $a\omega_1 + b\omega_2$ and $c\omega_1 + d\omega_2$ is the same as the lattice generated by ω_1 and ω_2.*

Exercise 10.2.3. *If ω_1, ω_2 generate the lattice L and ω_1', ω_2' also generate the same lattice L, show that there is a matrix $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$ such that $\omega_1' = a\omega_1 + b\omega_2$ and $\omega_2' = c\omega_1 + d\omega_2$.*

Exercise 10.2.4. *Let L be the lattice spanned by ω_1, ω_2 with associated functions g_2, g_3. If λ is a non-zero scalar, and L' is the lattice spanned by $\lambda\omega_1$, $\lambda\omega_2$, show that the corresponding elliptic curve for L' is*

$$y^2 = 4x^3 - \lambda^{-4}g_2x - \lambda^{-6}g_3.$$

Given any elliptic curve E

$$y^2 = 4x^3 - g_2 x - g_3,$$

we define

$$j(E) = \frac{1728 g_2^3}{g_2^3 - 27 g_3^2}.$$

One can show that $j(E)$ is isomorphism invariant. In fact $E_1 \cong E_2$ over $\mathbb{C}$ if and only if $j(E_1) = j(E_2)$.

In Theorem 4.5.1 we proved that the map $z \mapsto j(z)$ is a bijection from the standard fundamental domain $\mathcal{F}$ onto the complex numbers $\mathbb{C}$. This fact allows us to prove the following *uniformization theorem*.

Theorem 10.2.5. *Given any two A, $B \in \mathbb{C}$ with $A^3 - 27B^2 \neq 0$, there exists a lattice L such that its associated Weierstrass function satisfies*

$$\wp'(z)^2 = 4\wp(z)^3 - A\wp(z) - B.$$

Proof. We consider two cases. The first case is when $A = 0$. The second case is when $A \neq 0$.

In the first case $(A, B) = (0, B)$ with $B \neq 0$. We choose $z_0 \in \mathbb{H}$ so that $g_2(z_0) = 0$. This can be done since $g_2(z)$ is, up to a scalar multiple, $E_4(z)$ and by Exercise 4.3.5 $z = \rho$ works. Since $\Delta_0(z_0) \neq 0$, we have $g_3(z_0) \neq 0$. Choosing λ so that $\lambda^{-6} g_3(z_0) = B$, the lattice $L = [\lambda, \lambda z_0]$ has the required property, by Exercise 10.2.4.

Now consider the case when $A \neq 0$. Let $a = B^2/A^3$. Since $A^3 - 27B^2 \neq 0$, we have $a \neq 1/27$. By Theorem 4.5.1, we can choose $z_0 \in \mathbb{H}$ so that $j(z_0) = 1728/(1 - 27a)$. If $B = 0$, then we choose λ so that $\lambda^{-4} g_2(z_0) = A$ and we see that $L = [\lambda, \lambda z_0]$ works. If $B \neq 0$, we choose λ so that

$$\lambda^2 = \frac{A g_3(z_0)}{B g_2(z_0)}$$

and consider the lattice $L = [\lambda, \lambda z_0]$. From (10.2) we deduce that $a = g_3(z_0)^2/g_2(z_0)^3$. Thus for the lattice L, the associated values of g_2 and g_3 are A and B respectively. Indeed,

$$g_2 = \frac{g_2(z_0)}{\lambda^4} = \frac{B^2 g_2(z_0)^3}{A^2 g_3(z_0)^2} = A$$

and

$$g_3 = \frac{g_3(z_0)}{\lambda^6} = \frac{B^3 g_2(z_0)^3}{A^3 g_3(z_0)^2} = B$$

as desired. □

The previous theorem shows that *all* elliptic curves over $\mathbb{C}$ have associated with them a lattice L and a Weierstrass $\wp$-function such that $(\wp(z), \wp'(z))$

parametrizes all the complex points of the curve. By Exercise 10.2.2, we see that for any given elliptic curve E, any two lattices that give rise to E are $\mathrm{SL}_2(\mathbb{Z})$-equivalent. Since ω_1, ω_2 are linearly independent over $\mathbb{R}$, we can ensure that our lattice satisfies $\mathrm{Im}\,(\omega_2/\omega_1) > 0$. If $z = \omega_2/\omega_1$ with $\mathrm{Im}\,(\omega_2/\omega_1) > 0$, then the j-invariant of E is given by the modular invariant $j(z)$. Combined with the remark after Exercise 10.2.4 and Theorem 4.5.1, we see that to each point z in the standard fundamental domain $\mathcal{F}$, we can associate an elliptic curve E_z and this is a one-to-one correspondence. Thus, points of $\mathcal{F}$ parametrize *all* elliptic curves. (This is an example of a *moduli space*.)

Using our earlier notation, we see that

$$j(z) = \frac{E_4^3(z)}{\Delta(z)}.$$

Exercise 10.2.6. *Show that*

$$2E_4(z)E_6'(z) - 3E_6(z)E_4'(z)$$

is a cusp form of weight 12 *for the full modular group.*

Exercise 10.2.7. *Prove that* $E_4^2 E_6 = E_{14}$. *Deduce that*

$$\sum_{(m,n)\neq(0,0)} m/(m\rho + n)^{13} = 0,$$

where ρ *is a primitive cube root of unity.*

Exercise 10.2.8. *Show that*

$$j'(z) = -2\pi i \frac{E_{14}(z)}{\Delta(z)}.$$

10.3 Theta series and lattices

In this section, we describe a general method of constructing modular forms arising from theta series associated to lattices.

Let V be a vector space over $\mathbb{R}$ of dimension n equipped with an inner product. That is, we have a map

$$(\cdot,\cdot) : V \times V \to \mathbb{R}$$

which is bilinear, symmetric and positive definite.

Recall the *Kronecker delta* function

$$\delta_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

Let μ be an *invariant measure* on V. That is, for any measurable subset U of V, we have $\mu(v+U) = \mu(U)$ for all $v \in V$ (translation invariance) and $\mu(\lambda U) = \lambda^n \mu(U)$ for any $\lambda \in \mathbb{R}$. Also, if $e_1, \ldots, e_n$ is a basis of V such that $(e_i, e_j) = \delta_{ij}$ then the set

$$\left\{ \sum_{j=1}^{n} \lambda_j e_j : 0 \leq \lambda_j \leq 1, 1 \leq j \leq n \right\}$$

has measure 1.

A *lattice* L of V is a subset of V which can be written as

$$L = \omega_1 \mathbb{Z} + \cdots + \omega_n \mathbb{Z}$$

where $\{\omega_1, \ldots, \omega_n\}$ is a basis of V. The *volume* of L is defined as

$$\mu(V/L) := \mu\left(\left\{ \sum_{j=1}^{n} \lambda_j \omega_j : 0 \leq \lambda_j \leq 1, 1 \leq j \leq n \right\} \right)$$

Recall that the *dual vector space* $V^\star$ of V is the vector space of linear functionals on V. Since V is an inner product space these functionals are of the form $(\cdot, v_0)$ for some $v_0 \in V$ (Riesz representation theorem). We define the *dual lattice* $L^\star$ of L as the collection of elements $v^\star \in V^\star$ such that $v^\star(\omega) \in \mathbb{Z}$ for all $\omega \in L$.

Exercise 10.3.1. *If $L = \omega_1 \mathbb{Z} + \cdots \omega_n \mathbb{Z}$ and if $\omega_1^\star, \ldots, \omega_n^\star \in L^\star$ is the* dual basis, *that is $\omega_i^\star(\omega_j) = \delta_{ij}$, show that*

$$L^\star = \omega_1^\star \mathbb{Z} + \cdots + \omega_n^\star \mathbb{Z}$$

To each lattice L we associate a *theta series*

$$\theta_L(z) = \sum_{\omega \in L} e^{\pi i z(\omega, \omega)}$$

for $z \in \mathbb{H}$. We first address the question of convergence.

Since each $\omega \in L$ is of the form

$$\omega = \sum_{i=1}^{n} x_i \omega_i, \qquad x_i \in \mathbb{Z},$$

we see that

$$(\omega,\omega)=\sum_{i=1}^{n}\sum_{j=1}^{n}(\omega_i,\omega_j)x_ix_j$$

is a positive definite quadratic form. With respect to the basis $\omega_1,\ldots,\omega_n$, this quadratic form is represented by a positive definite real symmetric matrix. As such, it can be diagonalized by an orthogonal transformation (by a fundamental theorem in linear algebra). Moreover, the eigenvalues are real and positive (since the matrix is positive definite). In other words, there is $\lambda_{min}>0$ such that

$$(\omega,\omega)\geq\lambda_{min}\left(x_1^2+\cdots+x_n^2\right).$$

Thus if $z=u+iv$ with $v>0$, we have

$$\begin{aligned}|\theta_L(z)|&\leq\sum_{x_1,\ldots,x_n\in\mathbb{Z}}e^{-\pi v\lambda_{min}(x_1^2+\cdots+x_n^2)}\\&\leq\left(\sum_{x\in\mathbb{Z}}e^{-\pi\lambda_{min}vx^2}\right)^n<\infty.\end{aligned}$$

This proves:

Theorem 10.3.2. *The theta series $\theta_L(z)$ is a holomorphic function for $z\in\mathbb{H}$.*

Proof. From the proof immediately above, we easily see that the series representing $\theta_L(z)$ converges uniformly and absolutely on every compact subset of $\mathbb{H}$ (since for any compact set in $\mathbb{H}$, v is bounded from below.) Thus $\theta_L(z)$ is holomorphic there. □

The matrix $A=\Big((\omega_i,\omega_j)\Big)$ is called the *Gram matrix* associated to the lattice L.

Exercise 10.3.3. *Prove that $\mu(V/L)=\sqrt{\det A}$.*

Exercise 10.3.4. *With A as above, show that the matrix A^{-1} is the Gram matrix of the dual lattice $L^\star$.*

To derive the modular transformation law for $\theta_L(z)$, we need the following version of Poisson's summation formula for lattices.

Let V be as above a vector space over $\mathbb{R}$ of dimension n. We view V as being isomorphic to $\mathbb{R}^n$ with the usual Euclidean topology. A bounded function $f:V\to\mathbb{R}$ is called *rapidly decreasing* if it can be partially differentiated arbitrarily often and each of these partial derivatives are bounded.

The *Fourier transform* of f is a map $\hat{f}:V^\star\to\mathbb{R}$ defined by

$$\hat{f}(x)=\int_V f(u)e^{-2\pi i(u,x)}d\mu(x)$$

where (u, x) denotes $x(u)$. Once can show that $\hat{f}$ is again a rapidly decreasing function on $V^\star$.

Theorem 10.3.5. *For f rapidly decreasing,*

$$\sum_{\omega \in L} f(\omega) = \frac{1}{\mu(V/L)} \sum_{\omega^\star \in L^\star} \hat{f}(\omega^\star).$$

Proof. We use the basis of L to establish a topological isomorphism with $\mathbb{R}^n$ so that L is then mapped to $\mathbb{Z}^n$. This is a lattice of volume 1. Identifying μ with the product measure $dx_1 \cdots dx_n$, we reduce the identity to be proved to

$$\sum_{x \in \mathbb{Z}^n} f(x) = \sum_{x \in \mathbb{Z}^n} \hat{f}(x)$$

which is the classical Poisson summation formula essentially proved earlier in Chapter 9. □

This allows us to prove the following fundamental theorem.

Theorem 10.3.6. *Let L be a lattice as above. Then*

$$\theta_{L^\star}\left(-\frac{1}{z}\right) = \left(\frac{z}{i}\right)^{n/2} \mu\left(V/L\right) \theta_L(z)$$

for all $z \in \mathbb{H}$. (Here, the square root is chosen so that it is positive for any positive real number.)

Proof. By complex analysis, it suffices to prove the identity for purely imaginary z. In other words, it suffices to show that

$$\sum_{\omega^\star \in L^\star} e^{-\pi(\omega^\star, \omega^\star)/y} = y^{n/2} \mu\left(V/L\right) \sum_{\omega \in L} e^{-\pi y(\omega, \omega)}.$$

Now let $f : V \to \mathbb{R}$ be given by $f(x) = e^{-\pi(x,x)}$. Then f is rapidly decreasing and $\hat{f} = f$ once we identify V with $V^\star$ using the pairing provided by the inner product. Since the Fourier transform of $e^{-\pi|x|^2}$ is itself, we now proceed as follows. Let $\Omega = y^{1/2} L$. Then $\Omega^\star = y^{-1/2} L^\star$ so that $\mu\left(V/\Omega\right) = y^{n/2} \mu(V/L)$. The Poisson summation formula applied to Ω and f now yields

$$y^{n/2} \mu\left(V/L\right) \sum_{x \in y^{1/2} L} e^{-\pi(x,x)} = \sum_{x \in y^{-1/2} L^\star} e^{-\pi(x,x)}$$

and a simple change of variables in the summation now gives the desired result. □

A lattice L is said to be *self-dual* if $L^\star = L$. It is said to be *even* if $||\omega||^2 \in 2\mathbb{Z}$ for all $\omega \in L$.

Exercise 10.3.7. *If L is self-dual and even, show that $8 \mid n$.*

Exercise 10.3.8. *If L is self-dual show that the Gram matrix of L and $L^\star$ have determinant 1.*

Exercise 10.3.9. *Show that the theta function of a self-dual, even lattice is a modular form of weight $n/2$ for the full modular group $\mathrm{SL}_2(\mathbb{Z})$.*

10.4 Special values of zeta and L-functions

The logarithmic derivative of the Γ-function is called the *digamma function*. Higher derivatives of the digamma function give rise to the *polygamma functions*. More precisely, the digamma function is given by

$$\psi(z) = -\gamma - \frac{1}{z} - \sum_{n=1}^{\infty} \left(\frac{1}{n+z} - \frac{1}{n} \right),$$

so that the polygamma functions $\psi_k(z)$ are given by

$$\psi_k(z) = (-1)^{k-1} k! \sum_{n=0}^{\infty} \frac{1}{(n+z)^{k+1}}.$$

Exercise 10.4.1. *Prove that*

$$\psi_k(z+1) = \psi_k(z) + \frac{(-1)^k k!}{z^{k+1}}.$$

Exercise 10.4.2. *For every integer $k \geq 0$, prove that*

$$-\frac{d^k}{dz^k}(\pi \cot \pi z) = \psi_k(z) + (-1)^{k+1} \psi_k(-z) + (-1)^k \frac{k!}{z^{k+1}}.$$

Exercise 10.4.3. *Show that*

$$-\frac{d^k}{dz^k}(\pi \cot \pi z) = \psi_k(z) + (-1)^{k+1} \psi_k(1-z).$$

Let N be a natural number and let $\chi : (\mathbb{Z}/N\mathbb{Z})^* \to \mathbb{C}$ be a Dirichlet character and define the L-series $L(s, \chi)$ by

$$L(s, \chi) := \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s}.$$

Exercise 10.4.4. *Show that*

$$(k-1)!L(k,\chi) = (-N)^{-k}\sum_{a=1}^{N}\chi(a)\psi_{k-1}(a/N).$$

Exercise 10.4.5. *If χ is an even character (that is, $\chi(-1)=1$) and k is even, show that*

$$-2(k-1)!L(k,\chi) = (-N)^{-k}\sum_{a=1}^{N}\chi(a)\frac{d^k}{dz^k}(\pi\cot\pi z)\Big|_{z=a/N}.$$

Deduce that $L(k,\chi)$ is an algebraic multiple of π^k.

Exercise 10.4.6. *If χ is an odd character (that is, $\chi(-1)=-1$) and k is odd, show that*

$$-2(k-1)!L(k,\chi) = (-N)^{-k}\sum_{a=1}^{N}\chi(a)\frac{d^k}{dz^k}(\pi\cot\pi z)\Big|_{z=a/N}.$$

Deduce that $L(k,\chi)$ is an algebraic multiple of π^k.

The gist of the preceding exercises is that the value of $L(k,\chi)$ when k and χ have the same parity, is a non-zero algebraic multiple of π^k. This fact was first proved by Hecke [14] and he noted that this implies an interesting result for real quadratic fields. Namely, if F is a real quadratic field, then $\zeta_F(2m)$ is an algebraic multiple of π^{4m}. This motivated him to ask if such a result holds generally for any totally real field F. That is, if F is totally real of degree r over the rationals, then is it true that $\zeta_F(2m)$ is an algebraic multiple of π^{2rm}? This was anwered in the affirmative by Siegel and Klingen and is now known as the Siegel-Klingen theorem.

The proof of the Siegel-Klingen theorem makes use of the theory of classical modular forms and Hilbert modular forms. Since a detailed explanation of the proof is beyond the scope of this book, we content ourselves with a brief outline which should enable the interested student to make a more detailed study.

We summarise the contents of the appendix in [39]. For the relevant background in algebraic number theory, we refer the student to [25]. For a detailed study of Hilbert modular forms, we suggest the book by Garrett [10].

Let F be a totally real number field of degree r and discriminant D. Let $x \mapsto x^{(i)}$ be an indexing of real embeddings of F and let O_F be the ring of integers of O_F. The group $SL_2(O_F)$ is called the Hilbert modular group and it acts on $\mathbb{H}^r$ via the map:

$$g := \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(O_F), \qquad g\cdot(z_1,..,z_r) = \left(\frac{a^{(1)}z_1+b^{(1)}}{c^{(1)}z_1+d^{(1)}},...,\frac{a^{(r)}z_r+b^{(r)}}{c^{(r)}z_r+d^{(r)}}\right).$$

For $a,b \in F$ and $z \in \mathbb{H}^r$, it is convenient to use the notation

$$N(az+b) := \prod_{i=1}^{r}(a^{(i)}z_i + b^{(i)}).$$

Let k be a natural number (which is even if K has a unit of norm -1). For an ideal $\mathfrak{a}$, we define the analogue of the Eisenstein series as

$$F_k(\mathfrak{a}, z) = N(\mathfrak{a})^k \sum_{(\lambda,\mu)\neq(0,0)} N(\lambda z + \mu)^{-k},$$

where the sum runs over a complete system of pairs of numbers in the ideal $\mathfrak{a}$ different from $(0, 0)$ and not differing from one another by a factor which is a unit (that is, not associated). One can show that the series converges absolutely for $k > 2$ and for $k = 2$, one can apply a limit process (Hecke's trick). The function $F_k(\mathfrak{a}, z)$ is an example of a Hilbert modular form in the sense that

$$F_k(\mathfrak{a}, gz) = N(cz + d)^k F_k(\mathfrak{a}, z) \qquad \forall \quad g \in SL_2(O_F).$$

As O_F is a lattice and its dual is the inverse *different* $\mathfrak{d}^{-1}$ of O_F (see [25]), one has a Fourier expansion of $F_k(\mathfrak{a}, z)$ of the form

$$\zeta(\mathfrak{a}, k) + \left(\frac{(2\pi i)^k}{(k-1)!}\right)^r D^{1/2-k} \sum_{\nu\in\mathfrak{d}^{-1},\nu\gg 0} \sigma_{k-1}(\mathfrak{a}, \nu) e^{2\pi i Tr(z\nu)},$$

where ν runs over totally positive numbers in $\mathfrak{d}^{-1}$ and

$$\zeta(\mathfrak{a}, k) = N(\mathfrak{a})^k \sum_{\mathfrak{a}|(\mu)} N(\mu)^{-k},$$

$$\sigma_{k-1}(\mathfrak{a}, \nu) = \sum_{\mathfrak{d}^{-1}|(\alpha)\mathfrak{a}|\nu} \mathrm{sign}(N(\alpha)^k) N((\alpha)\mathfrak{a}\mathfrak{d})^{k-1}.$$

The summation is over principal ideals $(\mu), (\alpha)$ under the conditions given. If we set all the variables $z_1, ..., z_r$ to z, then $F_k(\mathfrak{a}, z)$ becomes a classical modular form of weight rk. By Exercises 5.5.18, 5.5.19, and 5.5.20, we deduce that $\zeta(\mathfrak{a}, k)$ is a an algebraic multiple of π^{kr}. Since the Dedekind zeta function $\zeta_F(k)$ can be written as a rational linear combination of the values $\zeta(\mathfrak{a}, k)$, the Siegel-Klingen theorem follows from this.

10.5 Supplementary problems

Exercise 10.5.1. *Let f be an elliptic function with period lattice L. Show that*

$$\sum_{z\in\mathbb{C}/L} \mathrm{ord}_z(f) z \in L$$

Exercise 10.5.2. *With $\wp$ denoting the Weierstrass $\wp$-function as in Section 10.1, prove that*

$$\wp''(z) = 6\wp(z)^2 - g_2/2.$$

Exercise 10.5.3. *With G_k as in Exercise 10.1.12, prove that for $m \geq 4$,*

$$(2m+1)(m-3)(2m-1)G_{2m} = 3\sum_{j=1}^{m-2}(2j-1)(2m-2j-1)G_{2j}G_{2m-2j}.$$

Exercise 10.5.4. *Show that for $k \geq 2$, each G_{2k} can be expressed as a polynomial in G_4 and G_6 with positive rational coefficients.*

Exercise 10.5.5. *Let $\vartheta(z)$ be given by the series*

$$\sum_{n=-\infty}^{\infty} e^{\pi i n^2 z}.$$

Show that

$$\vartheta(z) = \prod_{n=1}^{\infty}(1 - e^{2\pi i n z})(1 + e^{(2n-1)\pi i z})^2.$$

Deduce that $\vartheta(z)$ does not vanish in the upper half-plane.

Exercise 10.5.6. *With η being the Dedekind η-function and ϑ as in the previous exercise, show that*

$$\vartheta(z) = \eta^2((z+1)/2)/\eta(z+1).$$

Exercise 10.5.7. *Show that if j is the modular invariant, then $j'(z)$ satisfies*

$$j'\left(\frac{az+b}{cz+d}\right) = (cz+d)^2 j'(z) \qquad \forall \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$$

but that $j'(z)$ is not a modular form of weight 2 for the full modular group.

Part II
Solutions

Chapter 1
Jacobi's q-series

1.1 The q-exponential function

1.1.1. *Show that*

$$\mathcal{E}_q(x) - \mathcal{E}_q\left(\frac{x}{q}\right) = \frac{x}{q}\mathcal{E}_q\left(\frac{x}{q}\right).$$

Observe that

$$\begin{aligned}\mathcal{E}_q(x) - \mathcal{E}_q\left(\frac{x}{q}\right) &= \sum_{n=1}^{\infty} \frac{q^n x^n - x^n}{q^n(q^n-1)(q^{n-1}-1)\cdots(q-1)} \\ &= \sum_{n=1}^{\infty} \frac{(\frac{x}{q})^n}{(q^{n-1}-1)\cdots(q-1)} = \frac{x}{q}\mathcal{E}_q\left(\frac{x}{q}\right). \qquad \square\end{aligned}$$

1.1.2. *Prove that if* $|q| > 1$ *then*

$$\mathcal{E}_q(x) = \prod_{n=1}^{\infty}\left(1 + \frac{x}{q^n}\right).$$

From Exercise 1.1.1, we see that

$$\mathcal{E}_q(x) = \left(1 + \frac{x}{q}\right)\mathcal{E}_q\left(\frac{x}{q}\right).$$

Iterating this relation and using that $\mathcal{E}_q(\frac{x}{q^n}) \to 1$ as $n \to \infty$ gives the result. $\square$

1.1.3. *Show that if* $|q| < 1$ *then*

$$\prod_{n=0}^{\infty}(1 + q^n x) = \sum_{n=0}^{\infty} \frac{q^{\binom{n}{2}} x^n}{(1-q)\cdots(1-q^n)}.$$

© Springer Science+Business Media Singapore 2016 and Hindustan Book Agency 2016

M.R. Murty et al., *Problems in the Theory of Modular Forms*, HBA Lecture Notes in Mathematics, DOI 10.1007/978-981-10-2651-5_11

In Exercise 1.1.2, we replace q by $\frac{1}{q}$ to obtain

$$\mathcal{E}_{\frac{1}{q}}(x) = \prod_{n=1}^{\infty}(1+q^n x).$$

By the definition of $\mathcal{E}_q(x)$, we find

$$\prod_{n=0}^{\infty}(1+q^n x) = (1+x)\left(\sum_{n=0}^{\infty}\frac{q^{\frac{n(n+1)}{2}}x^n}{(1-q)\cdots(1-q^n)}\right),$$

on noting that $q^{1+2+\cdots+n} = q^{\frac{n(n+1)}{2}}$. Thus

$$\begin{aligned}\prod_{n=0}^{\infty}(1+q^n x) &= \sum_{n=0}^{\infty}\frac{q^{\frac{n(n+1)}{2}}x^n(1+x)}{(1-q)\cdots(1-q^n)}\\ &= 1+\sum_{n=1}^{\infty}\frac{x^n}{(1-q)\cdots(1-q^n)}\left(q^{\frac{n(n+1)}{2}}+q^{\frac{n(n-1)}{2}}(1-q^n)\right)\\ &= 1+\sum_{n=1}^{\infty}\frac{q^{\binom{n}{2}}x^n}{(1-q)\cdots(1-q^n)},\end{aligned}$$

as required. □

1.1.4. *Prove that if* $|q|>1$ *then*

$$(1+x)\mathcal{E}_{\frac{1}{q}}(x) = \frac{1}{\mathcal{E}_q(x)}.$$

Deduce that if $|q|<1$ *then*

$$\prod_{n=0}^{\infty}(1+q^n x)^{-1} = \sum_{n=0}^{\infty}\frac{x^n}{(q^n-1)\cdots(q-1)}.$$

Using the same method as in Exercise 1.1.1, we find that

$$\begin{aligned}\mathcal{E}_{\frac{1}{q}}(x) - \mathcal{E}_{\frac{1}{q}}\left(\frac{x}{q}\right) &= \sum_{n=1}^{\infty}\frac{x^n}{\left(\frac{1}{q^n}-1\right)\cdots\left(\frac{1}{q}-1\right)}\left(1-\frac{1}{q^n}\right)\\ &= -x\sum_{n=0}^{\infty}\frac{x^n}{\left(\frac{1}{q^n}-1\right)\cdots\left(\frac{1}{q}-1\right)}\\ &= -x\mathcal{E}_{\frac{1}{q}}(x).\end{aligned}$$

Therefore

$$\mathcal{E}_{\frac{1}{q}}(x) = \frac{1}{(1+x)}\mathcal{E}_{\frac{1}{q}}\left(\frac{x}{q}\right).$$

Iterating this gives the first result since as $n \to \infty$, we have $x/q^n \to 0$, and hence $\mathcal{E}_{1/q}(\frac{x}{q^n}) \to 1$. In particular

$$\mathcal{E}_{\frac{1}{q}}(x) = \prod_{n=0}^{\infty} \left(1 + \frac{x}{q^n}\right)^{-1}. \tag{1.1}$$

By Exercise 1.1.2, we have that

$$(1+x)\,\mathcal{E}_{\frac{1}{q}}(x) = \mathcal{E}_q(x)^{-1}.$$

Replace q with $1/q$ in (1.1) for the final required deduction. □

1.2 Jacobi's Triple Product Identity

1.2.2 (Euler's pentagonal number theorem). *Show that if $|q| < 1$ then*

$$\prod_{n=1}^{\infty}(1-q^n) = \sum_{k=-\infty}^{\infty}(-1)^k q^{\frac{k(3k-1)}{2}}.$$

In the Jacobi triple product identity

$$\prod_{n=1}^{\infty}(1-q^{2n})(1+q^{2n-1}x)\left(1+\frac{q^{2n-1}}{x}\right) = \sum_{k=-\infty}^{\infty} x^k q^{k^2},$$

we replace q by $q^{\frac{3}{2}}$ and x by $-q^{-\frac{1}{2}}$ to get

$$\prod_{n=1}^{\infty}(1-q^{3n})(1-q^{3n-2})(1-q^{3n-1}) = \sum_{k=-\infty}^{\infty}(-1)^k q^{\frac{(3k-1)k}{2}}.$$

But the left hand side is simply $\prod_{n=1}^{\infty}(1-q^n)$. □

1.2.4. *Prove that*

$$\sum_{n=-\infty}^{\infty}(4n+1)q^{2n^2+n} = \sum_{k=0}^{\infty}(-1)^k(2k+1)q^{\frac{k(k+1)}{2}}.$$

The left hand side can be written as

$$1 + \sum_{n=1}^{\infty}(4n+1)q^{2n^2+n} + \sum_{n=1}^{\infty}(1-4n)q^{2n^2-n}.$$

We compare coefficients with the right hand side. When $k = 0$, the summand is 1. When $k = 2n$, the summand is $(4n+1)q^{2n^2+n}$. When $k = 2n-1$, the summand is $-(4n-1)q^{2n^2-n}$. □

1.3 Jacobi's two-square theorem

1.3.3. *Let $x_4(n)$ be defined as follows:*

$$x_4(n) = \begin{cases} 0 & \text{if } n \text{ is even} \\ +1 & \text{if } n \equiv 1 \pmod 4 \\ -1 & \text{if } n \equiv 3 \pmod 4 \end{cases}$$

Prove that $x_4(n)$ is completely multiplicative. That is, show that $x_4(mn) = x_4(m)x_4(n)$.

If m or n is even, this is clear. If m and n are both $\equiv 1 \pmod 4$ or both $\equiv 3 \pmod 4$, this is also clear since their product is $\equiv 1 \pmod 4$. If $m \equiv 1 \pmod 4$ and $n \equiv 3 \pmod 4$ then their product is $\equiv 3 \pmod 4$ and the result is clear. □

1.3.4. *Show that*

$$r_2(n) = 4 \sum_{d|n} x_4(d).$$

By Corollary 1.3.2, we must show that the sum is equal to $d_1(n) - d_3(n)$, where $d_i(n)$ is the number of divisors of n congruent to $i \pmod 4$. By Exercise 1.3.3, the sum is precisely equal to the desired quantity. □

1.4 Jacobi's four square theorem

1.4.1. *Show that*

(a) $$\frac{1}{2}\left(\cot\frac{\theta}{2}\right)\sin n\theta = \frac{1}{2} + \cos\theta + \cos 2\theta + \cdots + \cos(n-1)\theta + \frac{1}{2}\cos n\theta,$$

(b) $$2(\sin m\theta)(\sin n\theta) = \cos(m-n)\theta - \cos(m+n)\theta.$$

Part (b) follows easily from two applications of the usual formula

$$\cos(x+y) = \cos x \cos y - \sin x \sin y.$$

Part (a) follows by an induction argument. Indeed, the result is clear if $n = 1$ since

$$\frac{1}{2}\cot\frac{\theta}{2}\sin\theta=\frac{1}{2}\frac{\cos\left(\frac{\theta}{2}\right)}{\sin\left(\frac{\theta}{2}\right)}2\cos\frac{\theta}{2}\sin\frac{\theta}{2}=\cos^2\frac{\theta}{2}$$

and as

$$\cos\theta=\cos^2\frac{\theta}{2}-\sin^2\frac{\theta}{2}=2\cos^2\frac{\theta}{2}-1,$$

we see that

$$\frac{1}{2}\cot\frac{\theta}{2}\sin\theta=\frac{1+\cos\theta}{2}.$$

Now suppose the result has been proved for n. Observe that

$$\begin{aligned}&\frac{1}{2}\cot\frac{\theta}{2}\sin n\theta+\frac{1}{2}\cos n\theta+\frac{1}{2}\cos(n+1)\theta\\=&\frac{\cos\frac{\theta}{2}\sin n\theta+\cos n\theta\sin\frac{\theta}{2}}{2\sin\frac{\theta}{2}}+\frac{1}{2}\cos(n+1)\theta\\=&\frac{\sin\left(n+\frac{1}{2}\right)\theta}{2\sin\frac{\theta}{2}}+\frac{1}{2}\cos(n+1)\theta.\end{aligned}$$

Writing $(n+\frac{1}{2})=(n+1)-\frac{1}{2}$, we see that this is

$$\begin{aligned}&\frac{\sin(n+1)\theta\cos\frac{\theta}{2}-\sin\frac{\theta}{2}\cos(n+1)\theta}{2\sin\frac{\theta}{2}}+\frac{1}{2}\cos(n+1)\theta\\=&\frac{1}{2}\cot\frac{\theta}{2}\sin(n+1)\theta,\end{aligned}$$

as desired. □

1.4.2. *For $|q|<1$, we let $u_r=\frac{q^r}{1-q^r}$ so that*

$$\frac{q^r}{(1-q^r)^2}=u_r(1+u_r).$$

Prove that

$$\sum_{m=1}^{\infty}u_m(1+u_m)=\sum_{n=1}^{\infty}nu_n.$$

Since

$$\frac{1}{1-q}=\sum_{n=0}^{\infty}q^n,$$

by differentiating both sides with respect to q and then multiplying both sides by q, we obtain

$$\frac{q}{(1-q)^2} = \sum_{n=1}^{\infty} nq^n.$$

Hence,

$$\sum_{m=1}^{\infty} \frac{q^m}{(1-q^m)^2} = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty} nq^{mn} = \sum_{n=1}^{\infty} n \sum_{m=1}^{\infty} q^{mn} = \sum_{n=1}^{\infty} \frac{nq^n}{1-q^n},$$

as required. □

1.4.3. *With notation as in the previous exercise, show that*

$$\sum_{m=1}^{\infty} (-1)^{m-1} u_{2m}(1+u_{2m}) = \sum_{n=1}^{\infty} (2n-1)u_{4n-2}.$$

As in the previous exercise, we have

$$\begin{aligned}
\sum_{m=1}^{\infty} (-1)^{m-1} \frac{q^{2m}}{(1-q^{2m})^2} &= \sum_{m=1}^{\infty} (-1)^{m-1} \sum_{r=1}^{\infty} rq^{2mr} \\
&= \sum_{r=1}^{\infty} r \sum_{m=1}^{\infty} (-1)^{m-1} q^{2mr} \\
&= \sum_{r=1}^{\infty} \frac{rq^{2r}}{1+q^{2r}} \\
&= \sum_{r=1}^{\infty} \left(\frac{rq^{2r}}{1-q^{2r}} - \frac{2rq^{4r}}{1-q^{4r}} \right) \\
&= \sum_{r=1}^{\infty} \frac{rq^{2r}}{1-q^{2r}} - \sum_{\substack{r=1 \\ r \text{ even}}}^{\infty} \frac{rq^{2r}}{1-q^{2r}} \\
&= \sum_{n=1}^{\infty} \frac{(2n-1)q^{4n-2}}{1-q^{4n-2}},
\end{aligned}$$

as required. □

1.5 Supplementary problems

1.5.1. *Define $\tau(n)$ by*

$$q \prod_{n=1}^{\infty} (1-q^n)^{24} = \sum_{n=1}^{\infty} \tau(n) q^n.$$

Show that $\tau(n)$ is odd if and only if $n = (2m+1)^2$ for some m.

We use the fact that for any prime p and for $1 \le j \le p^\alpha - 1$, the binomial coefficient

$$\binom{p^\alpha}{j} \equiv 0 \pmod{p}.$$

Thus $(1-q^n)^8 \equiv 1 - q^{8n} \pmod 2$ and hence

$$q\prod_{n=1}^{\infty}(1-q^n)^{24} \equiv q\prod_{n=1}^{\infty}(1-q^{8n})^3 \pmod 2.$$

From Jacobi's formula (Theorem 1.2.3), we deduce

$$\begin{aligned}\sum_{n=1}^{\infty}\tau(n)q^n &\equiv q\sum_{k=0}^{\infty}(2k+1)q^{4k^2+4k} \pmod 2\\ &\equiv \sum_{k=0}^{\infty}(2k+1)q^{(2k+1)^2} \pmod 2,\end{aligned}$$

from which the result follows immediately. □

1.5.2. *Let $r_k(n)$ be the number of ways of writing n as a sum of k squares. Show that*

$$r_k(n) = \sum_{a=0}^{n} r_i(a)r_{k-i}(n-a)$$

for any i satisfying $1 \le i \le k$.

This is clear from the identity

$$\left(\sum_{k=-\infty}^{\infty} q^{n^2}\right)^k = \left(\sum_{k=-\infty}^{\infty} q^{n^2}\right)^i \left(\sum_{k=-\infty}^{\infty} q^{n^2}\right)^{k-i}$$

on comparing the coefficients of both sides of the equation. □

1.5.3. *Define the q-logarithm function $\mathcal{L}_q(x)$ as*

$$\mathcal{L}_q(x) = \sum_{n=1}^{\infty}\frac{x^n}{q^n-1}.$$

Suppose $|x| < |q|$ and $|q| > 1$. Show that

$$\mathcal{L}_q(x) - \mathcal{L}_q\left(\frac{x}{q}\right) = \frac{x}{q-x}.$$

Deduce that

$$\mathcal{L}_q(x) = \sum_{n=1}^{\infty} \frac{x}{q^n - x}.$$

We have

$$\begin{aligned}\mathcal{L}_q(x) - \mathcal{L}_q\left(\frac{x}{q}\right) &= \sum_{n=1}^{\infty} \frac{x^n - \left(\frac{x}{q}\right)^n}{q^n - 1} \\ &= \sum_{n=1}^{\infty} \left(\frac{x}{q}\right)^n \\ &= \frac{x}{q-x}.\end{aligned}$$

Since $|q| > 1$, as $n \to \infty$ we have $x/q^n \to 0$ and thus $\mathcal{L}_q\left(x/q^n\right) \to 0$. Therefore, iteration leads to

$$\begin{aligned}\mathcal{L}_q(x) &= \frac{x}{q-x} + \frac{\frac{x}{q}}{q-\frac{x}{q}} + \mathcal{L}_q\left(\frac{x}{q^2}\right) \\ &= \cdots \\ &= \sum_{n=1}^{\infty} \frac{x}{q^n - x}.\end{aligned}$$

□

1.5.4. *Show that if $|x| < |q|$ and $|q| > 1$ then*

$$\mathcal{L}_q(x) = \frac{x\mathcal{E}_q'(-x)}{\mathcal{E}_q(-x)},$$

where the derivative is with respect to x.

From the product

$$\mathcal{E}_q(x) = \prod_{n=1}^{\infty} \left(1 + \frac{x}{q^n}\right),$$

we take the logarithmic derivative and use Exercise 1.5.3 to deduce the desired result. □

Chapter 2
The Modular Group

2.1 The full modular group

2.1.1. *Let R be a commutative ring with identity. Show that the set*

$$\mathrm{SL}_2(R) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : \begin{array}{c} a, b, c, d \in R \\ ad - bc = 1 \end{array} \right\}$$

forms a group under matrix multiplication.

It is easily checked that $\mathrm{SL}_2(R)$ is closed under multiplication and has an identity element. One can verify directly that

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

so that inverses also exist. □

2.1.3. *Show that S has order 4, ST has order 6, and T has infinite order.*

This is verified directly since $S^2 = -I$ and

$$ST = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}$$

so that $(ST)^3 = -I$. Finally, $T^n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$ for any $n \in \mathbb{Z}$ so T has infinite order. □

2.1.4. *Show that $\mathrm{SL}_2(\mathbb{Z})$ is generated by two elements of finite order, namely S and ST of order 4 and 6 respectively.*

Since S has order 4 and ST has order 6, we see from Theorem 2.1.2 that

$$\mathrm{SL}_2(\mathbb{Z}) = \langle S, T \rangle = \langle S, ST \rangle.$$ □

© Springer Science+Business Media Singapore 2016 and Hindustan Book Agency 2016

M.R. Murty et al., *Problems in the Theory of Modular Forms*, HBA Lecture Notes in Mathematics, DOI 10.1007/978-981-10-2651-5_12

2.1.5. *Show that any homomorphism*

$$\phi : \mathrm{SL}_2(\mathbb{Z}) \to \mathbb{C}^\times$$

has image contained in the finite subgroup of $\mathbb{C}^\times$ consisiting of 12th roots of unity.

By Exercise 2.1.4, $\mathrm{SL}_2(\mathbb{Z})$ is generated by S of order four and ST of order six. The function ϕ must send these elements to a fourth root of unity and a sixth root of unity, respectively. Thus the image is contained in the group generated by i and $\frac{1+\sqrt{-3}}{2}$, which is the group of 12th roots of unity. □

2.1.6. *Show that* $\mathrm{SL}_2(\mathbb{Z})$ *is generated by*

$$T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \text{ and } U = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

We need only observe that

$$T^{-1}UT^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} = S$$

to deduce the result. □

2.1.7. *Suppose that $(c, d, N) = 1$. Show that there are elements $c' = c + tN$ and $d' = d + sN$ for some integers s, t such that $(c', d') = 1$.*

Write $c = c_1c_2$ with $c_2 = (c, N)$. Thus $(c_1, N) = 1$, and so by the Extended Euclidean Algorithm we can find u, $v \in \mathbb{Z}$ such that $1 = c_1u + Nv$. Hence $1 \equiv Nv \pmod{c_1}$. Set $m = v(1 - d)$ so that

$$d + mN \equiv 1 \pmod{c_1}. \tag{2.1}$$

Take $c' = c$ and $d' = d + mN$. We claim $(c', d') = 1$. Indeed, if p is a prime such that $p|c'$ and $p|d'$ then $p|c$ and $p|d + mN$. Since $c = c_1c_2$, we have $p|c_1$ or $p|c_2$. If $p|c_1$, we get a contradiction to (2.1). Thus $p|c_2$ which implies that $p|N$. From $p|d + mN$, we deduce $p|d$ and so $p|(c, d, N)$, a contradiction. □

2.2 Subgroups of the modular group

2.2.1. *Show that the natural map*

$$\begin{aligned} \mathrm{SL}_2(\mathbb{Z}) &\to \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z}) \\ g &\mapsto g \pmod{N} \end{aligned}$$

is a surjective homomorphism.

The map is obviously a homomorphism. Let

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z}).$$

We must lift this to an element of $\mathrm{SL}_2(\mathbb{Z})$. By Exercise 2.1.7, since $(c, d, N) = 1$, we can find c', d' such that

$$\begin{aligned} c' &\equiv c \pmod{N} \\ d' &\equiv d \pmod{N} \end{aligned}$$

and $(c', d') = 1$. We need to find $k, l \in \mathbb{Z}$ so that

$$\begin{pmatrix} a + kN & b + lN \\ c' & d' \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}).$$

We know that $ad' - bc' = 1 + jN$ for some j so the problem reduces to solving

$$jN + kd'N - lc'N = 0,$$

or equivalently,

$$j = lc' - kd'.$$

Since $(c', d') = 1$, the extended Euclidean Algorithm produces $k, l \in \mathbb{Z}$ satisfying this requirement. □

2.2.2. *Prove that* $\Gamma(N)$ *is a normal subgroup of* $\mathrm{SL}_2(\mathbb{Z})$ *of finite index.*

By Exercise 2.2.1 we have $\Gamma(N)$ is the kernel of the natural map

$$\mathrm{SL}_2(\mathbb{Z}) \to \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})$$

and consequently normal. The image of this map is a finite set and the result follows from the first isomorphism theorem. □

2.2.3. *For any commutative ring* R *with identity, let* $\mathrm{GL}_2(R)$ *be the set of* 2×2 *invertible matrices with entries in* R*. Show that* $\mathrm{GL}_2(R)$ *is a group.*

A matrix A is invertible if and only if $\det(A) \in R^\times$ and the result is immediate from the equality $\det(AB) = \det(A)\det(B)$. □

2.2.4. *Let* p *be a prime. Show that the order of* $\mathrm{GL}_2(\mathbb{Z}/p\mathbb{Z})$ *is given by* $(p^2 - 1)(p^2 - p)$.

Each element of $\mathrm{GL}_2(\mathbb{Z}/p\mathbb{Z})$ gives rise to a basis of $(\mathbb{Z}/p\mathbb{Z})^2$. Conversely, given a basis, we can construct an element of $\mathrm{GL}_2(\mathbb{Z}/p\mathbb{Z})$ by taking the basis elements as columns of the matrix. Thus, to count the size of $\mathrm{GL}_2(\mathbb{Z}/p\mathbb{Z})$, we need only count the number of bases of $(\mathbb{Z}/p\mathbb{Z})^2$. For the first column, we can choose any non-zero vector, so we have $p^2 - 1$ choices. For the second column, we can choose any vector which is not a scalar multiple of the first

column. Thus, there are $p^2 - p$ choices. Hence, the group $\mathrm{GL}_2(\mathbb{Z}/p\mathbb{Z})$ has $(p^2-1)(p^2-p)$ elements. □

2.2.5. *Let p be a prime. Show that $\mathrm{SL}_2(\mathbb{Z}/p\mathbb{Z})$ is a normal subgroup of $\mathrm{GL}_2(\mathbb{Z}/p\mathbb{Z})$ of index $p-1$. Deduce that $\mathrm{SL}_2(\mathbb{Z}/p\mathbb{Z})$ has order $p(p^2-1)$.*

The map

$$\begin{aligned} \mathrm{GL}_2(\mathbb{Z}/p\mathbb{Z}) &\to (\mathbb{Z}/p\mathbb{Z})^\times \\ A &\mapsto \det A \end{aligned}$$

is a homomorphism whose kernel is $\mathrm{SL}_2(\mathbb{Z}/p\mathbb{Z})$. Consequently, $\mathrm{SL}_2(\mathbb{Z}/p\mathbb{Z})$ is a normal subgroup. Moreover, this homomorphism is surjective since

$$\begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix} \mapsto a \in (\mathbb{Z}/p\mathbb{Z})^\times,$$

so the image has size $(p-1)$. Thus $\mathrm{SL}_2(\mathbb{Z}/p\mathbb{Z})$ has index $p-1$ in $\mathrm{GL}_2(\mathbb{Z}/p\mathbb{Z})$. The cardinality is now deduced from Exercise 2.2.4. □

2.2.6. *Show that $\mathrm{SL}_2(\mathbb{Z}/p^2\mathbb{Z})$ has size $p^4(p^2-1)$.*

The natural map given by reduction modulo p

$$\mathrm{SL}_2(\mathbb{Z}/p^2\mathbb{Z}) \to \mathrm{SL}_2(\mathbb{Z}/p\mathbb{Z}) \tag{2.2}$$

can be seen to be surjective by first lifting a matrix in $\mathrm{SL}_2(\mathbb{Z}/p\mathbb{Z})$ to $\mathrm{SL}_2(\mathbb{Z})$ by Exercise 2.2.1 and then reducing modulo p^2. The kernel of the map in (2.2) consists of matrices of the form

$$\begin{pmatrix} 1+t_1p & t_2p \\ t_3p & 1+t_4p \end{pmatrix}$$

with $0 \le t_i \le p-1$ and determinant 1 modulo p^2. This last condition is equivalent to $t_1 + t_4 \equiv 0 \pmod{p}$. Hence, the kernel contains p^3 elements. From Exercise 2.2.5, we see that $\mathrm{SL}_2(\mathbb{Z}/p^2\mathbb{Z})$ has $p^4(p^2-1)$ elements. □

2.2.7. *Apply an induction argument to show that $\mathrm{SL}_2(\mathbb{Z}/p^n\mathbb{Z})$ has size $p^{3n-2}(p^2-1)$.*

The result is true for $n = 1$ and $n = 2$ by the previous exercises. Now consider the natural map

$$\mathrm{SL}_2(\mathbb{Z}/p^{n+1}\mathbb{Z}) \to \mathrm{SL}_2(\mathbb{Z}/p^n\mathbb{Z}),$$

which is seen to be surjective by first lifting a matrix in $\mathrm{SL}_2(\mathbb{Z}/p^n\mathbb{Z})$ to $\mathrm{SL}_2(\mathbb{Z})$ by Exercise 2.2.1 and then reducing modulo p^{n+1}. The kernel consists of matrices

$$\begin{pmatrix} 1+t_1p^n & t_2p^n \\ t_3p^n & 1+t_4p^n \end{pmatrix}$$

with $0 \leq t_i \leq p-1$ and determinant 1 modulo p^{n+1}. This last condition is equivalent to $t_1 + t_4 \equiv 0 \pmod{p}$. So again, there are only p^3 possibilities for the t_is. Hence

$$\left|\mathrm{SL}_2(\mathbb{Z}/p^{n+1}\mathbb{Z})\right| = p^3 \left|\mathrm{SL}_2(\mathbb{Z}/p^n\mathbb{Z})\right|$$

and the result follows by induction. □

2.2.8. *Show that*

$$|\mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})| = N^3 \prod_{p|N} \left(1 - \frac{1}{p^2}\right).$$

Deduce that

$$[\mathrm{SL}_2(\mathbb{Z}) : \Gamma(N)] = N^3 \prod_{p|N} \left(1 - \frac{1}{p^2}\right).$$

Suppose $N = p_1^{n_1} \cdots p_t^{n_t}$ where the p_i are unique primes. By the Chinese Remainder Theorem,

$$\mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z}) = \mathrm{SL}_2(\mathbb{Z}/p_1^{n_1}\mathbb{Z}) \times \cdots \times \mathrm{SL}_2(\mathbb{Z}/p_t^{n_t}\mathbb{Z}).$$

Exercise 2.2.7 now gives $|\mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})| = N^3 \prod_{p|N} \left(1 - \frac{1}{p^2}\right)$. As in the solution to Exercise 2.2.2, since $\Gamma(N)$ is the kernel of the natural map $\mathrm{SL}_2(\mathbb{Z}) \to \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})$, we deduce that $[\mathrm{SL}_2(\mathbb{Z}) : \Gamma(N)] = |\mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})|$. □

2.2.9. *Show that* $\mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})$ *is generated by*

$$T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \text{ and } U = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix},$$

both of order N.

It is clear that T and U have order N since

$$T^n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} \text{ and } U^n = \begin{pmatrix} 1 & 0 \\ n & 1 \end{pmatrix}.$$

The natural map

$$\mathrm{SL}_2(\mathbb{Z}) \to \mathrm{SL}_2(\mathbb{Z}/N\mathbb{Z})$$

is surjective and since $\mathrm{SL}_2(\mathbb{Z})$ is generated by T and U (by Exercise 2.1.6), we are done. □

2.2.10. *Show that* $\Gamma(2)$ *is generated by* $-I$, T^2, *and* U^2, *where*

$$T^2 = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \text{ and } U^2 = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}.$$

Clearly $-I$, T^2, and U^2 are in $\Gamma(2)$ so that the group generated by these three elements is contained in $\Gamma(2)$. We adapt the proof of Theorem 2.1.2 and use the following modified division algorithm: given a and b, with $b \neq 0$, there are q and r such that $a = bq + r$ with $|r| \leq |b|/2$. Observe that r can be negative. Now choose any

$$g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma(2).$$

If $c = 0$, then as before, $g = \pm \begin{pmatrix} 1 & m \\ 0 & 1 \end{pmatrix}$ for some $m \in \mathbb{Z}$. Since $g \in \Gamma(2)$, m is even and hence of the form $m = 2k$. We can thus write $g = \pm T^{2k} \in \langle -I, T^2 \rangle$.

Now suppose $c \neq 0$. If $|a| > |c|$, with $c \neq 0$, we write $a = (2c)q + r$ with $|r| \leq |c|$ and note that

$$T^{-2q} g = \begin{pmatrix} 1 & -2q \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} r & b - 2qd \\ c & d \end{pmatrix}.$$

As c is even, we must have r and c of opposite parity and so $|r| < |c|$. If $|a| < |c|$, then as a is odd, it is not zero and we can write $c = (2a)q + r$ with $|r| \leq |a|$. Thus

$$U^{-2q} g = \begin{pmatrix} 1 & 0 \\ -2q & 1 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & b \\ r & d - 2qb \end{pmatrix}.$$

As before, we must have $|r| < |a|$, as a and r are of opposite parity. Using T^2 and U^2 in this way, we can rearrange the lower left entry to be zero and thus we are done. □

2.2.11. *In $\mathrm{SL}_2(\mathbb{Z})$, show that the group Γ generated by S and T^2 is a congruence subgroup of index three.*

We first show that $\Gamma(2) \subset \Gamma$ and hence that Γ is a congruence subgroup. By Exercise 2.2.10, $\Gamma(2)$ is generated by $-I$, T^2, and U^2, and so it suffices to show that $-I$ and U^2 lie in Γ. Since $-I = S^2$ and $U^2 = ST^{-2}S^{-1}$, this is clear. To compute the index, we observe that

$$\mathrm{SL}_2(\mathbb{Z}) / \Gamma(2) \cong \mathrm{SL}_2(\mathbb{Z}/2\mathbb{Z})$$

and use the correspondence theorem and work in $\mathrm{SL}_2(\mathbb{Z}/2\mathbb{Z})$. The image of $\Gamma/\Gamma(2)$ in $\mathrm{SL}_2(\mathbb{Z}/2\mathbb{Z})$ is the group generated by $S \pmod 2$, which has order 2. Thus $[\Gamma : \Gamma(2)] = 2$. As $\Gamma(2)$ has index 6 in $\mathrm{SL}_2(\mathbb{Z})$, it follows that Γ has index 3 in $\mathrm{SL}_2(\mathbb{Z})$. □

2.3 The Hecke subgroups

2.3.1. *Show that the map* $\Gamma_0(N) \to (\mathbb{Z}/N\mathbb{Z})^\times$ *given by*

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto d \pmod{N}$$

is a surjective homomorphism.

The fact that the map is a homomorphism is easily checked. The surjectivity follows from the fact $(N, d) = 1$ implies there are integers a, b such that $ad - bN = 1$. Consequently, the matrix

$$\begin{pmatrix} a & b \\ N & d \end{pmatrix} \in \Gamma_0(N)$$

and has image d under this map. □

2.3.2. *Show that the map* $\Gamma_1(N) \to \mathbb{Z}/N\mathbb{Z}$ *given by*

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto b \pmod{N}$$

is a surjective homomorphism with kernel $\Gamma(N)$*.*

It is easily checked that this is a homomorphism. The matrix

$$T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

lies in $\Gamma_1(N)$. We see that $T^b = \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \in \Gamma_1(N)$ for any integer b, so the surjectivity is immediate. The fact that $\Gamma(N)$ is the kernel is immediate. □

2.3.3. *Show that* $\Gamma_1(N)$ *has index* $N^2 \prod_{p|N} \left(1 - \frac{1}{p^2}\right)$ *in* $\mathrm{SL}_2(\mathbb{Z})$*.*

By the previous exercise,

$$\Gamma_1(N)/\Gamma(N) \cong \mathbb{Z}/N\mathbb{Z}$$

so that from

$$[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_1(N)][\Gamma_1(N) : \Gamma(N)] = [\mathrm{SL}_2(\mathbb{Z}) : \Gamma(N)]$$

we deduce

$$[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_1(N)] = \frac{[\mathrm{SL}_2(\mathbb{Z}) : \Gamma(N)]}{N}.$$

The result now follows from Exercise 2.2.8. □

2.3.4. *Show that* $\Gamma_0(N)$ *has index* $N\prod_{p|N}\left(1+\frac{1}{p}\right)$ *in* $\mathrm{SL}_2(\mathbb{Z})$.

From Exercise 2.3.1, we have

$$\Gamma_0(N)/\Gamma_1(N) \cong (\mathbb{Z}/N\mathbb{Z})^\times$$

and so

$$[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(N)] = \frac{[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_1(N)]}{\phi(N)},$$

where ϕ denotes Euler's function. Since by Exercise 2.3.3 the numerator is

$$N^2 \prod_{p|N}\left(1-\frac{1}{p^2}\right)$$

and

$$\phi(N) = N\prod_{p|N}\left(1-\frac{1}{p}\right),$$

the result now follows. □

2.4 Groups acting on topological spaces

2.4.1. *If* G *acts on* X *and* $g \in G$, $x \in X$, *show that*

$$G_{gx} = gG_xg^{-1}.$$

For any $h \in G$, we have

$$\begin{aligned} h \in G_{gx} &\iff h(gx) = gx \iff (g^{-1}hg)x = x \\ &\iff g^{-1}hg \in G_x \iff h \in gG_xg^{-1}. \end{aligned}$$

□

2.4.2. *If* G *acts transitively on* X, *show that all stabilizers are conjugate.*

For any $x, y \in X$, by transitivity there exists $g \in G$ with $y = gx$. Hence by Exercise 2.4.1, we have $G_y = G_{gx} = gG_xg^{-1}$. □

2.4.3. *(Orbit-Stabilizer Formula) Suppose that* G *acts on* X.

(a) If $x \in X$ *and* $g, h \in G$, *show that* $gx = hx$ *if and only if* g *and* h *lie in the same left coset of* G_x.

(b) If G *and* X *are both finite, show that* $|Gx| = [G : G_x]$. *Deduce that*

$$|X| = \sum_i [G : G_{x_i}],$$

where the summation is over a complete set of representatives of the orbits of G.

(a) We have

$$gx = hx \iff (h^{-1}g)x = x \iff h^{-1}g \in G_x \iff g \in hG_x,$$

so that g and h lie in the same left coset of G_x.

(b) For a fixed x, the listing

$$gx : g \in G$$

contains $|G|$ elements. From (a), we see that any element that appears in the list is repeated $|G_x|$ times. Thus $|Gx| = [G : G_x]$. Since X is a disjoint union of its orbits, we deduce that

$$|X| = \sum_i [G : G_{x_i}]$$

where the sum is over a complete set of representatives of the orbits of G. □

2.4.6. *Let G be a group with subgroups H and K. Suppose that X is a subset of G which is invariant under the left action of H and the right action of K. Show that X is a disjoint union of double cosets HgK.*

Clearly, for any $x \in X$ we have $HxK \subseteq X$ and in fact

$$X = \bigcup_{x \in X} HxK.$$

But double cosets HxK and HyK are either disjoint or equal. To see this, suppose they share a common element $h_1xk_1 = h_2yk_2$. Then $x = h_1^{-1}h_2yk_2k_1^{-1}$. For any $h_3xk_3 \in HxK$, we have

$$h_3xk_3 = \left(h_3h_1^{-1}h_2\right) y \left(k_2k_1^{-1}k_3\right) \in HyK$$

and we deduce that $HxK \subseteq HyK$. The reverse inclusion is analogous. □

2.4.7. *Suppose a group H acts on a set Y and $|H\backslash Y| < \infty$. If $X \subseteq Y$ is invariant under the action of H and if H_1 is a finite-index subgroup of H, then prove that $|H_1\backslash X| < \infty$.*

Write $n = |H\backslash Y|$ and $m = [H : H_1]$. There exist $y_1, \ldots, y_n \in Y$ and $h_1, \ldots h_m \in H$ such that

$$Y = \coprod_{i=1}^{n} Hy_i$$

and

$$H = \coprod_{j=1}^{m} H_1h_i.$$

Every $x \in X \subseteq Y$ must be of the form $x = hy_i$ for some $h \in H$ and some $1 \leq i \leq n$. Now $Hx = Hy_i$. Thus $H\backslash X$ is a subset of $H\backslash Y$, and hence it must be finite. There is in index set $I \subseteq \{1, 2, \ldots, n\}$ such that

$$X = \coprod_{i \in I} Hy_i = \coprod_{i \in I} \coprod_{j=1}^{m} H_1 h_i y_i$$

and so $|H_1\backslash X| < \infty$. □

2.5 Supplementary problems

2.5.1. *If Γ is a congruence subgroup of level N and $\gamma \in \mathrm{SL}_2(\mathbb{Z})$, show that $\gamma\Gamma\gamma^{-1}$ is again a congruence subgroup of level N.*

By Exercise 2.2.2, we know that $\Gamma(N)$ is a normal subgroup of $\mathrm{SL}_2(\mathbb{Z})$. Hence $\Gamma(N) = \gamma\Gamma(N)\gamma^{-1} \subseteq \gamma\Gamma\gamma^{-1}$.

2.5.2. *Prove that*

$$\Gamma(M) \cap \Gamma(N) = \Gamma(\mathrm{lcm}(M, N)).$$

We know that

$$\begin{aligned} g \in \Gamma(M) \cap \Gamma(N) &\iff g \in \Gamma(M) \text{ and } g \in \Gamma(N) \\ &\iff g \equiv I \pmod{M} \text{ and } g \equiv I \pmod{N} \\ &\iff g \equiv I \pmod{\mathrm{lcm}(M, N)} \\ &\iff g \in \Gamma(\mathrm{lcm}(M, N)) \end{aligned}$$

where the penultimate equivalence follows from the Chinese Remainder Theorem and the definition of an integer congruence. □

2.5.3. *Prove that*

$$\Gamma(M)\Gamma(N) = \Gamma(\gcd(M, N)).$$

If $g \in \Gamma(M)\Gamma(N)$, then $g = ab$ with $a \in \Gamma(M)$ and $b \in \Gamma(N)$. In particular, $a \equiv I \pmod{M}$ and $b \equiv I \pmod{N}$ and thus $g = ab \equiv I \pmod{\gcd(M, N)}$. Consequently,

$$\Gamma(M)\Gamma(N) \subset \Gamma(\gcd(M, N)).$$

Thus, it suffices to show that $[\Gamma(1) : \Gamma(M)\Gamma(N)] = [\Gamma(1) : \Gamma(\gcd(M, N))]$.

By the Second Isomorphism Theorem in group theory, we have

$$\Gamma(M)\Gamma(N)/\Gamma(N) \cong \Gamma(M)/\Gamma(M) \cap \Gamma(N)$$

and hence $[\Gamma(M)\Gamma(N) : \Gamma(N)] = [\Gamma(M) : \Gamma(M) \cap \Gamma(N)]$. Thus, by the Tower Rule for the indices of groups, we deduce that

$$\begin{aligned}[\Gamma(1):\Gamma(M)\Gamma(N)] &= \frac{[\Gamma(1):\Gamma(N)]}{[\Gamma(M)\Gamma(N):\Gamma(N)]}\\ &= \frac{[\Gamma(1):\Gamma(N)]}{[\Gamma(M):\Gamma(M)\cap\Gamma(N)]}\\ &= \frac{[\Gamma(1):\Gamma(N)][\Gamma(1):\Gamma(M)]}{[\Gamma(1):\Gamma(M)\cap\Gamma(N)]}.\end{aligned}$$

By Exercise 2.5.2, $\Gamma(M)\cap\Gamma(N)=\Gamma(\mathrm{lcm}(M,N))$. Thus, by Exercise 2.2.8,

$$\begin{aligned}[\Gamma(1):\Gamma(M)\Gamma(N)] &= \gcd(M,N)^3\prod_{p\mid \gcd(M,N)}\left(1-\frac{1}{p^2}\right)\\ &= [\Gamma(1):\Gamma(\gcd(M,N))],\end{aligned}$$

as desired. □

2.5.4. *Define*

$$\Gamma^1(N)=\left\{g\in \mathrm{SL}_2(\mathbb{Z}) : g\equiv \begin{pmatrix}1&0\\ \star&1\end{pmatrix} \pmod N\right\}.$$

Show that $\Gamma^1(N)$ is a congruence subgroup of level N.

This is clear. □

2.5.5. *If $\mathrm{SL}_2(\mathbb{Z})$ has a non-congruence subgroup, show that it has infinitely many.* (Hint: consider $\Gamma\cap\Gamma(p)$ for a non-congruence subgroup Γ and rational primes p.)

It is clear that $\Gamma\cap\Gamma(p)$ is a non-congruence subgroup if Γ is a non-congruence subgroup since $\Gamma\supseteq\Gamma\cap\Gamma(p)$. Thus, we only need to show that as p varies, we produce infinitely many distinct groups. If $\Gamma\cap\Gamma(p)=H$ (say), for infinitely many rational primes p, then the index

$$[\mathrm{SL}_2(\mathbb{Z}):\Gamma(p)]=p(p^2-1)$$

divides $[\mathrm{SL}_2(\mathbb{Z}):H]$ for infinitely many primes p, which is a contradiction. □

2.5.6. *For each $g\in\mathrm{SL}_2(\mathbb{Z})$ and a subgroup Γ of finite index in $\mathrm{SL}_2(\mathbb{Z})$, we define $n_g(\Gamma)$ to be the least positive integer such that*

$$T^{n_g(\Gamma)}\in g\Gamma g^{-1},\qquad T=\begin{pmatrix}1&1\\0&1\end{pmatrix}\ \textit{as usual.}$$

We say Γ is of class N if the least common multiple of the set of numbers $\{n_g(\Gamma): g\in\mathrm{SL}_2(\mathbb{Z})\}$ equals N. Show that $\Gamma(N)$ is of class N. Also show that the normal closure of T^N, namely

$$D(N) := \langle gT^N g^{-1} : g \in \mathrm{SL}_2(\mathbb{Z}) \rangle$$

is of class N. (This exercise extends the notion of congruence level defined for congruence subgroups to all subgroups of finite index, and is due to Fricke and Wohlfahrt.)

Clearly, $n_I(\Gamma(N)) = N$ and as $\Gamma(N)$ is a normal subgroup of $\mathrm{SL}_2(\mathbb{Z})$, we see that $\Gamma(N)$ is of class N. Moreover, as $D(N) \subseteq \Gamma(N)$, and $T^N \in D(N)$, it follows that $D(N)$ is of class N. □

2.5.7.

(a) Using the Chinese remainder theorem (or otherwise), show that for given x, y, the following congruence

$$x(\alpha y + \beta) \equiv \gamma \mod m$$

has a solution in integers α, β, γ such that $(\alpha, \beta, m) = 1$.

(b) Using (a), show that given $g \in \Gamma(N)$, there exists

$$g' = \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix} \in \Gamma(N)$$

such that $b' \equiv c' \equiv 0 \mod mN$ and $g' \in D(N)gD(N)$ with $D(N)$ defined in Exercise 2.5.6. (Hint: consider conjugates of T by the transpose of T that is in $D(N)$.)

(a) If p is a prime dividing m, we may take $y_p = 0$ if $p|\alpha$ and choose y_p so that $\alpha y_p + \beta \equiv 1 \mod p$ when p is coprime to α. In either case, $\alpha y_p + \beta$ and p are relatively prime and so x_p can be chosen to satisfy the given congruence $\mod p^e$ where $p^e || m$. The proof is now completed using the Chinese remainder theorem.

(b) Let

$$g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

be given and let L be the transpose of T. Consider the elements of the form

$$g_1 = L^{-y} T^{nx} L^y g = \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} \qquad \text{(say)}$$

where $b_1 = b + nx(by + d)$. We choose x and y so that

$$b_1 = b + nx(by + d) \equiv 0 \mod mN$$

and as b is divisible by N, this is equivalent to solving $b/N + x(by + d) \equiv 0 \mod m$ which by (a) has a solution. Now consider the elements of the form

$$g_2 = g_1 L^{nz} = \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix} \qquad \text{(say)}$$

where $c_2 = c_1 + d_1 N z$. We observe that we may find z such that

$$c_2 = c_1 + dNz \equiv 0 \mod mN$$

which is equivalent to $c_1/N + d_1 z \equiv 0 \mod mN$ because $a_1 d_1 = 1 + b_1 c_1 \equiv 1 \mod mN$ or in other words, $a_1 d_1 \equiv 1 \mod m$. Thus, g_2 is the required modification.

□

2.5.8. *With notation as in Exercise 2.5.6, show that $D(N)\Gamma(MN) = \Gamma(N)$ for all positive integers M, N.*

Clearly, it suffices to prove the inclusion $\supseteq$ for $M > 1$. In view (b) of Exercise 2.5.7, it suffices to show that an element of the form

$$g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma(N)$$

with $b \equiv c \equiv 0 \mod MN$ belongs to $\Delta(N)\Gamma(MN)$. Indeed, consider the matrix

$$g' = \begin{pmatrix} a & ad-1 \\ -ad+1 & -d(ad-2) \end{pmatrix}.$$

The $g \equiv g' \mod MN$ so that there exists $g'' \in \Gamma(MN)$ such that $g = g'g''$. A simple calculation shows that (with L the transpose of T)

$$g' = L^{-d+1} T^{-1} L^{1-a} T^d$$

whence $g' \in D(N)$. This completes the proof. □

2.5.9. *Let Γ be a subgroup in $\mathrm{SL}_2(\mathbb{Z})$ of class N. Show that Γ is a congruence subgroup if and only if it is a congruence subgroup of level N. Deduce that if Γ is a subgroup of $\mathrm{SL}_2(\mathbb{Z})$ of class N and the index $[\mathrm{SL}_2(\mathbb{Z}) : \Gamma]$ does not divide $[\mathrm{SL}_2(\mathbb{Z}) : \Gamma(N)]$, then Γ is not a congruence subgroup.*

One direction is a tautology. For the other direction, it suffices to observe that $\Gamma(M) \supseteq \Gamma(\ell M)$ for any integer ℓ. Thus, if Γ is a congruence subgroup, then $\Gamma \supseteq \Gamma(N)$ for some N. Also, $\Gamma \supseteq \Delta(N)$ since Γ is of class N. Therefore, we have

$$\Gamma \supseteq \Gamma(MN)\Delta(N) = \Gamma(N).$$

Now, N is the least such is clear from the definition of the level. The conclusion is now evident. □

2.5.10. *Let F be a free group on n generators, say $x_1, ..., x_n$. For a word $w \in F$, define $\ell_i(w)$ to be the sum of all the exponents of x_i occuring in w. For a subset I of $\{1, ..., n\}$, and a positive integer ℓ, define*

$$\Gamma_\ell(I) = \{w \in F : \ell_i(w) \equiv 0 \mod \ell \quad \forall i \in I\}.$$

Show that $\Gamma_\ell(I)$ is a normal subgroup of F of index $\ell^{|I|}$.

Normality is evident since $\Gamma_\ell(I)$ is the kernel of the surjective homomorphism $F \to (\mathbb{Z}/\ell\mathbb{Z})^{|I|}$ sending w to $(\ell_i(w))_{i\in I}$. It is easily seen that

$$\{\prod_{i\in I} x_i^{e_i} : 0 \le e_i < \ell\}$$

is a set of (left) coset representatives of $\Gamma_\ell(I)$ in F, from which the result is evident. □

2.5.11. *Recall that $\Gamma(2)$ is generated by $-I, T^2, U^2$ (see Exercise 2.2.10) and that the subgroup F generated by $x_1 = T^2$ and $x_2 = U^2$ is free of index 2 in $\Gamma(2)$.*

(a) With notation as in the previous exercise, compute the class of the group $\Gamma_\ell(\{1\})$ and $\Gamma_\ell(\{1,2\})$.
(b) Show that $\Gamma_\ell(\{1,2\})$ is not a congruence subgroup if ℓ is not a power of 2.

(a) Let H denote any of the two groups $\Gamma_\ell(\{1\})$ or $\Gamma_\ell(\{1,2\})$. Then $T^k \in H$ if and only if $T^k \equiv I \mod 2$ and $\ell_1(T^k) \equiv 0 \mod \ell$. This happens if and only if $k \equiv 0 \mod 2$ and $k \equiv 0 \mod \ell$. That is, if and only if $2\ell | k$ if ℓ is odd and or $k|\ell$ if ℓ is even. Thus, H is of class 2ℓ if ℓ is odd and ℓ if ℓ is even.

(b) We have shown the following facts:

$$[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_\ell(\{1,2\})] = [\mathrm{SL}_2(\mathbb{Z}) : \Gamma(2)][\Gamma(2) : F][F : \Gamma_\ell(\{1,2\})] = 12\ell^2.$$

Since $\Gamma_p(\{1,2\}) \supseteq \Gamma_\ell(\{1,2\})$ for any prime p dividing ℓ, it follows that $\Gamma_p(\{1,2\})$ is a congruence subgroup if $\Gamma_\ell(\{1,2\})$ is a congruence subgroup. Also, $[\mathrm{SL}_2(\mathbb{Z}) : \Gamma(2p)] = 6p(p^2-1)$ if $p > 2$ and 48 if $p = 2$. Since ℓ is not a power of 2, let p be an odd prime that divides ℓ. If $\Gamma_\ell(\{1,2\})$ is a congruence subgroup, then so is $\Gamma_p(\{1,2\})$ by our remark. However, since $\Gamma_p(\{1,2\})$ is of level $2p$, it follows that if $\Gamma_p(\{1,2\})$ is a congruence subgroup, then $\Gamma_p(\{1,2\}) \supseteq \Gamma(2p)$ so that $12p^2 | 6p(p^2-1)$, a contradiction. Thus, $\Gamma_\ell(\{1,2\})$ is not a congruence subgroup.

. □

Chapter 3
The Upper Half-Plane

3.1 The group $\mathrm{GL}_n^+(\mathbb{R})$ and its subgroups

3.1.1. *Show that*

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} z = \frac{az+b}{cz+d}$$

defines an action of $\mathrm{GL}_2^+(\mathbb{R})$ *on* $\mathbb{H}$.

Let $g = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{GL}_2^+(\mathbb{R})$. Then it is easily checked that

$$\mathrm{Im}(gz) = \frac{(ad-bc)\,\mathrm{Im}(z)}{|cz+d|^2} > 0.$$

In addition, the identity I acts trivially (as well as $-I$). The other axiom is easily verified directly. □

3.1.2. *Show that the group* $\mathrm{PSL}_2(\mathbb{R}) = \mathrm{SL}_2(\mathbb{R})/\pm I$ *acts faithfully on* $\mathbb{H}$. *That is,* $gz = z$ *for all* $z \in \mathbb{H}$ *implies* $g = 1 \in \mathrm{PSL}_2(\mathbb{R})$.

Let $g = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ and suppose that for all $z \in \mathbb{H}$, we have

$$\frac{az+b}{cz+d} = z.$$

Then $cz^2 + (d-a)z - b = 0$ for all $z \in \mathbb{H}$ which implies $c = b = 0$ and $d = a$ since a quadratic polynomial has at most two roots. Since the determinant $a^2 = 1$, we have $a = \pm 1$ so that $g = \pm I$. □

3.1.3. *Show that the action of* $\mathrm{PSL}_2(\mathbb{R})$ *on* $\mathbb{H}$ *is transitive, i.e. there is only one orbit and for all* $z, w \in \mathbb{H}$, *there exists some* $g \in \mathrm{PSL}_2(\mathbb{R})$ *such that* $gz = w$.

It suffices to show that for any $z \in \mathbb{H}$, there is a $g \in \mathrm{PSL}_2(\mathbb{R})$ such that $gz = i$. Then given $w \in \mathbb{H}$, there is also an element $g_1 \in \mathrm{PSL}_2(\mathbb{R})$ such that $g_1(w) = i$. Thus $gz = g_1 w$ and $(g_1^{-1}g)z = w$.

© Springer Science+Business Media Singapore 2016 and Hindustan Book Agency 2016

M.R. Murty et al., *Problems in the Theory of Modular Forms*, HBA Lecture Notes in Mathematics, DOI 10.1007/978-981-10-2651-5_13

Now, as $z = x + iy \in \mathbb{H}$, we have $y > 0$ and it is immediate that

$$\begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \sqrt{y} & 0 \\ 0 & 1/\sqrt{y} \end{pmatrix} i = \begin{pmatrix} \sqrt{y} & x/\sqrt{y} \\ 0 & 1/\sqrt{y} \end{pmatrix} i = x + iy. \qquad \square$$

3.1.4. *Show that the map*

$$z \mapsto \frac{z - i}{z + i}$$

is an analytic isomorphism of the upper half-plane $\mathbb{H}$ *and the unit disc* $D = \{w \in \mathbb{C} : |w| < 1\}$.

Clearly the map is analytic since $-i \notin \mathbb{H}$. Moreover, $|z - i| < |z + i|$ for $z \in \mathbb{H}$ so that the image lies in D. Injectivity is easily checked. To prove surjectivity, we must show that for any $w \in D$, there is a z such that $\frac{z-i}{z+i} = w$. Solving for z, we obtain

$$z = \frac{i(1 + w)}{1 - w}.$$

We must ensure $\mathrm{Im}(z) > 0$. But this is clear since

$$\mathrm{Im}(z) = \mathrm{Im}\left(\frac{i(1 + w)(1 - \overline{w})}{|1 - w|^2}\right) = \frac{1 - |w|^2}{|1 - w|^2} > 0. \qquad \square$$

3.1.6. *Show that* $\mathrm{SL}_2(\mathbb{Z})$ *is a discrete subgroup of* $\mathrm{SL}_2(\mathbb{R}) \subset \mathbb{R}^4$, *where* $\mathbb{R}^4$ *is endowed with the Euclidean topology.*

In $\mathbb{R}^4$, convergence is component-wise convergence. Consequently, if $\gamma_n \to I$ where $\gamma_n \in \mathrm{SL}_2(\mathbb{Z})$, then for n sufficiently large, $\gamma_n = I$. $\square$

3.2 Fundamental domains

3.2.4. *Suppose that* $z, w \in \mathcal{F}$ *are distinct points which are* $\mathrm{SL}_2(\mathbb{Z})$*-equivalent. Show that* $\mathrm{Im}(z) = \mathrm{Im}(w)$ *and* $\mathrm{Re}(z) = -\mathrm{Re}(w)$.

Without loss of generality, we may suppose $\mathrm{Im}(z) \geq \mathrm{Im}(w)$ so that if $\gamma z = w$ and $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$, then $|cz + d|^2 \leq 1$. Thus $|c|\sqrt{3}/2 \leq 1$ implies $c \in \{0, \pm 1\}$.

If $c = 0$, then

$$\gamma = \begin{pmatrix} \pm 1 & b \\ 0 & \pm 1 \end{pmatrix}.$$

But then $\mathrm{Re}(w) = \mathrm{Re}(z) \pm b$ so that $|b| = 1$. Hence $\mathrm{Re}(w) = \pm 1/2$ so that $\mathrm{Re}(z) = -\mathrm{Re}(w)$. Also, $\mathrm{Im}(w) = \mathrm{Im}(z \pm b) = \mathrm{Im}(z)$.

If $|c| = 1$, then $|cz+d|^2 \leq 1$ becomes $|z \pm d|^2 \leq 1$. Thus, assuming $z = x+iy$, we have $(x \pm d)^2 + y^2 \leq 1$. As $y \geq \sqrt{3}/2$, this means $|x \pm d| \leq 1/2$, forcing

$|d| \leq 1$. If $|d| = 1$, we must have $z = \rho$ or ρ^2 (see Figure 3.1 on page 28). If $d = 0$, our condition becomes $|z| = 1$. Now

$$\operatorname{Im}(w) = \frac{\operatorname{Im}(z)}{|cz + d|^2} = \operatorname{Im}(z)$$

and

$$w = \begin{pmatrix} a & \mp 1 \\ \pm 1 & 0 \end{pmatrix} z = \pm a - \frac{1}{z}.$$

As $Sz = -1/z$ maps the arc of the unit circle with $\pi/3 \leq \arg z \leq \pi/2$ to the arc with $\pi/2 \leq \arg z \leq 2\pi/3$, we see that either $w = -1/z$ or that $\{z, w\} = \{\rho, \rho^2\}$. Hence $\operatorname{Re}(w) = -\operatorname{Re}(z)$, as required. □

3.2.5. *If $\mathcal{F}$ is the standard fundamental domain for $\mathrm{SL}_2(\mathbb{Z})$ and $\gamma \in \mathrm{SL}_2(\mathbb{Z})$, show that $\gamma\mathcal{F}$ is again a fundamental domain for $\mathrm{SL}_2(\mathbb{Z})$.*

The map $z \mapsto \gamma z$ is an analytic automorphism of $\mathbb{H}$. Consequently, the topology of $\mathcal{F}$ is preserved under the mapping. Thus $\gamma\mathcal{F}$ is again a closed set with connected interior and its boundary is again a finite union of smooth curves. Since every point $z \in \mathbb{H}$ is $\mathrm{SL}_2(\mathbb{Z})$-equivalent to some $w \in \mathcal{F}$, it is $\mathrm{SL}_2(\mathbb{Z})$-equivalent to $\gamma w \in \gamma\mathcal{F}$. If two interior points z, w of $\gamma\mathcal{F}$ are $\mathrm{SL}_2(\mathbb{Z})$-equivalent, then $\gamma^{-1}z$ and $\gamma^{-1}w$ would be interior points of $\mathcal{F}$, which are $\mathrm{SL}_2(\mathbb{Z})$-equivalent, a contradiction. □

3.2.7. *Show that I, S, $T^{-1}S$ are coset representatives for $\Gamma_0(2)$ in $\mathrm{SL}_2(\mathbb{Z})$. Deduce that a fundamental domain for $\Gamma_0(2)$ is given by $\mathcal{F} \cup S\mathcal{F} \cup ST\mathcal{F}$, where $\mathcal{F}$ is the standard fundamental domain for $\mathrm{SL}_2(\mathbb{Z})$.*

The group $\Gamma_0(2)$ has index three in $\mathrm{SL}_2(\mathbb{Z})$ and it is easily verified that $I, S, T^{-1}S$ are coset representatives for $\Gamma_0(2)$. To see that $\mathcal{F} \cup S\mathcal{F} \cup ST\mathcal{F}$ has a connected interior, we need only sketch the region explicitly as in Figure 3.1 on page 178. □

3.3 The extended upper-half plane

3.3.1. *Show that every cusp is $\mathrm{SL}_2(\mathbb{Z})$-equivalent to $i\infty$.*

Given a rational number $s \neq i\infty$, we need to find a matrix $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$ such that $cs + d = 0$. Writing $s = -d/c$ with $(c, d) = 1$, we can find by the Euclidean algorithm, integers a, b, such that $ad - bc = 1$. This gives the desired matrix. □

3.3.2. *If Γ is of finite index in $\mathrm{SL}_2(\mathbb{Z})$, show that the number of Γ-inequivalent cusps is $\leq [\mathrm{SL}_2(\mathbb{Z}) : \Gamma]$.*

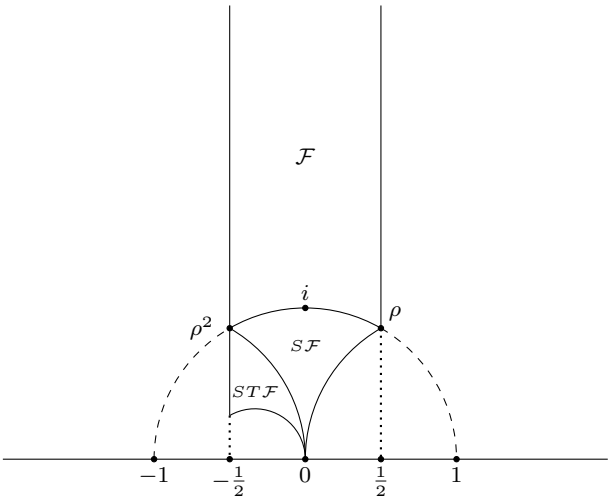


Fig. 3.1 Fundamental Domain for $\Gamma_0(2)$

Let $s \in \mathbb{Q} \cup \{i\infty\}$ be a cusp. By Exercise 3.3.1, there exists $\tau \in \mathrm{SL}_2(\mathbb{Z})$ such that $s = \tau(i\infty)$. If $g_1, \ldots, g_r$ are right coset representatives of Γ in $\mathrm{SL}_2(\mathbb{Z})$, then $s = \tau(i\infty) = \gamma g_j(i\infty)$ for some $\gamma \in \Gamma$. Hence the set of Γ-inequivalent cusps is contained in the set

$$\{g_1(i\infty), \ldots, g_r(i\infty)\}$$

whose size is at most $[\mathrm{SL}_2(\mathbb{Z}) : \Gamma]$. □

3.3.3. *Show that* $\{0, i\infty\}$ *is a complete set of inequivalent cusps of* $\Gamma_0(2)$.

For any $g = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$, since $g \in \Gamma_0(2)$ if and only if $c \equiv 0 \pmod 2$, we see that $i\infty$ is equivalent to a rational number a/c if and only if c is even and a is odd. Similarly, 0 is equivalent to all b/d with d odd. If 0 and $i\infty$ are equivalent, there is an element $g \in \Gamma_0(2)$ such that $g(i\infty) = 0$. This means $a = 0$ and the matrix g would then be singular modulo 2, a contradiction.

□

3.3.4. *With the action of* $\mathrm{SL}_2(\mathbb{Z})$ *on* $\mathbb{H}^\star$*, show that the stabilizer* $\Gamma_{i\infty}$ *of* $i\infty$ *is given by*

$$\left\{ \pm \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} : n \in \mathbb{Z} \right\}.$$

It is clear that if

$$\frac{a(1) + b(0)}{c(1) + d(0)} = \frac{a}{c} = \frac{1}{0}$$

then we must have $c = 0$ so that $ad = 1$ and hence $a = d = 1$ or $a = d = -1$.

□

3.3.5. *Let p be prime. Show that the matrices*

$$ST^k = \begin{pmatrix} 0 & -1 \\ 1 & k \end{pmatrix}, \qquad 0 \le k \le p-1,$$

along with $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$, give a complete set of right coset representatives for $\Gamma_0(p)$ in $\mathrm{SL}_2(\mathbb{Z})$.

It is easily checked that these matrices are distinct coset representatives. Indeed, $\left(\begin{smallmatrix} 0 & -1 \\ 1 & k \end{smallmatrix}\right) \notin \Gamma_0(p)$ and if

$$\Gamma_0(p) \begin{pmatrix} 0 & -1 \\ 1 & k \end{pmatrix} = \Gamma_0(p) \begin{pmatrix} 0 & -1 \\ 1 & k' \end{pmatrix},$$

then

$$\begin{pmatrix} 0 & -1 \\ 1 & k \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & k' \end{pmatrix}^{-1} = \begin{pmatrix} 0 & -1 \\ 1 & k \end{pmatrix} \begin{pmatrix} k' & 1 \\ -1 & 0 \end{pmatrix} \in \Gamma_0(p)$$

which means $k \equiv k' \pmod{p}$.

Now given $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$, if $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \notin \Gamma_0(p)$ then we want to show that there exists some k such that

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(p) \begin{pmatrix} 0 & -1 \\ 1 & k \end{pmatrix}.$$

This is equivalent to ensuring

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & k \end{pmatrix}^{-1} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} k & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} ak-b & a \\ ck-d & c \end{pmatrix} \in \Gamma_0(p),$$

so we need only find k so that $ck \equiv d \pmod{p}$. This is always possible when $p \nmid c$. □

3.3.6. *Let p be prime. Show that the set of matrices*

$$\begin{pmatrix} k & 1 \\ -1 & 0 \end{pmatrix}, \qquad 0 \le k \le p-1,$$

along with $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$, is a complete set of left coset representatives for $\Gamma_0(p)$.

If $g_1, \dots, g_r$ are right coset representatives of a subgroup H in G, then since

$$Hg_i \neq Hg_j \iff g_i g_j^{-1} \notin H \iff g_i^{-1} H \neq g_j^{-1} H,$$

we see that $g_1^{-1}, \dots, g_r^{-1}$ are left coset representatives and the result is immediate from Exercise 3.3.5. □

3.3.7. *Let p be prime. Show that a complete set of inequivalent cusps for $\Gamma_0(p)$ is given by $\{0, i\infty\}$.*

For $0 \leq k \leq p-1$, we have that $\left(\begin{smallmatrix} 0 & -1 \\ 1 & k \end{smallmatrix}\right)(i\infty) = 0$. Hence, by the solution to Exercise 3.3.2 and by an application of Exercise 3.3.5, the $\Gamma_0(p)$-inequivalent cusps are contained in the set containing $i\infty$ and 0. We need only show that 0 and $i\infty$ are $\Gamma_0(p)$-inequivalent. Indeed, if there is a matrix,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(p)$$

such that

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}(i\infty) = 0,$$

then the left hand side is a/c if $c \neq 0$ and $i\infty$ if $c = 0$. This means that our matrix is of the form

$$\begin{pmatrix} 0 & b \\ c & d \end{pmatrix},$$

which has determinant $-bc$ and is consequently divisible by p, a contradiction. □

3.4 The Poincaré metric

3.4.1. *Prove that $d(z, w)$ is $\mathrm{SL}_2(\mathbb{R})$-invariant. That is, show that $d(gz, gw) = d(z, w)$ for all $g \in \mathrm{SL}_2(\mathbb{R})$.*

It suffices to show that

$$\frac{|z-w|^2}{\mathrm{Im}(z)\,\mathrm{Im}(w)}$$

is $\mathrm{SL}_2(\mathbb{R})$-invariant. Indeed, let $g = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{R})$. Then by the solution to Exercise 3.1.1,

$$\mathrm{Im}(gz)\,\mathrm{Im}(gw) = \frac{\mathrm{Im}(z)\,\mathrm{Im}(w)}{|cz+d|^2|cw+d|^2}$$

and

$$\frac{az+b}{cz+d} - \frac{aw+b}{cw+d} = \frac{z-w}{(cz+d)(cw+d)},$$

since $ad - bc = 1$. The result is now immediate. □

3.4.2. *Show that for $x \geq 1$,*

$$\cosh^{-1} x = \log(x + \sqrt{x^2-1}).$$

We want y such that

$$\frac{e^y + e^{-y}}{2} = x.$$

This is a quadratic equation in e^y which is easily solved and we get

$$e^y = x + \sqrt{x^2 - 1}$$

since the inverse hyperbolic cosine function is non-negative (and hence we need $e^y \geq 1$). □

3.4.3. *Show that for $y > 0$,*

$$d(i, iy) = |\log y|.$$

We have

$$\begin{aligned} d(i, iy) &= \cosh^{-1}\left(1 + \frac{(y-1)^2}{2y}\right) \\ &= \cosh^{-1}\left(\frac{1}{2}\left(y + \frac{1}{y}\right)\right) \\ &= \log\left(\frac{1}{2}\left(y + \frac{1}{y}\right) + \sqrt{\frac{y^2}{4} + \frac{1}{4y^2} - \frac{1}{2}}\right) \\ &= \log\left(\frac{1}{2}\left(y + \frac{1}{y}\right) + \sqrt{\left(\frac{y}{2} - \frac{1}{2y}\right)^2}\right). \end{aligned}$$

Thus

$$d(i, iy) = \log\left(\frac{1}{2}\left(y + \frac{1}{y}\right) + \frac{1}{2}\left|y - \frac{1}{y}\right|\right).$$

If $y > \frac{1}{y}$, this is $\log y$. If $y < \frac{1}{y}$, this is $-\log y$. In either case, it can be written as $|\log y|$. □

3.4.5. *Let $\mathcal{F}$ be the standard fundamental domain for $\mathrm{SL}_2(\mathbb{Z})$. Show that*

$$\int\int_{\mathcal{F}} \frac{dxdy}{y^2} = \frac{\pi}{3}.$$

We compute easily

$$\int\int_{\mathcal{F}:y\geq 1} \frac{dxdy}{y^2} = 1.$$

The integral

$$\begin{aligned} \int\int_{\mathcal{F}:y\leq 1} \frac{dxdy}{y^2} &= 2\int_0^{\frac{1}{2}} \left(\int_{\sqrt{1-x^2}}^1 \frac{dy}{y^2}\right) dx \\ &= 2\int_0^{\frac{1}{2}} \left(-1 + \frac{1}{\sqrt{1-x^2}}\right) dx \\ &= -1 + 2\sin^{-1}\frac{1}{2} \\ &= -1 + \frac{\pi}{3}, \end{aligned}$$

from which the result follows. □

3.5 Supplementary problems

3.5.1. *Show that a set of coset representatives for Γ_∞ in $\mathrm{SL}_2(\mathbb{Z})$ is given by*

$$\left\{ \begin{pmatrix} \star & \star \\ c & d \end{pmatrix} : (c,d) = 1, c, d \in \mathbb{Z} \right\},$$

where the asterisks indicate that we can take any a, b such that $ad - bc = 1$.

The coset

$$\Gamma_\infty \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \left\{ \begin{pmatrix} a+nc & b+nd \\ c & d \end{pmatrix} : n \in \mathbb{Z} \right\}.$$

Thus, for fixed c, d with $(c,d) = 1$, the coset parametrizes all integer solutions of $ad - bc = 1$, by the Euclidean algorithm. □

3.5.2. *Let $\mathcal{F}$, Γ, $g_1, \ldots, g_r$ and $\mathcal{D}$ be as in Theorem 3.2.6. Show that*

$$\int\int_{\mathcal{D}} \frac{dxdy}{y^2} = [\mathrm{SL}_2(\mathbb{Z}) : \Gamma] \frac{\pi}{3}.$$

Using the notation of Theorem 3.2.6, we have coset representatives $g_1, \ldots, g_r$ of Γ in $\mathrm{SL}_2(\mathbb{Z})$ such that

$$\mathcal{D} = \bigcup_{i=1}^{r} g_i^{-1} \mathcal{F}$$

is a fundamental domain for Γ. Thus

$$\int\int_{\mathcal{D}} \frac{dxdy}{y^2} = \sum_{i=1}^{r} \int\int_{g_i^{-1}\mathcal{F}} \frac{dxdy}{y^2}.$$

Since the measure $dxdy/y^2$ is $\mathrm{SL}_2(\mathbb{Z})$-invariant, each integral is equal to $\pi/3$, the area of $\mathcal{F}$, by Exercise 3.4.5. □

3.5.3. *Let $G = \mathrm{SL}_2(\mathbb{R})$ act on $\mathbb{H}$. Show that K, the stabilizer of i, consists of matrices*

$$\begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \qquad \text{for } \theta \in \mathbb{R}.$$

Then show that the upper half-plane is homeomorphic to G/K.

If the matrix $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{R})$ fixes i, then we must have

$$ai + b = i(ci + d) = -c + di.$$

In other words, $b = -c$ and $a = d$. Since $ad - bc = 1$, this means $a^2 + b^2 = 1$. We can write $a = \cos\theta$ and $b = \sin\theta$ to deduce the result on K. Since $\mathrm{SL}_2(\mathbb{R})$ acts transitively on $\mathbb{H}$, we deduce that the upper half-plane is homeomorphic to G/K by Theorem 2.4.4. □

Chapter 4
Modular Forms of Level One

4.1 Fourier expansions and q-series

4.1.1. *Show that the punctured fundamental neighborhood*

$$U_c = \{z \in \mathbb{H} : \mathrm{Im}(z) > c\}$$

of $i\infty$ is mapped via $z \mapsto e^{2\pi iz}$ into the punctured open disc centered at zero, of radius $e^{-2\pi c}$.

If $z = x + iy$, then $e^{2\pi iz} = e^{-2\pi y}e^{2\pi ix}$. For fixed y and $x \in \mathbb{R}$, the map $z \mapsto e^{2\pi iz}$ takes the line $\mathrm{Im}(z) = y$ in $\mathbb{H}$ to the circle centered at zero of radius $e^{-2\pi y}$. □

4.1.2. *Let $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{GL}_2^+(\mathbb{R})$ and $z \in \mathbb{H}$. Define $j(\gamma, z) = cz + d$. Show that*

$$j(\gamma_1\gamma_2, z) = j(\gamma_1, \gamma_2 z)j(\gamma_2, z).$$

This is just a brute force calculation. Let

$$\gamma_1 = \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix}, \quad \gamma_2 = \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix},$$

so that

$$j(\gamma_1\gamma_2, z) = (c_1a_2 + d_1c_2)z + (c_1b_2 + d_1d_2).$$

Now $j(\gamma_2, z) = c_2z + d_2$ and

$$\begin{aligned} j(\gamma_1, \gamma_2 z) &= c_1\left(\frac{a_2z + b_2}{c_2z + d_2}\right) + d_1 \\ &= \frac{(c_1a_2 + d_1c_2)z + (c_1b_2 + d_1d_2)}{c_2z + d_2} \end{aligned}$$

from which the result follows immediately. □

© Springer Science+Business Media Singapore 2016 and Hindustan Book Agency 2016

M.R. Murty et al., *Problems in the Theory of Modular Forms*, HBA Lecture Notes in Mathematics, DOI 10.1007/978-981-10-2651-5_14

4.1.3. *Show that* $f|(\gamma_1\gamma_2) = (f|\gamma_1)|\gamma_2$ *for all* $\gamma_1, \gamma_2 \in \mathrm{GL}_2^+(\mathbb{R})$.

Since $(f|\gamma_1)(z) = (\det\gamma_1)^{k/2} j(\gamma_1, z)^{-k} f(\gamma_1 z)$, we have

$$\begin{aligned}\Big((f|\gamma_1)\Big|\gamma_2\Big)(z) &= (\det\gamma_2)^{k/2} j(\gamma_2, z)^{-k} (f|\gamma_1)(\gamma_2 z)\\ &= (\det\gamma_2)^{k/2} j(\gamma_2, z)^{-k} \Big((\det\gamma_1)^{k/2} j(\gamma_1, \gamma_2 z)^{-k} f(\gamma_1(\gamma_2 z))\Big)\\ &= f|(\gamma_1\gamma_2),\end{aligned}$$

by the multiplicativity of determinants and by Exercise 4.1.2. □

4.1.4. *Prove that if* $f : \mathbb{H} \to \mathbb{C}$ *is holomorphic, satisfies*

(i) $f(z+1) = f(z)$
(ii) $f\left(-\frac{1}{z}\right) = z^k f(z)$

for all $z \in \mathbb{H}$, *and is holomorphic at* $i\infty$, *then* f *is a modular form of weight* k *for* $\mathrm{SL}_2(\mathbb{Z})$.

Using the slash notation, we have $f|T = f$ and $f|S = f$ by (i) and (ii). By Exercise 4.1.3, we see that $f|\gamma = f$ for all γ in the group generated by S, T, which is $\mathrm{SL}_2(\mathbb{Z})$. □

4.1.5. *Let* k_1, $k_2 \in \mathbb{Z}$. *Show that:*

(a) Both $M_{k_1}(\mathrm{SL}_2(\mathbb{Z}))$ *and* $S_{k_1}(\mathrm{SL}_2(\mathbb{Z}))$ *are* $\mathbb{C}$*-vector spaces.*
(b) If $f \in M_{k_1}(\mathrm{SL}_2(\mathbb{Z}))$ *and* $g \in M_{k_2}(\mathrm{SL}_2(\mathbb{Z}))$, *then* $fg \in M_{k_1+k_2}(\mathrm{SL}_2(\mathbb{Z}))$.
(c) If $f \in M_{k_1}(\mathrm{SL}_2(\mathbb{Z}))$ *and* $g \in S_{k_2}(\mathrm{SL}_2(\mathbb{Z}))$, *then* $fg \in S_{k_1+k_2}(\mathrm{SL}_2(\mathbb{Z}))$.

Suppose $f, g \in M_{k_1}(\mathrm{SL}_2(\mathbb{Z}))$, $\alpha, \beta \in \mathbb{C}$, and $\gamma = \left(\begin{smallmatrix} a & b \\ c & d\end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$. Then

$$\begin{aligned}(\alpha f + \beta g)(\gamma z) &= \alpha f(\gamma z) + \beta g(\gamma z)\\ &= \alpha(cz+d)^{k_1} f(z) + \beta(cz+d)^{k_1} g(z)\\ &= (cz+d)^{k_1}(\alpha f + \beta g)(z).\end{aligned}$$

Consider the q-expansions $f = \sum a_n q^n$ and $g = \sum b_n q^n$. We clearly have $\alpha f + \beta g = \sum(\alpha a_n + \beta b_n) q^n$ so that $v_{i\infty}\Big(\alpha f + \beta g\Big) \geq \min\{v_{i\infty}(f), v_{i\infty}(g)\}$. We conclude that $\alpha f + \beta g \in M_{k_1}(\mathrm{SL}_2(\mathbb{Z}))$. It is now clear that $M_{k_1}(\mathrm{SL}_2(\mathbb{Z}))$ and $S_{k_1}(\mathrm{SL}_2(\mathbb{Z}))$ are vector spaces.

Now for $f \in M_{k_1}(\mathrm{SL}_2(\mathbb{Z}))$ and $g \in M_{k_2}(\mathrm{SL}_2(\mathbb{Z}))$, if $\gamma = \left(\begin{smallmatrix} a & b \\ c & d\end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$, then

$$\begin{aligned}(fg)(\gamma z) &= f(\gamma z) \cdot g(\gamma z)\\ &= (cz+d)^{k_1} f(z) \cdot (cz+d)^{k_2} g(z)\\ &= (cz+d)^{k_1+k_2}(fg)(z).\end{aligned}$$

Moreover, $v_{i\infty}(fg) = v_{i\infty}(f) + v_{i\infty}(g)$. This proves (b) and (c).

4.1.6. *Show that $M_k(\mathrm{SL}_2(\mathbb{Z})) = \{0\}$ for k odd.*

For $\gamma = -I$, the modular condition becomes $f(z) = (-1)^k f(z)$ from which we deduce $f(z) \equiv 0$ since k is odd. □

4.2 Eisenstein series for $\mathrm{SL}_2(\mathbb{Z})$

4.2.1. *Show that the series for $G_k(z)$ converges absolutely for all $z \in \mathbb{H}$ and for $k > 2$.*

We note that for $z = x + iy$, $x, y \in \mathbb{R}$, we have

$$\begin{aligned} |mz + n|^2 &= (mx + n)^2 + y^2 m^2 \\ &= \left(x^2 + y^2\right) m^2 + 2xmn + n^2 \end{aligned}$$

which is a binary quadratic form corresponding to the matrix

$$\begin{pmatrix} x^2 + y^2 & x \\ x & 1 \end{pmatrix}.$$

For $z \in \mathbb{H}$, we have $y \neq 0$ and so this is positive definite. By our earlier remark, there is a constant $\lambda > 0$ such that

$$|mz + n|^2 \geq \lambda(m^2 + n^2).$$

We are reduced to studying the convergence of

$$\sum_{m,n}{}' \frac{1}{(m^2 + n^2)^{k/2}} = \sum_{t=1}^{\infty} \frac{r_2(t)}{t^{k/2}},$$

where $r_2(t)$ is the number of ways of writing t as a sum of two squares of integers. Since $r_2(t) = O(t^\epsilon)$ (see Exercise 1.3.4) for any $\epsilon > 0$, the series converges for $k > 2$. □

4.2.4. *Define the Bernoulli numbers B_k by the power series*

$$\frac{x}{e^x - 1} = \sum_{k=0}^{\infty} \frac{B_k x^k}{k!},$$

so that $B_0 = 1$, $B_1 = -\frac{1}{2}$, $B_2 = \frac{1}{6}$, $B_3 = 0$, $B_4 = -\frac{1}{30}$, ... Show that

$$2\zeta(2k) = -\frac{(2\pi i)^{2k} B_{2k}}{(2k)!}, \qquad k \geq 1.$$

From (4.4) on page 39,

$$\pi z \cot \pi z = \pi i z + \frac{2\pi i z}{e^{2\pi i z} - 1}.$$

But the right hand side is

$$\pi i z + \sum_{k=0}^{\infty} B_k \frac{(2\pi i z)^k}{k!}. \tag{4.1}$$

On the other hand, from (4.3) on page 39, we have

$$\pi z \cot \pi z = 1 + 2\sum_{n=1}^{\infty} \frac{z^2}{z^2 - n^2} = 1 - 2\sum_{n=1}^{\infty}\sum_{k=1}^{\infty} \left(\frac{z}{n}\right)^{2k},$$

upon writing

$$\frac{z^2}{n^2 - z^2} = \frac{(z/n)^2}{1 - (z/n)^2} = \sum_{k=1}^{\infty} \left(\frac{z}{n}\right)^{2k}.$$

Thus

$$\pi z \cot \pi z = 1 - 2\sum_{k=1}^{\infty} \zeta(2k) z^{2k}.$$

Combining this with (4.1) gives

$$\pi i z + \sum_{k=0}^{\infty} B_k \frac{(2\pi i z)^k}{k!} = 1 - 2\sum_{k=1}^{\infty} \zeta(2k) z^{2k} \tag{4.2}$$

and the desired result follows by comparing coefficients. □

4.2.5. *Prove that the Bernoulli numbers satisfy the recursion formula:*

$$(m+1)B_m = -\sum_{k=0}^{m-1} \binom{m+1}{k} B_k.$$

This is easily deduced by comparing coefficients of the identity

$$x = (e^x - 1)\sum_{k=0}^{\infty} \frac{B_k x^k}{k!},$$

upon writing $e^x - 1$ as a power series in x. □

4.2.6. *Define the n-th Bernoulli polynomial by*

$$B_n(x) = \sum_{k=0}^{n} \binom{n}{k} B_k x^{n-k}.$$

Prove that

$$\frac{te^{xt}}{e^t-1}=\sum_{n=0}^{\infty}B_n(x)\frac{x^n}{n!}.$$

This is immediate from comparing coefficients of the multiplication of power series for e^{xt} and $t/(e^t-1)$. □

4.2.8. *Prove that* $M_k\left(\mathrm{SL}_2(\mathbb{Z})\right)=\mathbb{C}E_k\oplus S_k\left(\mathrm{SL}_2(\mathbb{Z})\right)$ *for all* $k\geq 4$. *Deduce that* $\dim M_k\left(\mathrm{SL}_2(\mathbb{Z})\right)=1+\dim S_k\left(\mathrm{SL}_2(\mathbb{Z})\right)$.

Suppose $f\in M_k\left(\mathrm{SL}_2(\mathbb{Z})\right)$. Let $\lambda=f(i\infty)$ denote the constant term in the Fourier expansion. Clearly

$$f=\lambda E_k+\Big(f-\lambda E_k\Big)$$

where $f-\lambda E_k\in S_k\left(\mathrm{SL}_2(\mathbb{Z})\right)$ since it vanishes at the cusp $i\infty$. The conclusion follows. □

4.3 The valence formula

4.3.2. *If* $k<0$, *show that* $M_k(\mathrm{SL}_2(\mathbb{Z}))=\{0\}$.

By the valence formula, we deduce that such an f is identically zero. □

4.3.3. *Show that* $M_0(\mathrm{SL}_2(\mathbb{Z}))=\mathbb{C}$.

If $k=0$, the left hand side of the valence formula is zero. However, for any $f(z)\in M_0\left(\mathrm{SL}_2(\mathbb{Z})\right)$, the form $f(z)-f(i)\in M_0\left(\mathrm{SL}_2(\mathbb{Z})\right)$ clearly has a zero at $z=i$ and hence $f(z)$ must be constant. □

4.3.4. *Show that* $M_2(\mathrm{SL}_2(\mathbb{Z}))=\{0\}$.

The right hand side of the valence formula is $1/6$. This is impossible unless $f\equiv 0$. □

4.3.5. *Show that* $E_4(z)$ *has a simple zero at* $z=\rho$.

Applying the valence formula to $f=E_4$, of weight 4, we see the right hand side is $1/3$. This can only arise if E_4 has a simple zero at $z=\rho$. □

4.3.6. *Show that* $E_6(z)$ *has a simple zero at* $z=i$.

Applying the valence formula to $f=E_6$, of weight 6, we find the right hand side is $1/2$ and this can arise only if E_6 has a simple zero at $z=i$. □

4.3.7. *Show that* $S_k(\mathrm{SL}_2(\mathbb{Z}))=\{0\}$ *for* $k\leq 10$.

If the weight of a modular form f is ≤ 10, then the right hand side of the valence formula is $\leq 5/6 < 1$. If there were a non-trivial cusp form of weight ≤ 10, there would be a contribution of at least 1 from the zero at $i\infty$. □

4.3.8. *Show that* $M_k(\mathrm{SL}_2(\mathbb{Z})) = \mathbb{C}E_k$ *for* $k = 4, 6, 8,$ *and* 10.

This follows immediately from Exercises 4.2.8 and 4.3.7. □

4.3.9. *Show that* $E_4^2 = E_8$. *Deduce that*

$$\sigma_7(n) = \sigma_3(n) + 120 \sum_{m=1}^{n-1} \sigma_3(m)\sigma_3(n-m).$$

By the previous exercise, $M_8(\mathrm{SL}_2(\mathbb{Z}))$ is one-dimensional and spanned by E_8. Since E_4^2 has weight 8, it must be a scalar multiple of E_8. The scalar must be 1 since $E_4(i\infty) = 1$. The identity for $\sigma_7(n)$ is derived by comparing coefficients in the q-series expansions of both sides of the equation $E_8 = E_4^2$. □

4.3.10. *Show that* $E_4E_6 = E_{10}$. *Deduce that*

$$11\sigma_9(n) = 21\sigma_5(n) - 10\sigma_3(n) + 5040 \sum_{m=1}^{n-1} \sigma_3(n)\sigma_5(n-m).$$

As in Exercise 4.3.9, $E_4E_6 = E_{10}$ holds since $M_{10}(\mathrm{SL}_2(\mathbb{Z}))$ is one-dimensional. The identity for $\sigma_9(n)$ arises from comparing coefficients in the q-expansions of both sides of the equation $E_{10} = E_4E_6$. □

4.4 The dimension formula

4.4.3. *For* $k \geq 4$, *show that*

$$\dim(S_k(\mathrm{SL}_2(\mathbb{Z})) = \begin{cases} \lfloor \frac{k}{12} \rfloor - 1 & \text{if } k \equiv 2 \pmod{12} \\ \lfloor \frac{k}{12} \rfloor & \text{if } k \not\equiv 2 \pmod{12} \end{cases}$$

The result is immediate by Exercise 4.2.8 and Corollary 4.4.2. □

4.4.5. *Prove that* $\Delta = \dfrac{E_4^3 - E_6^2}{1728}$ *has integer Fourier coefficients.*

We can write

$$E_4 = 1 + 240A$$
$$E_6 = 1 - 504B$$

where A, B are q-series with integer coefficients. Now

$$E_4^3 - E_6^2 = (1+240A)^3 - (1-504B)^2$$
$$= 3(240A) + 3(240A)^2 + (240A)^3 + 2(504)B - (504B)^2.$$

Since $1728 = 12^3$, $240 = 12 \times 20$, and $504 = 12 \times 42$, we see that

$$3(240)^2 \equiv (240)^3 \equiv (504)^2 \equiv 0 \pmod{1728}.$$

Thus $E_4^3 - E_6^2 \equiv 144(5A + 7B) \pmod{1728}$. But

$$5A + 7B = \sum_{n=1}^{\infty} (5\sigma_3(n) + 7\sigma_5(n))q^n$$

and

$$5\sigma_3(n) + 7\sigma_5(n) = \sum_{d|n} (5d^3 + 7d^5).$$

Now $d^3(5+7d^2) \equiv d^3(d^2-1) \equiv 0 \pmod{3}$ and $d^3(5+7d^2) \equiv d^3(1-d^2) \equiv 0 \pmod{4}$ so that all the coefficients of $5A + 7B$ are divisible by 12 and hence all the coefficients of $E_4^3 - E_6^2$ are divisible by 1728. □

4.5 The j-function

4.5.2. *Show that $j(\rho) = 0$ and $j(i) = 1728$. Moreover, show that $v_\rho\Big(j(z)\Big) = 3$, that $v_i\Big(j(z) - 1728\Big) = 2$, and that $v_{z_0}\Big(j(z) - j(z_0)\Big) = 1$ for all $z_0 \in \mathbb{H}$ which are not $\mathrm{SL}_2(\mathbb{Z})$-equivalent to either ρ or i.*

By Exercises 4.3.5 and 4.3.6, we know that $E_4(\rho) = 0$ and $E_6(i) = 0$, and that these are simple zeros. By the valence formula, there are no other zeros of E_4 and E_6 in the fundamental domain $\mathcal{F}$. Now

$$j(\rho) = \frac{E_4(\rho)^3}{\Delta(\rho)} = 0$$

and

$$j(i) = \frac{1728E_4(i)^3}{E_4(i)^3 - E_6(i)^2} = 1728.$$

Moreover, it is clear that $v_\rho(j) = 3v_\rho(E_4) - v_\rho(\Delta) = 3$, and since

$$j(z) - 1728 = \frac{1728E_4(z)^3}{E_4(z)^3 - E_6(z)^2} - 1728 = \frac{1728E_6(z)^2}{E_4(z)^3 - E_6(z)^2},$$

we also have $v_i(j - 1728) = 2v_i(E_6) - v_i\left(E_4^3 - E_6^2\right) = 2$.

Suppose z_0 is $\mathrm{SL}_2(\mathbb{Z})$-inequivalent to ρ and to i. Then

$$j(z) - j(z_0) = \frac{E_4(z)^3 - j(z_0)\Delta(z)}{\Delta(z)}$$

vanishes at $z = z_0$. Since $\Delta(z_0) \neq 0$, we have $1 \leq v_{z_0}\Big(j(z) - j(z_0)\Big) = v_{z_0}\Big(E_4(z)^3 - j(z_0)\Delta(z)\Big)$. By the valence formula, this must be 1. □

4.5.3. *Show that any modular function f has a finite number of zeros and poles in $\mathcal{F}$.*

The set $\{z \in \mathbb{H} : \mathrm{Im}(z) > c\}$ being a fundamental neighborhood of $i\infty$, and as the zeros and poles of any meromorphic function are isolated, we can choose c sufficiently large such that $i\infty$ is the only zero or pole of f in this neighborhood. Now as the region

$$\mathfrak{F} \cap \{z \in \mathbb{H} : \mathrm{Im}(z) \leq c\},$$

is compact, it can only contain a finite number of zeros and poles. □

4.6 Supplementary problems

4.6.1. *Show that the Bernoulli numbers B_k vanish for odd $k \geq 3$.*

From the equation defining the Bernoulli numbers, we have

$$\frac{x}{e^x - 1} - 1 + \frac{x}{2} = \sum_{k=2}^{\infty} \frac{B_k x^k}{k!}.$$

Writing the left hand side as $f(x)$, we see that

$$f(-x) = \frac{xe^x}{e^x - 1} - 1 - \frac{x}{2} = \frac{x(e^x - 1) + x}{e^x - 1} - 1 - \frac{x}{2} = f(x)$$

is even. Hence $B_k = 0$ for odd $k \geq 3$.

Alternatively, we may conclude that the B_k vanish for odd $k \geq 3$ by comparing coefficients on both sides of equation (4.2) in the solution to Exercise 4.2.4. □

4.6.2. *Show that for k even and positive, $(-1)^{k/2} B_k < 0$.*

This follows from (4.2) in the solution to Exercise 4.2.4 since $\zeta(2k) > 0$. □

4.6.3. *Prove that*

$$1^k + 2^k + \cdots + n^k = \frac{B_{k+1}(n+1) - B_{k+1}}{k+1},$$

for every natural number k.

Consider the series

$$\sum_{k=0}^{\infty}\left(\sum_{j=0}^{n} j^k\right)\frac{t^k}{k!}$$

which is easily evaluated upon interchanging the summations. We get

$$\sum_{j=0}^{n} e^{jt} = \frac{e^{(n+1)t}-1}{e^t-1}$$

and the desired result now follows upon applying Exercise 4.2.6. □

4.6.4. *Prove that $\zeta(k)$ tends to 1 as k tends to infinity and use this to show that*

$$|B_{2k}| \sim 4\pi\sqrt{e}(k/\pi e)^{2k+1/2}$$

as k tends to infinity.

By the integral test, it is easily seen that $\zeta(k) = 1 + O(1/2^{k-1})$ so that $\zeta(k)$ tends to 1 as k tends to infinity. The asymptotic for the Bernoulli numbers follows upon using this together with Stirling's formula:

$$(k)! \sim (k/e)^k\sqrt{2\pi k}.$$

□

4.6.5. *Show that if the Fourier coefficients of $f \in M_k(\mathrm{SL}_2(\mathbb{Z}))$ are integers, then f can be written as a polynomial in E_4, E_6, and Δ with integer coefficients.*

We proceed by induction on the weight. Since $M_k(\mathrm{SL}_2(\mathbb{Z}))$ is one dimensional for $k \leq 10$ and spanned by E_k, the result is clear. If f is a cusp form, then f/Δ is a modular form of weight $k-12$ with integral coefficients since $1/\Delta$ has a q-expansion with $\mathbb{Z}$-coefficients. If f is not a cusp form, write $k = 4a + 6b$ and consider $g = f - \lambda E_4^a E_6^b$, where $f(i\infty) = \lambda$. Note that $\lambda \in \mathbb{Z}$ since λ is the constant term in the Fourier expansion of f at $i\infty$. Since $E_4(i\infty) = E_6(i\infty) = 1$. we see that g is a cusp form and we can apply induction to g/Δ and ultimately deduce the desired result. □

4.6.6. *Suppose $f = \sum_{n=0}^{\infty} a_n q^n \in M_k(\mathrm{SL}_2(\mathbb{Z}))$. Show that for any $x \in \mathbb{R}$ we have $\lim_{y\to\infty} f(x+iy) = a_0$. Show that for any $x \in \mathbb{Q}$, if f is not a cusp form, then $\lim_{y\to 0} f(x+iy) = \infty$, and if f is a cusp form, then $\lim_{y\to 0} f(x+iy) = 0$. Deduce that the real line is a natural boundary for any modular form.*

Write $z = x + iy \in \mathbb{H}$. Then

$$\begin{aligned}\lim_{y\to\infty}|f(x+iy)-a_0| &= \lim_{y\to\infty}\left|\sum_{n=1}^{\infty}a_n e^{2\pi i(x+iy)n}\right| \\ &\le \lim_{y\to\infty}\sum_{n=1}^{\infty}|a_n|\,e^{-2\pi yn} \\ &= 0.\end{aligned}$$

If $x\in\mathbb{Q}$, then there exists $\gamma=\left(\begin{smallmatrix}a & b\\ c & d\end{smallmatrix}\right)\in \mathrm{SL}_2(\mathbb{Z})$ such that $\gamma x = i\infty$. In particular, $cx+d=0$ and so

$$\gamma(x+iy)=\frac{ax+b+aiy}{cx+d+ciy}=\frac{ax+b+aiy}{ciy}=\frac{a}{c}-\left(\frac{ax+b}{c}\right)\frac{i}{y}.$$

Since $\gamma(x+iy)\in\mathbb{H}$, we must have $(ax+b)/c<0$. Thus we set $t:=\exp\left(2\pi(ax+b)/c\right)<1$. Writing $m=v_{i\infty}(f)$ now gives

$$\left|\frac{f\left(\gamma(x+iy)\right)}{a_m e^{2\pi i\gamma(x+iy)m}}-1\right|\le\sum_{n=m+1}^{\infty}\left|\frac{a_n}{a_m}e^{2\pi i\left(\frac{a}{c}-\frac{ax+b}{cy}i\right)(n-m)}\right|=\sum_{n=m+1}^{\infty}\left|\frac{a_n}{a_m}\right|t^{(n-m)/y}$$

which goes to 0 as y goes to 0. Hence $\lim_{y\to 0}\frac{f(\gamma(x+iy))}{a_m e^{2\pi i\gamma(x+iy)m}}=1$.

By modularity,

$$f(z)=(cz+d)^{-k}f\left(\gamma z\right)$$

and hence

$$\begin{aligned}\lim_{y\to 0}f(x+iy) &= \lim_{y\to 0}(cx+iyc+d)^{-k}f\left(\gamma(x+iy)\right) \\ &= \lim_{y\to 0}(iyc)^{-k}f\left(\gamma(x+iy)\right) \\ &= \lim_{y\to 0}\frac{a_m e^{2\pi i\gamma(x+iy)m}}{(iyc)^k}.\end{aligned}$$

If f is not a cusp form, then $m=0$ and we have

$$\lim_{y\to 0}f(x+iy)=\lim_{y\to 0}\frac{a_0}{(iyc)^k}=\infty.$$

If f is a cusp form, then

$$\left|\lim_{y\to 0}f(x+iy)\right|=\frac{a_m}{|c|^k}\lim_{y\to 0}\frac{t^{m/y}}{y^k}=0$$

by l'Hopital's rule. In either case, since $\mathbb{Q}$ is dense in $\mathbb{R}=\partial\mathbb{H}$ and since meromorphic functions have isolated zeros and poles, there can be no meromorphic continuation of f to a larger domain. □

4.6.7. *Prove that* $E_4^2E_6=E_{14}$.

The dimension formula shows that $M_{14}(SL_2(\mathbb{Z}))$ is one-dimensional and so is spanned by E_{14}. Since $E_4^2E_6$ also has weight 14, we have $E_4^2E_6 = cE_{14}$ for some constant c. A quick verification using Bernoulli numbers and Corollary 4.2.5 gives the result. □

4.6.8. *Let* $r_k = \dim M_k(SL_2(\mathbb{Z}))$. *Put* $\ell = k - 12r_k + 12$ *and* $E_0 = 1$. *Show that*

$$E_\ell E_{14-\ell} = E_{14}$$

for $\ell = 0, 4, 6, 8, 10$ *and* 14.

The result is clear for $\ell = 0$. For $\ell = 4$, we must show that $E_4E_{10} = E_{14}$ which again is checked as in the previous exercise. Alternately, by exercise 4.3.10, $E_4E_6 = E_{10}$ so that what we need to show is $E_4^2E_6 = E_{14}$ which is the previous exercise. For $\ell = 6$, we need to show $E_6E_8 = E_{14}$. But again by exercise 4.3.9, we have $E_4^2 = E_8$ and the question reduces to the previous exercise. Finally, for $\ell = 8, 10$, we are back to the above cases. This completes the proof. □

4.6.9. *Let* f *and* g *be modular forms of weight* k *and* ℓ *respectively, for the full modular group. Show that*

$$kfg' - \ell f'g$$

is a modular form of weight $k + \ell + 2$ *for the full modular group.*

It is easily checked that

$$f'\left(\frac{az+b}{cz+d}\right) = (cz+d)^{k+1}(kcf(z) + (cz+d)f'(z))$$

so that the desired result is easily deduced by direct verification. □

4.6.10. *Let* $j(z) = E_4^3/\Delta$. *Prove that*

$$\frac{dj}{dz} = -2\pi i E_{14}/\Delta.$$

We have

$$dj/dz = 3E_4^2E_4'\Delta^{-1} - \Delta^{-2}E_4^3\Delta'$$

so that

$$\Delta^2 dj/dz = 3E_4^2E_4'\Delta - E_4^3\frac{d}{dz}\left(\frac{E_4^3 - E_6^2}{1728}\right)$$

which equals

$$\frac{E_4^2E_6}{1728}\left(2E_4E_6' - 3E_6E_4'\right).$$

By the previous exercise, the expression in the brackets is a modular form of weight 12. A quick verification shows that it is a cusp form and so must be a suitable multiple of Δ. Comparing coefficients in the q-expansion immediately gives us the result. □

4.6.11. *Let $r_k = \dim M_k(SL_2(\mathbb{Z}))$ and $\ell = k - 12r_k + 12$. For any modular form $f \in M_k(SL_2(\mathbb{Z}))$, show that*

$$fE_\ell^{-1} \in M_{12r_k-12}(SL_2(\mathbb{Z})).$$

Since $f \in M_k(SL_2(\mathbb{Z}))$, it is a linear combination of

$$E_4^a E_6^b$$

such that $4a + 6b = k$. From the previous exercise, $E_\ell^{-1} = E_{14-\ell}/E_{14}$, so that fE_ℓ^{-1} is a linear combination of terms like

$$E_4^a E_6^b E_{14-\ell}/E_{14}. \tag{4.3}$$

Now $E_{14} = E_4^2 E_6$ and $E_{14-\ell}$ is one of $E_0, E_4, E_6, E_8, E_{10}$ or E_{14}. Recalling that $E_8^2 = E_4^2$ and $E_{10} = E_4 E_6$, we see that $E_{14-\ell}/E_{14}$ is one of $1/E_4^2 E_6$, $1/E_4 E_6$, $1/E_4^2$, $1/E_6$, or 1. If $a \geq 2, b \geq 1$, then the term in 4.3 is holomorphic. Thus, we need only consider the five cases

$$(a, b) \in \{(0, 1), (1, 0), (1, 1), (2, 0), (2, 1)\}.$$

But in all these cases, $r_k = 1$ so that $\ell = k \leq 14$ and the result is evident. □

4.6.12. *With notation as in the previous exercise, show that*

$$fE_\ell^{-1}\Delta^{1-r_k}$$

is a modular function and hence a polynomial in j.

By the previous exercise fE_ℓ^{-1} is a modular form of weight $12r_k - 12$ so that $fE_\ell^{-1}\Delta^{1-r_k}$ has weight zero and hence is a modular function. By Theorem 4.5.4, it is a rational function of j. However, since Δ does not vanish, it must be a polynomial in j. □

4.6.13. *Let f be a modular form of weight k for the full modular group whose Fourier expansion at $i\infty$ is given by*

$$f(z) = \sum_{n=0}^{\infty} a_n e^{2\pi i n z}, \qquad a_n \in \mathbb{C}.$$

Recall that by Corollary 4.4.4, there are unique c_{ab}'s such that

$$f = \sum_{4a+6b=k} c_{ab} E_4^a E_6^b, \qquad c_{ab} \in \mathbb{C}.$$

Let F be an algebraic extension of $\mathbb{Q}$. Suppose that

$$\#\{n : a_n \in F\} \geq \dim M_k(SL_2(\mathbb{Z})).$$

Then, show that c_{ab} belong to F, whence all the Fourier coefficients belong to F.

We begin by noting that the Fourier coefficients of E_4 and E_6 are rational (in fact, integral). Inserting the q-expansions of f and $E_4^a E_6^b$ in our expression, and comparing the coefficients of q^n on both sides for exactly $\dim M_k(SL_2(\mathbb{Z}))$ many values of n such that $a_n \in F$, we get a system of linear equations satisfied by the c_{ab} say, $A(c_{ab}) = v$ (with obvious notation), for some matrix A with rational entries and vector v with entries from F. Now the uniqueness (and the existence) of complex numbers c_{ab} imply the invertibility of A. Since A has rational entries, so does A^{-1}. But then, the c_{ab}'s belong to F. □

4.6.14. *If f is a modular form whose Fourier coefficients at $i\infty$ are algebraic numbers, then show that the smallest exension field of the rationals containing all the Fourier coefficients of f is a finite extension of $\mathbb{Q}$.*

Let F be the smallest field over $\mathbb{Q}$ containing the first d_k Fourier coefficients of f, where d_k is the dimension of $M_k(\mathrm{SL}_2(\mathbb{Z}))$ and k is the weight of f. Then, by the previous exercise, all the Fourier coefficients are in F, which is a finite extension of $\mathbb{Q}$ since it was obtained by adjoining finitely many algebraic numbers. □

4.6.15. *Suppose that f is a modular form of weight k whose Fourier expansion at $i\infty$ is given by*

$$f(z) = \sum_{n=0}^{\infty} a_n e^{2\pi i n z}, \qquad a_n \in \overline{\mathbb{Q}}.$$

Show that, for every $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, the series

$$f^{\sigma}(z) := \sum_{n=0}^{\infty} \sigma(a_n) e^{2\pi i n z},$$

defines a modular form of weight k.

Using Corollary 4.4.4 and the earlier exercises, there are unique algebraic numbers c_{ab} such that

$$f = \sum_{4a+6b=k} c_{ab} E_4^a E_6^b.$$

Since σ fixes the rationals pointwise, it follows that

$$f^{\sigma} = \sum_{4a+6b=k} \sigma(c_{ab}) E_4^a E_6^b$$

because E_4 and E_6 have rational Fourier coefficients. Thus, f^{σ} is a modular form of weight k. □

Chapter 5
The Ramanujan τ-function

5.1 The Eisenstein series E_2

5.1.2. *Show that $E_2(z)$ has the Fourier expansion at $i\infty$ given by*

$$E_2(z) = 1 - 24\sum_{n=1}^{\infty}\sigma_1(n)q^n,$$

where $q = e^{2\pi iz}$ and $\sigma_1(n) = \sum_{d|n} d$.

We first write

$$E_2(z) = 1 + \frac{1}{\zeta(z)}\sum_{m=1}^{\infty}\sum_{n=-\infty}^{\infty}\frac{1}{(mz+n)^2}.$$

Using the Lipschitz formula (Theorem 4.2.2) with $k = 2$ for the inner infinite series, we get

$$E_2(z) = 1 + \frac{(-2\pi i)^2}{\zeta(2)}\sum_{m=1}^{\infty}\sum_{d=1}^{\infty} dq^{md}.$$

As the double series is absolutely convergent, we may group together terms with $md = n$ to get

$$E_2(z) = 1 - \frac{4\pi^2}{\zeta(2)}\sum_{n=1}^{\infty}\sigma_1(n)q^n.$$

Finally, using the fact $\zeta(2) = \pi^2/6$ gives the result. □

5.1.3. *Show that for all $z \in \mathbb{H}$,*

$$E_2\left(-\frac{1}{z}\right) = z^2E_2(z) + \frac{6}{\pi i}z.$$

© Springer Science+Business Media Singapore 2016 and Hindustan Book Agency 2016

M.R. Murty et al., *Problems in the Theory of Modular Forms*, HBA Lecture Notes in Mathematics, DOI 10.1007/978-981-10-2651-5_15

Since $G_2 = 2\zeta(2)E_2$, this is immediate from Theorem 5.1.1. □

5.1.5. *For $z \in \mathbb{H}$, we define the* Dedekind η-function *by*

$$\eta(z) = e^{2\pi iz/24} \prod_{n=1}^{\infty} (1 - e^{2\pi inz}).$$

Show that

$$\eta\left(-\frac{1}{z}\right) = \sqrt{\frac{z}{i}}\eta(z),$$

where the square root is the branch with non-negative real part.

The product defining $\eta(z)$ clearly converges to a non-zero value for any $z \in \mathbb{H}$ and thus defines a holomorphic function on $\mathbb{H}$. Taking the logarithmic derivative, we find

$$\frac{\eta'(z)}{\eta(z)} = \frac{\pi i}{12}\left(1 - 24\sum_{n=1}^{\infty} \frac{nq^n}{1-q^n}\right),$$

where $q = e^{2\pi iz}$. As in the proof of Theorem 5.1.4, the right hand side is easily seen to be $\pi i E_2(z)/12$. Therefore

$$\frac{\eta'(-1/z)}{\eta(-1/z)} = \frac{\pi i}{12} E_2\left(-\frac{1}{z}\right) = \frac{\pi i}{12}\left(z^2 E_2(z) + \frac{6z}{\pi i}\right).$$

In other words

$$\frac{\eta'(-1/z)}{\eta(-1/z)} \cdot \frac{1}{z^2} = \frac{\pi i E_2(z)}{12} + \frac{1}{2z}.$$

so that

$$\frac{\eta'(-1/z)}{\eta(-1/z)} \cdot \frac{1}{z^2} = \frac{\eta'(z)}{\eta(z)} + \frac{1}{2z}.$$

But the right hand side is clearly the logarithmic derivative of $\sqrt{z}\eta(z)$. Therefore, $\eta(-1/z) = \lambda\sqrt{z}\eta(z)$ for some constant λ. Putting $z = i$ shows that $\lambda = 1/\sqrt{i}$. Finally, since $z \in \mathbb{H}$, it is clear that $\frac{\sqrt{z}}{\sqrt{i}} = \sqrt{\frac{z}{i}}$. □

5.1.6. *Show that*

$$\frac{\Delta'(z)}{\Delta(z)} = 2\pi i E_2(z).$$

Since

$$\Delta(z) = e^{2\pi iz} \prod_{n=1}^{\infty} (1 - e^{2\pi inz})^{24},$$

we take the logarithmic derivative of both sides and get

$$\frac{\Delta'(z)}{\Delta(z)} = 2\pi i\left(1 - 24\sum_{n=1}^{\infty} \frac{ne^{2\pi inz}}{1 - e^{2\pi inz}}\right).$$

Putting $q = e^{2\pi iz}$ and expanding

$$\sum_{n=1}^{\infty} \frac{nq^n}{1-q^n}$$

as in the proof of Theorem 5.1.4 gives us the result. □

5.1.7. *Prove that for all* $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$ *and all* $z \in \mathbb{H}$*, we have*

$$E_2\left(\frac{az+b}{cz+d}\right) = (cz+d)^2 E_2(z) + \frac{6c(cz+d)}{\pi i}.$$

Recall that for any modular form f in $M_k(\mathrm{SL}_2(\mathbb{Z}))$, we have

$$f\left(\frac{az+b}{cz+d}\right) = (cz+d)^k f(z) \qquad \forall \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}).$$

Differentiating this with respect to z gives

$$f'\left(\frac{az+b}{cz+d}\right)\frac{d}{dz}\left(\frac{az+b}{cz+d}\right) = (cz+d)^k f'(z) + k(cz+d)^{k-1} c f(z).$$

Now

$$\frac{d}{dz}\left(\frac{az+b}{cz+d}\right) = (cz+d)^{-2}$$

so that

$$f'\left(\frac{az+b}{cz+d}\right) = (cz+d)^{k+2} f'(z) + k(cz+d)^{k+1} c f(z).$$

We apply this to $\Delta \in M_{12}(\mathrm{SL}_2(\mathbb{Z}))$ to get

$$\Delta'\left(\frac{az+b}{cz+d}\right) = (cz+d)^{14}\Delta'(z) + 12c(cz+d)^{13}\Delta(z).$$

Dividing through by

$$\Delta\left(\frac{az+b}{cz+d}\right) = (cz+d)^{12}\Delta(z),$$

we obtain using Exercise 5.1.6,

$$2\pi i E_2\left(\frac{az+b}{cz+d}\right) = (cz+d)^2 2\pi i E_2(z) + 12c(cz+d),$$

which is the desired result. □

5.1.8. *The* Serre derivative *of a modular form* $f \in M_k(\mathrm{SL}_2(\mathbb{Z}))$ *is defined as*

$$\theta_k f := \frac{1}{2\pi i} f' - \frac{k}{12} E_2 f.$$

Prove that $\theta_k : M_k(\mathrm{SL}_2(\mathbb{Z})) \to M_{k+2}(\mathrm{SL}_2(\mathbb{Z}))$.

First, we must check that

$$(\theta_k f)\left(\frac{az+b}{cz+d}\right) = (cz+d)^{k+2}(\theta_k f)(z) \qquad \forall \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}).$$

Since $f \in M_k(\mathrm{SL}_2(\mathbb{Z}))$, for any $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$, we have

$$f\left(\frac{az+b}{cz+d}\right) = (cz+d)^k f(z).$$

Now from Exercise 5.1.7 and its solution,

$$E_2\left(\frac{az+b}{cz+d}\right) = (cz+d)^2 E_2(z) + \frac{6c(cz+d)}{\pi i},$$

and

$$f'\left(\frac{az+b}{cz+d}\right) = (cz+d)^{k+2} f'(z) + kc(cz+d)^{k+1} f(z).$$

Putting these together, we find

$$\begin{aligned}(\theta_k f)\left(\frac{az+b}{cz+d}\right) &= \frac{1}{2\pi i} f'\left(\frac{az+b}{cz+d}\right) - \frac{k}{12} E_2\left(\frac{az+b}{cz+d}\right) f\left(\frac{az+b}{cz+d}\right) \\ &= (cz+d)^{k+2}\left[\frac{1}{2\pi i} f'(z) - \frac{k}{12} E_2(z) f(z)\right] \\ &= (cz+d)^{k+2}(\theta_k f)(z)\end{aligned}$$

as required.

It remains to check that $\theta_k f$ is holomorphic at the cusp $i\infty$. In a neighborhood of $i\infty$,

$$f(z) = \sum_{n=0}^{\infty} a_n e^{2\pi i z n}$$

and so

$$\frac{1}{2\pi i} f'(z) = \sum_{n=0}^{\infty} a_n n e^{2\pi i z n}.$$

By Exercise 5.1.2, $E_2(z)$ also has a Fourier expansion at $i\infty$, and putting the expansions together easily shows that $\theta_k f$ is holomorphic at $i\infty$. □

5.2 Hecke operators of level one

5.2.3. *Show that*

$$\sum_{b=0}^{d-1} e^{2\pi i n b/d} = \begin{cases} d & \text{if } d \mid n \\ 0 & \text{otherwise} \end{cases}$$

If $d|n$, the result is clear. Otherwise the sum in question is a geometric sum

$$\sum_{b=0}^{d-1} r^b, \qquad r = e^{2\pi i n/d} \neq 1,$$

which is easily summed to be $(r^d - 1)/(r - 1) = 0$. □

5.2.5. *Prove that* $T_m(\Delta) = \tau(m)\Delta$. *Deduce that* $\tau(mn) = \tau(m)\tau(n)$ *for* $(m, n) = 1$.

By Theorem 5.2.2, $T_m(\Delta)$ is again a cusp form of weight 12. Since $S_{12}(\mathrm{SL}_2(\mathbb{Z}))$ is one-dimensional, and spanned by Δ, we must have $T_m(\Delta) = \lambda_m \Delta$ for some constant λ_m. On the other hand, by Theorem 5.2.4, the n-th coefficient in the q-expansion of $T_m(\Delta)$ is

$$\sum_{d|m,n} d^{11} \tau\left(\frac{mn}{d^2}\right) = \tau(mn)$$

whenever $(m, n) = 1$. But $T_m(\Delta) = \lambda_m \Delta$ and the right hand side of the equation shows that the n-th coefficient is $\lambda_m \tau(n)$. Thus $\tau(mn) = \lambda_m \tau(n)$. Putting $n = 1$ shows $\tau(m) = \lambda_m$ from which the result follows. □

5.2.6. *Prove that for any* $m, n \geq 1$,

$$\tau(m)\tau(n) = \sum_{d|m,n} d^{11} \tau\left(\frac{mn}{d^2}\right).$$

Again, comparing the n-th coefficient of both sides of the equation $T_m(\Delta) = \tau(m)\Delta$, we get

$$\tau(m)\tau(n) = \sum_{d|m,n} d^{11} \tau\left(\frac{mn}{d^2}\right),$$

by an application of Theorem 5.2.4. □

5.2.7. *Let* p *be prime and* $a \geq 1$. *Show that* $\tau(p^{a+1}) = \tau(p)\tau(p^a) - p^{11}\tau(p^{a-1})$.

We apply Exercise 5.2.6 with $m = p$ and $n = p^a$ to get

$$\tau(p)\tau(p^a) = \tau(p^{a+1}) + p^{11}\tau(p^{a-1}),$$

as required. □

5.2.8. *Let f be $E_4\Delta$, $E_6\Delta$, $E_8\Delta$, $E_{10}\Delta$, or $E_{14}\Delta$. Show that the Fourier coefficients of f are multiplicative.*

The weight of f is, respectively, 16, 18, 20, 22 or 26. As the space of cusp forms is one-dimensional in each of these cases, it follows that $T_m(f) = \lambda_m f$ for some constant λ_m. By Theorem 5.2.4, upon writing

$$f(z) = \sum_{n=1}^{\infty} a_f(n) q^n,$$

we have that the n-th Fourier coefficent of $T_m(f)$ is

$$\sum_{d|n,m} d^{k-1} a_f\left(\frac{mn}{d^2}\right),$$

where $k = 16$, 18, 20, 22, or 26, respectively. If $(m, n) = 1$, the sum becomes $a_f(mn)$. Hence $a_f(mn) = \lambda_m a_f(n)$, whenever $(m, n) = 1$. Choosing $n = 1$, we deduce $a_f(m) = \lambda_m$. The result now follows. □

5.2.9. *Let p be a prime number. Show that*

$$\sum_{n=0}^{\infty} \tau(p^n) x^n = \frac{1}{1 - \tau(p)x + p^{11}x^2}.$$

Equivalently, we must prove that

$$\left(1 - \tau(p)x + p^{11}x^2\right) \sum_{n=0}^{\infty} \tau(p^n) x^n = 1.$$

For $n \geq 2$, the coefficient of x^n in the product on the left hand side is $\tau(p^n) - \tau(p)\tau(p^{n-1}) + p^{11}\tau(p^{n-2})$ which is zero by Exercise 5.2.7. For $n = 1$, the coefficient of x is easily checked to be zero also, and as the coefficient of x^0 is one, we are done. □

5.2.10. *Prove that for $n > 1$,*

$$(1 - n)\tau(n) = 24 \sum_{j=1}^{n-1} \sigma_1(j)\tau(n - j).$$

By Exercise 5.1.6,

$$\frac{1}{2\pi i}\Delta'(z) = \Delta(z)E_2(z)$$

In other words,

$$\sum_{n=1}^{\infty} n\tau(n)q^n = \left(\sum_{n=1}^{\infty} \tau(n)q^n\right)\left(1 - 24\sum_{n=1}^{\infty} \sigma_1(n)q^n\right).$$

The result now follows upon comparing coefficients of q^n of both sides of the equation. □

5.2.11. *If n is even, prove that $8|\tau(n)$.*

This is evident from the previous exercise. □

5.2.12. *If p is prime, show that $\tau(p)$ is even.*

This follows from Exercise 1.5.1 where it was shown that $\tau(n)$ is odd if and only if n is an odd square. □

5.3 Modular forms and Dirichlet series

5.3.1. *Let p be prime. Prove that Ramanujan's conjecture*

$$|\tau(p)| \leq 2p^{11/2}$$

is equivalent to the assertion that the roots of the quadratic polynomial $1 - \tau(p)x + p^{11}x^2$ are non-real.

The discriminant of the quadratic is $\tau(p)^2 - 4p^{11}$ and this is < 0 if and only if $|\tau(p)| < 2p^{11/2}$. We have strict inequality here since $\tau(p)$ is an integer and $p^{11/2}$ is irrational. □

5.3.2. *Assuming Ramanujan's conjecture that $|\tau(p)| \leq 2p^{11/2}$, prove that $|\tau(n)| \leq n^{11/2}d(n)$ for every natural number n. Here $d(n) = \sigma_0(n)$ is the number of divisors of n.*

From Exercise 5.2.9, we have

$$\sum_{m=0}^{\infty} \tau(p^m)x^m = (1 - \tau(p)x + p^{11}x^2)^{-1}.$$

Writing $1 - \tau(p)x + p^{11}x^2 = (1 - \alpha_p x)(1 - \beta_p x)$, we see from Exercise 5.3.1 that Ramanujan's conjecture is equivalent to $|\alpha_p| = |\beta_p| = p^{11/2}$. Thus, we may write $\alpha_p = p^{11/2}e^{i\theta_p}$ and $\beta_p = p^{11/2}e^{-i\theta_p}$ so that we have

$$\sum_{m=0}^{\infty} \tau(p^m)x^m = \frac{1}{(\alpha_p - \beta_p)x}\left(\frac{1}{1 - \alpha_p x} - \frac{1}{1 - \beta_p x}\right).$$

Expanding both geometric series, we deduce that the mth coefficient is

$$\tau(p^m) = \frac{\alpha_p^{m+1} - \beta_p^{m+1}}{\alpha_p - \beta_p} = p^{\frac{11m}{2}} \frac{e^{i(m+1)\theta_p} - e^{-i(m+1)\theta_p}}{e^{i\theta_p} - e^{-i\theta_p}} = p^{\frac{11m}{2}} \frac{\sin(m+1)\theta_p}{\sin\theta_p}.$$

Now

$$\frac{\sin(m+1)\theta}{\sin\theta} = \frac{e^{i(m+1)\theta} - e^{-i(m+1)\theta}}{e^{i\theta} - e^{-i\theta}} = e^{im\theta} + e^{i(m-2)\theta} + \cdots + e^{-i(m-2)\theta} + e^{-im\theta}$$

so that

$$\left|\frac{\sin(m+1)\theta}{\sin\theta}\right| \leq (m+1).$$

Consequently, as $\tau(n)$ is multiplicative, we obtain

$$|\tau(n)| = \left|\prod_{p^\alpha || n} \tau(p^\alpha)\right| \leq n^{11/2} d(n). \qquad \square$$

5.3.3. *Let $f \in M_k(\mathrm{SL}_2(\mathbb{Z}))$. Show that $y^{k/2}|f(z)|$ is invariant under $\mathrm{SL}_2(\mathbb{Z})$, where $y = \mathrm{Im}(z)$.*

We have

$$f\left(\frac{az+b}{cz+d}\right) = (cz+d)^k f(z)$$

and

$$\mathrm{Im}\left(\frac{az+b}{cz+d}\right) = \frac{\mathrm{Im}(z)}{|cz+d|^2}$$

from which the result follows. $\square$

5.3.8. *Suppose that $f \in M_k(\mathrm{SL}_2(\mathbb{Z}))$ with Fourier expansion at $i\infty$ given by*

$$f(z) = \sum_{n=0}^{\infty} a_f(n) q^n,$$

and that $a_f(0) \neq 0$. Show that $L_f(s) = \sum_{n=1}^{\infty} \frac{a_f(n)}{n^s}$ has an analytic continuation to the entire complex plane, except for a simple pole at $s = 0$ and $s = k$ and satisfies the functional equation

$$(2\pi)^{-s}\Gamma(s)L_f(s) = i^k(2\pi)^{-(k-s)}\Gamma(k-s)L_f(k-s).$$

Put $a_0 = a_f(0)$. Let us note that $f(iy) - a_0$ has exponential decay and we can consider

$$\int_0^\infty (f(iy) - a_0) y^{s-1} dy = (2\pi)^{-s}\Gamma(s)L_f(s)$$

as in the proof of Theorem 5.3.7. We split the integral into two parts:

$$\int_0^1 (f(iy) - a_0) y^{s-1} dy + \int_1^\infty (f(iy) - a_0) y^{s-1} dy.$$

In the first integral, we change variables $y \mapsto 1/y$. Using the modular relation, we get

$$\int_1^\infty \{i^k y^k (f(iy) - a_0) + i^k a_0 y^k - a_0\} y^{-s-1} dy.$$

The integrals for $\mathrm{Re}(s) > k$, the integrals

$$\int_1^\infty y^{k-s} \frac{dy}{y} \text{ and } \int_1^\infty y^{-s} \frac{dy}{y}$$

are easily evaluated as $\frac{1}{s-k}$ and $\frac{1}{s}$, respectively. Therefore, for $\mathrm{Re}(s) > k$,

$$(2\pi)^{-s}\Gamma(s)L_f(s) = \frac{i^k a_0}{s-k} - \frac{a_0}{s} + \int_1^\infty (f(iy) - a_0)(y^s + y^{k-s} i^k) \frac{dy}{y},$$

from which the analytic continuation and functional equation are now immediate. □

5.4 Ramanujan congruences

5.4.1. *Prove that* $\tau(n) \equiv n\sigma_1(n) \pmod 5$.

Using the notation of Theorem 5.1.9, we have

$$1728 \sum_{n=1}^\infty \tau(n) q^n = Q^3 - R^2 \equiv Q - R^2 \pmod 5$$

since $Q \equiv 1 \pmod 5$. Now, $R \equiv P \pmod 5$ as $d^5 \equiv d \pmod 5$ so we obtain

$$3 \sum_{n=1}^\infty \tau(n) q^n \equiv Q - P^2 \pmod 5.$$

From Theorem 5.1.9, $Q - P^2 = -12DP$ so that

$$\sum_{n=1}^\infty \tau(n) q^n \equiv DP \pmod 5.$$

As $DP = -24 \sum_{n=1}^\infty n\sigma(n) q^n$, the result follows. □

5.4.3. *Prove that* $Q^2 \equiv P \pmod 7$ *and* $R \equiv 1 \pmod 7$.

The second congruence is obvious from the q-expansion of R. For the first, we note that $Q^2 = E_8$ and

$$E_8 = 1 + 480 \sum_{n=1}^{\infty} \frac{n^7 q^n}{1-q^n}.$$

Since $n^7 \equiv n \pmod 7$ by Fermat's little theorem, we conclude that

$$Q^2 = E_8 \equiv P \pmod 7. \qquad \square$$

5.4.4. *Prove that*

$$(Q^3 - R^2)^2 \equiv (QP-1)^2 \equiv -q\frac{dP^2}{dq} - 3q\frac{dQ}{dq} \pmod 7.$$

By the previous exercise,

$$(Q^3 - R^2)^2 \equiv (QP-1)^2 \pmod 7.$$

Now

$$\begin{aligned}(QP-1)^2 &\equiv Q^2P^2 - 2PQ + 1 \pmod 7\\ &\equiv P^3 - 2PQ + R \pmod 7\end{aligned}$$

by Exercise 5.4.3. Hence

$$(QP-1)^2 \equiv P(P^2 - Q) - PQ + R \pmod 7.$$

By Theorem 5.1.9, $P^2 - Q = 12DP$ and $PQ - R = 3DQ$. Thus

$$(Q^3 - R^2)^2 \equiv P(12DP) - 3DQ \pmod 7.$$

Now $q\frac{dP^2}{dq} = 2Pq\frac{dP}{dq}$ so we may rewrite this as

$$\begin{aligned}(Q^3 - R^2)^2 &\equiv 6q\frac{dP^2}{dq} - 3q\frac{dQ}{dq} \pmod 7\\ &\equiv -q\frac{dP^2}{dq} - 3q\frac{dQ}{dq} \pmod 7,\end{aligned}$$

as required. $\square$

5.4.5. *Prove that*

$$(Q^3 - R^2)^2 \equiv q^2 \left(\sum_{n=0}^{\infty} p(n)q^n\right) \prod_{n=1}^{\infty}(1-q^{49n}) \pmod 7.$$

Deduce that $p(7n+5) \equiv 0 \pmod 7$.

By Theorem 5.1.4,

$$\left(\frac{Q^3-R^2}{1728}\right)^2 = q^2\prod_{n=1}^{\infty}(1-q^n)^{48} = q^2\prod_{n=1}^{\infty}\frac{(1-q^n)^{49}}{1-q^n}.$$

Hence we obtain

$$(Q^3-R^2)^2 \equiv q^2\left(\sum_{n=0}^{\infty}p(n)q^n\right)\prod_{n=1}^{\infty}(1-q^{49n}) \pmod 7,$$

upon reducing modulo 7. Combining this with Exercise 5.4.4, we get

$$q^2\sum_{n=0}^{\infty}p(n)q^n \equiv \prod_{n=1}^{\infty}(1-q^{49n})^{-1}\left(-q\frac{dP^2}{dq}-3q\frac{dQ}{dq}\right) \pmod 7. \tag{5.1}$$

Noting that for any q-series J with integer coefficients, the nth coefficient of $q\frac{dJ}{dq}$ is always divisible by n, we need only compare coefficients of q^{7n} of both sides of the congruence (5.1) to deduce the result. □

5.5 Supplementary problems

5.5.1. *Prove that* $E_6^2 = E_{12} - \frac{762048}{691}\Delta$.

Since E_6^2 has weight 12, we can write it as $\alpha E_{12}+\beta\Delta$ because $M_{12}(\mathrm{SL}_2(\mathbb{Z}))$ is spanned by E_{12} and Δ. Computing the coefficients of q^0 and q^1 leads to the result. □

5.5.2. *Show that* $E_{12} = 1 + \frac{65520}{691}\sum_{n=1}^{\infty}\sigma_{11}(n)q^n$.

By Corollary 4.2.7, we need only compute the Bernoulli number B_{12}, which is easily done by directly computing it from the generating function or using the well-known recurrence $B_0 = 1$, $B_m = -\sum_{k=0}^{m-1}\binom{m}{k}\frac{B_k}{m-k+1}$. □

5.5.3. *Prove that* $\tau(n) \equiv \sigma_{11}(n) \pmod{691}$.

By Exercises 5.5.1 and 5.5.2, we find

$$65520\sigma_{11}(n) \equiv 762048\tau(n) \pmod{691},$$

from which the result immediately follows. □

5.5.4. *Prove that* $E_{14} \equiv E_2 \pmod{13}$.

A quick computation of the Bernoulli number B_{14} shows that $E_{14} \equiv 1 - 24\sum_{n=1}^{\infty}\sigma_{13}(n)q^n \pmod{13}$. Since $d^{13} \equiv d \pmod{13}$, the result follows. □

5.5.5. *Prove that* $\tau(n) \equiv n\sigma_3(n) \pmod 7$.

We have $Q^3 - R^2 = 1728\Delta \equiv -\Delta \pmod 7$. Now $Q^2 \equiv P \pmod 7$ and $R \equiv 1 \pmod 7$ by Exercise 5.4.3. Thus $QP - R \equiv -\Delta \pmod 7$. By Theorem 5.1.9, $3DQ = QP - R$. Thus $-3DQ \equiv \Delta \pmod 7$. Now

$$Q = 1 + 240 \sum_{n=1}^{\infty} \sigma_3(n) q^n,$$

so that $DQ = 240 \sum_{n=1}^{\infty} n\sigma_3(n)q^n$, from which the result follows. □

5.5.6. *Prove that*

$$\sum_{j=1}^{n-1} \sigma_5(j)\sigma_5(n-j) = \frac{65}{(252)(691)}\sigma_{11}(n) + \frac{\sigma_5(n)}{252} - \frac{3\tau(n)}{691}.$$

Deduce that

$$\sum_{j=1}^{n-1} \sigma_5(j)\sigma_5(n-j) = \frac{65}{(252)(691)}\sigma_{11}(n) + O\big(n^6\big).$$

This is immediate from Exercise 5.5.1 on comparing coefficients of q^n of both sides of the equation. The error term arises from using the estimate $\tau(n) = \mathcal{O}(n^6)$ and $\sigma_5(n) = n^{5+\epsilon}$. The error term can be sharpened using Deligne's result to $\mathcal{O}(n^{11/2+\epsilon})$. □

5.5.7. *If $k_1 > k_2 \geq 4$ are even, prove that for any $\epsilon > 0$,*

$$\sum_{j=1}^{n-1} \sigma_{k_1-1}(j)\sigma_{k_2-1}(n-j) = c(k_1,k_2)\sigma_{k_1+k_2-1}(n) + O\big(n^{k_1-1+\epsilon}\big),$$

where $c(k_1, k_2) \in \mathbb{Q}$ is a constant depending on k_1 and k_2.

If we consider the form $E_{k_1}E_{k_2}$ of weight $k_1 + k_2$, we may write it as $c_1E_{k_1+k_2} + c_2 f$, where f is a cusp form of weight $k_1 + k_2$. The result follows upon comparing coefficients of q^n in the q-expansions of both sides and using the estimate of Hecke given in Theorem 5.3.4 and the estimate $\sigma_{k-1} = O\big(n^{k-1+\epsilon}\big)$. If $k_1 = k_2 \geq 4$ is even, then the error term is $O\big(n^{k_1}\big)$, as in Exercise 5.5.6 above. □

5.5.8. *Let f_k be a sequence of complex numbers satisfying the recurrence with $f_0 = 1$, and*

$$f_{k+1} = ff_k - gf_{k-1}, \qquad k \geq 1$$

for some constants f and g. Write

$$1 - fx + gx^2 = (1 - \alpha x)(1 - \beta x).$$

Show that

$$f_k = \frac{\alpha^{k+1} - \beta^{k+1}}{\alpha - \beta}.$$

If $\alpha = \beta$, the right hand side is viewed as $\alpha^k + \alpha^{k-1}\beta + \cdots + \alpha\beta^{k-1} + \beta^k = (k+1)\alpha^k$.

The recurrence for f_k implies that we have the equality of power series

$$\sum_{k=0}^{\infty} f_k x^k = \left(1 - fx - gx^2\right)^{-1}.$$

By partial fractions, we can write this as

$$\frac{1}{x(\alpha - \beta)} \left(\frac{1}{1 - \alpha x} - \frac{1}{1 - \beta x} \right).$$

Expanding the right hand side as a power series gives the desired result. □

5.5.9. *Let g be a completely multiplicative function. An arithmetical function f is called* g-multiplicative *if the equation*

$$f(m)f(n) = \sum_{d|m,n} g(d) f\left(\frac{mn}{d^2}\right)$$

holds for all natural numbers m and n. Show that f is g-multiplicative if and only if f is multiplicative and for all primes p and all natural numbers $k \geq 1$, we have

$$f\left(p^{k+1}\right) = f(p)f(p^k) - g(p)f(p^{k-1}).$$

If f is g-multiplicative, then setting $m = p$ and $n = p^k$ in (5.8) from page 70 gives

$$f(p)f(p^k) = f(p^{k+1}) + g(p)f(p^{k-1})$$

which is (5.9) from page 70. For the converse direction, it suffices to observe that we need only establish (5.8) for m, n powers of a fixed prime since f is a multiplicative function. We write

$$1 - f(p)x + g(p)x^2 = (1 - \alpha_p x)(1 - \beta_p x)$$

so that by an application of the previous exercise, we obtain

$$f\left(p^k\right) = \frac{\alpha_p^{k+1} - \beta_p^{k+1}}{\alpha_p - \beta_p}. \tag{5.2}$$

Now since g is completely multiplicative, (5.8) on page 70 reduces to showing

$$f(p^j)f(p^k) = \sum_{i=0}^{\min(j,k)} g(p)^i f(p^{j+k-2i}).$$

From (5.2), we have to show for $j \leq k$,

$$\frac{\alpha_p^{j+1} - \beta_p^{j+1}}{\alpha_p - \beta_p} \cdot \frac{\alpha_p^{k+1} - \beta_p^{k+1}}{\alpha_p - \beta_p} = \sum_{i=0}^{j} (\alpha_p\beta_p)^i \left(\frac{\alpha_p^{j+k-2i+1} - \beta_p^{j+k-2i+1}}{\alpha_p - \beta_p} \right)$$

since $g(p) = \alpha_p\beta_p$. The right hand side of the above equation is easily summed as

$$\frac{\alpha_p^{k+1}}{\alpha_p - \beta_p} \sum_{i=0}^{j} \alpha_p^{j-i}\beta_p^i - \frac{\beta_p^{k+1}}{\alpha_p - \beta_p} \sum_{i=0}^{j} \alpha_p^i \beta_p^{j-i}$$

from which the desired result is immediate. □

5.5.10. *Let $g_k(n) = n^k$. Show that the Ramanujan τ-function is g_{11}-multiplicative.*

This is immediate from the identity

$$\tau(m)\tau(n) = \sum_{d|m,n} d^{11}\tau\left(\frac{mn}{d^2}\right)$$

established in Exercise 5.2.6.

5.5.11. *Let $g_s(n) = n^s$. Show that $\sigma_s(n) := \sum_{d|n} d^s$ is g_s-multiplicative.*

By Exercise 5.5.9, it suffices to show that for primes p,

$$\sigma_s\left(p^{k+1}\right) = \sigma_s(p)\sigma_s\left(p^k\right) - p^s\sigma_s\left(p^{k-1}\right)$$

for $k \geq 1$. But this is easily verified upon noting that

$$\sigma_s\left(p^\alpha\right) = 1 + p^s + \cdots + p^{s\alpha} = \frac{p^{s(\alpha+1)} - 1}{p^s - 1}.$$

5.5.12. *Let F and G be two complex-valued functions defined on $\mathbb{N} \times \mathbb{N}$. Suppose that*

$$F(m,n) = \sum_{d|n} G(m/d, n/d).$$

Show that

$$G(m,n) = \sum_{d|m,n} \mu(d)F(m/d, n/d)$$

and conversely. (Here μ denotes the Möbius function.)

This is easily deduced using the Möbius inversion formula or by the defining property of the Möbius function:

$$\sum_{d|n} \mu(d) = 0$$

unless $n = 1$, in which case it is equal to 1. Thus,

$$\sum_{d|m,n} \mu(d)F(m/d, n/d) = \sum_{d|m,n} \mu(d) \sum_{e|m/d,n/d} G(m/de, n/de)$$

which is equal to (on setting $r = de$):

$$\sum_{r|m,n} G(m/r, n/r) \sum_{de=r} \mu(d) = G(m, n),$$

as desired. The converse is similar.

5.5.13. *Show that the Ramanujan τ-function satisfies*

$$\tau(mn) = \sum_{d|m,n} d^{11}\mu(d)\tau(m/d)\tau(n/d)$$

for any two natural numbers m, n.

Since

$$\frac{\tau(m)}{m^{11/2}} \frac{\tau(n)}{n^{11/2}} = \sum_{d|m,n} \tau(mn/d^2) \frac{d^{11}}{(mn)^{11/2}},$$

we can apply the previous exercise to the functions $F(m, n) = \tau(m)\tau(n)/(mn)^{11/2}$ and $G(m, n) = \tau(mn)/(mn)^{11/2}$ to deduce the result. □

5.5.14. *Let A be the $n \times n$ matrix whose (i, j) entry is $\tau(ij)$. Show that the determinant of A is given by $n!^{11}\mu(1)\mu(2)\cdots\mu(n)$. In particular, the determinant is zero if $n \geq 4$.*

By Exercise 5.5.13, we can write our matrix A as a product LDL^t, where D is a diagonal matrix whose i-th diagonal entry is $i^{11}\mu(i)$ and L is the lower triangular matrix whose (i, j) entry is given by $\tau(j/i)$ is $i|j$ and zero otherwise. The result is now immediate since the determinant of U is 1. □

5.5.15. *Show that $\tau(n)$ is odd if and only if n is an odd square.*

We have

$$\sum_{n=1}^{\infty} \tau(n)q^n = q \prod_{n=1}^{\infty} (1 - q^n)^{24}$$

which modulo 2 is congruent to

$$q \prod_{n=1}^{\infty} (1 - q^{8n})^3.$$

By Jacobi's identity (Theorem 1.2.3), this is congruent (modulo 2) to

$$q\sum_{n=0}^{\infty} q^{4n^2+4n} = \sum_{n=0}^{\infty} q^{(2n+1)^2}.$$

□

5.5.16. *Let $r = \dim M_k(SL_2(\mathbb{Z}))$. Put*

$$F_k := E_{12r-k+2}\Delta^{-r}.$$

Show that F_k has q-expansion

$$C_{kr}q^{-r} + \cdots + C_{k1}q^{-1} + C_{k0} + \cdots$$

with $C_{kr} = 1$ and $C_{kj} \in \mathbb{Z}$ for all j.

Since

$$\Delta^{-1} = q^{-1}\prod_{n=1}^{\infty}(1+q^n+q^{2n}+\cdots)^{24},$$

the Fourier coefficients in the q-expansion of any integer power of Δ are integers. Writing $k = 12t + \delta$ with $0 \leq \delta \leq 10$, we have by the dimension formula that $r = t + 1$ or t according as $\delta \not\equiv 2 (\mathrm{mod}\, 12)$ or $\delta \equiv 2 (\mathrm{mod}\, 12)$ respectively. Thus, $12r - k + 2 = 14 - \delta$ or $2 - \delta$ according as $\delta \not\equiv 2 (\mathrm{mod}\, 12)$ or $\delta \equiv 2 (\mathrm{mod}\, 12)$. The Eisenstein series $E_{12r-k+2}$ has accordingly integer coefficients. Thus, the product $E_{12r-k+2}\Delta^{-r}$ also has integer coefficients. Moreover, it is easily checked that the leading coefficient is 1. □

5.5.17. *Let j denote the j-function. Show that for any integer $m \geq 0$,*

$$j^m\frac{dj}{dz}$$

has a q-expansion without a constant term.

For $m = 0$, this is clear. But since

$$j(z) = \frac{1}{q} + \cdots,$$

and

$$j^m\frac{dj}{dz} = \frac{1}{m+1}\frac{dj^{m+1}}{dz},$$

the right hand side has no constant term from which the result is immediate. □

5.5.18. *Let f be a modular form of weight k for the full modular group with q-expansion*

$$f(z) = a_0 + a_1 q + a_2 q^2 + \cdots$$

With C_{kj} defined as in Exercise 5.5.16, show that

$$C_{k0}a_0 + C_{k1}a_1 + \cdots + C_{kr}a_r = 0.$$

By Exercise 5.5.16, the q-expansion of

$$F_k = E_{12r-k+2}\Delta^{-r}$$

has integer coefficients. On the other hand, by Exercise 4.6.12,

$$W := fE_\ell^{-1}\Delta^{1-r}, \qquad \ell = k - 12r + 12,$$

is a modular function (without poles except at infinity) and hence a polynomial in j. Moreover,

$$W\frac{dj}{dz}$$

has a q-expansion with zero constant term (by Exercise 5.5.17. By Exercise 4.6.10,

$$\frac{dj}{dz} = -2\pi i E_{14}/\Delta$$

so that

$$-\frac{1}{2\pi i}E\frac{dj}{dz} = fE_\ell^{-1}\Delta^{1-r}E_{14}/\Delta.$$

From Exercise 4.6.8, $E_\ell E_{14-\ell} = E_{14}$ so that

$$-\frac{1}{2\pi i}W\frac{dj}{dz} = fE_{14-\ell}\Delta^{-r} = fF_k$$

(using Exercise 5.5.16). Comparing the constant term of both sides gives us the desired result. □

5.5.19. *Suppose that $k \equiv 2 \pmod 4$. With notation as in the previous exercise, show that $C_{k0} \neq 0$.*

C_{k0} is the constant term of $F_k = E_{12r-k+2}\Delta^{-r}$ with $r = \dim M_k(SL_2(\mathbb{Z}))$. Since $k \equiv 2 \pmod 4$, we have $k \equiv 2t \pmod{12}$ with $t = 1, 3$ or 5. Then $12r = k-2, k+6, k+2$ respectively, so that $12r - k + 2 = 0, 8, 4$ respectively. Thus,

$$E_{12r-k+2} = E_0, E_4^2, E_4$$

respectively. As all the Fourier coefficients of E_4 are positive and the same is true for Δ^{-r}, we deduce that all $C_{k0}, C_{k1}, ..., C_{kr}$ are positive and so $C_{k0} > 0$. □

5.5.20. *Suppose that $k \equiv 0 \pmod 4$. With notation as in the previous exercise, show that $C_{k0} \neq 0$.*

Since $k \equiv 0 \pmod 4$, we have $k \equiv 4t \pmod{12}$, with $t = 0, 1, 2$. Then $12r = k - 4t + 12$ so that

$$E_{k-12r+12} = E_{4t} = E_4^t$$

since $t = 0, 1$ or 2. Moreover,

$$F_k = E_{12r-k+2}\Delta^{-r} = -E_{12r-k+2}\Delta^{1-r}E_{14}^{-1}\frac{dj}{2\pi i dz},$$

by Exercise 4.6.10. Again, by Exercise 4.6.8, $E_\ell E_{14-\ell} = E_{14}$ with $\ell = k - 12r + 12$ so that

$$F_k = -E_{14-\ell}\Delta^{1-r}E_{14}^{-1}\frac{dj}{2\pi i dz} = -E_\ell^{-1}\Delta^{1-r}\frac{dj}{2\pi i dz}.$$

As $\ell = 4t$, we can rewrite this as

$$F_k = -E_4^{-t}\Delta^{1-r}\frac{dj}{2\pi i dz}.$$

Since $E_4^3/\Delta = j$, it is not difficult to see that

$$F_k = \frac{3}{t-3}\Delta^{1-r-t/3}\frac{dj^{1-t/3}}{2\pi i dz} = \frac{3}{t-3}\frac{d(E_4^{3-t}\Delta^{-r})}{2\pi i dz} + \frac{3r+t-3}{(3-t)r}E_4^{3-t}\frac{d\Delta^{-r}}{2\pi i dz}$$

so that C_{k0} is the constant term of the Fourier expansion of

$$V_k := \frac{3r+t-3}{(3-t)r}E_4^{3-t}\frac{d\Delta^{-r}}{2\pi i dz}.$$

Since $k > 2$, we have $3r + t - 3 > 0$ and the series E_4^{3-t} begins with 1. Moreover, it has all strictly positive coefficients. Furthermore, the coefficients of Δ^{-r} are all positive and when we differentiate and divide by $2\pi i$, the coefficients are all strictly negative, while the constant term is zero. Hence the constant term of V_k is negative. That is, $C_{k0} < 0$ and hence $C_{k0} \neq 0$. □

Chapter 6
Modular Forms of Higher Level

6.1 Modular forms for congruence subgroups

6.1.1. *Suppose that $f : \mathbb{H} \to \mathbb{C}$ is holomorphic and satisfies (6.1) for a congruence subgroup Γ of level N. Suppose $\gamma, \tau \in \mathrm{SL}_2(\mathbb{Z})$ and that $\gamma(i\infty)$ and $\tau(i\infty)$ are Γ-equivalent. Prove that $v_{\gamma(i\infty)}(f) = v_{\tau(i\infty)}(f)$. That is, the order of f at a Γ-equivalence class of cusps is well-defined.*

By assumption, there exists $\delta \in \Gamma$ such that $\delta\tau(i\infty) = \gamma(i\infty)$. In particular, $\gamma^{-1}\delta\tau$ stabilizes $i\infty$. Hence, by Exercise 3.3.4 we must have

$$\gamma^{-1}\delta\tau = \pm \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}$$

for some $b \in \mathbb{Z}$. Suppose that the expansions at $\gamma(i\infty)$ and $\tau(i\infty)$ are

$$f|\tau = \sum b_n q^{n/N}$$
$$f|\gamma = \sum c_n q^{n/N}.$$

Then since $f|\delta = f$, we have

$$\begin{aligned} f|\tau &= f|\delta\tau \\ &= f|\gamma| \pm \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \\ &= (\pm 1)^k \sum c_n e^{2\pi i(z+b)n/N} \\ &= (\pm 1)^k \sum c_n e^{2\pi i bn/N} q^{n/N}. \end{aligned}$$

In particular, for all n we have $b_n = \pm c_n e^{2\pi i bn/N}$. Thus, b_n and c_n are either both zero or both non-zero. Hence the order of f is well-defined. □

© Springer Science+Business Media Singapore 2016 and Hindustan Book Agency 2016

M.R. Murty et al., *Problems in the Theory of Modular Forms*, HBA Lecture Notes in Mathematics, DOI 10.1007/978-981-10-2651-5_16

6.2 Eisenstein series for $\Gamma(N)$

6.2.1. *Let $k \geq 3$ and N a natural number. Let $g \in (\mathbb{Z}/N\mathbb{Z})^2$. Define the Eisenstein series*

$$G_{k,g}(z) = \sum_{(m,n)\equiv g \pmod N}{}' \frac{1}{(mz+n)^k}, \qquad z \in \mathbb{H}$$

where the summation is over all integers m, n satisfying the congruence condition $(m,n) \equiv g \pmod N$ and the dash on the summation means we exclude $(m,n) = (0,0)$ in the case $g = (0,0)$. For any $\gamma \in \mathrm{SL}_2(\mathbb{Z})$, show that $G_{k,g}|\gamma = G_{k,g\gamma}$. Conclude that $G_{k,g} \in M_k(\Gamma(N))$.

The summands of $G_{k,g}(z)$ are a proper subset of the summands of $G_k(z)$. We have already seen (Exercise 4.2.1) that series of this form converge absolutely and uniformly on compact subsets of $\mathbb{H}$ for $k > 2$. Thus, $G_{k,g}(z)$ is holomorphic on the upper half-plane. Moreover, if $g = (a_1, a_2)$ with $a_1 \not\equiv 0 \pmod N$, then $\lim_{\mathrm{Im}\, z\to\infty} G_{k,g}(z) = 0$. On the other hand, if $a_1 \equiv 0 \pmod N$, then

$$\lim_{\mathrm{Im}\, z\to\infty} G_{k,g}(z) = \sum_{n\equiv a_2 \pmod N} n^{-k}.$$

In either case, $G_{k,g}(z)$ is holomorphic at $i\infty$.

Now let $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$. Then a direct calculation shows that

$$\begin{aligned} G_{k,g}\left(\frac{az+b}{cz+d}\right) &= (cz+d)^k \sum_{(m,n)\equiv g \pmod N}{}' \left(\frac{1}{(ma+nc)z+(mb+nd)}\right)^k \\ &= (cz+d)^k G_{k,g\gamma}(z) \end{aligned}$$

since

$$(\, m \; n \,)\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \big(\, ma+nc \;\; mb+nd \,\big)$$

and if we write $g\gamma = (b_1, b_2)$ the map $(m,n) \mapsto (m,n)\gamma$ sets up a one-to-one correspondence between pairs of integers $(m,n) \equiv g \pmod N$ and pairs $(m',n') \equiv g\gamma \pmod N$. This proves $G_{k,g}|\gamma = G_{k,g\gamma}$ for all $\gamma \in \mathrm{SL}_2(\mathbb{Z})$. Since $G_{k,g}(z)$ is holomorphic at $i\infty$, it follows that it is holomorphic at all cusps. Finally, if $\gamma \equiv I \pmod N$, then $g\gamma \equiv g \pmod N$ so that $G_{k,g}(z) \in M_k(\Gamma(N))$. □

6.2.2. *Show that $G_{k,g} = (-1)^k G_{k,-g}$. Deduce that if k is odd and $2g \equiv (0,0) \pmod N$, then $G_{k,g} = 0$.*

Putting $\gamma = -I$ in Exercise 6.2.1, we see that $G_{k,g}(z) = (-1)^k G_{k,-g}(z)$. To say $2g \equiv (0,0) \pmod N$ means $g \equiv -g \pmod N$ so that if k is odd, $G_{k,g}(z) \equiv 0$. □

6.2.3. *Let Γ be a congruence subgroup of $\mathrm{SL}_2(\mathbb{Z})$ of level N. If $\gamma_1, \ldots, \gamma_r$ are distinct right coset representatives of $\Gamma(N)$ in Γ, show that*

$$f := \sum_{j=1}^{r} G_{k,g\gamma_j} \in M_k(\Gamma).$$

First observe that f is independent of the choice of coset representatives: If $\Gamma(N)\delta = \Gamma(N)\delta'$, then $\delta'\delta^{-1} \in \Gamma(N)$ and hence $G_{k,g}|\delta'\delta^{-1} = G_{k,g}$ which implies that $G_{k,g\delta'} = G_{k,g}|\delta' = G_{k,g}|\delta = G_{k,g\delta}$.

By Exercise 6.2.1, we have for $\gamma \in \Gamma$,

$$f|\gamma = \sum_{j=1}^{r} G_{k,g\gamma_j\gamma}.$$

As the γ_j run through a complete set of coset representatives of $\Gamma(N)$ in Γ, so do the $\gamma_j\gamma$ and so $f|\gamma = f$, for all $\gamma \in \Gamma$. □

6.2.4. *Let Γ_1, Γ_2 be congruence subgroups of $\mathrm{SL}_2(\mathbb{Z})$ such that $\Gamma_1 \subset \Gamma_2$. If $f \in M_k(\Gamma_1)$ and $\{\gamma_1, \ldots, \gamma_r\}$ is a complete set of right coset representatives of Γ_1 in Γ_2, show that*

(a) $\displaystyle\sum_{j=1}^{r} f|\gamma_j \in M_k(\Gamma_2)$,

(b) $\displaystyle\prod_{j=1}^{r} (f|\gamma_j) \in M_{kr}(\Gamma_2)$.

The essential idea is the same as in Exercise 6.2.3. We have

$$\left(\sum_{j=1}^{r} f|\gamma_j\right)|\gamma = \sum_{j=1}^{r} f|(\gamma_j\gamma)$$

and $\{\gamma_1\gamma, \ldots, \gamma_r\gamma\}$ is again a complete set of coset representatives of Γ_1 in Γ_2. Thus the function is invariant upon 'slashing' and this proves (a).

Noting that $(f_1 f_2)|\gamma = (f_1|\gamma)(f_2|\gamma)$, we see that (b) follows similarly. □

6.2.5. *Show that $\Gamma_0(4)$ is generated by $T = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, by $\left(\begin{smallmatrix} 1 & 0 \\ 4 & 1 \end{smallmatrix}\right)$, and by $\left(\begin{smallmatrix} -1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$.*

Let $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma_0(4)$. Then a is odd. If $b = 0$, then $\gamma = \pm \left(\begin{smallmatrix} 1 & 0 \\ 4 & 1 \end{smallmatrix}\right)^k$ for some k. If $b \neq 0$, then noting that $\gamma T^k = \left(\begin{smallmatrix} a & b+ka \\ c & d+kc \end{smallmatrix}\right)$, and choosing k appropriately, we may suppose that $|b| < \frac{|a|}{2}$ (as a is odd). Noting $\gamma \left(\begin{smallmatrix} 1 & 0 \\ 4 & 1 \end{smallmatrix}\right)^k = \left(\begin{smallmatrix} a+4kb & b \\ c+4kd & d \end{smallmatrix}\right)$, we may assume $|a| < 2|b|$. Iteration eventually gives $b = 0$. □

6.2.6. *Show that there are exactly three inequivalent cusps for the group $\Gamma_0(4)$. Find the widths of each cusp.*

A complete set of inequivalent cusps is contained in $\{\gamma(i\infty)\}$ as γ runs through the coset representatives of (6.4) on page 76. In particular, there are

three cusps: $i\infty$, $1/2$, and 0. To determine the width of $\gamma(i\infty)$, we look for $h \in \mathbb{Z}$ such that

$$\gamma(\pm I)\begin{pmatrix} 1 & h \\ 0 & 1 \end{pmatrix}\gamma^{-1} \in \Gamma_0(4).$$

Since $-I \in \Gamma_0(4)$, we consider $\gamma\left(\begin{smallmatrix} 1 & h \\ 0 & 1 \end{smallmatrix}\right)\gamma^{-1}$ for each of the coset representatives γ. We first consider $\gamma = \left(\begin{smallmatrix} 1 & 0 \\ 2 & 1 \end{smallmatrix}\right)$. Now

$$\gamma\begin{pmatrix} 1 & h \\ 0 & 1 \end{pmatrix}\gamma^{-1} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}\begin{pmatrix} 1 & h \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} 1-2h & h \\ -4h & 1+2h \end{pmatrix}.$$

It is clear that this matrix is in $\Gamma_0(4)$ for all $h \in \mathbb{Z}$. Hence, the cusp $\left(\begin{smallmatrix} 1 & 0 \\ 2 & 1 \end{smallmatrix}\right) i\infty = \frac{1}{2}$ has width 1.

Now consider $\gamma = \left(\begin{smallmatrix} 0 & -1 \\ 1 & m \end{smallmatrix}\right)$ for $m = 1$, 2, 3, or 4. We have

$$\gamma\begin{pmatrix} 1 & h \\ 0 & 1 \end{pmatrix}\gamma^{-1} = \begin{pmatrix} 0 & -1 \\ 1 & m \end{pmatrix}\begin{pmatrix} 1 & h \\ 0 & 1 \end{pmatrix}\begin{pmatrix} m & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ h & 1 \end{pmatrix},$$

which is in $\Gamma_0(4)$ if and only if $4 \mid h$. Hence, the cusp $\left(\begin{smallmatrix} 0 & -1 \\ 1 & m \end{smallmatrix}\right) i\infty = 0$ has width 4. Finally, it is clear that the cusp $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right) i\infty = i\infty$ has width 1.

6.3 Fourier expansions of higher level Eisenstein series

6.3.2. *For $g = (a_1, 0) \in (\mathbb{Z}/N\mathbb{Z})^2$ and $n \neq 0$, show that*

$$b_{k,g}(n) = \frac{(-2\pi i)^k}{N^k(k-1)!} \sum_{\substack{d \mid n \\ n/d \equiv a_1 \pmod N}} d^{k-1}(\operatorname{sgn} d).$$

This is immediate from Theorem 6.3.1 since $a_2 = 0$. □

6.3.3. *If $g = (0, a_2) \in (\mathbb{Z}/N\mathbb{Z})^2$, show that $b_{k,g}(n) = 0$ if $N \nmid n$, and*

$$b_{k,g}(Nn) = \frac{(-2\pi i)^k}{N^k(k-1)!} \sum_{d \mid n} d^{k-1}(\zeta^{da_2} + (-1)\zeta^{-da_2}),$$

where $\zeta = e^{2\pi i/N}$. Deduce that $G_{k,g}(z)$, with $g = (0, a_2)$, has a q-expansion of the form

$$\sum_{n=0}^{\infty} a_n q^n, \qquad q = e^{2\pi i z}.$$

Again from Theorem 6.3.1, we see that the condition $n/d \equiv a_1 \equiv 0 \pmod N$ implies that summation is empty, and hence that $b_{k,g}(n) = 0$, whenever $N \nmid n$.

If $N \mid n$, we see that the summation formula for $b_{k,g}(n)$ in Theorem 6.3.1 shows that if we write $n = Nn'$, the summation over divisors of Nn' can be re-written as a sum over divisors of n' which reduces to the stated formula. Finally, since the Fourier expansion is only supported on multiples of N, we deduce that $G_{k,g}$ has a traditional q-expansion, as given. □

6.3.4. *Prove that if $g = (0, a_2) \in (\mathbb{Z}/N\mathbb{Z})^2$, then $G_{k,g} \in M_k(\Gamma_1(N))$ for $k \geq 3$.*

By Exercise 6.2.1, we have

$$G_{k,g}|\gamma = G_{k,g\gamma} \qquad \forall \gamma \in \Gamma_1(N)$$

and we need only observe that $(0, a_2)\gamma \equiv (0, a_2) \pmod{N}$, to deduce the desired result. □

6.3.5. *Let*

$$E_{2,N}(z) := E_2(z) - NE_2(Nz).$$

Show that $E_{2,N} \in M_2(\Gamma_0(N))$.

It is clear that $E_{2,N}(z)$ is a holomorphic function on $\mathbb{H}$. Now let $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma_0(N)$. Then

$$E_{2,N}\left(\frac{az+b}{cz+d}\right) = E_2\left(\frac{az+b}{cz+d}\right) - NE_2\left(\frac{aNz+bN}{cz+d}\right).$$

By Exercise 5.1.7 (the transformation law for E_2), we have

$$E_2\left(\frac{az+b}{cz+d}\right) = (cz+d)^2 E_2(z) + \frac{6c(cz+d)}{\pi i}.$$

Also,

$$E_2\left(\frac{a(Nz)+bN}{\frac{c}{N}(Nz)+d}\right) = (cz+d)^2 E_2(Nz) + \frac{6c(cz+d)}{N\pi i}$$

so that

$$\begin{aligned} E_{2,N}\left(\frac{az+b}{cz+d}\right) &= (cz+d)^2(E_2(z) - NE_2(Nz)) \\ &= (cz+d)^2 E_{2,N}(z). \end{aligned}$$

We must check that $E_{2,N}$ is holomorphic at every cusp. This will be a more general argument that will obviate the calculation above. Suppose now that $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$. We must determine a Fourier expansion of $E_{2,N}(z)$ at $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)(i\infty)$. Equivalently, we determine an expansion at $i\infty$ of $E_{2,N}(z)|\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. The heart of this calculation deals with

$$NE_2(Nz)|\begin{pmatrix} a & b \\ c & d \end{pmatrix} = (cz+d)^{-2} NE_2\left(\begin{pmatrix} N & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix} z\right).$$

Recalling Theorem 5.2.1, since

$$\begin{pmatrix} N & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} Na & Nb \\ c & d \end{pmatrix} \in X_N,$$

we know that there exist

$$\begin{pmatrix} r & s \\ t & u \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}) \text{ and } \begin{pmatrix} x & y \\ 0 & w \end{pmatrix} \in X_N$$

such that

$$\begin{pmatrix} N & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} r & s \\ t & u \end{pmatrix} \begin{pmatrix} x & y \\ 0 & w \end{pmatrix}.$$

In particular, some computations as in the proof of Theorem 5.2.1 easily show that we must have

$$\begin{aligned} x &= \gcd(c, N) \\ c &= tx \\ Na &= ux \\ xw &= N \\ d &= yt + uw. \end{aligned}$$

Now

$$\begin{aligned} NE_2(Nz)\Big|\begin{pmatrix} a & b \\ c & d \end{pmatrix} &= (cz+d)^{-2} NE_2\left(\begin{pmatrix} r & s \\ t & u \end{pmatrix}\begin{pmatrix} x & y \\ 0 & w \end{pmatrix} z\right) \\ &= (cz+d)^{-2} NE_2\left(\frac{r\sigma + s}{t\sigma + u}\right) \end{aligned}$$

where

$$\sigma := \begin{pmatrix} x & y \\ 0 & w \end{pmatrix} z = \frac{x}{w} z + \frac{y}{w}.$$

Observe that

$$w(t\sigma + u) = xtz + (yt + uw) = cz + d.$$

Moreover, by Exercise 5.1.7,

$$E_2\left(\frac{r\sigma + s}{t\sigma + u}\right) = (t\sigma + u)^2 E_2(\sigma) + \frac{6t(t\sigma + u)}{\pi i}.$$

Thus,

$$\begin{aligned} NE_2(Nz)|\begin{pmatrix} a & b \\ c & d \end{pmatrix} &= (cz+d)^{-2}N\left[(t\sigma+u)^2E_2(\sigma)+\frac{6t(t\sigma+u)}{\pi i}\right] \\ &= (cz+d)^{-2}N\left[\frac{(cz+d)^2}{w^2}E_2(\sigma)+\frac{6t(cz+d)}{w\pi i}\right] \\ &= \frac{N}{w^2}E_2(\sigma)+\frac{6c}{\pi i(cz+d)}. \end{aligned}$$

Therefore,

$$\begin{aligned} E_{2,N}(z)|\begin{pmatrix} a & b \\ c & d \end{pmatrix} &= E_2(z)|\begin{pmatrix} a & b \\ c & d \end{pmatrix} - NE_2(Nz)|\begin{pmatrix} a & b \\ c & d \end{pmatrix} \\ &= E_2(z)-\frac{x}{w}E_2(\sigma). \end{aligned}$$

Now

$$\begin{aligned} E_2(\sigma) = E_2\left(\frac{x}{w}z+\frac{y}{w}\right) &= 1-24\sum_{n=1}^{\infty}\sigma_1(n)e^{2\pi in\left(\frac{x}{w}z+\frac{y}{w}\right)} \\ &= 1-24\sum_{n=1}^{\infty}\sigma_1(n)e^{2\pi iny/w}q^{nx/w}. \end{aligned}$$

In particular, we see that $E_{2,N}(z)|\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ is holomorphic at $i\infty$. Also, observe that if $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma_0(N)$, then $N|c$ and so $x = N$ and $w = 1$. In particular, this shows that $E_{2,N}(z)|\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) = E_2(z) - \frac{x}{w}E_2(z+y) = E_{2,N}(z)$ is modular. □

6.3.6. *If $f \in M_k(\mathrm{SL}_2(\mathbb{Z}))$, show that*

$$f_N(z) := f(z) - Nf(Nz) \in M_k(\Gamma_0(N)).$$

In particular, deduce that for $k \geq 3$,

$$E_{k,N}(z) := E_k(z) - NE_k(Nz) \in M_k(\Gamma_0(N)).$$

It is clear that f_N is holomorphic on $\mathbb{H}$ since f is. The modular transformation rule is immediate from

$$f_N\left(\frac{az+b}{cz+d}\right) = f\left(\frac{az+b}{cz+d}\right) - Nf\left(\frac{a(Nz)+bN}{(\frac{c}{N})(Nz)+d}\right) = (cz+d)^k f_N(z)$$

which holds for all $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma_0(N)$.

We must check that f_N is holomorphic at every cusp. Suppose now that $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$. As in the solution to Exercise 6.3.5, we use Theorem 5.2.1 to deduce the existence of $\left(\begin{smallmatrix} r & s \\ t & u \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$ and $\left(\begin{smallmatrix} x & y \\ 0 & w \end{smallmatrix}\right) \in X_N$ such that

$$\begin{pmatrix} N & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} r & s \\ t & u \end{pmatrix}\begin{pmatrix} x & y \\ 0 & w \end{pmatrix}.$$

Let $\sigma := \left(\begin{smallmatrix} x & y \\ 0 & w \end{smallmatrix}\right) z = \frac{x}{w}z + \frac{y}{w}$. Now $w(t\sigma + u) = xtz + yt + uw = cz + d$ as in the solution to Exercise 6.3.5. Thus

$$\begin{aligned} f_N(z)\left|\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right. &= f(z)\left|\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right. - Nf(Nz)\left|\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right. \\ &= f(z) - N(cz+d)^{-k} f\left(\begin{pmatrix} r & s \\ t & u \end{pmatrix}\sigma\right) \\ &= f(z) - N(cz+d)^{-k}(t\sigma+u)^k f(\sigma) \\ &= f(z) - N(cz+d)^{-k}\left(\frac{cz+d}{w}\right)^k f(\sigma) \\ &= f(z) - \frac{N}{w^k} f\left(\frac{x}{w}z + \frac{y}{w}\right). \end{aligned}$$

If we assume that $f(z) = \sum_{n=0}^{\infty} a_n q^n$ is the Fourier expansion of f at $i\infty$, then

$$\frac{N}{w^k} f\left(\frac{x}{w}z + \frac{y}{w}\right) = \frac{N}{w^k}\sum_{n=0}^{\infty} a_n e^{2\pi i n y/w} q^{xn/w}$$

is clearly holomorphic at $i\infty$ and hence so is $f_N(z)\left|\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)\right.$. □

6.4 The valence and dimension formulas

6.4.1. *Show that the only elliptic points for* $\mathrm{SL}_2(\mathbb{Z})$ *lying in the standard fundamental domain* $\mathcal{F}$ *are* i, $\rho = e^{\pi i/3}$, *and* ρ^2.

Suppose that $z \in \mathcal{F}$ and $\pm I \neq \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$ are such that

$$z = \frac{az+b}{cz+d}. \tag{6.1}$$

We must have $c \neq 0$ since otherwise $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) = \pm\left(\begin{smallmatrix} 1 & b \\ 0 & 1 \end{smallmatrix}\right)$ is a translation which will only fix z when $b = 0$. Now by (6.1) above, we have

$$cz^2 + (d-a)z - b = 0$$

which implies that

$$z = \frac{a - d \pm \sqrt{(d-a)^2 + 4bc}}{2c}.$$

Since $\det\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) = 1$, we have $bc = ad - 1$ and hence

$$z = \frac{a - d \pm \sqrt{(a+d)^2 - 4}}{2c}.$$

Now $z \in \mathbb{H}$ and so $(a+d)^2 - 4 < 0$ which can only happen when $a+d = 0, \pm 1$.

If $a + d = 0$ then $a = -d$ and $z = \frac{a}{c} \pm \frac{i}{c} \in \mathcal{F}$. Thus $\frac{1}{|c|} = \operatorname{Im} z \geq \frac{\sqrt{3}}{2}$ and hence $|c| = 1$. Now $-\frac{1}{2} \leq \operatorname{Re} z \leq \frac{1}{2}$ implies that $a = 0$. Hence $z = i$.

If $a + d = \pm 1$, then $\operatorname{Im} z = \frac{\sqrt{3}}{2|c|}$. But $z \in \mathcal{F}$ implies that $\frac{\sqrt{3}}{2|c|} \geq \frac{\sqrt{3}}{2}$ which forces $|c| = 1$ and $\operatorname{Im} z = \frac{\sqrt{3}}{2}$. Thus $z = \rho$ or ρ^2. □

6.4.2. *Find the stabilizer subgroups* $\mathrm{SL}_2(\mathbb{Z})_i$ *and* $\mathrm{SL}_2(\mathbb{Z})_\rho$ *in* $\mathrm{SL}_2(\mathbb{Z})$ *and show that* $\mathrm{SL}_2(\mathbb{Z})_i / \{\pm I\}$ *and* $\mathrm{SL}_2(\mathbb{Z})_\rho / \{\pm I\}$ *have orders two and three, respectively.*

Suppose $\pm I \neq \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})_i$. From the solution to Exercise 6.4.1, we know that $0 = a = -d$ and $|c| = 1$. Since $ad - bc = 1$, we deduce that $b = -c$ and hence that $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) = \pm \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$. Therefore $\mathrm{SL}_2(\mathbb{Z})_i = \left\{\pm I, \pm \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)\right\}$ and so $|\mathrm{SL}_2(\mathbb{Z})_i / \pm I| = 2$.

Now suppose that $\pm I \neq \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})_\rho$. From the solution to Exercise 6.4.1, we know that $|a + d| = 1$ and $|c| = 1$. We also know that

$$-\frac{1}{2} + \frac{\sqrt{3}}{2} i = \frac{a - d \pm \sqrt{(a+d)^2 - 4}}{2c}.$$

Hence $\operatorname{Re} \rho = -1/2 = (a - d)/2c$. Thus $c = d - a$. It is easy to determine the four possible combinations of a, c and d which satisfy these requirements. In each case, b can be determined from $ad - bc = 1$. We may conclude that $\mathrm{SL}_2(\mathbb{Z})_\rho = \left\{\pm I, \pm \left(\begin{smallmatrix} 0 & -1 \\ 1 & 1 \end{smallmatrix}\right), \pm \left(\begin{smallmatrix} 1 & 1 \\ -1 & 0 \end{smallmatrix}\right)\right\}$. In particular, $|\mathrm{SL}_2(\mathbb{Z})_\rho / \pm I| = 3$. □

6.4.3. *If* Γ *has finite index in* $\mathrm{SL}_2(\mathbb{Z})$*, show that there are only finitely many* Γ*-inequivalent elliptic points for* Γ*.*

Suppose that $z \in \mathbb{H}$ is an elliptic point for Γ. By the definition of a fundamental domain, there exists $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ such that $\gamma z \in \mathcal{F}$, where $\mathcal{F}$ is the standard fundamental domain for $\mathrm{SL}_2(\mathbb{Z})$. It is easily verified that $\Gamma_{\gamma z} = \gamma \Gamma_z \gamma^{-1} \neq \{\pm I\}$. Hence γz is an elliptic point for Γ and hence also for $\mathrm{SL}_2(\mathbb{Z})$. By Exercise 6.4.1, we see that z is $\mathrm{SL}_2(\mathbb{Z})$-equivalent to either i or ρ (or to $\rho^2 = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right) \rho$ and hence to ρ).

Let $\gamma_1, \ldots, \gamma_d$ be a complete set of coset representatives for Γ in $\mathrm{SL}_2(\mathbb{Z})$. We deduce that z is Γ-equivalent to an element of the finite set

$$\{\gamma_j(i), \gamma_j(\rho) : 1 \leq j \leq d\}.$$ □

6.4.5. *Verify that Theorem 6.4.4 reduces to Theorem 4.3.1 in the case* $\Gamma = \mathrm{SL}_2(\mathbb{Z})$*.*

By Exercises 6.4.1 and 6.4.2, the only $\mathrm{SL}_2(\mathbb{Z})$-inequivalent elliptic points are i and ρ and the appropriate stabilizers have orders 2 and 3 respectively. This tallies with the left hand side of the formula in Theorem 4.3.1. Since $\epsilon_2 = 1$, $\epsilon_3 = 1$, $g = 0$, and $\epsilon_\infty = 1$, this gives us $k/12$, which agrees with Theorem 4.3.1. □

6.4.8. *Let* $f = \sum_{n=0}^{\infty} a_n q^n$ *and* $g = \sum_{n=0}^{\infty} b_n q^n \in M_k(\Gamma)$ *for some congruence subgroup* Γ *of* $\mathrm{SL}_2(\mathbb{Z})$. *Suppose that* $a_n = b_n$ *for all* $n \leq k[\mathrm{SL}_2(\mathbb{Z}) : \{\pm\}\Gamma]/12$. *Show that* $f = g$.

Apply Theorem 6.4.7 to $f - g \in M_k(\Gamma)$. □

6.4.9. *For any congruence subgroup* Γ *of* $\mathrm{SL}_2(\mathbb{Z})$, *prove that*

$$\dim M_k(\Gamma) \leq \frac{k}{12}[\mathrm{SL}_2(\mathbb{Z}) : \{\pm\}\Gamma] + 1.$$

Consider the linear map $M_k(\Gamma) \to \mathbb{C}^{r+1}$, where $r = k[\mathrm{SL}_2(\mathbb{Z}) : \{\pm\}\Gamma]/12$, given by $f = \sum_{n=0}^{\infty} a_n q^n \mapsto (a_0, a_1, \ldots, a_r)$. Sturm's bound (Theorem 6.4.7) says that this map is injective. Hence $\dim M_k(\Gamma) \leq r + 1$. □

6.4.10. *Prove that*

$$\dim M_4(\Gamma_0(4)) = 3.$$

(Hint: consider $E_4, E_{4,2}$, *and* $E_{4,4}$ *from Exercise 6.3.6.)*

By Exercises 2.3.4 and 6.4.9, we have

$$\dim M_4(\Gamma_0(4)) \leq \frac{1}{3}[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(4)] + 1 = 3.$$

Now E_4, $E_{4,2}$, $E_{4,4} \in M_4(\Gamma_0(4))$ and a quick check of their q-expansions at $i\infty$ shows that

$$\begin{aligned} E_4 &= 1 + 240q + 2160q^2 + \cdots \\ E_{4,2} &= 1 - 16q + 112q^2 + \cdots \\ E_{4,4} &= 1 - 16q^2 + 112q^4 + \cdots \end{aligned}$$

and these are easily verified to be linearly independent over $\mathbb{C}$. Thus, the forms are a basis for $M_4(\Gamma_0(4))$. □

6.4.11. *Prove that*

$$M_2(\Gamma_0(2)) = \mathbb{C}E_{2,2},$$

where $E_{2,2}(z) = E_2(z) - 2E_2(2z)$.

By Exercise 6.4.9,

$$\dim M_2(\Gamma_0(2)) \leq 3/2,$$

so that $\dim M_2(\Gamma_0(2)) \leq 1$. Since $E_{2,2} \in M_2(\Gamma_0(2))$ by Exercise 6.3.5, and as $E_{2,2} \neq 0$, we are done. □

6.4.12. *Show that* $\dim M_2(\Gamma_0(4)) = 2$ *and that* $\{E_{2,2}, E_{2,4}\}$ *is a basis for* $M_2(\Gamma_0(4))$.

By Exercise 6.4.9,

$$\dim M_2(\Gamma_0(4)) \leq 2.$$

Now $E_{2,2}$ and $E_{2,4} \in M_2(\Gamma_0(4))$ by Exercise 6.3.5. It is easy to check that

$$E_{2,2} = -1 - 24q + \cdots$$
$$E_{2,4} = -3 - 24q + \cdots$$

Clearly, these elements are linearly independent over $\mathbb{C}$. □

6.5 The Jacobi four-square theorem revisited

6.5.1. *Let $\eta(z)$ denote the Dedekind η-function. Prove that*

$$\theta(z) := \sum_{n=-\infty}^{\infty} q^{n^2} = \frac{\eta^5(2z)}{\eta^2(z)\eta^2(4z)},$$

where $q = e^{2\pi iz}$.

Set $x = 1$ in Jacobi's Triple Product Identity (Theorem 1.2.1) to get

$$\sum_{n=-\infty}^{\infty} q^{n^2} = \prod_{n=1}^{\infty}(1-q^{2n})(1+q^{2n-1})^2,$$

which can be rewritten as

$$\sum_{n=-\infty}^{\infty} q^{n^2} = \prod_{n=1}^{\infty} \frac{(1-q^{2n})(1+q^n)^2}{(1+q^{2n})^2}.$$

Now $1+q^n = (1-q^{2n})/(1-q^n)$ and $1+q^{2n} = (1-q^{4n})/(1-q^{2n})$, so this becomes

$$\sum_{n=-\infty}^{\infty} q^{n^2} = \prod_{n=1}^{\infty} \frac{(1-q^{2n})^5}{(1-q^n)^2(1-q^{4n})^2} = \frac{\eta^5(2z)}{\eta^2(z)\eta^2(4z)}.$$ □

6.6 Supplementary problems

6.6.1. *Show that $\eta^{12}(2z) \in S_6(\Gamma_0(4))$.*

By the proof of Theorem 6.5.2, we see that

$$\eta\left(\frac{2z}{4z+1}\right) = e^{\frac{-\pi i}{6}}\sqrt{i\left(\frac{4z+1}{2z}\right)}\sqrt{\frac{2z}{i}}\eta(2z)$$

so that $\eta^{12}(2z)$ transforms with weight 6 using Exercise 6.2.5. Now $\Gamma_0(4)$ has three inequivalent cusps, 0, $1/2$ and $i\infty$. Clearly

$$\eta^{12}(2z) = q\prod_{n=1}^{\infty}\left(1-q^{2n}\right)^{12} = q+\cdots$$

has a zero of order 1 at the cusp $i\infty$.

To check the behaviour of $\eta^{12}(2z)$ at 0, we need to consider $\eta^{12}(2z)\left|\left(\begin{smallmatrix}0&-1\\1&0\end{smallmatrix}\right)\right.$ at $i\infty$. But

$$\begin{aligned}\eta^{12}(2z)\left|\begin{pmatrix}0&-1\\1&0\end{pmatrix}\right. &= z^{-6}\eta^{12}\left(2\left(-\frac{1}{z}\right)\right) = z^{-6}\eta^{12}\left(-\frac{1}{(z/2)}\right)\\ &= z^{-6}\left(\sqrt{\frac{(z/2)}{i}}\right)^{12}\eta^{12}\left(\frac{z}{2}\right)\\ &= -\frac{1}{2^6}\eta^{12}\left(\frac{z}{2}\right)\\ &= -\frac{1}{2^6}q^{1/4}+\cdots.\end{aligned}$$

In particular, $\eta^{12}(2z)$ has a zero of order 1 at the cusp 0. (Recall that the cusp 0 has width 4).

Finally, to check the behaviour at $1/2$, we need to consider $\eta^{12}(2z)\left|\left(\begin{smallmatrix}1&0\\2&1\end{smallmatrix}\right)\right.$ at $i\infty$. Now

$$\begin{aligned}\eta^{12}(2z)\left|\begin{pmatrix}1&0\\2&1\end{pmatrix}\right. &= (2z+1)^{-6}\eta^{12}\left(\frac{2z}{2z+1}\right)\\ &= (2z+1)^{-6}\eta^{12}\left(1-\frac{1}{2z+1}\right)\\ &= -(2z+1)^{-6}\eta^{12}\left(-\frac{1}{2z+1}\right)\\ &= \eta^{12}(2z+1)\\ &= -\eta^{12}(2z)\\ &= -q+\cdots.\end{aligned}$$

In particular, $\eta^{12}(2z)$ has a zero of order 1 at the cusp $1/2$. We conclude that $\eta^{12}(2z)\in S_6(\Gamma_0(4))$. □

6.6.2. *Show that* $S_6(\Gamma_0(4)) = \eta^{12}(2z)\mathbb{C}$.

If $f \in S_6(\Gamma_0(4))$, we must have a zero of order at least 1 at each of $i\infty$, 0, and $1/2$. Then for some λ, $f(z) - \lambda\eta^{12}(2z)$ has a zero of order 2 at $i\infty$. By Sturm's bound (Theorem 6.4.7), we deduce $f(z) = \lambda\eta^{12}(2z)$. □

6.6.3. *Let $k \geq 6$ be even. Show that $S_k(\Gamma_0(4)) = \eta^{12}(2z)M_{k-6}(\Gamma_0(4))$.*

From the solution to Exercise 6.6.1, $\eta^{12}(2z)$ has a zero of order 1 at every cusp. Moreover, $\eta^{12}(2z)$ does not vanish for $z \in \mathbb{H}$. Hence, for any $f \in S_k(\Gamma_0(4))$, we deduce that $f/\eta^{12}(2z)$ is holomorphic in $\mathbb{H}$ and at all cusps of $\Gamma_0(4)$. Hence $f/\eta^{12}(2z) \in M_{k-6}(\Gamma_0(4))$. □

6.6.4. *For even $k \geq 0$, prove that $\dim M_k(\Gamma_0(4)) = \frac{k}{2} + 1$.*

By Exercise 6.4.9, we have $\dim M_k(\Gamma_0(4)) \leq \frac{k}{2} + 1$. Hence, it suffices to find enough linearly independent forms. We modify the basis from Exercise 6.4.12 by setting

$$F := -E_{2,2}$$
$$G := \frac{E_{2,4} - 3E_{2,2}}{48}$$

so that $F = 1 + 24q + \cdots$ and $G = q + \cdots \in M_2(\Gamma_0(4))$ are linearly independent. Now

$$F^{\frac{k}{2}-i}G^i = q^i + \cdots$$

for $i = 0, 1, \ldots, \frac{k}{2}$, are clearly linearly independent.

6.6.5. *Let $r_{12}(n)$ be the number of ways of writing n as a sum of 12 squares and let $\eta^{12}(2z) = \sum_{n=1}^{\infty} a_n q^n$. Show that*

$$r_{12}(n) = 8\sigma_5(n) - 512\sigma_5\left(\frac{n}{4}\right) + 16a_n.$$

By Exercise 6.6.4 applied to $k = 6$, we have $\dim M_6(\Gamma_0(4)) = 4$. Moreover,

$$E_6(z), E_{6,2}(z), E_{6,4}(z), \eta^{12}(2z)$$

comprise a basis. Now $\theta^{12}(z) \in M_6(\Gamma_0(4))$ and a quick comparison of the q-expansions immediately leads to the result. □

6.6.6. *Let $f(z) \in M_k(\Gamma_1(N))$ and suppose $r \in \mathbb{Z}$ is positive. Show that $f(rz) \in M_k(\Gamma_1(rN))$. Moreover, if $f(z)$ is a cusp form, then so is $f(rz)$.*

By Equation (4.2) on page 36, $f(rz) = r^{-k/2}f(z)|\left(\begin{smallmatrix} r & 0 \\ 0 & 1 \end{smallmatrix}\right)$. Now the cusp conditions (holomorphy/vanishing) follow from Exercises 7.1.6 and 7.1.7. It remains to show that $f(z)|\left(\begin{smallmatrix} r & 0 \\ 0 & 1 \end{smallmatrix}\right)$ is invariant under the modular transformation. For any $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma_1(rN)$, we have

$$\left(f(z)\left|\begin{pmatrix} r & 0 \\ 0 & 1 \end{pmatrix}\right.\right)\left|\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right. = f(z)\left|\begin{pmatrix} ra & rb \\ c & d \end{pmatrix}\right. = \left(f(z)\left|\begin{pmatrix} a & rb \\ c/r & d \end{pmatrix}\right.\right)\left|\begin{pmatrix} r & 0 \\ 0 & 1 \end{pmatrix}\right..$$

Since $rN|c$, we see that $\left(\begin{smallmatrix} a & rb \\ c/r & d \end{smallmatrix}\right) \in \Gamma_1(N)$. Hence

$$\left(f(z)\left|\begin{pmatrix} a & rb \\ c/r & d \end{pmatrix}\right.\right)\left|\begin{pmatrix} r & 0 \\ 0 & 1 \end{pmatrix} = f(z)\right|\begin{pmatrix} r & 0 \\ 0 & 1 \end{pmatrix}.$$

We conclude that $f(rz) \in M_k(\Gamma_1(rN))$. □

Chapter 7
The Petersson Inner Product

7.1 The Hilbert space of cusp forms

7.1.1. *Let $f, g \in M_k(\Gamma)$ and let $\mathcal{F}$ be any fundamental domain for the congruence subgroup Γ. Show that the integral*

$$\iint_{\mathcal{F}} y^k f(z)\overline{g(z)}\frac{dxdy}{y^2} < \infty$$

as long as one of $f, g \in S_k(\Gamma)$.

The integral is

$$\sum_j \iint_{\alpha_j^{-1}(\mathcal{D})} y^k f(z)\overline{g(z)}\frac{dxdy}{y^2},$$

where $\mathcal{D}$ is the standard fundamental domain for $\mathrm{SL}_2(\mathbb{Z})$ and

$$\mathrm{SL}_2(\mathbb{Z}) = \bigcup_j \alpha_j \overline{\Gamma}$$

is a disjoint coset decomposition of $\overline{\Gamma}$ in $\mathrm{SL}_2(\mathbb{Z})$. Each of the integrals is

$$\iint_{\mathcal{D}} \left(\mathrm{Im}(\alpha_i^{-1}z)\right)^k f(\alpha_i^{-1}z)\overline{g(\alpha_i^{-1}z)}\frac{dxdy}{y^2}.$$

Since one of the $f, g \in S_k(\Gamma)$, we see that the integral is $O\left(y^k e^{-2\pi y}\right)$ and hence the integral converges. □

7.1.2. *Prove that*

(i) (f, g) *is linear in* f,
(ii) (f, g) *is conjugate symmetric. That is,* $(f, g) = \overline{(g, f)}$,
(iii) $(f, f) > 0$ *for* $f \neq 0$.

© Springer Science+Business Media Singapore 2016 and Hindustan Book Agency 2016

M.R. Murty et al., *Problems in the Theory of Modular Forms*, HBA Lecture Notes in Mathematics, DOI 10.1007/978-981-10-2651-5_17

Therefore, the Petersson inner product defines a Hermitian inner product on $S_k(\Gamma)$.

This is immediate from the definitions. □

7.1.4. *Suppose $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$ has integer entries. Let $D = \det \alpha$. If $\Gamma(N) \subseteq \Gamma$ for some N, show that $\Gamma(ND) \subseteq \alpha^{-1}\Gamma\alpha$ and $\Gamma(ND) \subseteq \alpha\Gamma\alpha^{-1}$.*

Note that α is not necessarily in $\mathrm{SL}_2(\mathbb{Z})$ and so Exercise 2.2.2 is not applicable. The inverse α^{-1} does not necessarily have integer entries.

Let $\gamma \in \Gamma(ND)$. Write $\gamma = I + ND\beta$ for some 2×2 matrix β with integer entries. We want to show $\gamma \in \alpha^{-1}\Gamma\alpha$, which is equivalent to showing $\alpha\gamma\alpha^{-1} \in \Gamma$. Now

$$\alpha\gamma\alpha^{-1} = \alpha(I + ND\beta)\alpha^{-1} = I + N\alpha\beta(D\alpha^{-1})$$

and $D\alpha^{-1}$ has integer entries. Therefore $\alpha\gamma\alpha^{-1} \in \Gamma(N)$. A similar argument shows $\alpha^{-1}\gamma\alpha \in \Gamma(N)$. □

7.1.5. *Let $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$ have integer entries. Show that there exists $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ such that*

$$\gamma^{-1}\alpha = \begin{pmatrix} a & b \\ 0 & d \end{pmatrix},$$

where a and d are positive integers and $b \in \mathbb{Z}$.

Let

$$\alpha = \begin{pmatrix} r & t \\ s & u \end{pmatrix}$$

and suppose that $\delta = (r, s)$. Put $r_1 = r/\delta$ and $s_1 = s/\delta$. Since $(r_1, s_1) = 1$, we can find integers x, y such that $r_1 x + s_1 y = 1$. Now,

$$\begin{pmatrix} x & y \\ -s_1 & r_1 \end{pmatrix} \begin{pmatrix} r & t \\ s & u \end{pmatrix} = \begin{pmatrix} \delta & \star \\ 0 & \star \end{pmatrix}$$

has the desired form since the determinants of both matrices on the left hand side of the equation are positive. □

7.1.6. *Let $f \in M_k(\Gamma)$ with Γ a congruence subgroup of $\mathrm{SL}_2(\mathbb{Z})$. Let $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$. Show that $f|\alpha$ is holomorphic at every cusp $s \in \mathbb{Q} \cup \{i\infty\}$.*

A typical cusp $s \in \mathbb{Q} \cup \{i\infty\}$ is of the form $s = \beta(i\infty)$ for some $\beta \in \mathrm{SL}_2(\mathbb{Z})$. We must show that for every $\beta \in \mathrm{SL}_2(\mathbb{Z})$, the function $(f|\alpha)|\beta = f|(\alpha\beta)$ is holomorphic at $i\infty$.

From (4.2) on page 36, we see that $\alpha\beta$ can be multiplied by a positive scalar without affecting $f|\alpha\beta$. Hence we may suppose that $\alpha\beta$ has integer entries. By Exercise 7.1.5, we can find $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ such that

$$\alpha\beta = \gamma \begin{pmatrix} a & b \\ 0 & d \end{pmatrix},$$

with a, d positive integers and $b \in \mathbb{Z}$. Now

$$f|\alpha\beta = f|\gamma \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} = (f|\gamma) \left| \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \right. .$$

Since $f \in M_k(\Gamma)$ is holomorphic at every cusp, $f|\gamma$ has a Fourier expansion $(f|\gamma)(z) = \sum_{n=0}^{\infty} a_n q^{n/h}$. Consequently,

$$\begin{aligned} (f|\gamma) \left| \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \right. &= \left(\sum_{n=0}^{\infty} a_n q^{n/h} \right) \left| \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \right. \\ &= \left(\frac{a}{d}\right)^{k/2} \sum_{n=0}^{\infty} a_n e^{2\pi i n(az+b)/dh} \\ &= \left(\frac{a}{d}\right)^{k/2} \sum_{n=0}^{\infty} a_n e^{2\pi i b/dh} q^{an/dh} \end{aligned}$$

is holomorphic. □

7.1.7. *Let $f \in S_k(\Gamma)$ with Γ a congruence subgroup of $\mathrm{SL}_2(\mathbb{Z})$. Let $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$. Show that $f|\alpha$ vanishes at every cusp $s \in \mathbb{Q} \cup \{i\infty\}$.*

This solution is identical to the previous solution, except that all Fourier expansions now start at $n = 1$ instead of $n = 0$. □

7.1.9. *Let Γ' be a subgroup of $\mathrm{GL}_2^+(\mathbb{R})$ and $\mathcal{F}'$ a fundamental domain for the action of Γ' on $\mathbb{H}$. If $\alpha \in \mathrm{GL}_2^+(\mathbb{R})$, show that $\alpha^{-1}\mathcal{F}'$ is a fundamental domain for the action of $\alpha^{-1}\Gamma'\alpha$ on $\mathbb{H}$.*

It suffices to show that every point in $\mathbb{H}$ is $(\alpha^{-1}\Gamma'\alpha)$-equivalent to some point in $\alpha^{-1}\mathcal{F}'$ and that no two interior points of $\alpha^{-1}\mathcal{F}$ are $(\alpha^{-1}\Gamma'\alpha)$-equivalent since the map $z \mapsto \alpha^{-1}z$ is an analytic map. Indeed, if $z \in \mathbb{H}$, then $\alpha z \in \mathbb{H}$, and there is a $\gamma \in \Gamma'$ such that $\gamma(\alpha z) = w \in \mathcal{F}'$ since $\mathcal{F}'$ is a fundamental domain for Γ'. Hence $(\alpha^{-1}\gamma\alpha)z = \alpha^{-1}w$, which means that every point in $\mathbb{H}$ is $(\alpha^{-1}\Gamma'\alpha)$-equivalent to a point in $\alpha^{-1}\mathcal{F}'$.

Now, if $w_1, w_2 \in \alpha^{-1}\mathcal{F}'$ are $(\alpha^{-1}\Gamma'\alpha)$-equivalent, then there is a $\gamma_1 \in \Gamma'$ such that $(\alpha^{-1}\gamma_1\alpha)w_1 = w_2$. Since $w_i \in \alpha^{-1}\mathcal{F}'$ for $i = 1, 2$, we can write $w_i = \alpha^{-1}v_i$ with $v_i \in \mathcal{F}'$. Thus, $\alpha^{-1}\gamma_1 v_1 = \alpha^{-1}v_2$ which implies v_1 is Γ' equivalent to v_2. This means that v_1, v_2 lie on the boundary of $\mathcal{F}'$, and so w_1 and w_2 also lie on the boundary of $\alpha^{-1}\mathcal{F}'$. □

7.1.11. *Let $\alpha \in \mathrm{GL}_2^+(\mathbb{Q})$ have integer entries and determinant D. Let $\alpha' = D\alpha^{-1}$. Show that for $f, g \in S_k(\Gamma)$, with Γ a congruence subgroup of $\mathrm{SL}_2(\mathbb{Z})$, we have*

$$(f|\alpha, g) = (f, g|\alpha').$$

From (4.2) on page 36, we see that $g|\alpha' = g|\alpha^{-1}$. Thus, we need only replace g with $g|\alpha^{-1}$ in Theorem 7.1.10 to deduce the result. □

7.2 Commutativity of the Hecke operators

7.2.3. *Let p be prime. Show that for all $r, s \geq 1$, we have*

$$T_{p^r} T_{p^s} = T_{p^s} T_{p^r}.$$

Without loss of generality, $r \leq s$. We induct on s. For the base case $s = 1$, there is nothing to show. Assume $T_{p^r} T_{p^s} = T_{p^s} T_{p^r}$ whenever $1 \leq r \leq s \leq t$. We consider $T_{p^r} T_{p^{t+1}}$ where $1 \leq r \leq t+1$. If $r = t+1$, then there is nothing to show. Otherwise, $r \leq t$ and by Theorem 7.2.2 and the inductive hypothesis, we have

$$T_{p^r} T_{p^{t+1}} = T_{p^r} \left(T_{p^t} T_p - p^{k-1} T_{p^t} \right) = \left(T_{p^t} T_p - p^{k-1} T_{p^t} \right) T_{p^r} = T_{p^{t+1}} T_{p^r}. \quad \square$$

7.2.4. *Let p be prime. Show that there is a polynomial $P_s(x) \in \mathbb{Z}[x]$ such that $T_{p^s} = P_s(T_p)$.*

For $s = 0$ and $s = 1$, we take $P_0(x) = 1$ and $P_1(x) = x$, respectively. We proceed by induction on s. From Theorem 7.2.2, we have

$$\begin{aligned} T_{p^{s+1}} &= T_{p^s} T_p - p^{k-1} T_{p^{s-1}} \\ &= P_s(T_p) T_p - p^{k-1} P_{s-1}(T_p), \end{aligned}$$

which is a polynomial in T_p with integer coefficients by the induction hypothesis. $\square$

7.2.5. *Prove that for all $m, n \geq 1$, the Hecke operators commute:*

$$T_n T_m = T_m T_n.$$

Take prime power decompositions $n = p_1^{s_1} \cdots p_a^{s_a}$ and $m = q_1^{t_1} \cdots q_b^{t_b}$ so that by Theorem 7.2.1 we have

$$T_n T_m = T_{p_1^{s_1}} \cdots T_{p_a^{s_a}} T_{q_1^{t_1}} \cdots T_{q_b^{t_b}}.$$

When p_i and q_j are distinct primes, $T_{p_i^{s_i}} T_{q_j^{t_j}} = T_{q_j^{t_j}} T_{p_i^{s_i}}$ by Theorem 7.2.1. When $p_i = q_j$, we have $T_{p_i^{s_i}} T_{q_j^{t_j}} = T_{q_j^{t_j}} T_{p_i^{s_i}}$ by Exercise 7.2.3. Hence

$$T_n T_m = T_{q_1^{t_1}} \cdots T_{q_b^{t_b}} T_{p_1^{s_1}} \cdots T_{p_a^{s_a}} = T_m T_n,$$

as desired. $\square$

7.3 Hecke operators as Hermitian operators

7.3.3. *Suppose $f = \sum_{n=1}^{\infty} a_n q^n \in S_k(\mathrm{SL}_2(\mathbb{Z}))$ is an eigenfunction for all T_m. Prove that $a_1 \neq 0$ and that the eigenvalue of T_m is a_m/a_1.*

Suppose $T_m(f) = \lambda_m f$. By Theorem 5.2.4, the coefficient of q^1 in $T_m(f)$ is a_m. Hence $a_m = \lambda_m a_1$. If $a_1 = 0$, then $a_m = 0$ for all m and $f = 0$, contradicting the fact that f is an eigenvector. Thus $a_1 \neq 0$. □

7.3.4. *Let $f, g \in S_k(\mathrm{SL}_2(\mathbb{Z}))$ be normalized eigenfunctions for all the Hecke operators. Show that $f = g$ or $(f, g) = 0$.*

Write $f(z) = \sum_{n=1}^{\infty} a_n q^n$ and $g(z) = \sum_{n=1}^{\infty} b_n q^n$. Since f and g are normalized, the Fourier coefficients are the eigenvalues of the T_n's. Thus,

$$a_n(f, g) = (T_n(f), g) = (f, T_n(g)) = (f, b_n(g)) = \overline{b_n}(f, g).$$

The b_n's are real by Corollary 7.3.2 so that if $(f, g) \neq 0$, then $a_n = \overline{b_n} = b_n$ for all n. Hence $f = g$. □

7.4 Basis of eigenforms

7.4.3. *Show that E_k is an eigenfunction for all the Hecke operators acting on $M_k(\mathrm{SL}_2(\mathbb{Z}))$.*

It suffices to show this for the T_p's. Moreover, by linearity it suffices to show that $f = -\frac{B_k}{2k} E_k$ is an eigenfunction of T_p. We write $f = \sum_{n=0}^{\infty} a(n) q^n$ so that $T_p f = \sum_{n=0}^{\infty} b(n) q^n$ where

$$b(n) = \sum_{d|(n,p)} d^{k-1} a\left(\frac{np}{d^2}\right). \tag{7.1}$$

Notice that

$$b(0) = p^{k-1} a(0) + a(0) = \left(p^{k-1} + 1\right) a(0),$$

and so, by comparing the constant terms, we see that the only possible eigenvalue is $p^{k-1} + 1$. Thus, we need to check $T_p(f) = (p^{k-1} + 1) f$, but this is directly verified by comparing the Fourier coefficients of both sides and using the multiplicativity of $a(n) = \sigma_{k-1}(n)$ as follows.

If $(n, p) = 1$, then by (7.1)

$$b(n) = a(np) = a(p)a(n) = \left(p^{k-1} + 1\right) a(n).$$

If $(n, p) = p$, then write $n = p^t m$ where $(m, p) = 1$. Now (7.1) gives

$$\begin{aligned} b(n) &= p^{k-1}a(p^{t-1}m) + a(p^{t+1}m) \\ &= \Big(p^{k-1}a(p^{t-1}) + a(p^{t+1})\Big)a(m). \end{aligned}$$

Since $a(p^{t+1}) = \sum_{r=0}^{t+1} p^{r(k-1)} = p^{(t+1)(k-1)} + a(p^t)$, we have

$$\begin{aligned} b(n) &= \Big(p^{k-1}a(p^{t-1}) + p^{(k-1)}p^{t(k-1)} + a(p^t)\Big)a(m) \\ &= \Big(p^{k-1}a(p^t) + a(p^t)\Big)a(m) \\ &= \left(p^{k-1}+1\right)a(p^t m) \\ &= \left(p^{k-1}+1\right)a(n), \end{aligned}$$

as desired. □

7.4.4. *Show that* $(E_k, f) = 0$ *for any* $f \in S_k(\mathrm{SL}_2(\mathbb{Z}))$.

It suffices to prove this assertion when f is a Hecke eigenform since $S_k(\mathrm{SL}_2(\mathbb{Z}))$ has a basis of eigenforms by Theorem 7.4.2. Observe that Theorem 7.3.1 actually holds when one of the forms is a cusp form and the other is non-vanishing at $i\infty$. Thus we have

$$(p^{k-1}+1)(E_k, f) = (T_p(E_k), f) = (E_k, T_p(f)) = \lambda_p(E_k, f),$$

where $T_p(f) = \lambda_p f$. If $(E_k, f) \neq 0$, we have $p^{k-1} + 1 = \lambda_p$ for all primes p. But by Theorem 5.3.4 , $\lambda_p = O\big(p^{k/2}\big)$ and this is a contradiction. □

7.4.5. *Show that* Δ^k *is not a Hecke eigenform for any* $k \geq 2$.

The coefficient of q in the q-expansion of Δ^k is zero if $k \geq 2$. By Exercise 7.3.3, Δ^k is not an eigenform. □

7.5 Supplementary problems

7.5.1. *Let* $12 \mid k$ *and* $d = \dim S_k(\mathrm{SL}_2(\mathbb{Z}))$. *For each* $1 \leq j \leq d$, *define*

$$f_j := \Delta^j E_6^{2(d-j)} = \sum_{n=1}^{\infty} a_n^{(j)} q^n.$$

Verify that $a_n^{(j)} = 0$ *for* $n < j$ *and* $a_j^{(j)} = 1$. *Conclude that the* f_j *form a basis for* $S_k(\mathrm{SL}_2(\mathbb{Z}))$.

From the q-expansion $\Delta^j = q^j + \cdots$, it is evident that $a_n^{(j)} = 0$ for $n < j$ and $a_j^{(j)} = 1$. From this, we immediately see that the f_j's are linearly independent over $\mathbb{C}$. Moreover, the weight of f_j is $12d$ which equals k by Exercise 4.4.3. □

7.5.2. *Let $k \geq 4$ be even and $d = \dim S_k(\mathrm{SL}_2(\mathbb{Z}))$. Choose non-negative integers a, b such that $12 \neq 4a + 6b \leq 14$ and $4a + 6b \equiv k \pmod{12}$. For each $1 \leq j \leq d$, define*

$$f_j := \Delta^j E_6^{2(d-j)+b} E_4^a = \sum_{n=1}^{\infty} a_n^{(j)} q^n.$$

Verify that $a_n^{(j)} = 0$ for $n < j$ and $a_j^{(j)} = 1$. Conclude that the f_j form a basis for $S_k(\mathrm{SL}_2(\mathbb{Z}))$. (This basis is called the Miller basis *in the literature.)*

As in the previous exercise, the assertion about the vanishing of the Fourier coefficients is immediate. This implies that the f_j's are linearly independent. Moreover, the weight of f_j is $12d + 4a + 6b$ which is k, by an application of Exercise 4.4.3. □

7.5.3. *Let k, d, a, and b be as in Exercise 7.5.2. Verify that there exists a basis g_0, $g_1, \ldots, g_d$ for $M_k(\mathrm{SL}_2(\mathbb{Z}))$ such that*

$$g_i = q^i + \sum_{n=d+1}^{\infty} b_n^{(i)} q^n$$

with $b_n^{(i)} \in \mathbb{Z}$ for all n and all i. Conclude that if

$$f = \sum_{n=0}^{\infty} c_n q^n \in M_k(\mathrm{SL}_2(\mathbb{Z}))$$

has $c_0, c_1, \ldots, c_d \in \mathbb{Z}$, then $c_n \in \mathbb{Z}$ for all n.

It is straightforward to use linear algebra to explicitly produce $g_1, g_2, \ldots, g_d$ from the basis of f_i from Exercise 7.5.2 as follows. First of all, it is clear that all the f_j have integral coefficients and so we may set $g_d = f_d$. Then we use g_d to clear out the coefficient of q^d in f_{d-1} by setting

$$\begin{aligned} g_{d-1} &:= f_{d-1} - a_d^{(d-1)} g_d \\ &= \left(q^{d-1} + a_d^{(d-1)} q^d + O\left(q^{d+1}\right)\right) - a_d^{(d-1)} \left(q^d + O\left(q^{d+1}\right)\right) \\ &= q^{d-1} + O\left(q^{d+1}\right). \end{aligned}$$

In general, we recursively set

$$g_{d-j} = f_{d-j} - \sum_{s=0}^{j-1} a_{d-s}^{(d-j)} g_{d-s}.$$

In this way we produce $g_1, g_2, \ldots, g_d$. To create g_0, observe that $E_6^{2d+b} E_4^a = 1 + O(q) \in M_k(\mathrm{SL}_2(\mathbb{Z}))$ has integral coefficients. Use the g_i to clear out the coefficients of $q, q^2, \ldots, q^d$ as before.

Now since we have a basis for $M_k(\mathrm{SL}_2(\mathbb{Z}))$, any modular form $f = \sum c_n q^n \in M_k(\mathrm{SL}_2(\mathbb{Z}))$ can be expressed as a linear combination of $g_0, \ldots, g_d$. In fact, the weights are exactly the initial coefficients $c_0, \ldots, c_d$. If these are all integers, then clearly f is an integer linear combination of forms with integral coefficients. □

7.5.4. *Let $d = \dim S_k(\mathrm{SL}_2(\mathbb{Z}))$ and fix a positive integer m. Suppose*

$$f = \sum_{n=0}^{\infty} c_n q^n \in M_k(\mathrm{SL}_2(\mathbb{Z}))$$

is such that for $i = 0, 1, \ldots, d$, we have that $c_i \in \mathbb{Z}$ and that $c_i \equiv 0 \pmod{m}$. Prove that $c_i \equiv 0 \pmod{m}$ for all $i \geq 0$.

This is a straightforward generalization from Exercise 7.5.3. It is clear that

$$f = c_0 g_0 + c_1 g_1 + \cdots + c_d g_d$$

where the g_i are the basis vectors from Exercise 7.5.3. Since $m \mid c_i$ for $i = 0, 1, \ldots, d$, we have that

$$f = m\left(\frac{c_0}{m} g_0 + \cdots \frac{c_d}{m} g_d\right) \in m\mathbb{Z}[\![q]\!]$$

as desired. □

7.5.5. *Let f be a Hecke eigenform for the full modular group. Show that the Fourier coefficients of f are algebraic integers.*

We show that the eigenvalues of the nth Hecke operator T_n are algebraic integers. Indeed, let $f_1, \ldots, f_d$ be the Miller basis constructed in Exercise 7.5.2. The f_j's have $\mathbb{Z}$-coefficients and $T_n(f_j)$ can be written as a $\mathbb{Z}$-linear combination of the f_j's. Thus, the matrix representing T_n with respect to the Miller basis has $\mathbb{Z}$-coefficients. Consequently, the characteristic polynomial of T_n is monic with coefficients from $\mathbb{Z}$, so the eigenvalues of T_n are algebraic integers. □

Chapter 8
Hecke Operators of Higher Level

8.1 Hecke operators for congruence subgroups

8.1.1. *Let p be prime and $\Gamma = \mathrm{SL}_2(\mathbb{Z})$. Let $\mathbf{X}_p$ be the set of matrices with integer entries of determinant p. Show that*

$$\mathbf{X}_p = \Gamma \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} \Gamma.$$

Clearly $\Gamma \left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right) \Gamma \subseteq \mathbf{X}_p$. To show the reverse inclusion, suppose $\sigma \in \mathbf{X}_p$. By Theorem 5.2.1, there exists $\gamma_1 \in \Gamma$ such that

$$\sigma = \gamma_1 \begin{pmatrix} a & b \\ 0 & d \end{pmatrix}$$

with $ad = p$ and $0 \leq b \leq d-1$. If $a = p$, then $d = 1$, $b = 0$ and

$$\sigma = \gamma_1 \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \in \Gamma \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} \Gamma.$$

Otherwise $a = 1$ and $d = p$. Observe that

$$\begin{pmatrix} 1 & b \\ 0 & p \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}$$

and

$$\begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

so that

$$\sigma = \gamma_1 \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \in \Gamma \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix} \Gamma. \qquad \square$$

8.1.4. *For fixed N, S^+, and $S^\times$, show that*

© Springer Science+Business Media Singapore 2016 and Hindustan Book Agency 2016

M.R. Murty et al., *Problems in the Theory of Modular Forms*, HBA Lecture Notes in Mathematics, DOI 10.1007/978-981-10-2651-5_18

$$X_m X_n \subseteq X_{mn}.$$

Clearly, the determinant of any element of $X_m X_n$ is mn. Moreover, since

$$\begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix} = \begin{pmatrix} a_1a_2 + b_1c_2 & a_1b_2 + b_1d_2 \\ c_1a_2 + d_1c_2 & c_1b_2 + d_1d_2 \end{pmatrix},$$

we see that if $N|c_1$ and $N|c_2$, then $N|(c_1a_2 + d_1c_2)$. Also, $a_1, a_2 \in S^\times$ implies $a_1a_2 \in S^\times$. Thus $a_1a_2 + b_1c_2 \in S^\times$ because $N|c_2$. Finally, $a_1b_2 + b_1d_2 \in S^+$. □

8.1.5. *Show that*

$$X_1(N, S^\times, S^+)$$

is a congruence subgroup of $\mathrm{SL}_2(\mathbb{Z})$.

By Exercise 8.1.4, we see that X_1 is closed under matrix multiplication. Moreover, X_1 clearly contains the identity element and is a subgroup of $\mathrm{SL}_2(\mathbb{Z})$. Finally, if we write $S^+ = M\mathbb{Z}$, then $X_1 \supseteq \Gamma(NM)$. □

8.2 Nebentypus

8.2.1. *Let χ be a Dirichlet character modulo N. For each $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma_0(N)$, define $\psi(\gamma) = \chi(d)$. Show that $\psi(\gamma_1\gamma_2) = \psi(\gamma_1)\psi(\gamma_2)$.*

Suppose $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right), \left(\begin{smallmatrix} r & s \\ t & u \end{smallmatrix}\right) \in \Gamma_0(N)$. Then $c \equiv 0 \pmod{N}$ and so

$$\psi\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} e & f \\ g & h \end{pmatrix}\right) = \psi\left(\begin{pmatrix} \star & \star \\ \star & cf + dh \end{pmatrix}\right) = \chi(cf + dh) = \chi(dh) = \chi(d)\chi(h),$$

since χ is a homomorphism. □

8.2.2. *Show that the set of Dirichlet characters modulo N forms a group under multiplication of order $\phi(N)$, where ϕ denotes the Euler function.*

By the Chinese remainder theorem $(\mathbb{Z}/N\mathbb{Z})^\times \cong \prod_p (\mathbb{Z}/p^\alpha\mathbb{Z})^\times$ where $N = \prod_p p^\alpha$ is the unique factorization of N into distinct prime powers. Thus any Dirichlet character factors as a product of Dirichlet characters modulo p^α for $p|N$. For p odd, $(\mathbb{Z}/p^\alpha\mathbb{Z})^\times$ is a cyclic group of order $\phi(p^\alpha)$ (see for example Exercise 2.2.4 of [24]). Thus, a Dirichlet character modulo p^α is completely determined once it is defined on a generator and there are $\phi(p^\alpha)$ such Dirichlet characters for each odd prime p.

If $p = 2$ and $\alpha \geq 3$, then $(\mathbb{Z}/2^\alpha\mathbb{Z})^\times \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2^{\alpha-2}\mathbb{Z}$ (see Exercise 2.2.6 of [24]) which has $2 \cdot 2^{\alpha-2} = \phi(2^\alpha)$ elements. Since $(\mathbb{Z}/2\mathbb{Z})^\times$ and $(\mathbb{Z}/4\mathbb{Z})^\times$ are cyclic of orders 1 and 2 respectively, we see that in all cases there are $\phi(2^\alpha)$ Dirichlet characters modulo 2^α. □

8.2.3. *Show that*

$$\sum_{\chi} \chi(d) = \begin{cases} \phi(N) & \text{if } d \equiv 1 \pmod{N} \\ 0 & \text{otherwise} \end{cases}$$

where the sum is over all Dirichlet characters modulo N.

If $d = 1$, the result is clear by Exercise 8.2.2. Therefore, suppose $d \not\equiv 1 \pmod{N}$. Then there is a character ψ such that $\psi(d) \neq 1$. Thus

$$T = \sum_{\chi} \chi(d) = \sum_{\chi} (\chi\psi)(d) = \psi(d)T$$

because as χ ranges over all Dirichlet characters, so does $\chi\psi$. Hence $T = 0$ since $\psi(d) \neq 1$. □

8.2.4. *If $\chi \neq 1$ is a Dirichlet character modulo N, show that*

$$\sum_{a \in (\mathbb{Z}/N\mathbb{Z})^\times} \chi(a) = 0.$$

Since χ is not trivial, there is a $b \neq 1$ such that $\chi(b) \neq 1$ with $(b, N) = 1$. Then

$$S = \sum_{a \in (\mathbb{Z}/N\mathbb{Z})^\times} \chi(a) = \sum_{a \in (\mathbb{Z}/N\mathbb{Z})^\times} \chi(ab) = \chi(b)S$$

because as a ranges over coprime residue classes, so does ab. Therefore $s = 0$ since $\chi(b) \neq 1$. □

8.2.6. *Let $f \in M_k(\Gamma_0(N), \chi)$. If $f(z) = \sum_{m=0}^{\infty} \lambda(m) q^m$, then prove that*

$$T_n(f) = \sum_{m=0}^{\infty} \left(\sum_{a \mid m,n} \chi(a) a^{k-1} \lambda\left(\frac{mn}{a^2}\right) \right) q^m.$$

We proceed exactly as in the proof of Theorem 5.2.4 to derive this result. □

8.2.7. *If $f = \sum_{m=0}^{\infty} \lambda(m) q^m \in M_k(\Gamma_0(N), \chi)$ is an eigenfunction for all T_n's, show that $\lambda(1) \neq 0$ and $\lambda(mn)\lambda(1) = \lambda(m)\lambda(n)$ for $(m, n) = 1$.*

Let $T_n(f) = \mu_n f$. Then by Exercise 8.2.6 when $(m, n) = 1$ the coefficient of q^m is $\lambda(mn) = \mu_n \lambda(m)$. Putting $m = 1$, we get $\lambda(n) = \mu_n \lambda(1)$. Plugging this into the previous equation, we get $\lambda(mn)\lambda(1) = \lambda(m)\lambda(n)$ as desired. If $\lambda(1) = 0$, then all coefficients $\lambda(n) = 0$, contrary to the definition of an eigenvector. □

8.2.8. *For $p|N$, and $f \in M_k(\Gamma_0(N), \chi)$, show that*

$$T_p(f) = p^{\frac{k}{2}-1} \sum_{b=0}^{p-1} f \left| \begin{pmatrix} 1 & b \\ 0 & p \end{pmatrix} \right. .$$

In the sum defining T_n, we see from Equation (8.3) on page 100 that this is immediate. □

8.2.9. *Let $f = \sum_{n=0}^{\infty} a_n q^n \in M_k(\Gamma_0(N), \chi)$ and then write $T_p(f) = \sum_{n=0}^{\infty} b_n q^n$. Show that*

$$b_n = a_{pn} + \chi(p) p^{k-1} a_{n/p},$$

where we interpret $a_{n/p} = 0$ when $p \nmid n$.

This is clear from Exercise 8.2.6. □

8.2.10. *Let $f = \sum_{m=0}^{\infty} \lambda(m) q^m \in M_k(\Gamma_0(N), \chi)$ be a normalized eigenfunction for all the T_n's. Show that for p prime,*

$$\lambda(p^{a+1}) = \lambda(p)\lambda(p^a) - \chi(p) p^{k-1} \lambda(p^{a-1})$$

for $a \geq 1$.

This again is immediate from Exercise 8.2.6 and the solution to Exercise 8.2.7. We have $T_n(f) = \lambda(n) f$. When $n = p^a$, we compare coefficients of q^p of both sides. On the right hand side, we have $\lambda(p^a)\lambda(p)$. On the left hand side, we have $\lambda(p^{a+1}) + \chi(p) p^{k-1} \lambda(p^{a-1})$. □

8.2.13. *Show that $S_k(\Gamma_0(N), \chi)$ has a basis of eigenforms for all the T_n with $(n, N) = 1$.*

This is immediate from Theorem 7.4.1, Corollary 8.2.12, and the fact that the algebra of the T_n's is a commutative algebra. □

8.3 Oldforms and newforms

8.3.2. *Show that $\Delta(z)$ and $\Delta(2z)$ are both eigenforms for the T_n's (n odd) acting on $S_{12}(\Gamma_0(2))$, with the same eigenvalues.*

This is clear from Example 8.3.1 applied to $k = 12$, $\chi = 1$ and $N = 2$. □

8.3.3. *Show that $\Delta(z)$ and $\Delta(2z)$ are linearly independent in $S_{12}(\Gamma_0(2))$.*

We only need to look at the q-expansions:

$$\Delta(z) = q - 24q^2 + \cdots$$
$$\Delta(2z) = q^2 - 24q^4 + \cdots$$

which are clearly linearly independent. □

8.4 Supplementary problems

8.4.1. *Let* $w_N = \left(\begin{smallmatrix} 0 & -1 \\ N & 0 \end{smallmatrix}\right)$. *Prove that* $w_N \Gamma_0(N) w_N^{-1} \subset \Gamma_0(N)$.

Let $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma_0(N)$. Then $w_N \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) w_N^{-1} = \left(\begin{smallmatrix} d & -c/N \\ -bN & a \end{smallmatrix}\right) \in \Gamma_0(N)$. □

8.4.2. *Show that if* $f \in M_k(\Gamma_0(N))$, *then* $f|w_N \in M_k(\Gamma_0(N))$.

If $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \Gamma_0(N)$, then by Exercise 8.4.1, we have

$$(f|w_N)|\gamma = \left(f \left| \begin{pmatrix} d & -c/N \\ -bN & a \end{pmatrix}\right.\right) |w_N = f|w_N.$$

The form $f|w_N$ is holomorphic at the cusps by Exercise 7.1.6. □

8.4.3. *Show that if* $f \in S_k(\Gamma_0(N))$, *then* $f|w_N \in S_k(\Gamma_0(N))$.

The solution is identical to that of Exercise 8.4.2 except that now we appeal to Exercise 7.1.7 to deduce that $f|w_N$ vanishes at every cusp. □

8.4.4. *Show that if* $f \in M_k(\Gamma_0(N))$, *then* $(f|w_N)\,|w_N = f$. *Thus* $|w_N$ *is, in fact, an involution.*

We compute $w_N^2 = \left(\begin{smallmatrix} -N & 0 \\ 0 & -N \end{smallmatrix}\right)$. By Equation (4.2) on page 36, we see that $f|w_N^2 = f$. □

Chapter 9
Dirichlet Series and Modular Forms

9.1 General Dirichlet series

9.1.1. *Show that the series (9.3) on page 106 converges absolutely for* $\mathrm{Re}(s) > 1$.

Writing $s = \sigma + it$, with $\sigma > 1$, and $t \in \mathbb{R}$, we have

$$\left| \sum_{n=1}^{\infty} \frac{1}{n^s} \right| \leq \sum_{n=1}^{\infty} \frac{1}{n^\sigma} < \infty$$

for $\sigma > 1$, as is easily shown by a simple application of the integral test. □

9.1.2. *Given a sequence of complex numbers* a_n*, we suppose that for any* $\epsilon > 0$*, there is a constant* $C(\epsilon)$ *such that* $|a_n| \leq C(\epsilon) n^\epsilon$ *for all natural numbers* n*. Show that*

$$\sum_{n=1}^{\infty} \frac{a_n}{n^s}$$

converges absolutely for $\mathrm{Re}(s) > 1$.

Let $s = \sigma + it$, with $t \in \mathbb{R}$ and $\sigma > 1$. Choose $\epsilon > 0$ so that $\epsilon < \sigma - 1$. Then

$$\left| \sum_{n=1}^{\infty} \frac{a_n}{n^s} \right| \leq C(\epsilon) \sum_{n=1}^{\infty} \frac{1}{n^{\sigma - \epsilon}} < \infty,$$

as in the previous exercise. □

9.1.3. *Prove the identity (9.4) on page 106 for* $\mathrm{Re}(s) > 1$.

Since every natural number can be written as a product of powers of distinct primes, the result is immediate upon noting that

$$\prod_p \left(1 - \frac{1}{p^s}\right)^{-1} = \prod_p \left(1 + \frac{1}{p^s} + \frac{1}{p^{2s}} + \cdots\right),$$

© Springer Science+Business Media Singapore 2016 and Hindustan Book Agency 2016

M.R. Murty et al., *Problems in the Theory of Modular Forms*, HBA Lecture Notes in Mathematics, DOI 10.1007/978-981-10-2651-5_19

where the product is over distinct prime numbers p. A typical term in the Dirichlet series obtained by expanding the product on the right hand side is

$$\frac{1}{(p_1^{a_1}\cdots p_k^{a_k})^s}.$$

Since every natural number n can be written uniquely as $n = p_1^{\alpha_1}\cdots p_k^{\alpha_k}$ with distinct prime numbers $p_1,\dots,p_k$, we immediately deduce (9.4) of page 106. □

9.1.4. *Let $f(n)$ be a multiplicative function. That is, $f(mn) = f(m)f(n)$ for $(m,n) = 1$. Assume that for any $\epsilon > 0$, there is a constant $C(\epsilon)$ such that $|f(n)| \le C(\epsilon)n^\epsilon$. Show that for $\mathrm{Re}(s) > 1$, we have*

$$\sum_{n=1}^{\infty}\frac{f(n)}{n^s} = \prod_p\left(1+\frac{f(p)}{p^s}+\frac{f(p^2)}{p^{2s}}+\cdots\right),$$

where the product is over the prime numbers.

By Exercise 9.1.2, the Dirichlet series converges absolutely and so does each factor of the product. A typical term of the expanded product has the form

$$\frac{f(p_1^{a_1})\cdots f(p_k^{a_k})}{(p_1^{a_1}\cdots p_k^{a_k})^s}.$$

If $n = p_1^{\alpha_1}\cdots p_k^{\alpha_k}$ is the unique factorization of the natural number n, then we see that this is $f(n)/n^s$ since f is multiplicative. □

9.1.6. *Suppose that $\{a_n\}_{n=1}^\infty$ is a sequence of complex numbers and $f(u)$ is a continuously differentiable function on the interval $[1,x]$. Let $A(t) = \sum_{n\le t} a_n$. Prove that*

$$\sum_{n\le x} a_n f(n) = A(x)f(x) - \int_1^x A(t)f'(t)dt.$$

First suppose that x is a natural number. We apply Abel's lemma (Proposition 9.1.5) with $b_n = f(n)$, $m = 1$, $r = x$ to get

$$\sum_{n\le x} a_n f(n) = A(x)f(x) - \sum_{n\le x-1} A(n)\big(f(n+1)-f(n)\big).$$

Writing

$$f(n+1)-f(n) = \int_n^{n+1} f'(t)dt,$$

we obtain

$$\sum_{n\le x} a_n f(n) = A(x)f(x) - \sum_{n\le x-1} A(n)\int_n^{n+1} f'(t)dt.$$

Since $A(t)$ is a step function, we see that the latter sum can be written as

$$\sum_{n\le x-1}\int_n^{n+1} A(t)f'(t)dt = \int_1^x A(t)f'(t)dt$$

which proves the result if x is a natural number. If x is not a natural number, let us write $\lfloor x\rfloor$ for the greatest integer $\le x$ and note that

$$\sum_{n\le x} a_n f(n) = \sum_{n\le \lfloor x\rfloor} a_n f(n).$$

To complete the proof, we need to show that

$$A(x)\left(f(x)-f(\lfloor x\rfloor)\right) - \int_{\lfloor x\rfloor}^x A(t)f'(t)dt = 0,$$

which is evident since $A(t)$ is a step function. □

9.1.7. *Let $\alpha > \beta > 0$. Write $z = x + iy$, with $x, y \in \mathbb{R}$, $i = \sqrt{-1}$, and $x > 0$. Show that*

$$\left|e^{-\alpha z} - e^{-\beta z}\right| \le \frac{|z|}{x}\left(e^{-\alpha x} - e^{-\beta x}\right).$$

We have

$$e^{-\alpha z} - e^{-\beta z} = z\int_\alpha^\beta e^{-tz}dt$$

so that

$$\begin{aligned}\left|e^{-\alpha z} - e^{-\beta z}\right| &\le |z|\int_\alpha^\beta \left|e^{-tz}\right| dt\\ &= |z|\int_\alpha^\beta e^{-tx}dt\\ &= \frac{|z|}{x}\left(e^{-\alpha x} - e^{-\beta x}\right).\end{aligned}$$

□

9.1.9. *Suppose that*

$$A(x) = \sum_{n\le x} a_n = O(x^\delta)$$

for some $\delta > 0$. Show that

$$\sum_{n=1}^\infty \frac{a_n}{n^s} = s\int_1^\infty \frac{A(t)dt}{t^{s+1}}$$

and that the integral converges absolutely for $\mathrm{Re}(s) > \delta$.

We apply Exercise 9.1.6 with $f(n) = n^{-s}$ to get

$$\sum_{n\le x} a_n n^{-s} = \frac{A(x)}{x^s} + s\int_1^x \frac{A(t)}{t^{s+1}}dt.$$

For any s with $\mathrm{Re}(s) > \delta$, we have

$$\lim_{x\to\infty} \frac{A(x)}{x^s} = 0$$

since $A(x) = O(x^\delta)$. Thus

$$\sum_{n=1}^{\infty} \frac{a_n}{n^s} = s\int_1^\infty \frac{A(t)dt}{t^{s+1}},$$

and the integral converges absolutely for $\mathrm{Re}(s) > \delta$. □

9.1.10. *Show that*

$$\zeta(s) = \frac{s}{s-1} - s\int_1^\infty \frac{\{t\}}{t^{s+1}}dt,$$

where $\{t\} = t - \lfloor t\rfloor$ *and the integral converges absolutely for* $\mathrm{Re}(s) > 0$. *Deduce that* $\lim_{s\to 1^+}(s-1)\zeta(s) = 1$.

By the previous exercise,

$$\zeta(s) = s\int_1^\infty \frac{\lfloor t\rfloor}{t^{s+1}}dt.$$

Since $\lfloor t\rfloor = t - \{t\}$, we can rewrite the integral as

$$s\int_1^\infty \frac{t-\{t\}}{t^{s+1}}dt = \frac{s}{s-1} - s\int_1^\infty \frac{\{t\}}{t^{s+1}}dt.$$

Since $0 \le \{t\} < 1$, the integral converges absolutely for $\mathrm{Re}(s) > 0$. Multiplying by $(s-1)$ and taking limits as $s \to 1^+$, we deduce

$$\lim_{s\to 1^+}(s-1)\zeta(s) = 1.$$ □

9.1.11. *Show that the series*

$$1 - \frac{1}{2^s} + \frac{1}{3^s} - \frac{1}{4^s} + \frac{1}{5^s} - \cdots$$

has abscissa of convergence $\sigma_0 = 0$ *and abscissa of absolute convergence* $\sigma_1 = 1$.

The fact that $\sigma_1 = 1$ is clear by the divergence of the harmonic series. To see that $\sigma_0 = 0$, we apply Exercise 9.1.9 in which the corresponding sum $A(x) = O(1)$ so that the series converges for any s with $\mathrm{Re}(s) > 0$. Clearly, for $s = 0$, the series does not converge and so $\sigma_0 = 0$. □

9.1.12. *Show that*

$$f(s) = \sum_{n=1}^{\infty} a_n e^{-\lambda_n s}$$

can be identically zero only if all its coefficients a_n are zero.

Multiplying $f(s)$ by $e^{\lambda_1 s}$, we get

$$e^{\lambda_1 s} f(s) = a_1 + \sum_{n=2}^{\infty} a_n e^{-(\lambda_n - \lambda_1)s}.$$

Now, assuming s is real, let $s \to +\infty$. By the uniform convergence, we see that $e^{\lambda_1 s} f(s)$ tends to a_1. Since $f(s)$ is identically zero, $a_1 = 0$. Proceeding inductively, we deduce $a_n = 0$ for all n. □

9.1.14. *Prove the identity*

$$\sum_{n=0}^{\infty} \left(\frac{\alpha^{n+1} - \beta^{n+1}}{\alpha - \beta} \right) \left(\frac{\gamma^{n+1} - \delta^{n+1}}{\gamma - \delta} \right) T^n = \frac{1 - \alpha\beta\gamma\delta T^2}{(1 - \alpha\gamma T)(1 - \alpha\delta T)(1 - \beta\gamma T)(1 - \beta\delta T)}.$$

The series on the left hand side is easily seen to be

$$(\alpha - \beta)^{-1}(\gamma - \delta)^{-1} \left(\frac{\alpha\gamma}{1 - \alpha\gamma T} - \frac{\alpha\delta}{1 - \alpha\delta T} - \frac{\beta\gamma}{1 - \beta\gamma T} + \frac{\beta\delta}{1 - \beta\delta T} \right),$$

which simplifies to the expression on the right hand side upon choosing a common denominator. This example is due to S. Ramanujan. □

9.1.15. *A function $a : \mathbb{N} \to \mathbb{C}$ is called an* arithmetical function. *Given two arithmetical functions a and b, we define the* Dirichlet convolution *$a * b$, to be the arithmetical function given by*

$$(a * b)(n) = \sum_{d|n} a(d) b\left(\frac{n}{d}\right)$$

where the sum is over positive divisors d of n. Show that

$$\sum_{n=1}^{\infty} \frac{(a * b)(n)}{n^s} = \left(\sum_{n=1}^{\infty} \frac{a(n)}{n^s} \right) \left(\sum_{n=1}^{\infty} \frac{b(n)}{n^s} \right).$$

Expanding the product on the right hand side, we obtain

$$\sum_{d,e|n} \frac{a(d)b(e)}{d^s e^s}.$$

Collecting terms with $de = n$ gives the identity. The identity should be viewed in the ring of formal Dirichlet series so that questions of convergence need not be considered here. □

9.1.16. *An arithmetical function $a : \mathbb{N} \to \mathbb{C}$ is said to be* completely multiplicative *if $a(mn) = a(m)a(n)$ for all m, n. Show that*

$$\sum_{n=1}^{\infty} \frac{a(n)}{n^s} = \prod_p \left(1 - \frac{a(p)}{p^s}\right)^{-1}.$$

This is evident from Exercise 9.1.4. Indeed,

$$\sum_{n=1}^{\infty} \frac{a(n)}{n^s} = \prod_p \left(1 + \frac{a(p)}{p^s} + \frac{a(p^2)}{p^{2s}} + \cdots\right).$$

Since a is completely multiplicative, $a(p^m) = a(p)^m$ so that each Euler factor in the product is a geometric series that is easily summed to give the desired result. Again, the identity should be viewed in the ring of formal Dirichlet series. □

9.1.17. *Given any arithmetical function $a : \mathbb{N} \to \mathbb{C}$, we will write*

$$L(s, a) := \sum_{n=1}^{\infty} \frac{a(n)}{n^s}.$$

If a, b, c, d are completely multiplicative functions, show that

$$L(s, (a * b)(c * d)) = \frac{L(s, ac)L(s, ad)L(s, bc)L(s, bd)}{L(2s, abcd)}.$$

Here ac denotes the arithmetical function given by $(ac)(n) := a(n)c(n)$.

Since a, b, c, d are completely multiplicative, $a * b$ and $c * d$ are multiplicative. By Exercise 9.1.4, the left hand side has an Euler product whose p-th Euler factor is given by

$$\sum_{n=0}^{\infty} \left(\frac{a(p)^{n+1} - b(p)^{n+1}}{a(p) - b(p)}\right) \left(\frac{c(p)^{n+1} - d(p)^{n+1}}{c(p) - d(p)}\right) p^{-ns}.$$

By Exercise 9.1.14, with $T = p^{-s}$, we see that this simplifies to

$$\frac{1 - a(p)b(p)c(p)d(p)T^2}{(1 - a(p)c(p)T)(1 - a(p)d(p)T)(1 - b(p)c(p)T)(1 - b(p)d(p)T)}$$

from which the result is now evident. □

9.1.18. *Let*

$$\sigma_u(n) = \sum_{d|n} d^u.$$

Show that for any complex numbers a, b,

$$\sum_{n=1}^{\infty} \frac{\sigma_a(n)\sigma_b(n)}{n^s} = \frac{\zeta(s)\zeta(s-a)\zeta(s-b)\zeta(s-a-b)}{\zeta(2s-a-b)}.$$

The function σ_u is a Dirichlet convolution of two completely multiplicative arithmetical functions so that the previous exercise is applicable. The result is now immediate. We leave the details to the student. □

9.1.19. *Let $t_0 \in \mathbb{R}$. With $a = it_0$, $b = -it_0$ in the previous exercise, show that*

$$\sum_{n=1}^{\infty} \frac{(\sigma_{it_0}(n))^2}{n^s} = \frac{\zeta(s)^2\zeta(s-it_0)\zeta(s+it_0)}{\zeta(2s)}.$$

Using Landau's theorem (Theorem 9.1.13), show that $\zeta(1+it_0) \neq 0$.

The identity is immediate from the previous exercise. If $\zeta(1+it_0) = 0$, then $\zeta(1-it_0) = 0$, as can be deduced from the expression for $\zeta(s)$ given in Exercise 9.1.10. Thus, the double pole at $s = 1$ of $\zeta(s)^2$ is canceled by the double zero of $\zeta(s-it_0)\zeta(s+it_0)$ at $s = 1$. Therefore, the right hand side of the identity is analytic for $\mathrm{Re}(s) > \frac{1}{2}$. By Landau's theorem, the Dirichlet series has a singularity at its abscissa of convergence which must be $\leq \frac{1}{2}$. However, the right hand side has a zero at $s = 1/2$ (since the denominator has a pole there). However, the left hand side is ≥ 1 (coming from $n = 1$) as $s \to 1/2^+$. This contradiction shows $\zeta(1+it_0) \neq 0$. □

9.2 The Poisson summation formula

9.2.2. *If $F \in S$, show that*

$$\left| \int_{-\infty}^{\infty} t^m F^{(n)}(t) dt \right| < \infty$$

for all pairs of integers $m, n \geq 0$.

Since $F \in S$,

$$\sup_{t \in \mathbb{R}} \left| t^{m+2} F^{(n)}(t) \right| < \infty.$$

Thus

$$\left| \int_1^{\infty} t^m F^{(n)}(t) dt \right| \ll \int_1^{\infty} \frac{dt}{t^2} < \infty.$$

Similarly,

$$\left|\int_{-\infty}^{1} t^m F^{(n)}(t)dt\right| < \infty.$$

Finally

$$\left|\int_{-1}^{1} t^m F^{(n)}(t)dt\right| < \infty$$

and this completes the proof. □

9.2.3. *If $F \in S$, show that $\hat{F} \in S$.*

Since $F \in S$, the integral

$$\int_{-\infty}^{\infty} F(t)e^{-2\pi itx}dt$$

is easily shown to be absolutely convergent.

Differentiating n times gives

$$\hat{F}^{(n)}(x) = \int_{-\infty}^{\infty} (2\pi it)^n F(t)e^{-2\pi itx}dt.$$

Keeping in mind that $F \in S$, we integrate the right hand side by parts m times to get

$$\hat{F}^{(n)}(x) = \frac{1}{(2\pi ix)^m}\int_{-\infty}^{\infty} \frac{d^m}{dt^m}\left((2\pi it)^n F(t)\right)e^{-2\pi itx}dt.$$

Thus

$$\left|x^m \hat{F}^{(n)}(x)\right| \leq (2\pi)^{n-m}\left|\int_{-\infty}^{\infty} \frac{d^m}{dt^m}\left(t^n F(t)\right)e^{-2\pi itx}dt\right|.$$

The integral is a sum of terms of the form

$$\int_{-\infty}^{\infty} t^r F^{(s)}(t)e^{-2\pi itx}dt.$$

By Exercise 9.2.2, these integrals are absolutely convergent. This completes the proof. □

9.2.4. *Show that*

$$\int_{-\infty}^{\infty} e^{-\pi x^2}dx = 1.$$

Let

$$I = \int_{-\infty}^{\infty} e^{-\pi x^2}dx.$$

Then

$$I^2 = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-\pi(x^2+y^2)}dxdy.$$

Switching to polar co-ordinates, we put $x = r\cos\theta$, $y = r\sin\theta$, $0 \le \theta \le 2\pi$, $r > 0$. We find

$$\begin{aligned} I^2 &= \int_0^\infty \int_0^{2\pi} e^{-\pi r^2} r d\theta dr \\ &= \int_0^\infty e^{-\pi r^2}(2\pi r) dr = 1. \end{aligned}$$

Since $I > 0$, we conclude $I = 1$. □

9.2.5. *For $F(x) = e^{-\pi x^2}$, show that $\hat{F}(u) = e^{-\pi u^2}$. In other words, F is its own Fourier transform.*

We must prove that

$$\int_{-\infty}^{\infty} e^{-\pi x^2} e^{-2\pi i x u} dx = e^{-\pi u^2}.$$

This is equivalent to

$$J(u) := \int_{-\infty}^{\infty} e^{-\pi(x+iu)^2} dx = 1.$$

For $u = 0$, this follows from Exercise 9.2.4.
For the general case, we observe that

$$\begin{aligned} \frac{\partial}{\partial u}\left(\int_{-\infty}^{\infty} e^{-\pi(x+iu)^2} dx\right) &= \int_{-\infty}^{\infty} \left(\frac{\partial}{\partial u} e^{-\pi(x+iu)^2}\right) dx \\ &= 2\pi i \int_{-\infty}^{\infty} (x + iu e^{-\pi(x+iu)^2} dx \\ &= i \int_{-\infty}^{\infty} \left(\frac{\partial}{\partial x} e^{-\pi(x+iu)^2}\right) dx \\ &= \left[ie^{-\pi(x+iu)^2}\right]_{-\infty}^{\infty} = 0. \end{aligned}$$

Thus $J'(u) = 0$ so that $J(u)$ is constant. Since $J(0) = 1$, we deduce $J(u) = 1$, as desired. □

9.2.6. *Let $F \in S$ and define $G(t) = F(At + B)$ for real constants A, B, with $A \neq 0$. Show that $A\hat{G}(t) = e^{2\pi i B t/A}\hat{F}(t/A)$.*

This is immediate from the definition of the Fourier transform and an appropriate change of variables. □

9.2.8. *Show that if $F(x) = e^{-(Ax+B)^2\pi}$ with $A, B \in \mathbb{R}$ and $A \neq 0$, then*

$$A\hat{F}(x) = e^{2\pi i B x/A} e^{-\pi(x+B)^2}.$$

This is immediate from Exercises 9.2.5 and 9.2.6. □

9.2.9. *For any $a \in \mathbb{R}$, show that*

$$\sum_{n\in\mathbb{Z}} e^{-\pi(n+a)^2/x} = x^{1/2} \sum_{n\in\mathbb{Z}} e^{-\pi n^2 x + 2\pi i n a}.$$

We apply the Poisson summation formula to the function $F(t) = e^{-\pi(t+a)^2/x}$. By Exercise 9.2.8, the Fourier transform of F is

$$\sqrt{x} e^{2\pi i t a} e^{-\pi x t^2}.$$

Applying the Poisson summation formula gives the desired result. □

9.2.10. *Show that for $x > 0$,*

$$\sum_{n\in\mathbb{Z}} e^{-\pi n^2 x} = x^{-1/2} \sum_{n\in\mathbb{Z}} e^{-\pi n^2/x}.$$

Put $a = 0$ in Exercise 9.2.9. □

9.2.12. *Show that $(s-1)\zeta(s)$ extends to an entire function.*

Theorem 9.2.11 shows that

$$s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$$

is entire. Since $s\Gamma(s) = \Gamma(s+1)$, we see that $s\Gamma(s/2) = 2\Gamma(\frac{s}{2}+1)$ is analytic at $s = 0$. Therefore $\zeta(s)$ is analytic at $s = 0$. Since Theorem 9.2.11 gives an expression for $\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s)$ which is valid for all s and as $\Gamma(s)$ is analytic at $s = 1$, we see that $\zeta(s)$ has a simple pole at $s = 1$ which is also evident from Exercise 9.1.10. □

9.2.13. *Show that $\zeta(s)$ has simple zeros at $s = -2n$, for each positive integer n.*

By Theorem 9.2.11,

$$\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s)$$

is analytic for all $s \in \mathbb{C}, s \neq 0, 1$. Since $\Gamma(s)$ has simple poles at $s = 0, -1, -2, \ldots$ we deduce that $\zeta(s)$ vanishes for $s = -2, -4, \ldots$. □

9.2.14. *Show that $\zeta(0) = -1/2$.*

Since

$$\pi^{-s/2}\Gamma\left(s/2\right)\zeta(s) = \frac{1}{s-1} - \frac{1}{s} + \frac{1}{2}\int_1^\infty (\vartheta(it) - 1)(t^{s/2} + t^{\frac{1-s}{2}})\frac{dt}{t},$$

we multiply this expression by s and then set $s = 0$ to obtain

$$\lim_{s\to 0} \pi^{-s/2} s\Gamma\left(\frac{s}{2}\right)\zeta(s) = -1.$$

But $\frac{s}{2}\Gamma\left(\frac{s}{2}\right) = \Gamma\left(\frac{s}{2}+1\right)$ so that $\lim_{s\to 0} s\Gamma\left(\frac{s}{2}\right) = 2$. Thus $\zeta(0) = -1/2$. □

9.2.15. *Let $f \in S$. For $x > 0$ show that*

$$\sum_{n\in\mathbb{Z}} f(nx)$$

converges absolutely and satisfies

$$x\sum_{n\in\mathbb{Z}} f(nx) = \sum_{n\in\mathbb{Z}} \hat{f}\left(\frac{n}{x}\right).$$

Since $f \in S$, $x^m f^{(n)}(x)$ is bounded for all $x \in \mathbb{R}$ and each choice of integers $m, n \geq 0$. In particular, $x^2 f(x)$ is bounded from which we easily deduce the absolute convergence of $\sum_{n\in\mathbb{Z}} f(nx)$. By Exercise 9.2.6 for the function $G(t) = f(xt)$, we have $x\hat{G}(t) = \hat{f}(t/x)$. Applying the Poisson summation formula to G gives

$$\sum_{n\in\mathbb{Z}} G(n) = \sum_{n\in\mathbb{Z}} \hat{G}(n).$$

In other words,

$$x\sum_{n\in\mathbb{Z}} f(nx) = \sum_{n\in\mathbb{Z}} \hat{f}\left(\frac{n}{x}\right),$$

as desired. □

9.2.16. *Let $f \in S$ be an even function. That is, $f(x) = f(-x)$. For any $x > 0$, show that*

$$\sum_{n=1}^{\infty} f\left(\frac{n}{x}\right) = x\sum_{n=1}^{\infty} \hat{f}(nx) + \frac{1}{2}\left(x\hat{f}(0) - f(0)\right).$$

By the previous exercise, we have

$$x\sum_{n\in\mathbb{Z}} f(nx) = \sum_{n\in\mathbb{Z}} \hat{f}\left(\frac{n}{x}\right).$$

Since f is even, we have

$$xf(0) + 2x\sum_{n=1}^{\infty} f(nx) = \hat{f}(0) + 2\sum_{n=1}^{\infty} \hat{f}\left(\frac{n}{x}\right).$$

Changing x to $1/x$ and multiplying by $x/2$, we obtain the desired result after rearranging terms. □

9.2.17. *Let $f \in S$. Set $\zeta(s, f) = \int_0^\infty f(x)x^s \frac{dx}{x}$. Show that the integral converges for all s with $\mathrm{Re}(s) > 1$.*

Since f is bounded, the integral

$$\int_0^1 f(x)x^s\frac{dx}{x}$$

is convergent. Take an integer $m > \mathrm{Re}(s) > 1$. Then as $f \in S$, $|x^m f(x)|$ is bounded for all $x \in \mathbb{R}$. Thus, the integral

$$\left|\int_1^\infty f(x)x^s\frac{dx}{x}\right| \ll \int_1^\infty x^{\mathrm{Re}(s)-m-1}dx < \infty. \qquad \square$$

9.2.18. *Let $f \in S$ be an even function. With $\zeta(s,f)$ defined as in the previous exercise, show that $\zeta(s,f)$ extends for all $s \in \mathbb{C}$, apart from possible simple poles at $s = 0, 1$ and satisfies the functional equation*

$$\zeta^\star(s,f) = \zeta^\star(1-s,\hat{f}),$$

where $\zeta^\star(s,f) = \zeta(s)\zeta(s,f)$.

In the integral definition of $\zeta(s,f)$, we change x to nx:

$$\zeta(s,f) = n^s\int_0^\infty f(nx)x^s\frac{dx}{x}.$$

By Exercise 9.2.16, the second sum is

$$\sum_{n=1}^\infty f\left(\frac{n}{x}\right) = x\sum_{n=1}^\infty \hat{f}(nx) + \frac{1}{2}\left(x\hat{f}(0) - f(0)\right)$$

so that

$$\zeta(s)\zeta(s,f) = \int_1^\infty \left(\sum_{n=1}^\infty f(nx)x^s + \sum_{n=1}^\infty \hat{f}(nx)x^{1-s}\right)\frac{dx}{x} - \frac{1}{2}\left(\frac{f(0)}{s} + \frac{\hat{f}(0)}{1-s}\right).$$

The integral converges absolutely for all $s \in \mathbb{C}$ and thus defines an entire function. Moreover, the right hand side is invariant under the maps $s \to 1-s, f \to \hat{f}$ from which the result follows. $\square$

9.2.19. *Show that the symbol $\left(\frac{c}{d}\right)$ is completely multiplicative in each of the variables.*

This follows easily from the definition and the multiplicativity of the Legendre symbol. $\square$

9.2.20. *Let $d|N$ and let ψ be a character modulo d. Prove that*

$$L(s,\chi_0\psi) = L(s,\psi)\prod_{p|N}\left(1 - \frac{\psi(p)}{p^s}\right).$$

We have

$$\begin{aligned} L(s,\chi_0\psi) &= \prod_p \left(1-\frac{\chi_0(p)\psi(p)}{p^s}\right)^{-1} \\ &= \prod_{p\nmid N}\left(1-\frac{\psi(p)}{p^s}\right)^{-1} \\ &= \prod_{p\nmid d}\left(1-\frac{\psi(p)}{p^s}\right)^{-1}\prod_{\substack{p\nmid d\\ p\mid N}}\left(1-\frac{\psi(p)}{p^s}\right) \\ &= L(s,\psi)\prod_{p\mid N}\left(1-\frac{\psi(p)}{p^s}\right) \end{aligned}$$

since if $p|N$ and $p|d$ then $\psi(p)=0$. □

9.2.21. *If $(n,N)=1$, show that*

$$\chi(n)\tau(\overline{\chi}) = \sum_{m=1}^{N}\overline{\chi}(m)e\left(\frac{mn}{N}\right),$$

where $\overline{\chi}$ denotes the complex conjugate of χ.

We have

$$\chi(n)\tau(\overline{\chi}) = \sum_{m=1}^{N}\overline{\chi}(m)\chi(n)e\left(\frac{m}{N}\right).$$

Since $(n,N)=1$, let n' be such that $nn'\equiv 1 \pmod N$ so that setting $h\equiv mn' \pmod N$, we get

$$\chi(n)\tau(\overline{\chi}) = \sum_{h=1}^{N}\overline{\chi}(h)e\left(\frac{nh}{N}\right).$$

Changing h to m in the sum gives the desired expression. □

9.2.23. *If χ is a primitive character modulo N, then $|\tau(\chi)|^2 = N$.*

By Theorem 9.2.22

$$\chi(n)\tau(\overline{\chi}) = \sum_{m=1}^{N}\overline{\chi}(m)e\left(\frac{mn}{N}\right)$$

so that

$$|\chi(n)|^2|\tau(\chi)|^2 = \sum_{m_1=1}^{N}\sum_{m_2=1}^{N}\overline{\chi}(m_1)\chi(m_2)e\left(\frac{n(m_1-m_2)}{N}\right)$$

and summing over $n, 1 \le n \le N$ to deduce

$$\phi(N)|\tau(\chi)|^2 = N\phi(N)$$

using the orthogonality relations (see Exercise 8.2.3). □

9.2.26. *Prove that*

$$\sum_{n=-\infty}^{\infty} ne^{-n^2\pi x/N + 2\pi imn/N} = i\left(\frac{N}{x}\right)^{3/2} \sum_{n=-\infty}^{\infty} \left(n + \frac{m}{N}\right) e^{-\pi N(n+\frac{m}{N})^2/x}.$$

By Exercise 9.2.9,

$$\sum_{n=-\infty}^{\infty} e^{-n^2\pi y + 2\pi in\alpha} = y^{-1/2} \sum_{n=-\infty}^{\infty} e^{-\pi(n+\alpha)^2/y}.$$

We differentiate both sides with respect to α to get

$$2\pi i \sum_{n=-\infty}^{\infty} ne^{-\pi n^2 y + 2\pi in\alpha} = -2\pi y^{-3/2} \sum_{n=-\infty}^{\infty} (n+\alpha)e^{-\pi(n+\alpha)^2/y}.$$

Setting $y = x/N$, and $\alpha = m/N$ gives the result. □

9.2.27. *Let χ be an odd Dirichlet character modulo N. Let*

$$\vartheta_1(x,\chi) = \sum_{n=-\infty}^{\infty} n\chi(n)e^{-\pi n^2 x/N}.$$

Show that

$$2\pi^{-(\frac{s+1}{2})}\Gamma\left(\frac{s+1}{2}\right) N^{\frac{s+1}{2}} L(s,\chi) = \int_0^\infty \vartheta_1(x,\chi) x^{\frac{s+1}{2}} \frac{dx}{x}.$$

In the formula

$$\pi^{-s/2}\Gamma\left(\frac{s}{2}\right) N^{s/2} n^{-s} = \int_0^\infty e^{-n^2\pi x/N} x^{s/2} \frac{dx}{x},$$

we change s to $s+1$ to get

$$\pi^{-\frac{s+1}{2}} N^{\frac{s+1}{2}} \Gamma\left(\frac{s+1}{2}\right) n^{-s} = \int_0^\infty ne^{-\pi n^2 x/N} x^{\frac{s+1}{2}} \frac{dx}{x}.$$

Multiplying by $\chi(n)$ and summing over $n \ge 1$ gives

$$\pi^{-\frac{s+1}{2}} N^{\frac{s+1}{2}} \Gamma\left(\frac{s+1}{2}\right) L(s,\chi) = \int_0^\infty \vartheta_1(x,\chi) x^{\frac{s+1}{2}} \frac{dx}{x},$$

because χ is odd and so

$$2\sum_{n=1}^{\infty} n\chi(n)e^{-\pi n^2 x/N} = \sum_{n=-\infty}^{\infty} n\chi(n)e^{-\pi n^2 x/N}. \qquad \square$$

9.2.28. *Prove that if χ is an odd primitive Dirichlet character modulo N, then*

$$\vartheta_1\left(\frac{1}{x},\chi\right) = x^{3/2}\left(\frac{\tau(\chi)}{iN^{1/2}}\right)\vartheta_1(x,\overline{\chi})$$

By Theorem 9.2.22, we have

$$\tau(\overline{\chi})\chi(n) = \sum_{m=1}^{N} \overline{\chi}(m)e\left(\frac{mn}{N}\right).$$

Thus,

$$\tau(\overline{\chi})\vartheta_1(x,\chi) = \sum_{n=-\infty}^{\infty} n\left(\sum_{m=1}^{N} \overline{\chi}(m)e\left(\frac{mn}{N}\right)\right)e^{-\pi n^2 x/N}.$$

This is equal to

$$\sum_{m=1}^{N} \overline{\chi}(m)\sum_{n=-\infty}^{\infty} ne^{-\pi n^2 x/N + 2\pi imn/N}.$$

By Exercise 9.2.26, the inner sum is

$$i\left(\frac{N}{x}\right)^{3/2}\sum_{n=-\infty}^{\infty}\left(n+\frac{m}{N}\right)e^{-\pi N(n+m/N)^2/x},$$

so that

$$\begin{aligned}\tau(\overline{\chi})\vartheta_1(x,\chi) &= i\left(\frac{N}{x}\right)^{3/2}\sum_{m=1}^{N}\overline{\chi}(m)\sum_{n=-\infty}^{\infty}\left(n+\frac{m}{N}\right)e^{-\pi N(n+m/N)^2/x}\\ &= \left(\frac{iN^{1/2}}{x^{3/2}}\right)\sum_{n=-\infty}^{\infty}\overline{\chi(n)}ne^{-\pi n^2/Nx}\\ &= \left(\frac{iN^{1/2}}{x^{3/2}}\right)\vartheta_1\left(\frac{1}{x},\overline{\chi}\right).\end{aligned}$$

In other words, by changing χ to $\overline{\chi}$ in the equation, we get

$$\vartheta_1\left(\frac{1}{x},\chi\right) = x^{3/2}\left(\frac{\tau(\chi)}{iN^{1/2}}\right)\vartheta_1(x,\overline{\chi}),$$

as required. □

9.2.29. *Let χ be an odd primitive Dirichlet character modulo N. Let*

$$\xi(s,\chi) := \pi^{-s/2} N^{s/2} \Gamma\left(\frac{s+1}{2}\right) L(s,\chi).$$

Then $\xi(s,\chi)$ is entire and satisfies the functional equation

$$\xi(s,\chi) = w_\chi \xi(1-s,\overline{\chi}),$$

where $w_\chi = \tau(\chi)/iN^{1/2}$.

By Exercise 9.2.27,

$$2\pi^{-\left(\frac{s+1}{2}\right)} N^{\frac{s+1}{2}} L(s,\chi) = \int_0^\infty \theta_1(x,\chi) x^{\frac{s+1}{2}} \frac{dx}{x}.$$

This is equal to

$$\int_1^\infty \theta_1(x,\chi) x^{\frac{s+1}{2}} \frac{dx}{x} + \int_1^\infty \theta_1\left(\frac{1}{x},\chi\right) x^{-\frac{s+1}{2}} \frac{dx}{x}$$

after a suitable change of variables. By the previous exercise,

$$\theta_1\left(\frac{1}{x},\chi\right) = w_\chi x^{3/2} \theta_1(x,\overline{\chi})$$

so that our expression is

$$\int_1^\infty \theta_1(x,\chi) x^{\frac{s+1}{2}} \frac{dx}{x} + w_\chi \int_1^\infty \theta_1(x,\overline{\chi}) x^{\frac{2-s}{2}} \frac{dx}{x}$$
$$= \int_1^\infty \theta_1(x,\chi) x^{s/2} \frac{dx}{\sqrt{x}} + w_\chi \int_1^\infty \theta_1(x,\overline{\chi}) x^{\frac{1-s}{2}} \frac{dx}{\sqrt{x}}.$$

Both integrals are absolutely convergent for all $s \in \mathbb{C}$. The functional equation is easily deduced by noting that changing s to $1-s$, changing χ to $\overline{\chi}$, and multiplying by w_χ, the expression remains invariant. □

9.3 *L*-functions attached to modular forms

9.3.2. *Let $f \in M_k(\Gamma_0(N),\chi)$ and $g = f|w$ with $w = \begin{pmatrix} 0 & -1 \\ N & 0 \end{pmatrix}$. Supposing that*

$$f(z) = \sum_{n=0}^{\infty} a_n e^{2\pi i n z}$$
$$g(z) = \sum_{n=0}^{\infty} b_n e^{2\pi i n z}$$

are the Fourier expansions of f and g at $i\infty$, prove that

$$\Lambda_f(s) := \left(\frac{\sqrt{N}}{2\pi}\right)^s \Gamma(s) L_f(s)$$
$$\Lambda_g(s) := \left(\frac{\sqrt{N}}{2\pi}\right)^s \Gamma(s) L_g(s),$$

where $L_f(s) = \sum_{n=1}^{\infty} \frac{a_n}{n^s}$ and $L_g(s) = \sum_{n=1}^{\infty} \frac{b_n}{n^s}$ extend analytically to the entire complex plane with poles at $s = 0$ and $s = k$. Moreover,

$$\Lambda_f(s) + \frac{a_0}{s} + \frac{b_0 i^k}{k-s}$$

and

$$\Lambda_g(s) + \frac{b_0}{s} + \frac{a_0 i^k}{k-s}$$

are entire and satisfy the functional equation

$$\Lambda_f(s) = i^k \Lambda_g(k-s).$$

As in the proof of Theorem 9.3.1 and Theorem 9.2.1, we have the easily verified equality:

$$\Lambda_f(s) + \frac{a_0}{s} + \frac{b_0 i^k}{k-s} = \int_1^{\infty} \left(f\left(\frac{it}{\sqrt{N}}\right) - a_0\right) t^{s-1} dt + i^k \int_1^{\infty} \left(g\left(\frac{it}{\sqrt{N}}\right) - b_0\right) t^{k-s-1} dt.$$

Both integrals converge for all $s \in \mathbb{C}$ since the a_n's and b_n's have polynomial growth. Similarly,

$$\Lambda_g(s) + \frac{b_0}{s} + \frac{a_0 i^{-k}}{k-s} = \int_1^{\infty} \left(g\left(\frac{it}{\sqrt{N}}\right) - b_0\right) t^{s-1} dt + i^{-k} \int_1^{\infty} \left(f\left(\frac{it}{\sqrt{N}}\right) - a_0\right) t^{k-s-1} dt.$$

If we change s to $k-s$ in the integral expression we deduce the desired functional equation, since the polar part also satisfies the same functional equation. □

9.3.3. *With f, g as in the previous exercise, show that $L_f(0) = -a_0$, $L_g(0) = -b_0$,*

$$\lim_{s\to k}(s-k)L_f(s) = b_0\left(\frac{3\pi i}{\sqrt{N}}\right)^k/\Gamma(k)$$

and

$$\lim_{s\to k}(s-k)L_g(s) = a_0\left(\frac{2\pi i}{\sqrt{N}}\right)^k/\Gamma(k).$$

Since $\Gamma(s)$ has a simple pole at $s=0$ with residue 1,

$$\Lambda_f(s) = \left(\frac{\sqrt{N}}{2\pi}\right)^s \Gamma(s)L_f(s)$$

has a simple pole at $s=0$ with residue $L_f(0)$. By Exercise 9.3.2, the expression

$$\Lambda_f(s) + \frac{a_0}{s} + \frac{b_0 i^k}{k-s}$$

is entire. Therefore $L_f(0)+a_0 = 0$. Similarly $L_g(0) = -b_0$. From the same expression, $\Lambda_f(s)$ has a simple pole at $s=k$ with residue $b_0 i^k$. Thus

$$\lim_{s\to k}(s-k)\Lambda_f(s) = b_0 i^k$$

from which we deduce $\lim_{s\to k}(s-k)L_f(s) = b_0\left(2\pi i/\sqrt{N}\right)^k\Gamma(k)^{-1}$, as required. The result for $L_g(s)$ is similarly derived. □

9.3.5. *Suppose that $f(s)$ is analytic in the vertical strip defined by $a \le \mathrm{Re}(s) \le b$ and that for some $\alpha \ge 1$,*

$$f(s) = O\left(e^{|t|^\alpha}\right)$$

as $|t|\to\infty$. If for some $A>0$, $f(s) = O\left(|t|^A\right)$ on $\mathrm{Re}(s)=a$ and $\mathrm{Re}(s)=b$, then show that $f(s) = O\left(|t|^A\right)$ for all s in the vertical strip $a \le \mathrm{Re}(s) \le b$.

We apply the Phragmén-Lindelöf theorem to the function $g(s) = f(s)/(s-u)^A$ where $u > b$. Then g is anlytic in the strip and bounded on $\mathrm{Re}(s)=a$ and $\mathrm{Re}(s)=b$. By Theorem 9.3.4, $g(s)$ is bounded throughout the vertical strip which completes the proof. □

9.3.6. *Show that for any $\sigma > 0$,*

$$e^{-x} = \frac{1}{2\pi i}\int_{\sigma-i\infty}^{\sigma+i\infty}\Gamma(s)x^{-s}ds.$$

We consider the oriented rectangular contour $\mathcal{C}_T$ given by vertices $\sigma - iT$, $\sigma + iT$, $-U + iT$, $-U - iT$, where U is positive and not an integer and $T > 0$. Since the Γ-function has simple poles at $s = 0, -1, -2, \dots$ and $\mathrm{Res}_{s=-j} \Gamma(s) = (-1)^j/j!$, we find by Cauchy's Theorem that

$$\frac{1}{2\pi i} \int_{\mathcal{C}_T} \Gamma(s) x^{-s} ds = \sum_{0 \le j < U} \frac{(-1)^j}{j!} x^j.$$

By Stirling's formula, the contribution of the horizontal integrals goes to 0 as $T \to \infty$. Letting $U \to \infty$, we obtain

$$\frac{1}{2\pi i} \int_{\sigma - i\infty}^{\sigma + i\infty} \Gamma(s) x^j ds = \sum_{j=0}^{\infty} \frac{(-1)^j x^j}{j!} = e^{-x}. \qquad \square$$

9.3.7. *With f as in Exercise 9.3.2, show that for $\sigma > 0$ we have*

$$f\left(\frac{iy}{\sqrt{N}}\right) - a_0 = \frac{1}{2\pi i} \int_{\sigma - i\infty}^{\sigma + i\infty} \Lambda_f(s) y^{-s} ds.$$

The integral is

$$\begin{aligned} \frac{1}{2\pi i} \int_{\sigma - i\infty}^{\sigma + i\infty} \left(\frac{\sqrt{N}}{2\pi}\right)^s \Gamma(s) L_f(s) y^{-s} ds & \\ &= \sum_{n=1}^{\infty} a_n \left(\frac{1}{2\pi i} \int_{\sigma - i\infty}^{\sigma + i\infty} \Gamma(s) \left(\frac{2\pi n y}{\sqrt{N}}\right)^{-s} ds \right) \\ &= \sum_{n=1}^{\infty} a_n e^{-2\pi n y/\sqrt{N}} \\ &= f\left(\frac{iy}{\sqrt{N}}\right) - a_0. \qquad \square \end{aligned}$$

9.4 Twists of L-series attached to modular forms

9.4.1. *Verify the matrix identity*

$$\begin{pmatrix} 1 & u/r \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & -d^2 u/r \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} a + uc/r & b - bcdu/r - cd^2u^2/r^2 \\ c & d - cd^2 u/r \end{pmatrix}$$

where $ad - bc = 1$.

This is straightforward matrix multiplication. $\square$

9.4.3. *Let $(q,r)=1$ and $(u,r)=1$. Let d and w be integers such that $dr-quw=1$. Verify the matrix identity with $N=qr^2$,*

$$\begin{pmatrix}1 & u/r\\ 0 & 1\end{pmatrix}\begin{pmatrix}0 & -1\\ N & 0\end{pmatrix}=r\begin{pmatrix}0 & -1\\ q & 0\end{pmatrix}\begin{pmatrix}r & -w\\ -qu & d\end{pmatrix}\begin{pmatrix}1 & w/r\\ 0 & 1\end{pmatrix}.$$

The left hand side is $\left(\begin{smallmatrix}uN/r & -1\\ N & 0\end{smallmatrix}\right)$. Using $dr-quw=1$, we see that the right hand side is

$$r\begin{pmatrix}0 & -1\\ q & 0\end{pmatrix}\begin{pmatrix}r & 0\\ -qu & 1/r\end{pmatrix}=\begin{pmatrix}0 & -1\\ q & 0\end{pmatrix}\begin{pmatrix}r^2 & 0\\ -qru & 1\end{pmatrix}=\begin{pmatrix}qru & -1\\ qr^2 & 0\end{pmatrix}$$

which is the left hand side since $N=qr^2$. □

9.4.4. *Let*

$$w_N=\begin{pmatrix}0 & -1\\ N & 0\end{pmatrix},\qquad w_q=\begin{pmatrix}0 & -1\\ q & 0\end{pmatrix},$$

and $f\in S_k(\Gamma_0(q),\chi)$ where χ is a Dirichlet character modulo q. Let ψ be a primitive Dirichlet character modulo r with $(q,r)=1$. With f_ψ as in Theorem 9.4.2, $N=qr^2$, $g=f|w_q$, show that

$$f_\psi|w_N=w(\psi)g_{\overline{\psi}}$$

where

$$w(\psi)=\chi(r)\psi(q)\tau(\psi)^2/r.$$

By a simple modification of Exercise 8.4.3, we see that $g\in S_k(\Gamma_0(q),\overline{\chi})$. Since $(q,r)=1$, we have for any $(u,r)=1$, integers d and w such that $dr-quw=1$. By Exercise 9.4.3, with $N=qr^2$, we have the identity

$$\begin{pmatrix}1 & u/r\\ 0 & 1\end{pmatrix}\begin{pmatrix}0 & -1\\ N & 0\end{pmatrix}=r\begin{pmatrix}0 & -1\\ q & 0\end{pmatrix}\begin{pmatrix}r & -w\\ -qu & d\end{pmatrix}\begin{pmatrix}1 & w/r\\ 0 & 1\end{pmatrix}$$

so that

$$f\left|\begin{pmatrix}1 & u/r\\ 0 & 1\end{pmatrix}\right|w_N=\overline{\chi}(d)g\left|\begin{pmatrix}1 & w/r\\ 0 & 1\end{pmatrix}\right.=\chi(r)g\left|\begin{pmatrix}1 & w/r\\ 0 & 1\end{pmatrix}\right..$$

By (9.5) on page 129, since ψ is primitive,

$$\begin{aligned}\tau(\overline{\psi})f_\psi|w_N&=\sum_{u\,(\mathrm{mod}\,r)}\overline{\psi}(u)f\left|\begin{pmatrix}1 & u/r\\ 0 & 1\end{pmatrix}\right.w_N\\&=\sum_{u\,(\mathrm{mod}\,r)}\overline{\psi}(u)\chi(r)g\left|\begin{pmatrix}1 & w/r\\ 0 & 1\end{pmatrix}\right.\end{aligned}$$

where $quw\equiv -1\pmod r$. As u ranges over coprime residue classes modulo r, so does w. Since $quw\equiv -1\pmod r$, we have $\overline{\psi}(quw)=\psi(-1)$ so that

$\overline{\psi}(u) = \psi(-q)\psi(w)$. Thus

$$\begin{aligned}\tau(\overline{\psi})f_\psi|w_N &= \psi(-q)\chi(r)\sum_{w \pmod r}\psi(w)g\left|\begin{pmatrix}1 & w/r\\ 0 & 1\end{pmatrix}\right.\\ &= \psi(-q)\chi(r)\tau(\psi)g_{\overline{\psi}}.\end{aligned}$$

It is easy to see that (see Exercise 9.5.1)

$$\overline{\tau(\psi)} = \psi(-1)\tau(\overline{\psi}).$$

Together with Exercise 9.2.23, we obtain

$$\overline{\tau(\psi)}f_\psi|w_N = \psi(q)\chi(r)\tau(\psi)g_{\overline{\psi}}$$

so that

$$rf_\psi|w_N = \psi(q)\chi(r)\tau(\psi)^2 g_{\overline{\psi}}$$

which is the desired result. □

9.5 Supplementary problems

9.5.1. *If $\tau(\chi)$ denotes the Gauss sum, prove that*

$$\overline{\tau(\chi)} = \chi(-1)\tau(\overline{\chi}).$$

We have

$$\begin{aligned}\overline{\tau(\chi)} &= \sum_{a \pmod N}\overline{\chi(a)}e\left(-\frac{a}{N}\right)\\ &= \sum_{b \pmod N}\overline{\chi(-b)}e\left(\frac{b}{N}\right)\\ &= \chi(-1)\tau(\overline{\chi}).\end{aligned}$$

□

9.5.2. *For $\mathrm{Re}(c) > 0$, let $F(x) = e^{-c|x|}$. Show that*

$$\hat{F}(u) = \frac{2c}{c^2 + 4\pi^2u^2}.$$

We have

$$\begin{aligned}\hat{F}(u) &= \int_{-\infty}^{\infty} e^{-c|x|} e^{-2\pi i u x} dx \\ &= \int_{0}^{\infty} e^{-(c+2\pi i u)x} dx + \int_{-\infty}^{0} e^{(c-2\pi i u)x} dx \\ &= \int_{0}^{\infty} e^{-(c+2\pi i u)x} dx + \int_{0}^{\infty} e^{-(c-2\pi i u)x} dx.\end{aligned}$$

Since

$$\int_{0}^{\infty} e^{-vx} dx = \frac{1}{v},$$

for $\mathrm{Re}(v) > 0$, we get

$$\hat{F}(u) = \frac{1}{c+2\pi i} + \frac{1}{c-2\pi i u} = \frac{2c}{c^2 + 4\pi^2 u^2},$$

as required. □

9.5.3. *Show that for* $\mathrm{Re}(c) > 0$,

$$\frac{e^c + 1}{e^c - 1} = \sum_{n=-\infty}^{\infty} \frac{2c}{c^2 + 4\pi^2 n^2}.$$

Using Exercise 9.5.2, we apply the Poisson summation formula to get

$$\sum_{n\in\mathbb{Z}} e^{-c|n|} = \sum_{n\in\mathbb{Z}} \frac{2c}{c^2 + 4\pi^2 n^2}.$$

The left hand side is essentially a geometric series and easily summed to be

$$\begin{aligned}1 + 2\sum_{n=1}^{\infty} e^{-cn} &= 1 + \frac{2e^{-c}}{1 - e^{-c}} \\ &= 1 + \frac{2}{e^c - 1} = \frac{e^c + 1}{e^c - 1}.\end{aligned}$$

□

9.5.4. *From the previous exercise, deduce Euler's formula that*

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

From the previous exercise, we have

$$\frac{e^c + 1}{e^c - 1} \cdot \frac{1}{2c} - \frac{1}{c^2} = 2\sum_{n=1}^{\infty} \frac{1}{c^2 + 4\pi^2 n^2}.$$

If we set $c = 0$ in the right hand side, we get

$$\frac{1}{2\pi^2}\sum_{n=1}^{\infty}\frac{1}{n^2}.$$

On the left hand side, we have

$$\lim_{c\to 0}\left[\frac{e^c+1}{e^c-1}\cdot\frac{1}{2c}-\frac{1}{c^2}\right]=\frac{1}{12}$$

by an easy computation. For example, invert the power series expansion of e^c-1 at $c=0$ to get

$$\frac{1}{e^c-1}=\left(\frac{1}{c}-\frac{1}{2}+\frac{c}{12}+O\big(c^2\big)\right)$$

and simplify the expression in the limit above. □

9.5.5. *By differentiating the identity in Exercise 9.5.3, deduce that*

$$\sum_{n=1}^{\infty}\frac{1}{n^{2k}}\in\pi^{2k}\mathbb{Q}.$$

By Exercise 9.5.3,

$$\left(1+\frac{2}{e^c-1}\right)\frac{1}{2c}-\frac{1}{c^2}=2\sum_{n=1}^{\infty}\frac{1}{c^2+4\pi^2n^2}.$$

The left hand side is easily seen to be a power series in c with rational coefficients. Differentiating both sides k times and then setting $c=0$ gives the desired result. □

9.5.6. *For* $\mathrm{Re}(c)>0$, *and* $0\le v<1$, *show that*

$$\sum_{n\in\mathbb{Z}}\frac{e^{2\pi inv}}{c^2+4\pi^2n^2}=\frac{1}{2c}\left(e^{-cv}+\frac{2\cosh cv}{e^c-1}\right).$$

From the proof of Theorem 9.2.7, we can see that for any Schwartz function F, we have

$$\sum_{n\in\mathbb{Z}}F(n+v)=\sum_{n\in\mathbb{Z}}\hat{F}(n)e^{2\pi inv}.$$

Applying this generalized Poisson summation formula to $F(x)=e^{-c|x|}$, we have

$$\sum_{n\in\mathbb{Z}}e^{-c|n+v|}=\sum_{n\in\mathbb{Z}}\frac{2c}{c^2+4\pi^2n^2}e^{2\pi inv}.$$

The sum on the left hand side is summed easily as

$$e^{-cv} + \sum_{n=1}^{\infty} e^{-c(n+v)} + \sum_{n=1}^{\infty} e^{c(-n+v)}$$

since $0 \le v < 1$. The two sums above are geometric sums and the result is

$$e^{-cv} + \frac{1}{e^c - 1}\left(e^{cv} + e^{-cv}\right)$$

from which the result is immediate. □

9.5.7. *Prove that for* $0 \le v < 1$,

$$\sum_{n \neq 0} \frac{e^{2\pi i n v}}{n^2} = 2\left(v^2 - v + \frac{1}{6}\right)\pi^2.$$

By Exercise 9.5.6,

$$\frac{1}{c^2} + \sum_{n \neq 0} \frac{e^{2\pi i n v}}{c^2 + 4\pi^2 n^2} = \frac{1}{2c}\left(e^{-cv} + \frac{2\cosh cv}{e^c - 1}\right).$$

We need to compute

$$\lim_{c \to 0}\left[\frac{e^{-cv}}{2c} - \frac{1}{c^2} + \frac{\cosh cv}{e(e^c - 1)}\right].$$

Using the fact that

$$\cosh cv = 1 + \frac{c^2 v^2}{2!} + \frac{c^4 v^4}{4!} + \cdots$$

and that

$$\frac{1}{e^c - 1} = \frac{1}{c}\left(1 - \frac{c}{2} + \frac{1}{12}c^2 + \cdots\right)$$

we see that the limit in question is

$$\begin{aligned}
&\lim_{c \to 0}\left[\frac{e^{-cv}}{2c} - \frac{1}{c^2} + \frac{1}{c^2}\left(1 - \frac{c}{2} + \frac{1}{12}c^2 + \frac{c^2 v^2}{2} + \cdots\right)\right] \\
=&\lim_{c \to 0}\left[\frac{1}{2c}\left(1 - cv + \frac{c^2 v^2}{2} + \cdots\right) - \frac{1}{2c} + \frac{1}{12} + \frac{v^2}{2}\right] \\
=&\frac{1}{2}\left(v^2 - v + \frac{1}{6}\right),
\end{aligned}$$

as required. □

Chapter 10
Special Topics

10.1 Elliptic functions

10.1.1. *If f is a holomorphic elliptic function, then f is constant.*

This is a consequence of Liouville's theorem which says that any bounded entire function is constant. Since f is holomorphic and completely determined by its values on the compact region $\overline{P}$, we deduce that f is bounded and hence constant. □

10.1.2. *If two elliptic functions (for a fixed lattice L) have the same poles with the same principal parts, show that they differ by a constant.*

The difference of the two elliptic functions is again elliptic and it is clearly holomorphic. By the previous exercise, this is constant. □

10.1.3. *If f is an elliptic function with no poles on the boundary of the fundamental parallelogram P, show that the sum of the residues of f in P is zero.*

This is an application of Cauchy's Residue Theorem. The sum in question is

$$\frac{1}{2\pi i}\int_{\partial P} f(z)dz,$$

where ∂P is the oriented boundary of P, traversing from 0 to ω_1, then to $\omega_1+\omega_2$, then to ω_2 and then back to 0. But this integral is the sum of four line integrals, each corresponding to a side of the parallelogram P. The line integrals corresponding to opposite sides cancel because of the double periodicity of f and because of the oriented direction of the sides of ∂P. □

10.1.4. *Let f be an elliptic function with respect to a lattice L. Show that there is an $\alpha \in \mathbb{C}$ such that f is holomorphic on the boundary of the translate $\alpha + P$, where P is the fundamental parallelogram of L.*

© Springer Science+Business Media Singapore 2016 and Hindustan Book Agency 2016

M.R. Murty et al., *Problems in the Theory of Modular Forms*, HBA Lecture Notes in Mathematics, DOI 10.1007/978-981-10-2651-5_20

Any meromorphic function has only a finite number of poles in a bounded subset of $\mathbb{C}$. Thus if f has poles on the boundary of P, there are only finitely many. Thus we can find α such that f has no poles on the boundary of $\alpha+P$. □

10.1.5. *Show that there is no non-constant elliptic function having exactly one simple pole in the fundamental parallelogram P.*

Suppose there is such a function f. By Exercise 10.1.4, we can find α such that the translate $\alpha+P$ has no poles of f on the boundary. By Exercise 10.1.3, the sum of the residues of f is zero. This means that f is holomorphic and hence constant, a contradiction. □

10.1.6. *Let f be an elliptic function. Show that the number of zeros of f in the fundamental parallelogram P is equal to the number of poles of f (counted with multiplicity).*

If f is elliptic, then so is its derivative f'. Thus f'/f is elliptic. By choosing an appropriate translate $\alpha+P$ of P we can ensure there are no zeros or poles of f on the boundary of $\alpha + P$. By the same reasoning as in the solution to Exercise 10.1.3, we see that

$$\frac{1}{2\pi i}\int_{\partial(\alpha+P)} \frac{f'(z)}{f(z)}dz = 0.$$

The result now follows immediately from Cauchy's Argument Principle. □

10.1.7. *If f is an elliptic function of order m, then for any $c \in \mathbb{C}$, show that the number of solutions of $f(z) = c$ with $z \in P$ is exactly m (counted with multiplicities).*

We apply the previous exercise to the elliptic function $f(z) - c$. □

10.1.8. *For any $s > 2$, show that the series*

$$\sum_{\substack{\omega\in L\\ \omega\neq 0}} \frac{1}{\omega^s}$$

is absolutely convergent.

The sum is

$$\sum_{(m,n)\neq(0,0)} \frac{1}{(m\omega_1+n\omega_2)^s} = \sum_{(m,n)\neq(0,0)} \frac{1}{\omega_2^s\left(m\frac{\omega_1}{\omega_2}+n\right)^s}.$$

Since ω_1, ω_2 are linearly independent over $\mathbb{R}$, we have $\omega_1/\omega_2 \notin \mathbb{R}$. Now we appeal to the solution of Exercise 4.2.1 to deduce that there exists $\lambda > 0$ such that

$$\left| m\frac{\omega_1}{\omega_2} + n \right| \geq \lambda \left(m^2 + n^2 \right).$$

Thus,

$$\sum_{(m,n)\neq(0,0)} \frac{1}{\left| m\frac{\omega_1}{\omega_2} + n \right|^s} \leq \frac{1}{\lambda} \sum_{(m,n)\neq(0,0)} \frac{1}{(m^2+n^2)^{s/2}}$$

and the solution now proceeds as in Exercise 4.2.1. □

10.1.9. *Show that the sum*

$$\sum_{0\neq\omega\in L} \left[\frac{1}{(z-\omega)^2} - \frac{1}{\omega^2} \right]$$

is absolutely convergent for any $z \in \mathbb{C} \setminus L$.

Now

$$\frac{1}{(z-\omega)^2} - \frac{1}{\omega^2} = \frac{2\omega z - z^2}{\omega^2 (z-\omega)^2}.$$

Let $R = |z|$ and exclude the finite number of $\omega \in L$ with $|\omega| \leq 2R$. Then

$$|\omega - z| \geq |\omega| - |z| = |\omega| - R > R.$$

It is now clear that

$$\left| \frac{1}{(z-\omega)^2} - \frac{1}{\omega^2} \right| \leq \frac{R\,(2|w| + R)}{|\omega^2|\,(|\omega| - R)^2} = O\big(|\omega|^{-3}\big),$$

so that our series converges absolutely by Exercise 10.1.8. □

10.1.12. *Let*

$$G_k = \sum_{\substack{\omega\in L \\ \omega\neq 0}} \omega^{-k}$$

Prove that

$$\wp(z) = z^{-2} + \sum_{\substack{k=2 \\ k\ even}}^{\infty} (k+1) G_{k+2} z^k$$

is the Laurent expansion of $\wp(z)$ at $z = 0$.

For $|z| < |\omega|$, we have

$$\begin{aligned} (z-\omega)^{-2} - \omega^{-2} &= \omega^{-2} \Big[(1 - z/\omega)^{-2} - 1 \Big] \\ &= \sum_{k=1}^{\infty} (k+1) \frac{z^k}{\omega^{k+2}} \end{aligned} \tag{10.1}$$

using the familiar identity

$$\frac{1}{(1-x)^2} = \sum_{k=1}^{\infty} kx^{k-1}$$

obtained easily by differentiating the power series for $(1-x)^{-1}$. We insert (10.1) into the series defining $\wp(z)$. Reversing the order of summation (which we can do by the absolute convergence of the series) yields

$$\wp(z) = z^{-2} + \sum_{k=1}^{\infty}(k+1)G_{k+2}z^k.$$

It is clear that $G_k = 0$ whenever k is odd since the ω and $-\omega$ terms in the definition of G_k cancel. □

10.1.13. *With notation as in the previous exercise, show that for $z \in \mathbb{C} \setminus L$,*

$$\wp'(z)^2 = 4\wp(z)^3 - 60G_4\wp(z) - 140G_6.$$

From the previous exercise we have

$$\wp(z) = z^{-2} + 3G_4z^2 + 5G_6z^4 + \cdots$$

and so

$$\wp(z)^3 = z^{-6} + 9G_4z^{-2} + 15G_6 + \cdots.$$

Differentiating we obtain

$$\wp'(z) = -2z^{-3} + 6G_4z + 20G_6z^3 + \cdots$$

and so

$$\wp'(z)^2 = 4z^{-6} - 24G_4z^{-2} - 80G_6 + \cdots.$$

Hence the elliptic function $\wp'(z)^2 - 4\wp(z)^3 + 60G_4\wp(z) + 140G_6$ is holomorphic at $z = 0$ and vanishes there. By periodicity, this function vanishes for all $z \in L$. By Exercise 10.1.9, this function is holomorphic for all $z \in \mathbb{C} \setminus L$. Finally, by Exercise 10.1.1 this function is constant (and in fact zero since it vanishes at $z = 0$). □

10.1.14. *Let L be a lattice and g_2, g_3 defined as above. Show that all of the complex solutions of the equation*

$$y^2 = 4x^3 - g_2x - g_3$$

are given by $\big(\wp(z), \wp'(z)\big)$ where $\wp(z)$ is the Weierstrass $\wp$-function attached to L and z ranges over all complex numbers in $\mathbb{C} \setminus L$.

Let $(x, y) \in \mathbb{C}^2$ lie on the curve. We want to find a $z \in \mathbb{C} \setminus L$ such that $x = \wp(z)$ and $y = \wp'(z)$. If the equation $\wp(z) = x$ has no solution,

then $1 \Big/ \Big(\wp(z) - x\Big)$ is an elliptic function which is holomorphic. By Exercise 10.1.1, it must be a constant. This is a contradiction since $\wp(z)$ is not a constant function. Therefore, there must be a $z \in \mathbb{C} \setminus L$ such that $x = \wp(z)$.

By Exercise 10.1.13 we deduce that $y = \pm\wp'(z)$. As $\wp(z)$ is an even function and $\wp'(z)$ is odd, we may adjust the sign of z appropriately to ensure that $(x, y) = \Big(\wp(z), \wp'(z)\Big)$. □

10.1.16. *Let L be the lattice spanned by ω_1, ω_2 and let $\wp(z)$ be the associated Weierstrass function. Show that $\wp\left(\frac{\omega_1}{2}\right)$, $\wp\left(\frac{\omega_2}{2}\right)$, and $\wp\left(\frac{\omega_1+\omega_2}{2}\right)$ are the roots of the cubic $4x^3 - g_2x - g_3$ with g_2 and g_3 defined by (10.1).*

Since $\wp'(z)$ is an odd function, $\wp'(\omega_1/2) = -\wp'(-\omega_1/2)$. But $\wp'(z)$ is elliptic and so $\wp'(-\omega_1/2) = \wp'(-\omega_1/2 + \omega_1) = \wp'(\omega_1/2)$. Thus $\wp'(\omega_1/2) = 0$. Similarly, $\wp'(\omega_2/2) = 0$ and $\wp'((\omega_1 + \omega_2)/2) = 0$. By Exercise 10.1.13, we deduce that $\wp\left(\frac{\omega_1}{2}\right)$, $\wp\left(\frac{\omega_2}{2}\right)$, and $\wp\left(\frac{\omega_1+\omega_2}{2}\right)$ are the roots of the associated cubic. □

10.1.17. *Suppose that ω_1, ω_2 are linearly independent over $\mathbb{R}$ and that L is the lattice spanned by ω_1, ω_2. Let $\wp(z)$ be the associated $\wp$-function. Show that $\wp\left(\frac{\omega_1}{2}\right)$, $\wp\left(\frac{\omega_2}{2}\right)$, and $\wp\left(\frac{\omega_1+\omega_2}{2}\right)$ are all distinct.*

Consider the function

$$f_1(z) = \wp(z) - \wp(\omega_1/2).$$

Since (by the solution of the previous exercise) $\wp'(\omega_1/2) = 0$, we see that $f_1(z)$ has a double order zero at $z = \omega_1/2$. As $\wp(z)$ has a double order pole at $z = 0$ and as there are no other poles in the fundamental parallelogram, this accounts for all of the zeros of $f_1(z)$ by Exercise 10.1.6. If $\wp(\omega_1/2) = \wp(\omega_2/2)$, then $z = \omega_2/2$ would be a zero of $f_1(z)$ and hence $\omega_1 \equiv \omega_2 \pmod{L}$, which contradicts the linear independence of ω_1 and ω_2 over $\mathbb{R}$. Thus, $\wp(\omega_1/2)$ and $\wp(\omega_2/2)$ are distinct.

A similar argument applies for the other 2-division points, namely $\wp(\omega_2/2)$ and $\wp((\omega_1 + \omega_2)/2)$. □

10.2 The modular invariant

10.2.2. *If ω_1, ω_2 are linearly independent over $\mathbb{R}$ and $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$, show that the lattice generated by $a\omega_1 + b\omega_2$ and $c\omega_1 + d\omega_2$ is the same as the lattice generated by ω_1 and ω_2.*

The lattice L generated by ω_1 and ω_2 consists of all linear combinations $m\omega_1 + n\omega_2$ with $m, n \in \mathbb{Z}$. The lattice L' generated by $a\omega_1 + b\omega_2$ and $c\omega_1 + d\omega_2$ consists of all linear combinations

$$x(a\omega_1 + b\omega_2) + y(c\omega_1 + d\omega_2) = (xa + yc)\omega_1 + (xb + yd)\omega_2$$

with x, $y \in \mathbb{Z}$. Clearly $L' \subseteq L$. But given any pair of integers m, n we can find integers x, y such that

$$\begin{pmatrix} a & c \\ b & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} m \\ n \end{pmatrix}$$

since $ad - bc = 1$. Thus $L \subseteq L'$ and this completes the proof. □

10.2.3. *If ω_1, ω_2 generate the lattice L and ω_1', ω_2' also generate the same lattice L, show that there is a matrix $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{Z})$ such that $\omega_1' = a\omega_1 + b\omega_2$ and $\omega_2' = c\omega_1 + d\omega_2$.*

Since ω_1', ω_2' generate the same lattice as ω_1, ω_2, there are integers a, b, c, d such that

$$\begin{aligned} \omega_1' &= a\omega_1 + b\omega_2 \\ \omega_2' &= c\omega_1 + d\omega_2. \end{aligned}$$

Similarly there are integers e, f, g, h such that

$$\begin{aligned} \omega_1 &= e\omega_1' + f\omega_2' \\ \omega_2 &= g\omega_1' + h\omega_2'. \end{aligned}$$

Thus

$$\begin{pmatrix} \omega_1 \\ \omega_2 \end{pmatrix} = \begin{pmatrix} e & f \\ g & h \end{pmatrix} \begin{pmatrix} \omega_1' \\ \omega_2' \end{pmatrix} = \begin{pmatrix} e & f \\ g & h \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} \omega_1 \\ \omega_2 \end{pmatrix}$$

and because ω_1, ω_2 are linearly independent,

$$\begin{pmatrix} e & f \\ g & h \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

from which we derive the result. □

10.2.4. *Let L be the lattice spanned by ω_1, ω_2 with associated functions g_2, g_3. If λ is a non-zero scalar, and L' is the lattice spanned by $\lambda\omega_1$, $\lambda\omega_2$, show that the corresponding elliptic curve for L' is*

$$y^2 = 4x^3 - \lambda^{-4} g_2 x - \lambda^{-6} g_3.$$

Since $g_2 = 60G_4$ and $g_3 = 140G_6$ where

$$G_k = \sum_{(m,n)\neq(0,0)} \frac{1}{(m\omega_1 + n\omega_2)^k},$$

by Exercise 10.1.12 we see that changing ω_1 by $\lambda\omega_1$ and changing ω_2 by $\lambda\omega_2$ alters the sum by a factor of λ^{-k}. □

10.2.6. *Show that*

$$2E_4(z)E_6'(z) - 3E_6(z)E_4'(z)$$

is a cusp form of weight 12 *for the full modular group.*

Let $f(z) = 2E_4(z)E_6'(z) - 3E_6(z)E_4'(z)$. Now

$$E_4\left(\frac{az+b}{cz+d}\right) = (cz+d)^4 E_4(z)$$

and by differentiating both sides we get

$$E_4'\left(\frac{az+b}{cz+d}\right)(cz+d)^{-2} = 4(cz+d)^3 cE_4(z) + (cz+d)^4 E_4'(z).$$

Similarly,

$$E_6'\left(\frac{az+b}{cz+d}\right)(cz+d)^{-2} = 6(cz+d)^5 cE_6(z) + (cz+d)^6 E_6'(z).$$

Thus,

$$\begin{aligned} f\left(\frac{az+b}{cz+d}\right) &= 2E_4\left(\frac{az+b}{cz+d}\right)\left[6c(cz+d)^7 E_6(z) + (cz+d)^8 E_6'(z)\right] \\ &\quad - 3E_6\left(\frac{az+b}{cz+d}\right)\left[4c(cz+d)^5 E_4(z) + (cz+d)^6 E_4'(z)\right] \\ &= (cz+d)^{12} f(z). \end{aligned}$$

To see that it is a cusp form, we check its behaviour at $i\infty$ and see that from the q-expansion that it vanishes, since $E_4'(z)$ and $E_6'(z)$ both vanish there. □

10.2.7. *Prove that* $E_4^2 E_6 = E_{14}$. *Deduce that*

$$\sum_{(m,n)\neq(0,0)} m/(m\rho+n)^{13} = 0,$$

where ρ *is a primitive cube root of unity.*

This is easily deduced from the dimension formula (Corollary 4.4.2) or from Exercise 4.6.7. Indeed, $\dim M_{14}(\mathrm{SL}_2(\mathbb{Z})) = 1$ and comparing the constant terms of $E_4^2 E_6$ and E_{14} gives the result. For the second part, we see that $E_{14}(z)$ has a double zero at $z = \rho$ and so the derivative also vanishes at $z = \rho$ from which the second assertion follows. □

10.2.8. *Show that*

$$j'(z) = -2\pi i \frac{E_{14}(z)}{\Delta(z)}.$$

Since $j(z) = E_4^3(z)/\Delta(z)$, we see that

$$-j'(z) = \frac{\Delta'(z)E_4^3(z) - 3\Delta(z)E_4^2(z)E_4'(z)}{\Delta(z)}.$$

But $1728\Delta(z) = E_4^3(z) - E_6^2(z)$ so that

$$1728\Delta'(z) = 3E_4^2(z)E_4'(z) - 2E_6(z)E_6'(z).$$

Hence

$$-1728\Delta^2 j' = E_4^2 E_6 \left(2E_4E_6' - 3E_6E_4'\right).$$

By Exercise 10.2.7, $E_4^2E_6 = E_{14}$. By Exercise 10.2.6, the term in the bracket is a cusp form of weight 12. By considering the q-expansion, this cusp form is easily seen to be $2\pi i(1728)\Delta$ from which the result follows. □

10.3 Theta series and lattices

10.3.1. *If $L = \omega_1\mathbb{Z} + \cdots \omega_n\mathbb{Z}$ and if $\omega_1^\star, \ldots, \omega_n^\star \in L^\star$ is the* dual basis, *that is $\omega_i^\star(\omega_j) = \delta_{ij}$, show that*

$$L^\star = \omega_1^\star\mathbb{Z} + \cdots \omega_n^\star\mathbb{Z}$$

First, consider an element of the form $c_1\omega_1^\star + \cdots + c_n\omega_n^\star$ where $c_i \in \mathbb{Z}$. If $d_1\omega_1 + \cdots d_n\omega_n \in L$ with $d_j \in \mathbb{Z}$, then by linearity we clearly have

$$\begin{aligned}(c_1\omega_1^\star + \cdots + c_n\omega_n^\star)(d_1\omega_1 + \cdots d_n\omega_n) &= \sum_{i=1}^n \sum_{j=1}^n c_i d_j \omega_i^\star(\omega_j) \\ &= \sum_{i=1}^n \sum_{j=1}^n c_i d_j \delta_{ij} \\ &= \sum_{i=1}^n c_i d_i \in \mathbb{Z}.\end{aligned}$$

Thus $L^\star \supseteq \omega_1^\star\mathbb{Z} + \cdots \omega_n^\star\mathbb{Z}$.

Now given any $f^\star \in L^\star$, it suffices to show that we the equality of functions

$$f^\star = \sum_{j=1}^n f^\star(\omega_j)\omega_j^\star.$$

since by the definition of the dual lattice, we know $f^\star(\omega_j) \in \mathbb{Z}$ for $j = 1, \ldots, n$. It suffices to check this for each basis element $\omega_1, \ldots, \omega_n$. Indeed, by linearity we have

$$\left(\sum_{j=1}^{n} f^\star(\omega_j)\omega_j^\star\right)(\omega_i) = \sum_{j=1}^{n} f^\star(\omega_j)\omega_j^\star(\omega_i) = \sum_{j=1}^{n} f^\star(\omega_j)\delta_{ij} = f^\star(\omega_i).$$

Thus, $f^\star = \sum_{j=1}^{n} f^\star(\omega_j)\omega_j^\star$ and so $L^\star \subseteq \omega_1^\star\mathbb{Z} + \cdots \omega_n^\star\mathbb{Z}$. □

10.3.3. *Prove that* $\mu(V/L) = \sqrt{\det A}$.

V/L is a fundamental domain for the lattice. By linear algebra (and what we have said earlier), there is a basis $e_1, \ldots, e_n$ of V which is orthogonal with respect to $(\cdot,\cdot)$. In other words, we can write

$$\omega_j = \sum_{j=1}^{n} b_{ij}e_j$$

so that if $B = (b_{ij})$, the Gram matrix A is equal to B^tB. Thus $(\det B)^2 = \det A$ and as $\det A > 0$, we have $\det B = \sqrt{\det A}$. At the same time, having fixed our own basis $e_1, \ldots, e_n$, the lattice spanned by $\omega_1, \ldots, \omega_n$ is that lattice spanned by the columns of B and the fundamental domain for this lattice is $\left\{\sum_{j=1}^{n} \lambda_j e_j : 0 \leq \lambda_j \leq 1\right\}$ which has measure $\mu(V/L)$. Since this measure is the volume of the parallelepiped spanned by the columns of B, we are done. □

10.3.4. *With A as above, show that the matrix A^{-1} is the Gram matrix of the dual lattice $L^\star$.*

In Exercise 10.3.1, we have already seen that $\omega_1^\star, \ldots, \omega_n^\star$ is a basis of $L^\star$. The Gram matrix of $L^\star$ is therefore $\left(\left(\omega_i^\star, \omega_j^\star\right)\right)$. At the same time, L is spanned by $e_1, \ldots, e_n$ (using the notation of the previous exercise), and so if we put $e_j^\star = A^{-1}e_j$, we see that

$$\begin{aligned}(e_j^\star, e_k) &= \left(A^{-1}e_j\right)^t Ae_k \\ &= e_j^t A^{-1} A e_k \\ &= e_j^t e_k \\ &= \delta_{jk}\end{aligned}$$

so that $e_1^\star, \ldots, e_n^\star$ is the dual basis also. But

$$(e_j^\star, e_k^\star) = \left(A^{-1}e_j\right)^t A \left(A^{-1}e_k\right) = e_j A^{-1} e_k$$

so that A^{-1} is the Gram matrix of $L^\star$. □

10.3.7. *If L is self-dual and even, show that $8 \mid n$.*

Recall that the full modular group $\mathrm{SL}_2(\mathbb{Z})$ is generated by $T = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ and $S = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ by Theorem 2.1.2. We first consider how $\theta_L(z)$ transforms under T. Since L is even, we have

$$\theta_L(z+1) = \sum_{\omega \in L} e^{\pi i (z+1)(\omega,\omega)} = \sum_{\omega \in L} e^{\pi i z (\omega,\omega)} = \theta_L(z),$$

for all $z \in \mathbb{H}$. The transformation under S is the content of Theorem 10.3.6. In particular, since $Si = i$, we have

$$\theta_L(i) = \mu(V/L)\theta_L(i),$$

so that $\mu(V/L) = 1$ as $\theta_L(i) \neq 0$.

The fact that $(ST)^3 z = z$ inspires the following sequence of calculations. Start with

$$\theta_L(STz) = \left(\frac{Tz}{i}\right)^{n/2} \theta_L(Tz) = \left(\frac{z+1}{i}\right)^{n/2} \theta_L(z).$$

Replace z with STz to obtain

$$\begin{aligned} \theta_L\left((ST)^2 z\right) &= \left(\frac{STz+1}{i}\right)^{n/2} \theta_L(STz) \\ &= \left(\frac{STz+1}{i}\right)^{n/2} \left(\frac{z+1}{i}\right)^{n/2} \theta_L(z). \end{aligned}$$

Replace z with STz again to obtain

$$\begin{aligned} \theta_L\left((ST)^3 z\right) &= \left(\frac{(ST)^2 z+1}{i}\right)^{n/2} \left(\frac{STz+1}{i}\right)^{n/2} \theta_L(STz) \\ &= \left(\frac{(ST)^2 z+1}{i}\right)^{n/2} \left(\frac{STz+1}{i}\right)^{n/2} \left(\frac{z+1}{i}\right)^{n/2} \theta_L(z). \end{aligned}$$

Since $(ST)^3 z = z$, we clearly have

$$1 = \left(\frac{(ST)^2 z+1}{i}\right)^{n/2} \left(\frac{STz+1}{i}\right)^{n/2} \left(\frac{z+1}{i}\right)^{n/2}.$$

Now set $z = i$ so that $STz = -1/(i+1)$ and $(ST)^2 z = i - 1$. Now the previous equation becomes $i^{n/2} = 1$ which implies that $8 \mid n$. □

10.3.8. *If L is self-dual show that the Gram matrix of L and $L^\star$ have determinant 1.*

Since $L = L^\star$, $\mu(V/L) = \mu(V/L^\star)$ so that $\det A = \det A^{-1}$ by Exercise 10.3.4. Hence $\det A = \pm 1$. Since A is positive definite, $\det A = 1$. □

10.3.9. *Show that the theta function of a self-dual, even lattice is a modular form of weight $n/2$ for the full modular group $\mathrm{SL}_2(\mathbb{Z})$.*

From the solution to Exercise 10.3.7, we have that $\mu(V/L) = 1$ and that $\theta_L(z+1) = \theta_L(z)$. Now since $8 \mid n$ by Exercise 10.3.7, the result follows immediately from Theorem 10.3.6,

$$\theta_L\left(-\frac{1}{z}\right) = z^{n/2}\theta_L(z),$$

since S and T generate $\mathrm{SL}_2(\mathbb{Z})$.

10.4 Special values of zeta and L-functions

10.4.1. *Prove that*

$$\psi_k(z+1) = \psi_k(z) + \frac{(-1)^k k!}{z^{k+1}}.$$

This is immediate from the series expansion of $\psi_k(z)$. □

10.4.2. *For every integer $k \geq 0$, prove that*

$$-\frac{d^k}{dz^k}(\pi \cot \pi z) = \psi_k(z) + (-1)^{k+1}\psi_k(-z) + (-1)^k \frac{k!}{z^{k+1}}$$

From the partial fraction expansion of the cotangent function (see (4.3) in Section 4.2), we have

$$\pi \cot \pi z = \frac{1}{z} + \sum_{n=1}^{\infty}\left(\frac{1}{z-n} + \frac{1}{z+n}\right).$$

The result is now easily deduced by taking successive derivatives. □

10.4.3. *Show that*

$$-\frac{d^k}{dz^k}(\pi \cot \pi z) = \psi_k(z) + (-1)^{k+1}\psi_k(1-z).$$

This is clear from Exercises 10.4.2 and 10.4.1. □

10.4.4. *Show that*

$$(k-1)!L(k,\chi) = (-N)^{-k}\sum_{a=1}^{N}\chi(a)\psi_{k-1}(a/N).$$

We may write

$$L(k,\chi)=\sum_{a=1}^{N}\chi(a)\sum_{n\equiv a(\mathrm{mod}\,N)}n^{-k}=\sum_{a=1}^{N}\chi(a)\sum_{j=0}^{\infty}(Nj+a)^{-k}$$

from which the desired result is apparent. □

10.4.5. *If χ is an even character (that is, $\chi(-1)=1$) and k is even, show that*

$$-2(k-1)!L(k,\chi)=(-N)^{-k}\sum_{a=1}^{N}\chi(a)\frac{d^k}{dz^k}(\pi\cot\pi z)\Big|_{z=a/N}.$$

Deduce that $L(k,\chi)$ is an algebraic multiple of π^k.

From the previous exercise, we can write

$$2(k-1)!L(k,\chi)=(-N)^{-k}\sum_{a=1}^{N}\chi(a)\psi_{k-1}(a/N)+\chi(N-a)\psi_{k-1}(1-a/N)$$

which is

$$=(-N)^{-k}\sum_{a=1}^{N}\chi(a)(\psi_{k-1}(a/N)+\psi_{k-1}(1-a/N).$$

By exercise 10.4.3, the result follows. The last assertion is immediate upon noting that

$$\frac{d^k}{dz^k}(\pi\cot\pi z)\Big|_{z=a/N}$$

is an algebraic multiple of π^k. □

10.4.6. *If χ is an odd character (that is, $\chi(-1)=-1$) and k is odd, show that*

$$-2(k-1)!L(k,\chi)=(-N)^{-k}\sum_{a=1}^{N}\chi(a)\frac{d^k}{dz^k}(\pi\cot\pi z)\Big|_{z=a/N}.$$

Deduce that $L(k,\chi)$ is an algebraic multiple of π^k.

As before we can write

$$2(k-1)!L(k,\chi)=(-N)^{-k}\sum_{a=1}^{N}\chi(a)\psi_{k-1}(a/N)+\chi(N-a)\psi_{k-1}(1-a/N)$$

which is now equal to (since χ is odd)

$$=(-N)^{-k}\sum_{a=1}^{N}\chi(a)(\psi_{k-1}(a/N)-\psi_{k-1}(1-a/N).$$

By exercise 10.4.3, the result follows. The last assertion is also immediate upon noting (as before) that

$$\frac{d^k}{dz^k}(\pi \cot \pi z)\Big|_{z=a/N}$$

is an algebraic multiple of π^k. □

10.5 Supplementary problems

10.5.1. *Let f be an elliptic function with period lattice L. Show that*

$$\sum_{z \in \mathbb{C}/L} \mathrm{ord}_z(f) z \in L$$

We apply Cauchy's residue formula to obtain that the above sum is

$$\frac{1}{2\pi i} \int_{\partial P} \frac{w f'(w)}{f(w)} dw.$$

This is a sum of four line integrals corresponding to the oriented (counterclockwise) edges of the fundamental parallelogram.

Assuming first that f has no zeros or poles on the boundary of P, we have four oriented integrals: from 0 to ω_1; from ω_1 to $\omega_1 + \omega_2$; from $\omega_1 + \omega_2$ to ω_2; and from ω_2 to 0.

In the second integral, we change variables from w to $w - \omega_1$. In the third integral, change from w to $w - \omega_2$. Using the periodicity of f'/f, this simplifies to

$$\sum_{z \in \mathbb{C}/L} \mathrm{ord}_z(f) z = \frac{\omega_2}{2\pi i} \int_0^{\omega_1} \frac{f'(w)}{f(w)} dw + \frac{\omega_1}{2\pi i} \int_0^{\omega_2} \frac{f'(w)}{f(w)} dw.$$

But for any meromorphic function $g(z)$, the integral

$$\frac{1}{2\pi i} \int_a^b \frac{g'(z)}{g(z)} dz$$

is the winding number around zero of the path

$$\gamma : [0, 1] \to \mathbb{C}$$

given by

$$\gamma(t) = g\Big((1-t)a + tb\Big)$$

and in particular if $g(a) = g(b)$, the integral is an integer. Since f is elliptic, we indeed have $f(\omega_1) = f(0) = f(\omega_2)$. This completes the proof in the case that f has no zero or pole on the boundary of P. If it does, we translate P appropriately and proceed as above. □

10.5.2. *With $\wp$ denoting the Weierstrass $\wp$-function as in Section 10.1, prove that*

$$\wp''(z) = 6\wp(z)^2 - g_2/2.$$

This is easily seen upon differentiating the differential equation satisfied by the Weierstrass $\wp$-function. □

10.5.3. *With G_k as in Exercise 10.1.12, prove that for $m \geq 4$,*

$$(2m+1)(m-3)(2m-1)G_{2m} = 3\sum_{j=1}^{m-2}(2j-1)(2m-2j-1)G_{2j}G_{2m-2j}.$$

The result is immediate upon using Exercise 10.5.2 and comparing coefficients of the Laurent expansions about $z = 0$ for the differential equation satisfied by $\wp''(z)$. □

10.5.4. *Show that for $k \geq 2$, each G_{2k} can be expressed as a polynomial in G_4 and G_6 with positive rational coefficients.*

This is immediate from the preceding exercise and mathematical induction. □

10.5.5. *Let $\vartheta(z)$ be given by the series*

$$\sum_{n=-\infty}^{\infty} e^{\pi i n^2 z}.$$

Show that

$$\vartheta(z) = \prod_{n=1}^{\infty}(1 - e^{2\pi i n z})(1 + e^{(2n-1)\pi i z})^2.$$

Deduce that $\vartheta(z)$ does not vanish in the upper half-plane.

By Jacobi's triple product identity (Theorem 1.2.1), we have upon setting $q = e^{\pi i z}$ and $x = 1$,

$$\vartheta(z) = \prod_{n=0}^{\infty}(1 - e^{2(n+1)\pi i z})(1 + e^{(2n+1)\pi i z})^2,$$

from which the result is immediate upon changing variables appropriately. The non-vanishing of the product shows that $\vartheta(z)$ does not vanish in the upper half-plane. □

10.5.6. *With η being the Dedekind η-function and ϑ as in the previous exercise, show that*

$$\vartheta(z) = \eta^2((z+1)/2)/\eta(z+1).$$

By Exercises 5.1.5 and 10.5.5, the result is immediate. □

10.5.7. *Show that if j is the modular invariant, then $j'(z)$ satisfies*

$$j'\left(\frac{az+b}{cz+d}\right) = (cz+d)^2 j'(z) \qquad \forall \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$$

but that $j'(z)$ is not a modular form of weight 2 for the full modular group.

By Exercise 4.6.9 (or by direct verification), the first assertion is clear. $j'(z)$ is not a modular form since it is not holomorphic at infinity. □

A short guide for further reading

The readers will benefit much from reading Hardy's lectures [13] and especially the chapter relating to Ramanujan's work on the τ-function. Clearly, this inspired Rankin's doctoral thesis and his subsequent work on the topic. A good introduction to the "q-world" discussed in Chapter 1 can be found in [7].

The gentlest and most elegant introduction to the topic of modular forms is the last section of Serre's Course in Arithmetic [36]. There the student will find the rudimentary notions needed to enter the topic in the case of the full modular group. Another reasonably gentle introduction is [2]. Perhaps the earliest books on the subject were Gunning [12] and Ogg [27]. Both of these books are of reasonable size that the graduate student will not have difficulty in studying the material in one semester. Zagier's chapter in [5] is also a short introduction to some of the modern themes.

For an amplified study from an analytic and classical perspective, Rankin's book [31] is an excellent source of material. Schoenberg's book [35] also gives a classical treatment that is accessible. A more amenable treatment is Iwaniec [16] which assumes a bit more sophistication with analytic number theory. Lang's monograph [20] blends some of the analytic and algebraic themes and explores the theory of congruences from the standpoint of Galois representations. Miyake's book [22] is also detailed and reasonably self-contained for the advanced student. Sarnak [34] highlights some analytic applications of the theory of modular forms but assumes the reader has a modest understanding of the subject. Koblitz's book [19] is also readable and has the advantage of giving a quick introduction to the topic of half-integral weight modular forms, a topic that was untouched in this book.

A geometric approach to the subject of modular forms is highlighted in [9] as well as the online notes of Milne [21]. For the study of Hilbert modular forms, we recommend Garrett [10] as well as Bruinier's chapter in [5]. For the topic of Siegel modular forms, a topic which we have not touched in this book, we suggest the reader begin with van der Geer's chapter [5]. Also, Siegel's classic lectures notes from the Tata Institute [39] are an excel-

© Springer Science+Business Media Singapore 2016 and Hindustan Book Agency 2016

M.R. Murty et al., *Problems in the Theory of Modular Forms*, HBA Lecture Notes in Mathematics, DOI 10.1007/978-981-10-2651-5

lent repository of a wealth of topics that span several areas of mathematics. Shimura's new book [38] is for the advanced reader and blends the theory of half-integral weight forms with integral weight forms and also gives a good introduction to the theory of Siegel modular forms. The serious reader can also profit from a careful study of Shimura's classic [37].

References

1. G.E. Andrews, A simple proof of Jacobi's triple product identity. *Proc. Amer. Math. Soc.,* **16** (1965), 333-334.
2. T. Apostol, Modular functions and Dirichlet series in number theory, Springer-Verlag, 1976.
3. R. Askey, Ramanujan's extensions of the Gamma and Beta functions. *Amer. Math. Monthly,* **87** (1980), 346-359.
4. B. C. Berndt. Ramanujan's congruences for the partition function modulo 5, 7, and 11. *Int. J. Number Theory,* **3** (2007), 349–354.
5. J. H. Bruinier, G. van der Geer, G. Harder, and D. Zagier, 1-2-3 of Modular Forms, Springer, 2008.
6. J. H. Bruinier, K. Ono, and R. C. Rhoades. Differential operators for harmonic weak Maass forms and the vanishing of Hecke eigenvalues. *Math. Ann.*, 342(3):673–693, 2008.
7. Hei-Chi Chan, An invitation to q-series, World Scientific, 2011.
8. P. Deligne, La conjecture de Weil: I, *Publications Mathématiques de l'IHES,* **43** (1974), 273-307.
9. F. Diamond and J. Shurman. A first course in modular forms, Volume 228, *Graduate Texts in Mathematics*, Springer-Verlag, New York, 2005.
10. P. Garrett, Holomorphic Hilbert modular forms, Wadsworth and Brooks/Cole Mathematics Series, Chapman and Hall, CRC, 1989.
11. E. Grosswald, Representations of integers as sums of squares, Springer-Verlag, New York, 1985.
12. R. Gunning, Lectures on modular forms, Princeton University Press, 1962.
13. G. H. Hardy. Ramanujan. Twelve lectures on subjects suggested by his life and work, Cambridge University Press, Cambridge, England, 1940.
14. E. Hecke, Analytsche Arithmetik der Positiven Quadratischen Formen in Mathematische Werke, pp. 789-918.
15. M. D. Hirschhorn, A simple proof of Jacobi's two square theorem, *Amer. Math. Monthly,* **92** (1985), 579-80.
16. H. Iwaniec, Topics in classical automorphic forms, Graduate Studies in Mathematics, Volume 17, American Mathematical Society,1997.
17. H. Iwaniec, Spectral methods in automorphic forms, Graduate Studies in Mathematics, Volume 53, American Mathematical Society, 2002.
18. L. Kilford, Modular forms: a classical and computational introduction, Imperial College Press, 2008.
19. N. Koblitz, Introduction to elliptic curves and modular forms, Springer, 1984.
20. S. Lang, Introduction to modular forms, Springer-Verlag, 1976.
21. J. Milne, Modular functions and modular forms, online course notes available at http://www.jmilne.org/math/CourseNotes/mf.html.
22. T. Miyake, Modular forms, Springer-Verlag, Berlin, New York, 1989, 335 pp.
23. L.J. Mordell, On Mr. Ramanujan's empirical expansions of modular functions, *Proceedings of the Cambridge Philosophical Society,* **19** (1917), 117-124.

24. M. R. Murty. Problems in analytic number theory, Volume 206, *Graduate Texts in Mathematics*, Springer, New York, second edition, 2008. Readings in Mathematics.
25. M.R. Murty and J. Esmonde, Problems in algebraic number theory, Volume 190, Graduate Texts in Mathematics, 2nd edition, Springer, 2005.
26. V. Kumar Murty, Ramanujan and Harish-Chandra, *Math. Intelligencer,* **15** (1993), no. 2, 33-39.
27. A.P. Ogg, Modular forms and Dirichlet series, Benjamin, 1969.
28. H. Petersson. Über die Entwicklungskoeffizienten der automorphen Formen. *Acta Math.,* **58** (1932), 169–215.
29. S. Ramanujan, On certain arithmetical functions, *Trans. Cambridge Philos. Soc.,* **22** (1916), 159-184.
30. R. A. Rankin, The modular group and its subgroups, The Ramanujan Institute, Madras, 1969.
31. R.A. Rankin, Modular forms, Ellis Horwood, 1984.
32. M. Razar, Modular forms for $G_0(N)$ and Dirichlet series, *Trans. Amer. Math. Soc.,* **231** (1977), 489-495.
33. W. Rudin. Real and complex analysis, McGraw-Hill Book Co., New York, third edition, 1987.
34. P. Sarnak, Some applications of modular forms, Cambridge University Press, 1990.
35. B. Schoenberg, Elliptic modular functions, Springer 1974.
36. J.-P. Serre, A course in arithmetic, Springer, 1970.
37. G. Shimura, Introduction to the Arithmetic Theory of Automorphic Functions, Princeton University Press, 1971.
38. G. Shimura, Modular forms and beyond, Springer 2011.
39. C.L. Siegel, Advanced Analytic Number Theory, Tata Institute of Fundamental Research, Bombay, 1980.
40. J. Silverman, The arithmetic of elliptic curves, Springer 1986.
41. J. Silverman, Advanced topics in the arithmetic of elliptic curves, Springer 1994.
42. M. Spivak, Calculus on manifolds, A modern approach to classical theorems of advanced calculus, W. A. Benjamin, Inc., New York-Amsterdam, 1965.
43. W. Stein, Modular forms, a computational approach, Graduate Studies in Mathematics, Volume 79, American Mathematical Society,2007.
44. B. Sury, The Congruence Subgroup Problem, Vol. 24, Texts and Readings in Mathematics, Hindustan Book Agency, India, 2003.
45. H. P. F. Swinnerton-Dyer. Congruence properties of $\tau(n)$. in Ramanujan revisited (Urbana-Champaign, Ill., 1987), pages 289–311. Academic Press, Boston, MA, 1988.
46. A. Weil, Über die Bestimmung Dirichletscher Reihen durch Funkionalgleichungen, *Math. Annalen,* **168** (1967), 149-156.
47. X. Wang and D. Pei, Modular forms with integral and half-integral weights, Springer 2012.

Index

$G \setminus X$, 19
G-equivalent, 19
G_2, 56
G_x, 19
$HatF$, 114
$M_k(\Gamma)$, 75
$M_k(\Gamma_0(N), \chi)$, 99
$S_k(\Gamma)$, 75
$T_n(f)$, 98
X_n, 95
$X_n(N, S^\times, S^+)$, 95
δ_{ij}, 143
η, 58
θ_k, 58
$\Gamma(N)$, 16
$\Gamma_0(N)$, 18
Γ_z, 79
Γ_∞, 27
$f \mid \gamma(z)$, 36
$v_{z_0}(f)$, 41

Abel's lemma, 106, 246
abscissa of convergence, 109
action, 19–21
addition formula, 139
analytic automorphism, 26
analytic continuation, 66
arithmetical function, 70, 111, 211, 249
Atkin, xiv
Atkin-Lehner involution, 103

basis of eigenforms, 235
Bass, 18
Berndt, 68
Bernoulli numbers, 40, 187, 192
Bernoulli polynomial, 40, 188
Bruinier, 69

character, *see* Dirichlet character
Chinese remainder theorem, 22
completely multiplicative, 111
completely multiplicative function, 250
conductor of a character, 118
congruence subgroup, 17, 86
converse theory, 131
cotangent expansion, 55
cusp, 177
cusp form, 75
  on Γ, 75
  on full modular group, 36
cusps, 30

Dedekind, xiii
Dedekind η-function, 58, 81, 150, 200, 227, 283
Dedekind zeta function, 149
Deligne, xiii, 64
digamma function, 147
dimension formula, 47, 79, 190, 224
Dirichlet, 105
Dirichlet L-series, 147
  functional equation, 260
Dirichlet character, 99, 240
  conductor, 118
  even, 118
  imprimitive, 118
  odd, 118
  primitive, 118
  trivial, 118
Dirichlet convolution, 111, 249
Dirichlet series, 64, 105, 205
discrete subgroup, 27
double cosets, 21, 95, 169
doubly periodic, 135
dual basis, 144, 276

© Springer Science+Business Media Singapore 2016 and Hindustan Book Agency 2016

M.R. Murty et al., *Problems in the Theory of Modular Forms*, HBA Lecture Notes in Mathematics, DOI 10.1007/978-981-10-2651-5

dual lattice, 144
dual vector space, 144

Eisenstein, xiii
Eisenstein series, 54, 187
  higher level, 73, 76, 218
  on full modular group, 37
Eisenstein series E_2, 53, 199
elliptic curve, 139
elliptic function, 68, 135
elliptic functions, 269
elliptic point, 79
  order of, 79
Euler product, 246
Euler's pentagonal number theorem, 6, 155
even character, 148, 280
even lattice, 146
extended upper half-plane, 73
extended upper-half-plane, 177

faithful action, 25
Fermat's last theorem, xiv
field of elliptic functions, 138
fixed point, 19
Fourier expansion, 74
Fourier inversion formula, 114
Fourier transform, 114, 145
fractional linear transformations, 25
Frey, xiv
Fricke, 17, 22, 172
full modular group, 161
functional equation, 66, 114, 256
fundamental domain, 27, 176
fundamental parallelogram, 135, 269

g-multiplicative, 70, 211
Galois representations, xiv
Gamma function, 254
Gauss sum, 133, 257, 265
general Dirichlet series, 245
Gram matrix, 145, 277
Grothendieck, xiv

half-plane of absolute convergence, 109
half-plane of convergence, 109
Hardy, 64
Harish-Chandra, xiv
harmonic weak Maass forms, 69
Hausdorff space, 30
Hecke, xiv, 64, 66, 128
Hecke eigenforms, 92
Hecke eigenvalues, 91
Hecke operator
  general level, 96
  level one, 61
Hecke operators, 60, 63, 87, 90, 203, 234, 235
  higher level, 239
Hecke subgroups, 18, 167
Hecke's trick, 149
Hellegouarch, xiv
Hermitian, 90
Hermitian operators, 235
Hilbert modular form, 149
Hilbert modular group, 148
Hurwitz, xiii, 54
hyperbolic metric, 180

imprimitive Dirichlet character, 118
invariant measure, 144
irregular cusps, 75

j-function, 48, 141, 191, 214
Jacobi, xiii, 57
Jacobi triple product identity, 5, 7, 155
Jacobi's four square theorem, 9, 81, 83, 156, 227
Jacobi's two-square theorem, 7, 156

Khare, xiv
Klingen, 148
Kronecker, xiii
Kronecker delta, 143

L-series
  twist, 129, 131
Landau, 110
Langlands, xiv
lattice, 135, 144
  dual, 144
  even, 146
  self-dual, 146
lattices, 276
Lazard, 18
Legendre symbol, 117, 256
Lehmer's conjecture, 69
Lehner, xiv
level, 17
Lipschitz formula, 39, 77

Möbius function, 70, 212
Möbius inversion formula, 70, 212
Möbius transformations, 25
Maass, xiv
Miller basis, 93, 237, 238
Milnor, 18
modular form

on Γ, 75
on the full modular group, 36
of half-integral weight, 118
twist of, 129
modular function, 49
modular group, 15, 161
modular invariant, 71, 140, 214, 273, 283
moduli space, 143
Monster group, 48
Mordell, xiii
multiplicative, 106
completely, 111
multiplicity one, 103

Nebentypus, 99, 240
Nesterenko, 59
newforms, 102, 242
non-congruence subgroup, 22, 171, 173

odd character, 148, 280
oldforms, 102, 242
Ono, 69
orbit, 19
orbit-stabilizer formula, 20, 168
order
of f at $i\infty$, 36
of an elliptic point, 79
order at $i\gamma(i\infty)$, 75
order of a zero, 41

partial summation, 246
partition function, 67
Petersson, 48
Petersson inner product, 85, 231
Phragmén-Lindelöf theorem, 126
Pick, 17
Poincaré metric, 32, 180
Poisson summation formula, 112, 115, 251, 266
Poisson's summation formula for lattices, 145
polygamma function, 147
primitive Dirichlet character, 118
principal congruence subgroup, 16

q-expansion, 35
q-exponential, 3, 153
q-logarithm, 14, 159
q-series, 185
quasimodular form, 56
quotient space, 20

Ramanujan, xiii, 10, 58, 60, 64
Ramanujan τ-function, 53, 70, 199
Ramanujan congruences, 67, 207
Ramanujan conjectures, 53, 63, 65, 205
Ramanujan differential equations, 58
rapidly decreasing, 145
Rhoades, 69
Riemann, xiii, 105, 116
Riemann zeta function, 106, 109, 116, 248, 254
Riemann-Roch theorem, 79
Riesz representation theorem, 144

Schwartz function, 267
Schwartz space, 114
Schwarz lemma, 26
self-dual lattice, 146, 277
Serre, xiv, 18
Serre derivative, 58, 201
Siegel, 148
Siegel-Klingen theorem, 148
slash notation, 36
special values of zeta functions, 147, 279
stabilizer subgroup, 19
Stirling's formula, 126
Sturm's bound, 80, 226
Swinnerton-Dyer, 67

Taniyama's conjecture, xiv
theta function, 282
theta series, 144, 276
topological group, 20
totally real number field, 148
transitive action, 25
trivial character, 118
twist of L-series, 129, 131
twist of modular form, 129

uniformization theorem, 142, 272
upper half-plane, 25, 175

valence formula, 42, 50, 79, 189, 224

Weierstrass $\wp$-function, 137, 150, 272, 282
addition formula, 139
Weil, xiv
Weil conjectures, xiii, 64
Weil's theorem, 132
width of a cusp, 75
Wiles, xiv
Wilton, 64
Wintenberger, xiv
Wohlfahrt, 22, 172